Process Simulation Fundamentals

Dynamic Modeling using UniSim® Design

PDS-4528

4528.R440.01

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Module Index

1. Getting Started in Steady State

4528.01.R440.01

Workshop

The Getting Started in Steady State module introduces you to some of the basic concepts necessary for creating simulations in UniSim Design. Some of the things you will learn from this module are:

- Methods for moving through the different environments
- Selecting property packages and components
- Gaining familiarity with the property views of UniSim Design
- Adding streams
- Adding operations

You will use UniSim Design to define streams and unit operations to develop a flowsheet for an offshore platform.

Learning Objectives

After you have completed this section, you will be able to:

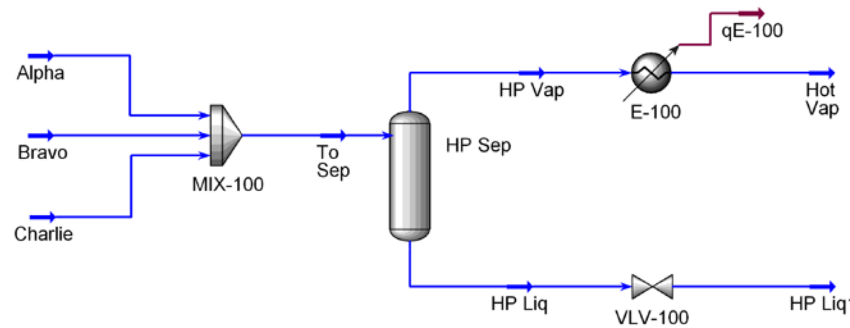
- Define a fluid package (property package, components)
- Add streams
- Add operations

For those who are familiar with UniSim Design, the steady state simulation can be built with the information listed on the following PFD. Step by step instructions follow the PFD.

Process Overview

For PFD Table				
		Alpha	Bravo	Charlie
Temperature	C	88.00	80.00	49.00
Pressure	kPa	6480	6650	6620
Molar Flow	kgmole/h	1.090e+004	9980	1.180e+004
Master Comp Mole Frac (Nitrogen)		0.0150	0.0050	0.0170
Master Comp Mole Frac (CO2)		0.0230	0.0100	0.0260
Master Comp Mole Frac (Methane)		0.7590	0.7190	0.7300
Master Comp Mole Frac (Ethane)		0.0660	0.0620	0.0740
Master Comp Mole Frac (Propane)		0.0340	0.0410	0.0390
Master Comp Mole Frac (i-Butane)		0.0060	0.0100	0.0060
Master Comp Mole Frac (n-Butane)		0.0130	0.0260	0.0140
Master Comp Mole Frac (i-Pentane)		0.0040	0.0100	0.0040
Master Comp Mole Frac (n-Pentane)		0.0050	0.0190	0.0060
Master Comp Mole Frac (n-Hexane)		0.0060	0.0100	0.0070
Master Comp Mole Frac (H2O)		0.0380	0.0400	0.0430
Master Comp Mole Frac (NBP200*)		0.0070	0.0100	0.0080
Master Comp Mole Frac (NBP280*)		0.0140	0.0210	0.0160
Master Comp Mole Frac (NBP425*)		0.0060	0.0100	0.0060
Master Comp Mole Frac (NBP630*)		0.0030	0.0050	0.0030
Master Comp Mole Frac (NBP920*)		0.0010	0.0020	0.0010

E-100		
Product Temperature	27.00	C
Pressure Drop	70.00	kPa



VLV-100		
Product Pressure	2690	kPa

Building a Steady State Simulation

Simulation Basis Manager

UniSim Design uses the concept of the fluid package to contain all necessary information for performing flash and physical property calculations. This approach allows you to define all information (property package, components, hypothetical components, tabular data, interaction parameters, reactions, etc.) inside a single entity.

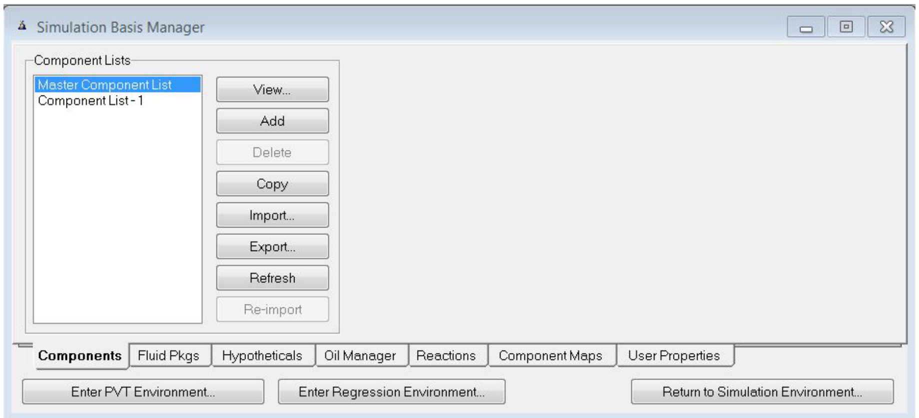
There are four key advantages to this approach:

- All associated information is defined in a single location, allowing for easy creation and modification of the data.
- Fluid packages can be saved as completely defined entities for use in any simulation.
- Component lists can be saved separately from the Fluid Packages as completely defined entities for use in any simulation.
- Multiple Fluid Packages can be used in the same simulation, however, are all defined inside the common Basis Manager.

The Simulation Basis Manager is a property view that allows you to create and manipulate multiple fluid packages or component lists in the simulation. The opening tab of the Simulation Basis Manager allows for the creation of component lists which are independent from fluid packages but can be associated with the individual fluid packages in the case.

The first tab of the Basis Manager allows you to manage the component list(s) used in your case. There are a number of buttons available:

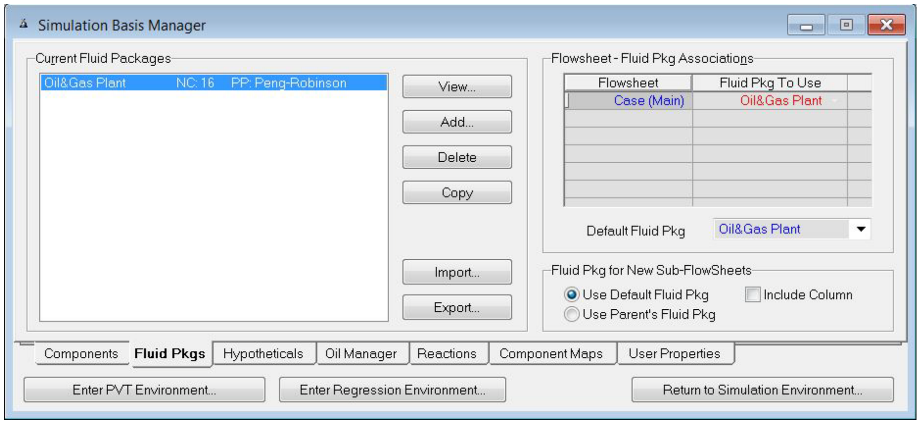
Figure 1



Button	Description
View	Allows you to access the property view for the selected Component List.
Add	Allows you to create a Component List. Note: Component Lists can be added via the Fluid Package property view.
Delete	Removes the selected Component List from the simulation.
Copy	Makes a copy of the selected Component List.
Import	Allows you to import a predefined Component List from disk. Component Lists have the file extension (.cml).
Export	Allows you to export the selected Component List to disk. The exported Component List can be retrieved into another case, by using the Import function.
Refresh	Updates the pure component properties from the database.
Re-import	Allows you to re-import a cml file.

The **Fluid Pkgs** tab allows you to access the fluid packages/flowsheet associations list as well as the fluid package definition. UniSim Design allows the user to use multiple fluid packages within a single simulation by associating the fluid packages with various flowsheets and linking the flowsheets together. In addition, UniSim Design allows the user to incorporate multiple fluid packages into a single flowsheet (by using a **Stream Cutter** unit operation).

Figure 2



Button	Description
View	This is only active when a fluid package exists in the case. It allows you to view the property view for the selected fluid package.
Add	Allows you to create and install a fluid package in the simulation.
Delete	Removes the selected Fluid Package from the simulation.
Copy	Makes a copy of the selected fluid package. Everything is identical in the copied version except the name. This is useful for modifying fluid packages.
Import	Allows you to import a predefined fluid package from disk. Fluid packages have the file extension (.fpk).
Export	Allows you to export the selected fluid package to a disk. The exported fluid package can be retrieved into another case by using the Import function.

Basis Environment icon

You can use the **CTRL B** hot key to re-enter the Simulation Basis Manager from any point in the simulation or click the Basis Environment icon from the tool bar. (In the Simulation Basis Manager, this is the **Home View** button.)

Defining the Simulation Basis

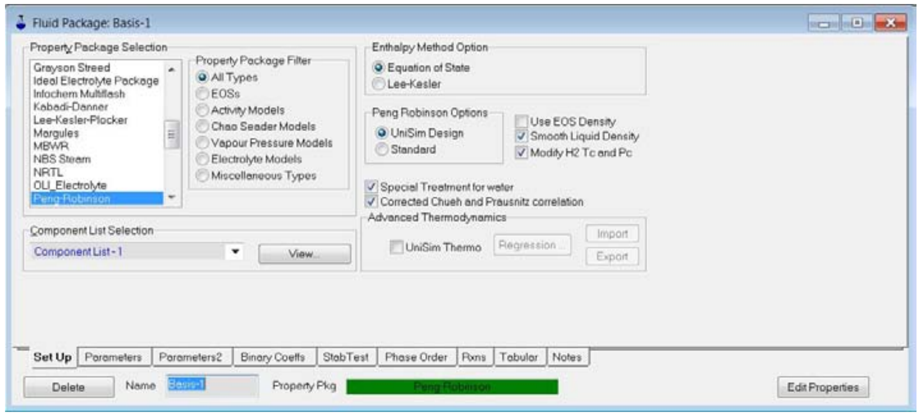
Add a Property Package

New Case icon

You can select the "**EOSs**" radio button in the Filter box, and then select Peng-Robinson; however, it is not necessary to select EOS after you have selected Peng-Robinson.

1. Start a new case by pressing the **New Case** button.
2. Go to the **Fluid Pkgs** tab and create a fluid package by clicking the **Add** button.
3. Scroll down the list and select the Peng-Robinson Equation of State model.

Figure 3



4. Change the **Name** from the default Basis-1 to **Oil&Gas Plant** using the Name input box at the bottom of the window.
5. Click the **View** button in the **Component List Selection** section of the **Set Up** tab. This will allow you to add components to the Component List that is now associated with the Oil&Gas Plant fluid package.

Add Components

You can select components for your simulation using several different methods.

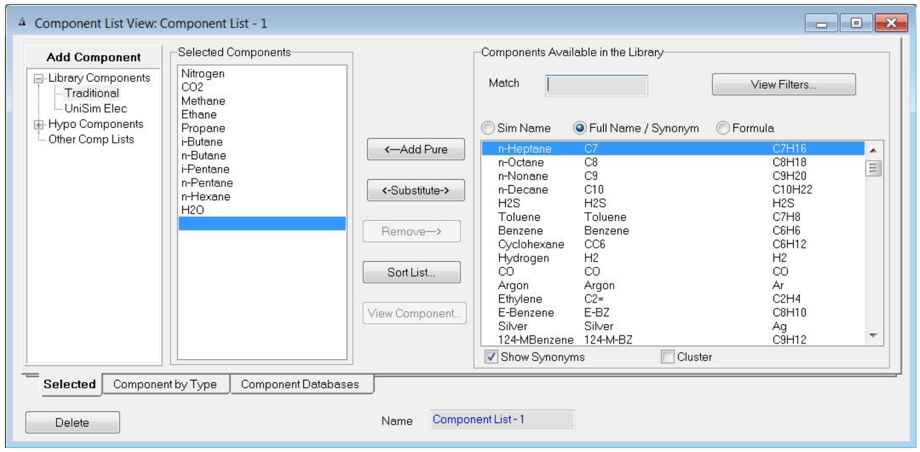
You can add a range of components by highlighting the entire range and clicking the **Add Pure** button.

To Use…	Do This…
Match Cell	1. Select one of the three name formats, SimName, Full Name/Synonym, or Formula by selecting the corresponding radio button. 2. Click on the **Match Cell** and enter the name of the component. As you start to type, the list will change to match what you have entered. 3. Once the desired component is highlighted, either: • Press the ENTER key. • Click the **Add Pure** button. • Double-click on the component to add it to your simulation. 4. You can add a range of components by highlighting the entire range and clicking the **Add Pure** button.
Component List	1. Using the scroll bar for the main component list, scroll through the list until you find the desired component. 2. To add the component, either: • Press the ENTER key. • Click the **Add Pure** button. 3. Double-click on the component to add it to your simulation.
Family Filter	1. Ensure the Match cell is empty, and click the **View Filters** button. 2. Check the Use Filter checkbox to display the various family filters. 3. Select the desired family (e.g. Hydrocarbons) from the list of Family Filters to display only that type of component. 4. Use either of the two previous methods to then select the desired component.

6. Select the library components:

- Nitrogen
- CO2
- Methane
- Ethane
- Propane
- i-Butane
- n-Butane
- i-Pentane
- n-Pentane
- n-Hexane
- H2O

Figure 4



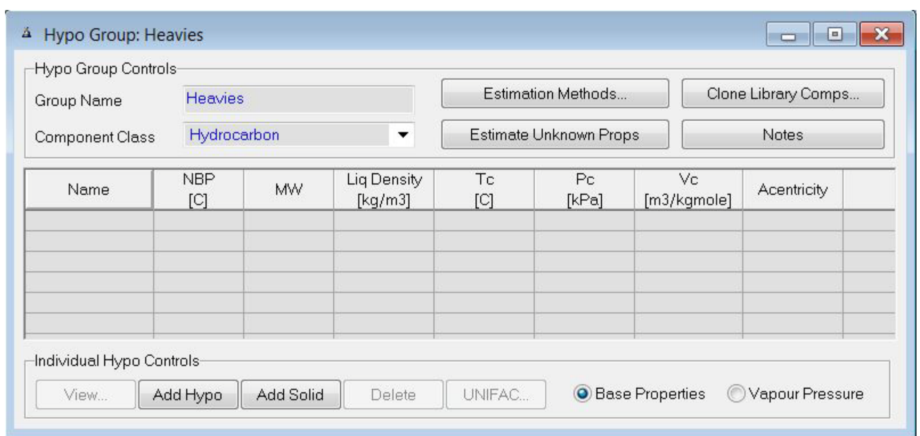
7. Select the Hypothetical branch in the Add Component tree to add hypothetical components to the Fluid Package.

8. Click on the **Hypo Manager** button.

9. Click the **Add** button to create a group of hypothetical components and rename the group to **Heavies**.

When you click the **Quick Create a Hypo Component** button, UniSim Design adds a hydrocarbon class hypo by default. If you want to add a hypo from another class, click the **Hypo Manager** button and then in the view that appears, click the **View Group** button. This will open the Tabular Hypothetical Input, where you can add non-hydrocarbon class hypothetical components.

Figure 5



A hypothetical component can be used to model non-library components, mixtures, or solids. You will be using hypothetical components to model five components in the gas mixture.

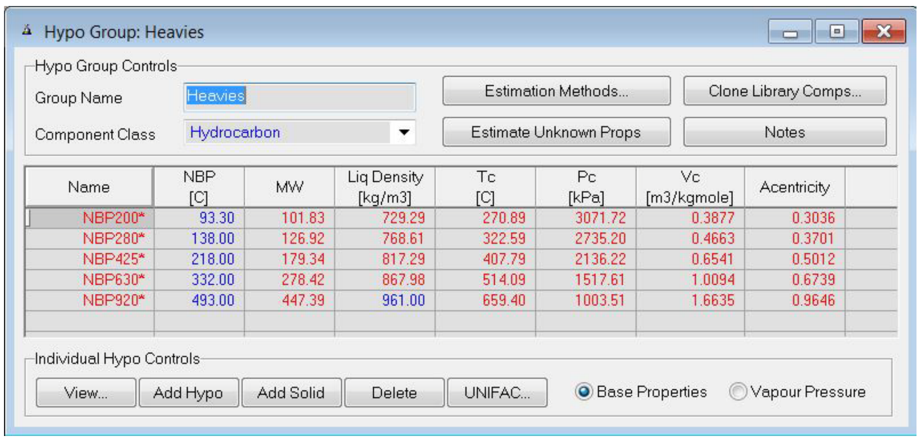
10. Click the **Add Hypo** button five times in order to add five hypothetical components. Use the following table to add Name, NBP and Liq Density information.

Name	NBP	Liq Density
NBP200	93.3 °C (200 °F)	<empty>
NBP280	138 °C (280 °F)	<empty>
NBP425	218 °C (425 °F)	<empty>
NBP630	332 °C (630 °F)	<empty>
NBP920	493 °C (920 °F)	961 kg/m3 (60 lb/ft3)

11. Select the **Estimate Unknown Props** button to estimate all of the unknown properties and close the window.

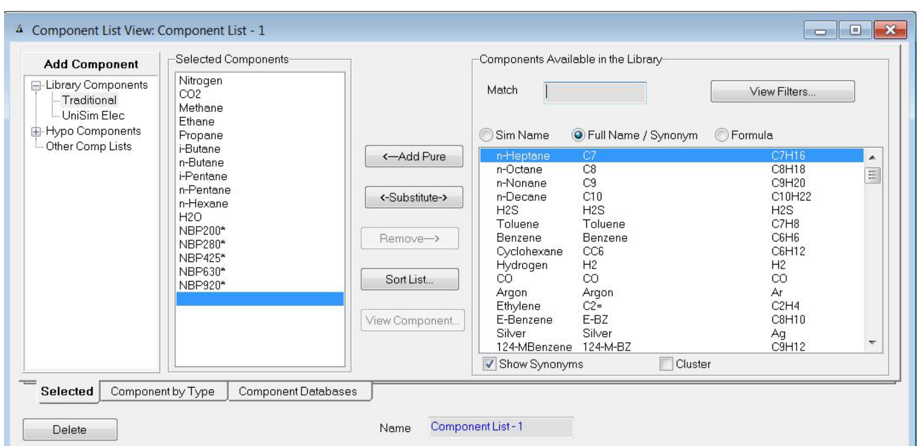
The minimum information required to define a hypo is the NBP or the density and MW.

Figure 6



12. On the component list view, **Selected** tab, click the **Add Group** button in the **Add Component** tree and add the hypo components to the Selected Components list.

Figure 7



13. You have now finished defining the fluid package. If desired, you can view the Peng-Robinson binary coefficients for your selected components by selecting the **Binary Coeffs** tab.
14. Click the **Enter Simulation Environment** button.

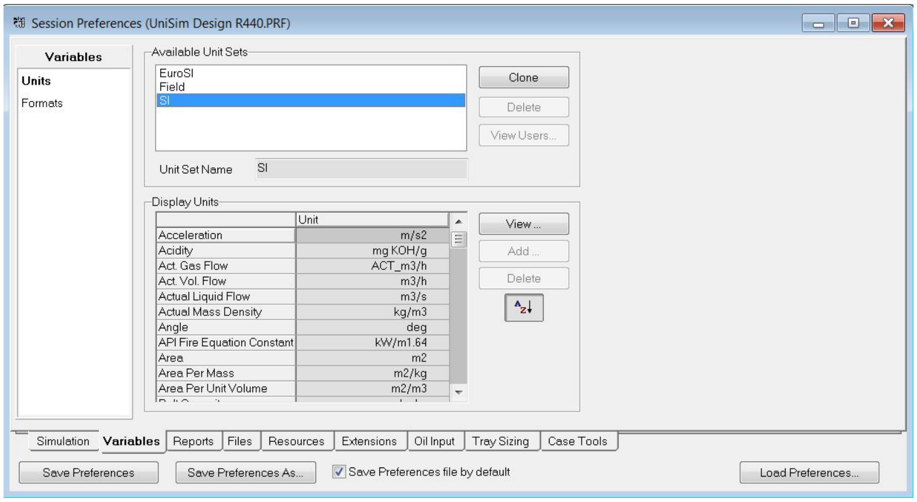
Building the Simulation

Selecting a Unit Set

In UniSim Design, it is possible to change the unit set used to display the different variables.

1. From the **Tools** menu, choose **Preferences**.
2. Click on the **Variables** tab.

Figure 8



3. Select the **SI** unit set.
4. Close this view to return to the simulation.

Adding Streams

In UniSim Design there are two types of streams, Material and Energy. Material streams have a composition and parameters such as temperature, pressure and flow rates. They are used to represent process streams. Energy streams have only one parameter: heat flow. They are used to represent the duty supplied to or by a unit operation.

There are a variety of ways to add Material streams in UniSim Design.

To Use...	Do This...
Menu Bar	• Select **Add Stream** from the **Flowsheet** menu. or • Press the **F11** Hot Key. The Stream property view opens.
Workbook	Open the Workbook and go to the **Material Streams** tab. Type a stream name into the ****New**** cell.
Object Palette	From the **Flowsheet** menu, select **Open Object Palette** or press **F4** to open the Object Palette. • Double-click on the **Material Stream** icon. or • Single-click on the **Material Stream** icon, then single click on the flowsheet where you want the stream to appear.

In this module, you will add three streams to represent three different gas wells. Each stream will be added using a different method of installation.

Adding a Stream from the Menu Bar

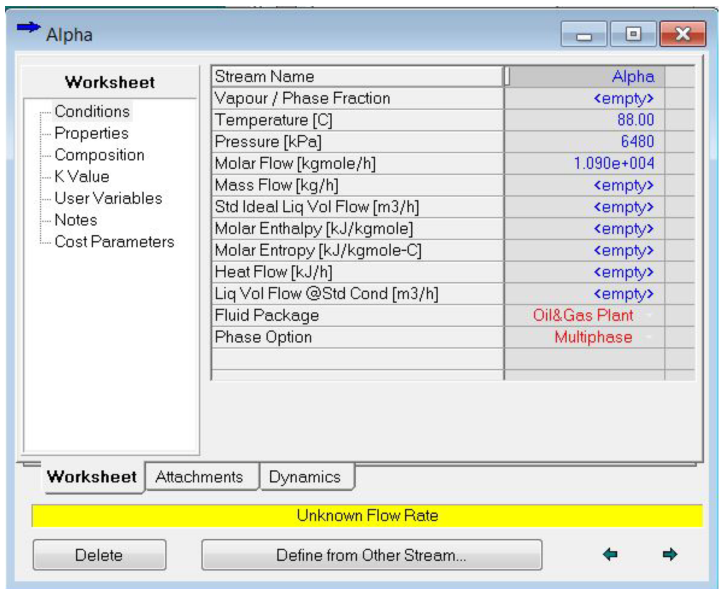
To add a stream using the **F11** hot key:

1. Press **F11**. The stream property view appears. If the stream property view is not displayed, double-click on the newly created stream (from the PFD) to bring up the property view.
2. Highlight the Stream Name cell and change the stream name to **Alpha**.

3. Enter the following information on the conditions page.

In this cell...	Enter...
Worksheet	
Temperature	88 °C (190 °F)
Pressure	6480 kPa (940 psia)
Flow Rate	10900 kgmole/h (24030 lbmole/hr)

Figure 9



Entering Stream Compositions

There are two different pages for entering stream composition:

To Use...	Do This...
Conditions Page	• Double-click on the Molar Flow cell to enter mole fractions. or • Double-click on the Mass Flow cell to enter mass fractions. or • Double-click on the Std Ideal Liq Vol Flow cell to enter volume fractions. The **Input Composition for Stream** view appears.
Composition Page	Click the **Edit** button. The **Input Composition for Stream** view appears.

1. If the Input Composition for Stream view is not already open, double-click on the Mole Flow cell to enter the mole fractions.
2. Enter the following compositions:

UniSim Design automatically normalizes the compositions once you click the **OK** button.

Ensure that you have entered the right composition values before you click **OK**.

If there are <empty> values, either enter 0 or click the **Normalize** button.

Component	Mole Fraction
Nitrogen	0.0150
CO2	0.0230
Methane	0.7590
Ethane	0.0660
Propane	0.0340
i-Butane	0.0060
n-Butane	0.0130
i-Pentane	0.0040
n-Pentane	0.0050
n-Hexane	0.0060
H2O	0.0380
NBP200*	0.0070
NBP280*	0.0140
NBP425*	0.0060
NBP630*	0.0030
NBP920*	0.0010

3. Click the **OK** button to close the Input Composition for Stream view.

Adding a Stream from the Workbook

Workbook Icon

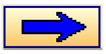
To open or display the Workbook, press the **Workbook** button on the button bar.

1. Enter the stream name, **Bravo** in the ****New**** cell.
2. Supply the following information and composition:

In this cell...	Enter...
Temperature	80 °C (176 °F)
Pressure	6650 kPa (964.5 psia)
Flow Rate	9980 kgmole/h (22000 lbmole/hr)
Component	**Mole Fraction**
Nitrogen	0.0050
CO2	0.0100
Methane	0.7190
Ethane	0.0620
Propane	0.0410
i-Butane	0.0100
n-Butane	0.0260
i-Pentane	0.0100
n-Pentane	0.0190
n-Hexane	0.0100
H2O	0.0400
NBP200*	0.0100
NBP280*	0.0210
NBP425*	0.0100
NBP630*	0.0050
NBP920*	0.0020

3. Close the stream property view.

Adding a Stream from the Object Palette



Material Stream Icon

4. If the Object Palette is not open, press the **F4** hot key to open it.
5. Double-click on the Material Stream icon, (the blue arrow on the Object Palette). The Stream property view displays.
6. Change the name of the stream to **Charlie** and supply the following information and composition:

In this cell...	Enter...
Workbook	
Temperature	49 °C (120 °F)
Pressure	6620 kPa (960 psia)
Flow Rate	11800 kgmole/h (26010 lbmole/hr)
Component	**Mole Fraction**
Nitrogen	0.0170
CO2	0.0260
Methane	0.7300
Ethane	0.0740
Propane	0.0390
i-Butane	0.0060
n-Butane	0.0140
i-Pentane	0.0040
n-Pentane	0.0060
n-Hexane	0.0070
H2O	0.0430
NBP200*	0.0080
NBP280*	0.0160
NBP425*	0.0060
NBP630*	0.0030
NBP920*	0.0010

7. Close the Stream property view.

Saving Your Case

You can use one of several different methods to save a case in UniSim Design:

- From the **File** menu, select **Save** to save your case with the same name.
- From the **File** menu, select **Save As** to save your case in a different location or with a different name.
- Click the **Save** icon on the toolbar to save your case with the same name.

Save your case as **4528.01.FS.usc**.

Save your case!

Adding Operations

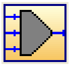
As with streams, there are various ways to install Unit Operations in UniSim Design. Some of these methods immediately open the operation property view, others do not.

The initial flowsheet for the platform will be built in Steady State mode. Generally, all of the necessary information required to specify the unit operation for steady state purposes is on the **Design** and on **Parameters** tabs. Once all the necessary information has been input, the status bar will display **OK**, and the color of the status bar will change to green.

To Use This...	Do This...
Menu Bar	• Open the **Flowsheet** menu bar and select **Add Operation**. or • Press the F12 hot key.
Workbook	From the Unit Ops page of the Workbook click the **Add Unit Op** button.
Object Palette	• Double-click on the **Operation** icon and the property view will appear. or • Single-click on the **Operation** icon, then single click on the flowsheet where you want the operation to appear.

Add the Operations

By choosing one of the three methods listed previously, add the following operations:



Mixer Icon

Add a Mixer

1. Add a mixer and select **Alpha, Bravo** and **Charlie** as the inlet streams and **To Sep** as the product stream.

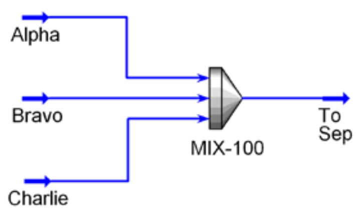
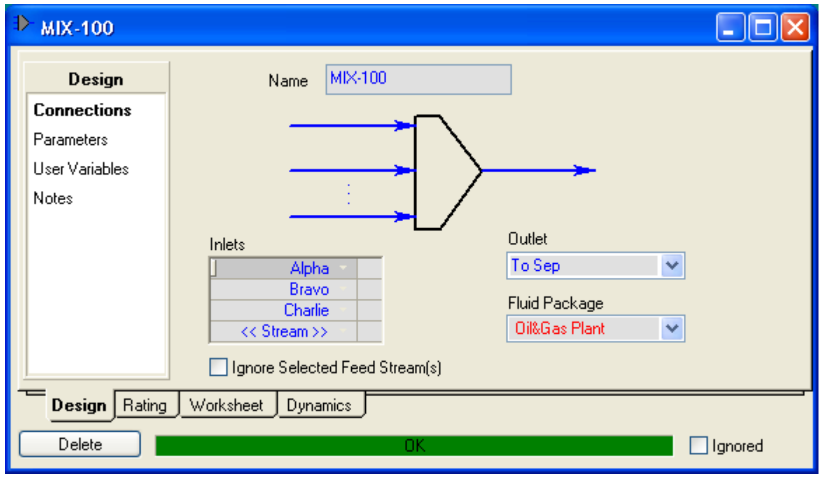
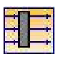


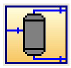
Figure 10



You can use the header unit operation instead of using mixers and tees.



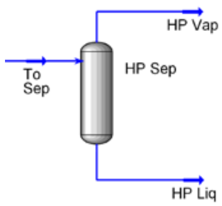
Add a Separator



Separator Icon

2. Add a separator and enter the following information:

In this cell...	Enter...
Connections	
Name	HP Sep
Inlet Stream	To Sep
Vapour Outlet Stream	HP Vap
Liquid Outlet Stream	HP Liq



Note that the Separator unit operation is completely defined in Steady State without having to specify a volume, however, as we will see in the following modules it is an important parameter for dynamic simulation.

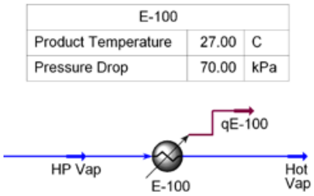
Add a Cooler

Cooler Icon

The vapor stream from the High Pressure Separator is cooled to 27 °C.

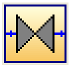
3. Add a cooler and enter the following information:

In this cell...	Enter...
Connections	
Name	E-100
Inlet Stream	HP Vap
Outlet Stream	Hot Vap
Energy Stream	qE-100
Parameters	
Delta P	70 kPa (10 psi)
Worksheet	
Hot Vap Temperature	27 °C (80 °F)



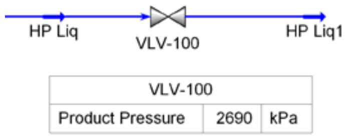
Add a Valve

The liquid stream from the High Pressure Separator is let down through a valve. The downstream pressure is 2690 kPa (390 psia).



Valve Icon

4. Add a valve with the following information:



In this cell...	Enter...
Connections	
Name	VLV-100
Inlet Stream	HP Liq
Outlet Stream	HP Liq1
Worksheet	
HP Liq1 Pressure	2690 kPa (390 psia)

The flowsheet should be completely solved.

Save your case as **4528.01.SS.usc**.

Save your case!

Undo/Redo/Recent Values

UniSim Design offers an undo/redo facility. It applies in the following circumstances:

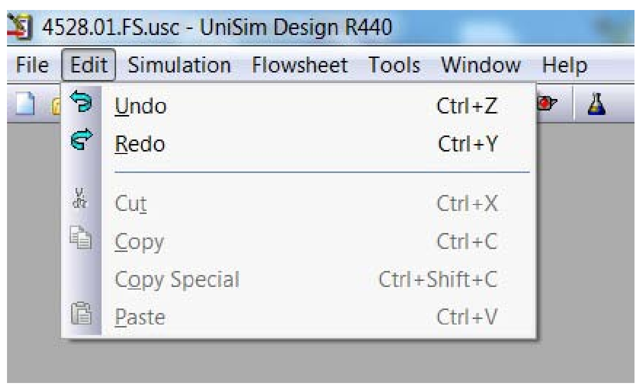
- Values entered into object property values (streams/unit operations)
- PFD object moves and resizing

Object Property Views

In any object property view (e.g. a material stream or any unit operation) UniSim Design has an unlimited undo/redo feature.

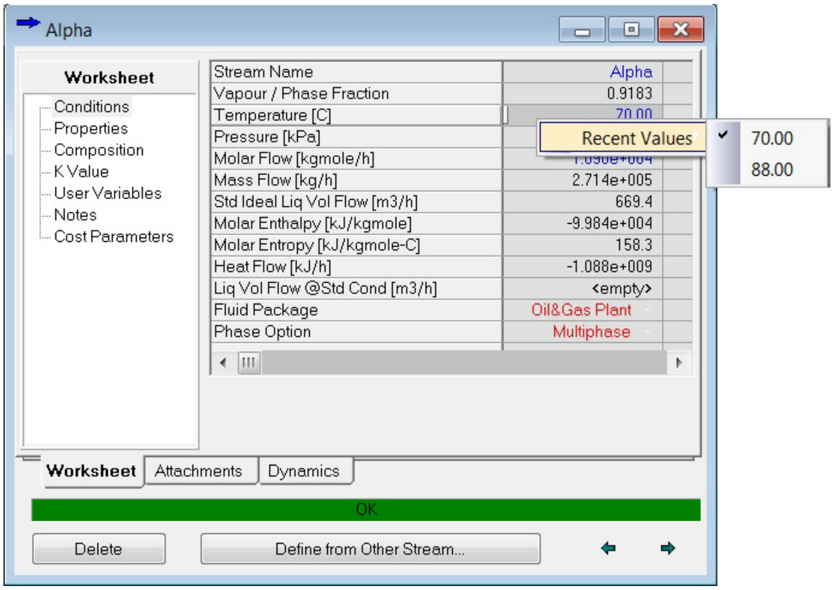
To use undo/redo use the **Undo** and **Redo** options on the **Edit** menu or the hot keys **CTRL Z** and **CTRL Y**.

Figure 11



Additionally by right clicking on any specified value a list of all recent entered values can be entered.

Figure 12



Note that any undo/redo/recent values information is lost when the object property view is closed. Each time an object property view is opened fresh undo/redo/recent values information is stored.

Note also that using the undo/redo/recent values facility has the same effect as typing the value in the cell; any other parts of the flowsheet that depend on the value will recalculate accordingly.

PFD object moves and resizing

UniSim Design also has an unlimited undo/redo facility for object moves and object resizing on the PFD. This is also accessed by the **Undo** and **Redo** options on the **Edit** menu or the hot keys **CTRL Z** and **CTRL Y.**

Note that any undo/redo information is lost when the PFD window is closed. Each time a PFD window is opened fresh undo/redo information is stored.

Note also that you cannot undo object deletion.

2. Pressure-Flow Theory

Workshop

The Pressure-Flow Theory module introduces you to the underlying concepts necessary for developing your own dynamic simulations with UniSim Design Dynamics.

Learning Objectives

The objectives of this module are to understand the basic principles of dynamic modeling and simulation using UniSim Design:

- Physical and mathematical background
- The pressure-flow solver
- Pressure-flow networks and specifications

Introduction

Dynamic simulation can be used to better design, optimize, and operate process plant and processing facilities. By their very nature, such processes never truly operate at steady state. Feed and environmental disturbances, flowline slugging, heat exchanger fouling, well work and export constraints continuously change the process conditions. The transient behavior of the process system can be studied using a dynamic simulation tool like UniSim Design.

The design and optimization of a process involves the study of both steady state and dynamic behavior. Steady state models can perform steady state energy and material balances and evaluate different plant scenarios. The design engineer can use steady state simulation to optimize the process by reducing capital and equipment costs while maximizing production.

Dynamic simulation can be used to confirm that a plant can produce the desired product in a manner that is safe and easy to operate. By defining detailed equipment specifications the dynamic simulation can be used to verify that the process equipment functions as expected in an actual plant situation.

Offline dynamic simulation can be used to optimize controller design without adversely affecting the profitability or safety of the plant. A variety of control strategies can be designed and test before choosing one that is suitable for implementation. The dynamic response to system disturbances can be examined and used to optimize the tuning of controllers.

Dynamic analysis provides feedback and improves the steady state model by identifying specific areas in a plant that have difficulty achieving the steady state objectives.

In UniSim Design, the dynamic analysis of a process system can provide insight into the process system which is not possible with steady state modeling. Dynamic simulation can be used to investigate:

- Process optimization
- Controller optimization
- Safety evaluation
- Transitions between operating conditions
- Start-up/Shutdown conditions

In UniSim Design, the dynamic model shares the same physical property packages as the steady state model. The dynamic model simulates the thermal, equilibrium and reactive behavior of the system in a similar manner to the steady state model.

Whilst the property methods are the same as used in steady state, the dynamic model uses a different set of conservation equations to account for changes occurring over time. The equations for material and energy conservation are differential equations and include transient accumulation terms to describe the rate of change of holdup of mass and energy with time.

Solution of a dynamic model requires solution of the differential equations which describe the underlying physical principles. However, the complexity of these equations means an analytical solution method does not exist. Instead numerical integration must be used which solves the equations at distinct time steps. In general, the smaller the time step, the more accurate the calculated solution. However, smaller time steps require more calculations for the same simulation time and hence take longer to solve. The aim of numerical integration is to use the largest possible time step size which maintains acceptable accuracy without becoming unstable.

The benefits of UniSim Design for dynamic modeling and simulation include:

- **Accuracy**. UniSim Design provides accurate results based on rigorous equilibrium, reaction, unit operations, and controller models.
- **Ease of Use.** UniSim Design uses the same intuitive and interactive graphical environment for dynamic modeling as used in steady state modeling. Streams and unit operations can be added to the dynamic simulation environment as easily as in steady state. Steady state models can be easily converted into dynamic models, allowing re-use of models for different applications.
- **Speed.** UniSim Design has been developed to provide a balance between accuracy and speed. UniSim Design uses the fixed step Implicit Euler integration method whilst pressure-flow, energy and composition balances are each solved at different frequencies to improve performance.
- **Detailed Design.** UniSim Design allows the user to provide specific rating details for each piece of equipment in the plant. Rating information includes the equipment size, geometry, nozzle placement and position relative to the ground. A comprehensive holdup model calculates levels, heat loss, static head contributions and product compositions based on the rating information of each piece of equipment.

- **Customizable.** UniSim Design is customizable. Many organizations have proprietary models and these can be integrated into UniSim Design using OLE/Automation.

Mathematical Modeling

Linear & Nonlinear Systems

The great majority of chemical engineering processes occurring in nature are nonlinear. Nonlinearity arises from equations describing equilibrium behavior, fluid flow behavior, or reaction rates of chemical systems. While a linear system of equations can be solved analytically using matrix algebra, the solution to a nonlinear set of equations usually requires a numerical (iterative) approach.

Distributed and Lumped Parameter Models

In most real-world process engineering systems, the process conditions (temperature, pressure, composition etc.) vary both with spatial position inside the process equipment and with time. To model such a system mathematically requires the use of partial differential equations (PDEs), which describe the rate of change (gradient) of the process variables with both position (**x**, **y**, **z**) and time (**t**). These kinds of models are known as distributed parameter models.

If the spatial variation of the process variables is ignored the system is described as '**lumped'**. In lumped parameter models, all physical properties are considered to be constant with spatial position and only the rate of change with time is considered. The resulting mathematical models can be described using only ordinary differential equations (ODEs) which are far less complex than PDEs and easier to solve. For most process engineering applications lumped parameter models give sufficient accuracy that distributed parameter models are not required.

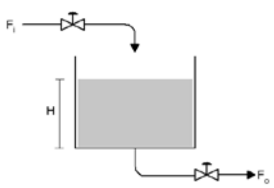
UniSim Design uses lumped parameter models for all unit operations. For example, in the Separator, it is assumed that there are no temperature or compositions gradients so that the temperature and composition at every point within each phase inside the Separator is assumed to be the same.

In some unit operations the spatial variance of the process variables cannot be neglected. For example, in a pipe or plug flow reactor the pressure, temperature and composition varies with length. In UniSim Design such unit operations are solved by dividing their overall volume into a number of sub-volumes which are each considered to be a lumped parameter model, so that the process variables are considered to be constant throughout each sub-volume. In essence, therefore, whilst such models are inherently distributed (with respect to the length of the vessel), they are still described using lumped parameter models.

Conservation Equations

The conservation equations are the basis of mathematical modeling in UniSim Design. The dynamic mass, component and energy balances that are derived in the following section are similar to the steady state balances, but with the inclusion of an accumulation term. This accumulation term allows the output variables from the system to vary with time.

Material Balance



Remember: in steady state accumulation is ignored; flow in equals flow out.

The conservation of mass is described by the following general principle:

Accumulation (holdup) = Flow In –Flow Out

$$\frac{dM}{dt} = \frac{d(\rho V)}{dt} = F_{in} - F_{out} \tag{1}$$

Where:	M	=	*total moles*
	F_{in}	=	*upstream flow*
	F_{out}	=	*downstream flow*
	ρ	=	*bulk molar density*
	V	=	*volume*
	t	=	*time*

Component Balance

In addition to the overall mass balance, for multi-component systems, component balances can be written for each component as follows:

Accumulation of i =

Flow in of i – Flow out of i + Generation rate of i

$$\frac{d(Mx_i)}{dt} = F_{in} x_{i,in} - F_{out} x_i + V \sum_{j=1}^{NR} \upsilon_{ij} r_j \tag{2}$$

Where: x_i = *mole fraction of component i*

υ_{ij} = *stoichiometric coefficient for generation of component i by reaction j*

r_j = *rate of reaction j*

NR = *total number of reactions*

For a system with NC components, there are NC component balances. However, the total mass balance and component balances are not independent (since the former is the sum of the latter). Therefore, in general, a model would include the total mass balance and (NC-1) component balances.

Energy Balance

The conservation of energy can be described is as follows:

Accumulation of energy =

+ inflow of energy (internal, kinetic, potential)

– outflow of energy (internal, kinetic, potential)

+ heat added to the system

– work done by system on surroundings

$$\frac{dU}{dt} = F_{in}h_{in} + Q - F_{out}h_{out} - W \tag{3}$$

Where: U = *internal energy*

Q = *heat added*

h = *enthalpy*

W = *work done by system on surroundings*

Note that, with the equation written in this form, the internal energy and enthalpies are mixture conditions relative to a defined reference state (**T**, **p**, $\mathbf{x_i}$).

In UniSim Design, the reference state is defined as the ideal gas enthalpy of formation at 25 °C, hence there is no explicit heat of reaction term in the equation above (it is included in the mixture enthalpies).

Solution Algorithm

Equations 1-3 are ordinary differential equations. In general, an ordinary differential equation can be written in the form:

$$\frac{dy}{dt} = y'(t) = f(t, y(t)), \quad y(t_0) = y_0 \tag{4}$$

To solve ordinary differential equations a numerical integration method is normally required. There are several numerical integrations schemes in existence.

The Explicit (or Forward) Euler Method

The simplest numerical integration method is Explicit Euler. Here the differential equation is replaced with a finite difference approximation:

$$y'(t) \approx \frac{y(t+h) - y(t)}{h} \qquad (5)$$

Where: h = step size (usually small)

Thus from equations (4) and (5):

$$y(t+h) \approx y(t) + hf(t, y(t)) \qquad (6)$$

This formula, equation (6) is usually applied by choosing a step size $\boldsymbol{h}$, and constructing the sequence of steps $\boldsymbol{t_0}$, $\boldsymbol{t_1 = t_0 + h}$, $\boldsymbol{t_2 = t_0 + 2h}$, **...** etc.

Denoting $\boldsymbol{y_n}$ as a numerical estimate of the exact solution $\boldsymbol{y(t_n)}$, and using equation (6), the estimates are computed from:

$$y_{n+1} = y_n + h \cdot f(t_n, y_n) \qquad (7)$$

in a stepwise manner, starting from a known initial value $\boldsymbol{y_0}$ in steps of size $\boldsymbol{h}$.

One of the main benefits of this method is that it is explicit (hence the name), so the results at step (n+1) can be calculated using only information already available from step n.

The main disadvantage of explicit methods is that for stiff equation systems[1], they can become unstable and fail to converge unless the step size $\boldsymbol{h}$ is very small. Since most engineering systems result in stiff differential equation sets, this usually prohibits the use of explicit methods in practical engineering applications.

The Implicit (or Backward) Euler Method

With Implicit Euler, a different finite difference approximation is used for the differential equation:

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[1] See http://en.wikipedia.org/wiki/Stiff_equation

$$y'(t) \approx \frac{y(t) - y(t-h)}{h} \tag{8}$$

Which is equivalent to:

$$y'(t+h) \approx \frac{y(t+h) - y(t)}{h} \tag{9}$$

Thus from equations (8) and (9):

$$y'(t+h) \approx \frac{y(t+h) - y(t)}{h} \tag{10}$$

$$y(t+h) \approx y(t) + hf(t+h, y(t+h)) \tag{11}$$

Again, this formula is usually applied by choosing a step size $\boldsymbol{h}$, and constructing the sequence of steps $\boldsymbol{t_0, t_1 = t_0 + h,}$ $\boldsymbol{t_2 = t_0 + 2h, \ldots}$ etc.

Again as before, denoting y_n as a numerical estimate of the exact solution $y(t_n)$ and using equation (10), the estimates are computed from:

$$y_{n+1} \approx y_n + hf(t_{n+1}, y_{n+1}) \tag{12}$$

in a stepwise manner, starting from a known initial value $\boldsymbol{y_0}$ in steps of size $\boldsymbol{h}$.

The main difference with this method is that it is implicit; so that an equation has to be solved for $\boldsymbol{y_{(n+1)}}$ at each step. This is usually done with a Newton-type nonlinear solver.

In general, the main advantage of implicit methods is that they are much more stable for stiff equation systems, meaning that a larger step size $\boldsymbol{h}$ can be used and hence the solution can be obtained faster with fewer steps.

UniSim Design uses the Implicit Euler Method and the integration parameters such as the time step size ($\boldsymbol{h}$) can be specified in the Integrator view from the Simulation menu in UniSim Design. The time step size can be adjusted to increase the accuracy, speed or stability of the system.

Integration Strategy

In UniSim Design, the dynamic equations are divided into categories:

- Pressure-Flow
- Energy
- Composition
- Logical Calculations (controller equations, spreadsheets etc.)

The rigorous equations in each category are not solved simultaneously at every time step. This would be computationally expensive. Instead, the equations are solved at different time step frequencies which are integer multiples of the integration time step. The default frequencies are one, one, two, and ten for the pressure-flow, logical, energy, and composition calculations respectively, but these can be customized by the user.

This means the pressure-flow equations are solved at every time step of the numerical integrator, but composition balances are only solved at every 10th time step. Since composition tends to change much more gradually than the pressure, flow, or energy in a system, the equations associated with composition can be solved less frequently and still maintain acceptable accuracy. Using this approach an approximate flash is used for each pressure-flow integration time step and a rigorous flash calculation is performed at every composition integration time step. A similar methodology is used for energy calculations.

Holdup Model

Dynamic behavior arises from the fact that many pieces of plant equipment have some sort of material inventory or holdup. A holdup model is necessary because changes in the composition, temperature, pressure or flow in an inlet stream to a vessel with volume (holdup) are not immediately seen in the exit stream. The model predicts how the holdup and exit streams of a piece of equipment respond to input changes to the holdup over time.

In some cases, the holdup model corresponds directly with a single piece of equipment in UniSim Design. For example, a separator is considered as a single holdup. In other cases, there are numerous holdups within a single piece of equipment. In the case of a distillation column, each tray can be considered as a single holdup. In a pipe or plug flow reaction, the overall volume is divided into a number of sub-volumes. Heat exchangers can also be split up into zones with each zone being a set of holdups.

Calculations included in the holdup model are:

- Material and energy accumulation.
- Thermodynamic equilibrium.
- Heat transfer.
- Chemical reaction.

In UniSim Design, unit operations with holdup use a special common enhanced holdup model, which has certain advantages:

- Rigorous flash calculations allow for accurate calculations of composition and vapour pressure effects.
- The concept of flash efficiencies is used to allow modeling of non-equilibrium behavior, both inside the holdup and between the feed phases of the holdup (to model a bypass, imperfect mixing etc.).
- The concept of feed and product nozzles is used to define the position of the connected streams relative to the holdup and the effect on the process. For example, if a product nozzle is placed below the liquid level in a separator, only liquid will exit from the nozzle until the vapour-liquid interface has fallen below the level of the nozzle.

Assumptions of Holdup Model

There are several underlying assumptions that are considered in the calculations of the holdup model:

- Each phase is assumed to be perfectly mixed (uniform temperature, pressure[2] and composition).
- Mass and heat transfer occur between feeds to the holdup and material already in the holdup.
- Mass and heat transfer occur between phases in the holdup.

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[2] with the exception of hydrostatic head contributions

Accumulation

The transient response that is observed in any unit operation is the result of the accumulation of material, energy, or composition in the holdup.

The holdup model is used to calculate material, energy, and composition accumulation. By default, the material accumulation calculations are performed at every integration time step, the rigorous energy calculations are performed at every 2nd time step and the rigorous composition calculations are performed every 10th time step. During interim steps, approximate calculations are used to increase solution speed.

Pressure-Flow Solver

The fundamental principle used in dynamic modeling in UniSim Design is the concept of the pressure-flow network and the pressure-flow solver.

All dynamic models are considered in terms of their pressure-flow network and at every time step of the numerical integration, the UniSim Design pressure-flow solver performs a simultaneous solution of all the pressure-flow network equations.

There are two basic equation types which define the pressure-flow network and these equations can considered in terms of only pressure and flow as variables:

- **Pressure nodes** – These are holdups of defined volume at which the pressure is calculated based on the material and energy flows in to and out of the holdup and the conditions in the holdup.
- **Resistance devices** – These are parts of the model which define the resistance to flow as a function of the pressure difference across the device. The most common resistance device is a valve.

Pressure Nodes

All unit operations (with holdup) represent pressure nodes. Some unit operations may contribute one or more pressure nodes. Examples of unit operations with more than one pressure node are:

- Heaters/Coolers with multiple zones.
- Heat Exchanger – shell side/tube side.

- Columns with multiple stages (trays).

Basic Pressure-Flow Equations

Volume Balance

For equipment with holdup, an underlying principle is that the physical volume of the vessel is constant and thus the volume of material in the vessel remains constant. Therefore, during calculations in dynamics, the change in volume of the material inside the vessel is zero:

$$\frac{dV}{dt} = 0 \tag{13}$$

The pressure inside a holdup (pressure node) is calculated as a function of the fixed volume, the holdup inventory and the thermal state (temperature/enthalpy).

$$\frac{dP}{dt} = f(V, M, T) \tag{14}$$

Where: V = *fixed volume*

P = *pressure*

M = *total holdup (inventory)*

T = *temperature (change in enthalpy)*

A volumetric flow balance around the vessel can be expressed using equations (13) and (14) as:

$$\frac{\Delta V_P}{\Delta t} + \frac{\Delta V_F}{\Delta t} + \frac{\Delta V_T}{\Delta t} = 0 \tag{15}$$

Where: ΔV_p = *volume change due to pressure change*

ΔV_F = *volume change due to flow change*

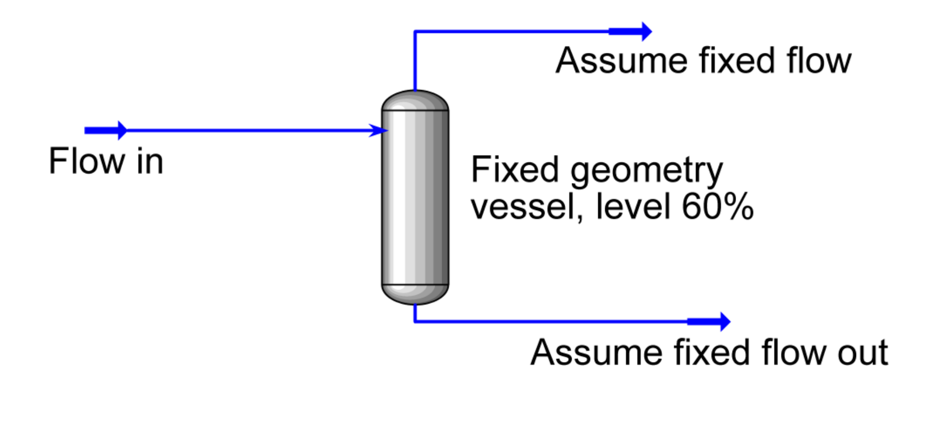
ΔV_T = *volume change due to temperature change*

Each vessel holdup contributes at least one volume balance equation to the pressure-flow matrix.

Example

The volume balance equation captures pressure effects in the vapour holdup due to disturbances in the feed. Consider a separator at constant temperature with a two-phase feed stream, where the feed flow is suddenly increased. Assume that the exit flows from the separator are fixed (Figure 1).

Figure 1



The increase in feed flow causes the vessel pressure to rise for the following reasons:

- The exit flows are fixed, hence an increase in the feed vapour flow would increase the vapour holdup. An increase in the holdup means that a larger amount of the vapour phase is compressed into the same vapour volume resulting in an increase in vessel pressure.
- The increase feed liquid flow results in an increase in the liquid level. This compresses the vapour holdup into a smaller volume within the vessel, causing the vessel pressure to rise further.

Resistance Devices and their Equations

In UniSim Design unit operations such as valves and heat exchangers act as resistance devices which calculate the flow rate through them based on the pressure drop across them using resistance equations. In UniSim Design, resistance equations are based on turbulent flow and have the form:

$$F = k\sqrt{\rho \Delta P} \tag{16}$$

Where:	F	=	*flow rate*
	ρ	=	*density*
	ΔP	=	*pressure gradient*
	k	=	*Conductance constant (reciprocal of resistance to flow)*

Equation (16) is a simplified form of the basic valve operation equation which uses the valve flow coefficient *Cv* to relate the flow rate through the valve to the frictional pressure drop across it:

$$F = f(Cv, P_1, P_2) \tag{17}$$

Where:	F	=	*flow rate*
	P_1	=	*upstream pressure*
	P_2	=	*downstream pressure*
	Cv	=	*the valve coefficient, UniSim Design will calculate this value on request*

The following unit operations have a resistance equation associated with them.

For more detailed description on the individual unit operations and the resistance equations associated with them, see the UniSim Design Operations Guide manual.

Unit Operation	Resistance Term
Valve	Using a pressure-flow specification, the user can specify conductance, (C_v,or k) on the **Specs** page of the **Dynamics** tab.
Pump	The heat flow and pump work define the pressure-flow equation of the pump. These parameters can be specified and/or calculated on the **Specs** page of the **Dynamics** tab.
Compressor/Expander	The heat flow and compressor work define the pressure-flow equations of the compressor. These parameters can be specified and/or calculated on the **Specs** page of the **Dynamics** tab.
Heater/Cooler/Heat Exchanger/ Air Cooler/LNG	With a pressure-flow specification, the user can specify the K-value on the **Specs** page of the **Dynamics** tab.
Tray Sections, Weir Equation	The weir equation determines the liquid flow rate from a tray as a function of the liquid level in the tray. Tray geometry can be specified on the **Sizing** page of the **Rating** tab.
Tray Sections, K-value	The K-value is used to determine vapour flow exiting from the tray as a function of the pressure difference between trays. K-values can either be calculated or specified on the **Specs** page of the **Dynamics** tab.

The Pressure-Flow Network

As described above, the basic pressure-flow equations in UniSim Design can be categorized as pressure nodes or resistance devices, and each unit operation provides one or more of these equations types.

The complete process model in UniSim Design is constructed by defining streams and unit operations, and connecting them according to the process flow diagram. The pressure-flow equations from each of these streams and unit operations are assembled together to form the complete pressure-flow network for the model. This pressure-flow network is solved at every time step of the numerical integration.

Simultaneous Solution of the Pressure-Flow Network

The relationship between pressure and flow in the pressure-flow network in UniSim Design is highly coupled:

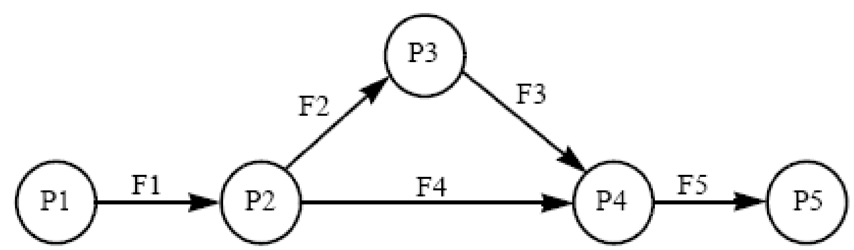
- The pressure node equations are a function of the flow rates into and out of the pressure nodes.
- The flow rates through resistance devices are functions of the upstream and downstream pressures.

In addition, in any given model there are usually multiple pressure nodes and multiple resistance devices, arising from the various unit operations in the model connected according to the process flow diagram.

To calculate the pressures and flows throughout the model in UniSim Design requires the solution of all the pressure-flow equations. This means that the entire pressure-flow network must be solved as a single (usually large) set of simultaneous equations (both linear and nonlinear). This is done using a Newton-type solver and matrix algebra with the Implicit Euler method.

Figure 2 below shows a generalized model where P# represents Pressure Nodes, or boundary pressures linked by F# resistance devices.

Figure 2



Unit Operation Pressure-Flow Specifications

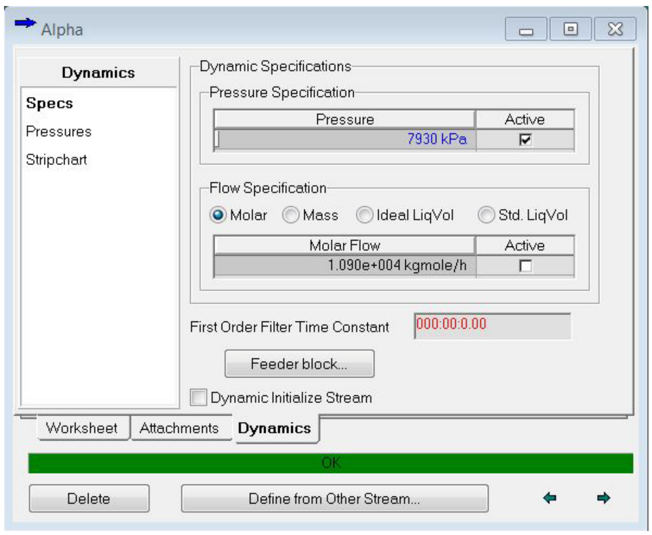
Each unit operation in a UniSim Design model contributes one or more pressure-flow equations to the pressure-flow network. Although the specific form of the various pressure-flow equations is dependent on the type of unit operation, in many cases the user is given a choice for which equation they would like to use. This information is given on the **Dynamics** tab.

For example, in the valve unit operation the user is given the choice of the standard pressure-flow relation, equation (17) or instead to use a fixed pressure drop.

Material Stream Pressure-Flow Specifications

In addition to the pressure-flow equations provided by the unit operations, UniSim Design allows the user to specify (or fix) the pressure and/or flow of each material streams in the model. The pressure-flow specifications are made on the **Dynamics** tab of the Material Stream property view (Figure 3).

Figure 3



The user is required to provide enough stream pressure-flow specifications to satisfy the degrees of freedom of the pressure-flow network. Normally the number of stream pressure-flow specifications required is the same as the number of boundary streams in the model (the “one pressure-flow specification per boundary stream” rule).

In addition to providing the correct number of specifications, the user must also ensure that the specifications they provide form a consistent set in order that the pressure-flow network can be solved. Failure to provide a consistent set of specifications can result in a singular pressure-flow network which cannot be solved.

Definition of the Pressure-Flow Network

Whilst UniSim Design provides the unit operations with their constituent pressure-flow equations, it is the user's responsibility to configure their model in such a way that solution of the pressure-flow network is possible and gives a realistic solution.

It is the responsibility of the user to:

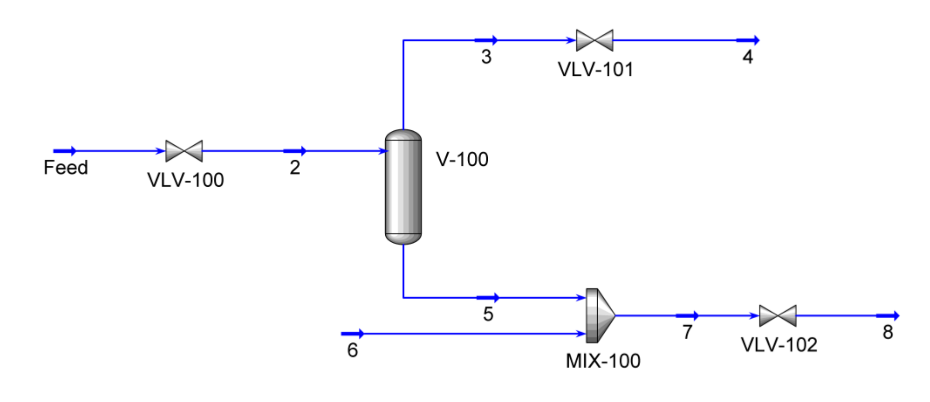
- Define their process model in such a way that the pressure-flow network can be solved. In practice, this usually means building the model as a sequence of alternating pressure nodes connected to resistance devices.
- Provide a consistent set of dynamic specifications on the material streams and unit operations, and sufficient in number to satisfy the overall degrees of freedom in the pressure-flow network.
- Configure the model is such a way that a realistic pressure gradient is defined in the model (no pressure = no flow, positive pressure gradient = negative flow!).

For inexperienced users UniSim Design provides a tool, the Dynamics Assistant, which can be used to check the model to ensure these conditions are satisfied. Further details are provided later in the module.

Example

As an example of the pressure-flow network, consider the flowsheet shown in Figure 4.

Figure 4



In this flowsheet there are eight material streams and one vessel with holdup. Each material stream has a pressure and a flow, and the holdup has a pressure. Thus the number of variables is:

$$8 \times 2 + 1 = 17 \tag{18}$$

Pressure-Flow Equation	Description	# of Eqns
Separator		
Volume Balance equation	The volume balance relates the pressure in V-101 with the flows in streams 2, 3 and 5: $\frac{d(P_{V-100})}{dt} = f(V_{V-100}, M_{V-100}, T_{V-100}, F_2, F_3, F_5)$	1
General Pressure relation	Static head option disabled: $P_{V-100} = P_2$ $P_{V-100} = P_3$ $P_{V-100} = P_5$	3
Valves		
Resistance equations	This is the general form of the valve resistance equation. The actual equations vary according to inlet stream conditions (single or two phase): $F_2 = k_{VLV-100}\sqrt{P_2 - P_1}$ $F_4 = k_{VLV-101}\sqrt{P_4 - P_3}$ $F_8 = k_{VLV-102}\sqrt{P_8 - P_7}$	3
General Flow relations	Since valves are usually not specified with holdup: $F_2 = F_1$ $F_4 = F_3$ $F_8 = F_7$	3
Mixer		
General Pressure relation	The **Equalize All** Pressure Assignment option is recommended for mixers in dynamics mode: $P_5 = P_7$ $P_6 = P_7$	2
General Flow relation	Since mixers do not have a holdup: $F_5 + F_6 = F_7$	1
	Total Number of Pressure-Flow Equations	**13**

The accumulation or change in amount of holdup is solved using material balances in the holdup model. Although the holdup is not solved by the pressure-flow solver, it is used by the volume balance equation to calculate the vessel pressure of the holdup which is a variable in the pressure-flow network.

With 17 variables to solve for in the network and 13 available equations, there are 4 degrees of freedom. Therefore, 4 variables need to be specified to define this system. This is the same as the number of flowsheet boundary streams. The user is free to specify pressure and/or flow in any stream in the model, as long as the overall specifications are consistent. Typical specifications would be material stream pressure specifications on streams **Feed**, **4** and **8**, and a flow specification of stream **6**. Specifications on internal streams are usually not advised and can result in an inconsistent specification and a singular pressure-flow network.

In addition to the number and of location of pressure-flow specifications, the user must also ensure that the pressure values specified are such that an appropriate pressure gradient exists to drive the flow through the model from inlet to outlet. If there is no pressure gradient there will be no flow. If there is a positive pressure gradient (outlet pressures are higher than inlet pressures) there will be reverse flow.

Summary

1. In dynamics mode UniSim Design solves a system of pressure-flow equations for the network defined in the model.
2. The basic pressure-flow equations in UniSim Design can be categorized as pressure nodes or resistance devices, and each unit operation provides one or more of these equations types. A Separator is a typical pressure node unit operation and a Valve is a typical resistance device.
3. Sufficient specifications are required so that there are zero degrees of freedom. It is the user's responsibility to define sufficient and appropriate specifications such that the pressure-flow equations can be solved. Typically one pressure or flow specification is required per boundary stream.
4. The flow through the model is driven by the pressure gradient. No pressure gradient means no flow.

Dynamic Tools in UniSim Design

The Dynamics Assistant

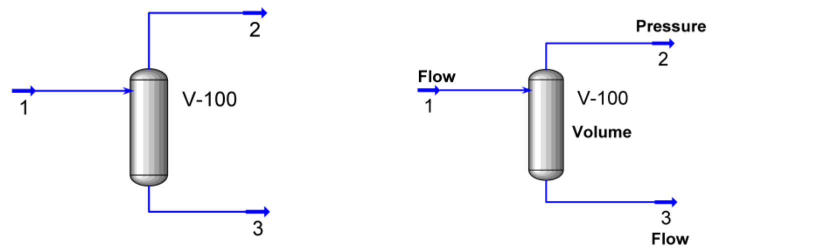
The Dynamics Assistant provides a quick method for ensuring that a correct set of pressure-flow specifications is used. The Dynamics Assistant can be used when initially preparing a case for dynamics, or when analyzing why a case modified in dynamics mode will not solve.

The Dynamics Assistant makes recommendations for specifying your model in dynamics mode. The user does not have to follow all the suggestions. It is strongly recommended that the user is aware of the effects of any changes made by the Dynamics Assistant. (It is advisable not to instruct the Dynamics Assistant to make changes without understanding what those changes will be.)

The Dynamics Assistant recommends a set of specifications which are reasonable and helps to ensure that the case is not over-specified, under-specified, or singular. It can be used for a quick examination of potential problems that can occur while moving from steady state to dynamics, or can be invoked each time the integrator is activated in dynamics.

In the case of a simple separator, UniSim Design adds pressure-flow specifications as shown in Figure 5.

Figure 5

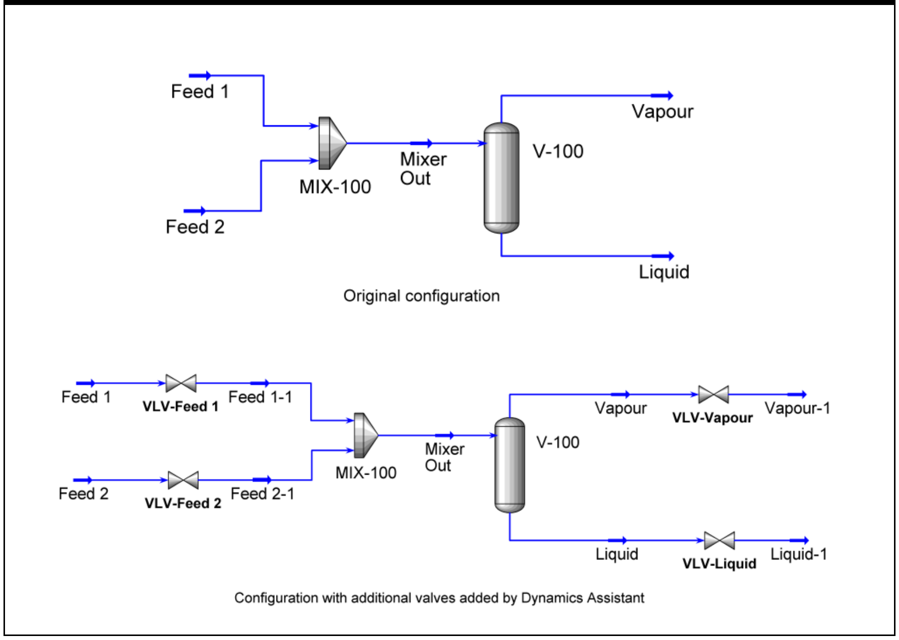


Original configuration

Configuration and pressure-flow specifications after running Dynamics Assistant

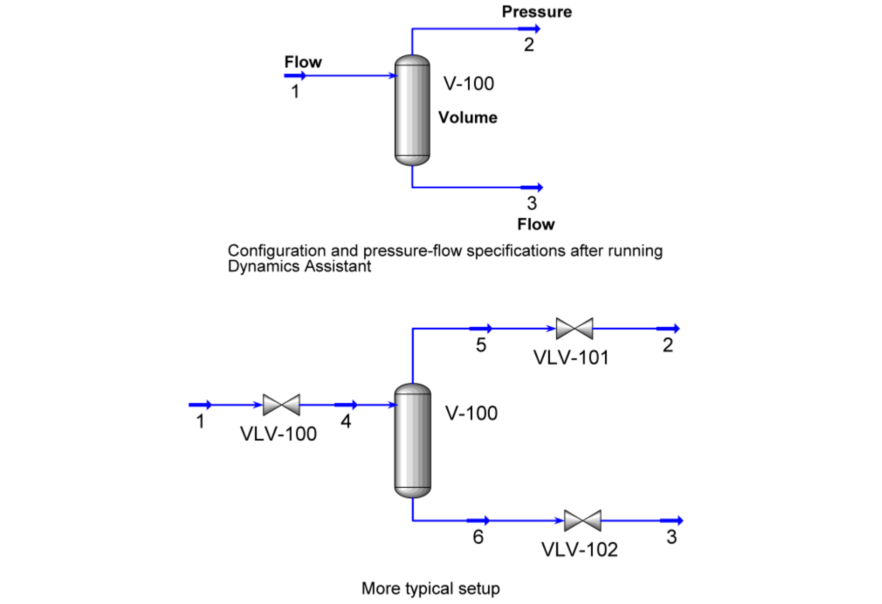
In more complicated models such as that in Figure 6, the Dynamics Assistant recommends the insertion of valves in some terminal streams.

Figure 6



In some situations a model is configured without the resistance devices (e.g. valves) necessary to calculate the flow rate from the surrounding pressures. In this situation the Dynamics Assistant may add a flow specification to fix the flows. Usually a better solution is to add an appropriate unit operations (e.g. pump, valve, etc.) to the model to provide the pressure-flow equations and hence a more realistic model. This can be seen with a simple separator as in Figure 7. Whilst fixing pressure-flow specs on the boundary streams is mathematically sound, a more realistic model would have valves on the terminal streams so that the flows can be determined independently. This approach is also more realistic when it comes to adding controllers which would manipulate the valve actuator positions to control the feed flow, vessel pressure and liquid level.

Figure 7



It should be noted that, although the pressure-flow specifications advised by the Dynamics Assistant are adequate for starting a case in dynamics, detailed dynamic modeling can require more advanced modifications.

In addition to ensuring that the correct pressure-flow specifications are used, the Dynamics Assistant can size all necessary equipment that has not yet been sized.

The parameters sized are:

- vessel volumes
- valve Cv or k
- k values (for equipment such as heaters, coolers, and heat exchangers)

The Dynamics Assistant performs the sizing based on the current (steady state) flow conditions and specified residence times. The Dynamics Assistant also checks the Tray Section pressure profile for both steady state and dynamic mode to ensure a smooth dynamics start. It also ensures that the attached streams have the same pressure as the tray they connect to in the tray section.

Although the Dynamics Assistant ensures that a steady state case will run in dynamics; it cannot guarantee to make all necessary changes for the case to be stable (or lined out) model once in dynamics. It is the user's responsibility to ensure that an adequate control scheme is added to the case and that their model is properly rated (i.e., existing vessels are adequately sized)

The Dynamics Assistant can be opened using any of the four following methods:

Dynamics Assistant icon

- Using the hot key combination: **CTRL Y**
- Selecting Dynamics Assistant from the **Tools** menu
- Clicking the Dynamics Assistant icon in the toolbar
- Clicking the Dynamics Assistant button from the Equation Summary View.

The Equation Summary View

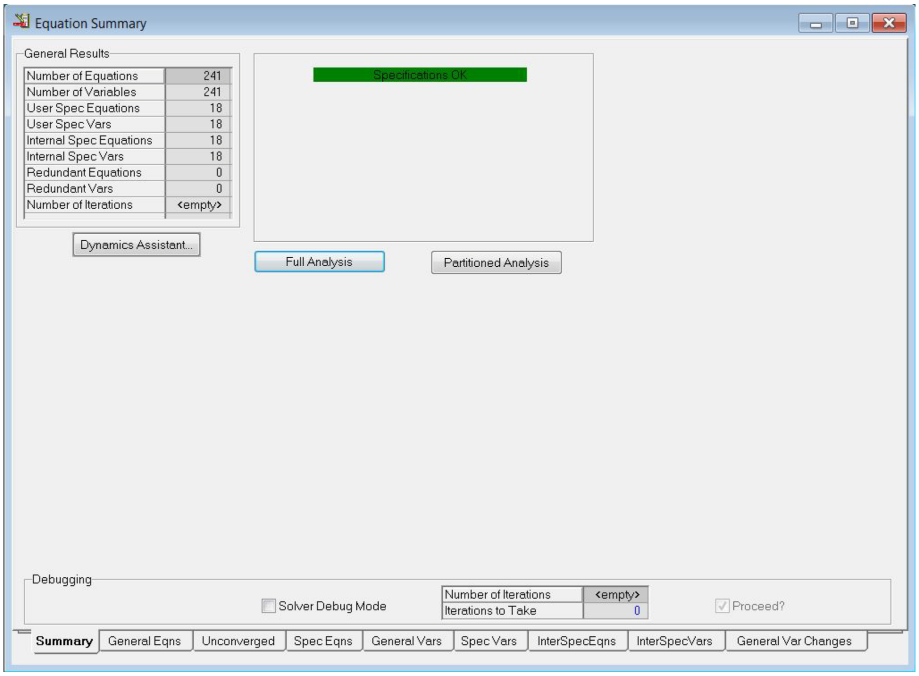
The **Equation Summary View** is accessed from the Simulation menu. (The **Equation Summary View** option only shows when UniSim Design is in Dynamics mode.)

It provides information that can help the user find specification problems in the case.

The **General Results** group contains a summary of the number of equations and variables that are in the case. The group on the right contains information regarding the general status of the case. If there are problems with the specifications, some basic information is provided.

If the Equation Summary View is opened from the menu bar, the **Summary** tab initially contains a single button. Clicking the **Full Analysis** button causes UniSim Design to analyze the pressure-flow parameters in order to determine if there are enough specifications for the problem.

Figure 8



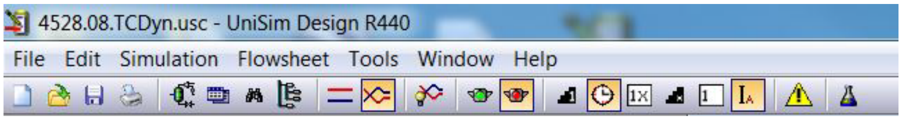
If UniSim Design determines a problem, the **Dynamics Assistant** and **Partitioned Analysis** buttons become visible, and the **Full Analysis** and **Partitioned Analysis** buttons can be used. These buttons invoke different methods of analyzing pressure-flow equations.

In addition, if UniSim Design detects any problems, the **Unconverged** tab is replaced with an **Extra Vars** or **Extra Specs** tab, depending on the nature of the specification problem.

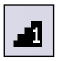
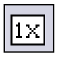
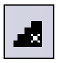
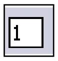
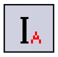
Dynamics Toolbar

In dynamics mode the **Dynamics Toolbar** becomes visible:

Figure 9



The **Dynamics Toolbar** allows single-click control of the Integrator functions in dynamics mode.

Icon on Dynamics Toolbar	Description
	Take one time step
	Run in real time (according to desired real time factor)
	Set desired real time factor
	Take multiple time steps (according to desired number of steps)
	Set desired number of time steps
	Integration Control in **Automatic/Manual** (Button pressed = **Automatic** mode)
	Open the Alarm Manager display

The Integrator

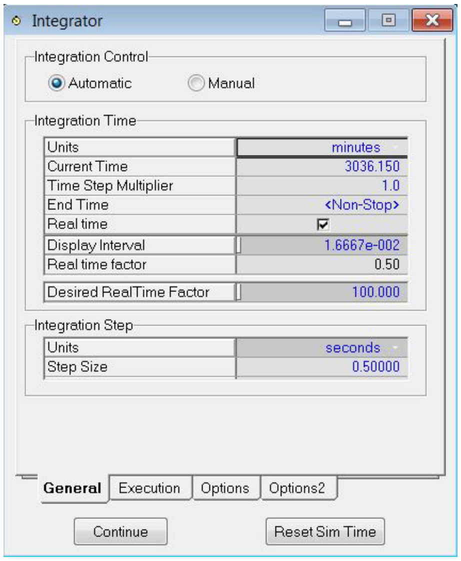
The Integrator is used when running a case in Dynamics mode. You can access the Integrator window from the **Simulation** menu or by using the **CTRL I** hot key.

UniSim Design solves all equations using the Implicit Euler integration method. On the **Integrator** view, the various integration parameters can be specified.

Dynamics Toolbar Integrator Auto/Manual mode icon

In **Automatic** mode the integrator keeps taking steps until told to stop or when the **End Time** is reached.

Figure 10



Note that the Integrator will only start if the **End Time** is greater than the **Current Time** (or set to **<Non-Stop>**).

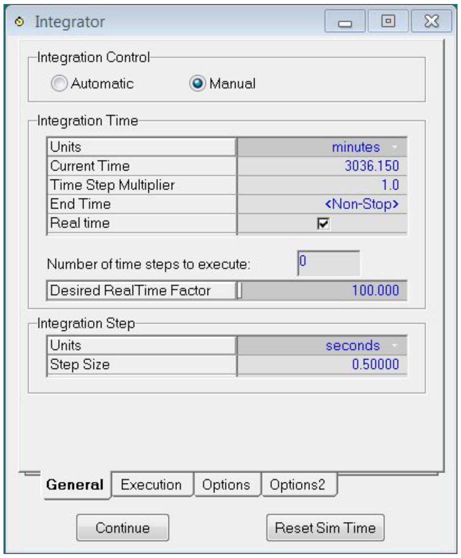
The **Reset Sim Time** button sets the current time to zero.

Dynamics Toolbar Integrator Take One Step icon

Manual integration lets the user specify the number of time steps which UniSim Design executes. It is usually used when troubleshooting or debugging a case because it allows the solver to be executed one time step at a time. Once the integrator has executed the specified number of time steps, the integrator does not stop, but remains in a holding mode. If additional time steps are entered, the integrator continues integration for the given number of time steps.

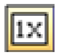
When running as a calculation engine under UniSim Operations the integrator is set to **Manual** mode. UniSim Operations instructs UniSim Design when to take steps.

Figure 11



It is **NOT** recommended to change the value of the **Time Step Multiplier** parameter from the default value of **1**. Since it is used as a multiplier value for the **Step Size** and hence changing the **Time Step Multiplier** is actually the same as changing the **Step Size**. The function of **Time Step Multiplier** can be substituted by using the **Desired Real Time Factor.** Speeding up a simulation is possible by increasing the Step Size. However, Step Size has a significant impact on the solution accuracy, so changing the value should only be done with caution. It is not recommended to change the value of the Time Step Multiplier from the default of 1. Changing the Time Step Multiplier is the same as changing the Step Size. The Desired Real Time Factor can be used to slow down a simulation (for example, to make it easier to watch a fast transient).

The **Real time factor** is calculated by dividing a change in simulation time by the change in actual time taken to simulate that period. For example a real time factor of 100 means that time in the UniSim Design model is running 100 times faster than real time.



Dynamics Toolbar Integrator Real time factor icon

Checking the **Real time** checkbox displays the **Desired Real Time Factor** parameter. By setting a value here the integrator can be made to run at whatever real time factor is required (as long as it is slower than the maximum achievable when the **Real time** checkbox is unchecked.)

The **Display Interval** option is visible only in the **Automatic** Integrator mode and is the time interval at which UniSim Design updates the user interface. The frequency of updating can have a significant impact on the speed at which the simulation runs.

While the Integrator is running the **Units** of the integration step can be changed but a change of **Step Size** is not allowed.

Execution Rates

The **Calculation Execution Rates** group on the **Execution** tab contains parameters that indicate the frequency at which the different balance equations are solved. The default values for Pressure-Flow equations, Control and Logic Ops, Energy, Composition and Flash and Utility Calculations are 1, 1, 2, 10 and 10 respectively. A value of 2 for the Energy Calculations means that a rigorous energy balance is performed every 2 time steps.

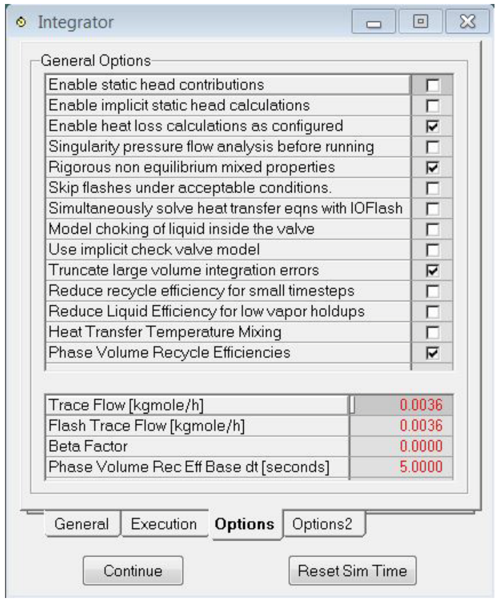
It is strongly recommended that the Calculation Execution Rates are not changed from their defaults without good reason.

Options

Integrator options are discussed in the Dynamics Tools chapter of the UniSim Design Dynamic Modeling Guide – sections 2.4.3 and 2.4.4.

The Integrator **Options** (Figure 12) and **Options2** (Figure 13) tabs allow advanced configuration settings to be made for the integrator.

Figure 12



Each setting of the **Options** tab is described below:

Option	Description
Enable Static head contributions	When checked UniSim Design calculates the static head considering the equipment hold up, the geometry, and the elevation of any attached nozzles. Default state is OFF.
Enable implicit static head calculations	If this option is active, UniSim Design will implicitly solve for static head contributions resulting from level changes inside the vessels. This means the levels and in turn the internal static heads (not those between unit operations due to elevation differences) are a function of the flows. This can increase the stability in cases where the levels in a vessel are closely tied with the flow rates and the liquid height in the vessel can change rapidly. Use this option only if you are experiencing stability problems related to the above setup. Default state is OFF.
Enable heat losses calculations as configured	When checked, heat loss model settings for unit operations (such as vessel and tray sections) are accounted for. If unchecked, all heat losses are zero irrespective of individual heat loss settings. Default state is ON.
Singularity pressure flow analysis before running	When checked, UniSim Design warns of a possible singular solution matrix before starting integration. For larger cases it is recommended that this option be unchecked to increase the overall start up speed. For cases where a singular solution is not considered to be a problem, this option can be disabled which increases the overall speed. Default state is OFF.
Rigorous non equilibrium mixed properties	It is recommended that this option remains active. Deactivating this option provides a slight speed increase when nozzle efficiencies are not 100%, although instabilities can occur. Default state is ON.
Skip flashes under acceptable conditions	It is recommended that this option remain inactive. Activating this option tells UniSim Design to skip flashes calculations under acceptable conditions (e.g. valves with zero pressure drop or mixers/tees with only one effective feed). This provides a slight speed increase, although instabilities can occur. Default state is OFF.
Simultaneously solve heat transfer eqns with IOFlash	When the checkbox is checked and IOFlash is the flash algorithm selected in the Basis environment, then UniSim Design tries to solve heat transfer equations (from heat exchangers) simultaneously with the flash, and potentially make the dynamics run faster. If you encounter inconsistencies with your heat transfer equipment, uncheck this checkbox. Default state is OFF.
Model choking of liquid inside the valve	When checked, this activates the model liquid choking option for all the valves. See also: Module 6 expanding the model Equation (7). Default state is OFF.
Use implicit check valve model	It is recommended to use this option if you are experiencing a delay in the opening and closing of check valves. Default state is OFF.

Option	Description
Truncate large volume integration errors	If there are large upsets or sudden severe changes in the system, they can result in a volume balance error during transients. This is where the volume of the material shown does not match the physical volume that it occupies. This may occur due to the hybrid solution nature of UniSim Design Dynamics whereby some variables and equations are solved in the pressure flow solution but others are solved in the sequential modular pass. In physical reality almost all problem variables are interdependent. In essence the solution of one or more variables is "torn" between the two solutions. In the case of UniSim Design this variable is the material volume - in some other dynamic simulators it may be pressure. If the error is large (greater than 10%), you can enable the **Truncate Large Volume Error** option and UniSim Design truncates the error and restores material inventory. However, truncating the error will violate the overall material balance. With this option on or off, UniSim Design will always slowly correct the error naturally over time. Note that there will never be a material imbalance when the model is at steady state. It is recommended that this option be turned on for dynamic models which must span a large operating range - for example operator training simulators. It is recommended that this option be turned off for depressuring utilities or other engineering studies where material balance accuracy is paramount. Default state is ON.
Reduce recycle efficiency for small time steps	For smaller integration step sizes (where the composition time step ends up being less than 5 seconds), you can enable the **Reduce recycle efficiency for small time steps** option and UniSim Design reduces the flash efficiency of material inside vessels. This option improves stability of the system, but in some cases can produce undesirable results. For example, the phases in a vessel may no longer be in equilibrium and can be at different temperatures. If you are reducing the integration step size or lower the composition period, you can turn this option off if you experience problems or unexpected results. Default state is OFF.

Option	Description
Reduce liquid efficiency for low vapour holdups	This option is provided for testing purposes. In the past there was a scheme to automatically reduce the liquid 1 and liquid 2 recycle efficiencies when the vapor holdup and/or sum of vapor inflow and outflow were small. This was to overcome an inaccuracy in the dynamic flash solution when such a small trace amount of vapor was present. Symptoms of this problem were undesirable oscillations in flow, pressure and holdup. However, this scheme also, at times, introduced its own oscillations. In summary, it is recommended to turn this option off and, where necessary, manually adjust the liquid recycle efficiencies to smaller values to obtain stability. A steam drum is a typical application which would benefit from this. The efficiencies for any operations that support them can always be accessed from the Dynamics tab, Holdup page and then the Advanced button. In the case of the Tray Section, just double-click on the row showing a particular stage. Default state is OFF.
Heat Transfer temperature mixing	Affects heat transfer unit operations (heat exchanger, LNG, etc) by using either of an averaged or weighted inlet temperature (checked) or the actual stream temperature (unchecked) in the heat transfer driving force delta temperature calculation. The averaged or weighted value is a blend of the holdup temperature and the stream inlet temperature and is preferentially weighted towards the holdup temperature as zero flow (infinite residence time) is approached. It is recommended that this only be checked on for backward compatibility. When checked on, this can produce different steady state results for different residence times, integration steps and holdup volumes which are not desired. Default state is OFF.
Phase Volume Recycle Efficiencies	When checked, holdup recycle efficiencies are based on volume percent. When unchecked they are based on mole percent. Recycle efficiency is the percentage of the holdup used in the equilibrium flash at each time step. Default state is ON.

Trace Flow

Any flows within the model that are less than the specified **Trace Flow** are set to zero. This change is performed automatically after the pressure flow solver converges.

This allows the simulation results to be more quickly reviewed and understandable when the small trace flows are viewed as zero. This may also stabilize some models when they are shut down.

Set the **Trace Flow** to zero to disable the feature.

Flash Trace Flow

Any streams which have a flow less than the specified **Flash Trace Flow** are not flashed.

This option conserves CPU usage and produces a larger potential integration **Real Time factor**.

Set the **Flash Trace Flow** to zero to disable the feature.

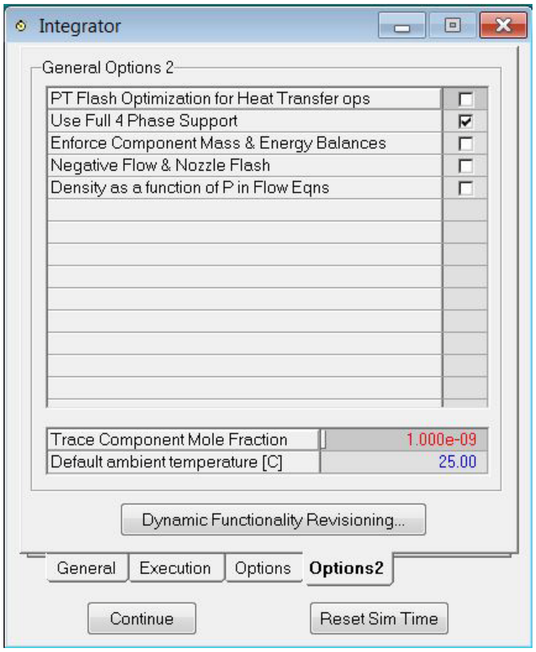
Beta Factor

The Beta Factor affects the calculation of product stream phase fractions for non-zero holdup volumes where the product streams are withdrawn at the homogenous mixture condition of the holdup (i.e. for holdups which do not use levels and nozzle calculations). A Beta Factor of 0.0 is recommended and this is the default for newly created cases. Consult the UniSim Design Dynamic Modeling Guide section 2.4.3 for model details.

Phase Volume Rec Eff Base dt[seconds]

This value is used when the Phase Volume Recycle Efficiencies option is checked. It is used to prorate recycle efficiencies in unit operations where the user has specified efficiency values lower than the default of 100% (used to model non-equilibrium behaviour).

Figure 13



Each setting of the **Options2** tab is described in detail in the table below.

Option	Description
PT Flash Optimization for Heat Transfer Ops	When this option is on, the model can integrate through time somewhat faster. This is because simpler constant Pressure-Temperature flashes are done rather than the more computationally intensive constant Enthalpy-Pressure (HP) flash. It has been found, however, that narrow boiling range mixtures and particularly pure components (water/steam) can experience inaccurate and unstable solution with this option on. If you have those types of systems within your model, you should try turning this off to stabilize an otherwise unstable model. This option affects all heat transfer based unit operations throughout your model. This includes the Heat Exchanger, Fired Heater, LNG, Kettle HX within a Separator and the Heater/Cooler. Default state is OFF.
Use Full 4 Phase Support	This option should only be turned off for old cases to preserve their original results. This option effects how the Cp/Cv ratio is calculated in the simulator. This will affect compressors and valves but only for cases where solid components were present. By turning this option on, you will get accurate and rigorous heat capacity ratio calculations even when solids are present. Default state is ON.
Enforce Component Mass & Energy Balances	This option implements a more rigorous semi-implicit formulation for the integration of component moles and energy. With this option also comes a revised formulation for the PF Solver Pressure Node balance to conserve total moles and solve for Pressure. To use this option the whole case, or optionally certain unit operations, need to have Energy and Composition Calculation Execution rates of 1. Any unit operation without execution rates of unity will use the previous implementation locally for its solution. Hence, you can mix the 2 formulations. Also, any unit operations with reactions, the Separator with PV Work Term Contribution less than 100 or a holdup with the No PNode Equation checkbox enabled will use the prior implementation. Default state is OFF

Option	Description
Negative Flow & Nozzle Flash	This option accommodates the situation where a stream leaves the holdup of a unit operation, but the stream pressure is different from that holdup's pressure. This primarily is intended for the situation where reverse flow occurs across a valve and the feed stream is at a lower pressure than the product stream - there could then be some temperature and flashing effect. Normally flows are positive out of the holdup and the holdup is always considered downstream of any integral hydraulic resistance (resistance k value). Additionally, static head contributions are not usually such that the stream's pressure is different than the holdup pressure that stream is leaving from, but if it is, this option has effect. The effect in the above situations is to do an adiabatic flash on the stream at the stream's pressure, hence giving thermodynamically correct stream conditions. This option will consume more CPU, depending on the number of streams requiring flashing. You may not want this option if you accept the fact that the next unit operation in the flow path would flash and make the stream thermodynamically consistent - energy conservation is always obeyed. Default State is OFF
Density as a function of P in Flow Eqns	Specific volume v = va + vb/P; va and vb are filtered.

The **Default Ambient temperature** is also specified on this view. This is used in heat loss calculations.

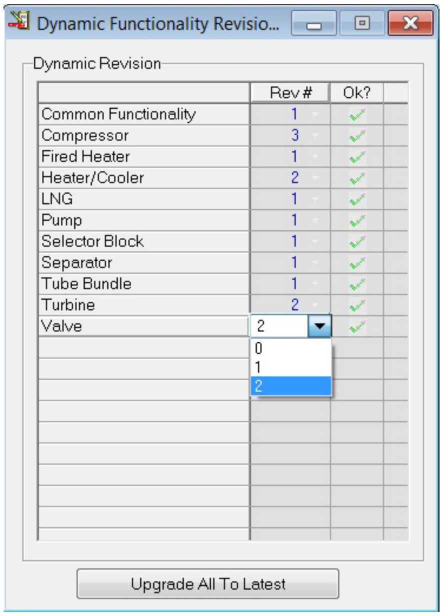
The **Trace Component Mole Fraction threshold value** is used in Dynamics to remove any small amounts of trace components throughout the flowsheet model which are not desired. This can, at times, assist with flash stability and convergence and also will present simulation results without the very small amounts of a component which truly would be zero or at least undetectable in a process. There is also provision within the implementation of this feature for reintroduction of a component back into the process. For this reason, you may at times see streams, coming only out of unit operations with non-zero volume of holdup, with less than the trace threshold. However this is only because the component holdup is accumulating in the overall holdup. The exact algorithm applied is:

1. For a feed stream (or product stream with reverse flow), if a mole fraction is less than the threshold, it is set to zero in the stream. This does mean that a Feeder Block can have a mole fraction below the threshold but when assigned to the stream the user will get a trace warning and a value of zero will be applied.

2. For zero holdup volume unit operations, if the inlet mole fraction is less than threshold, then the outlet is assigned a value of zero.
3. For non-zero holdup volume unit operations, if the holdup mole fraction is less than threshold and all the inflowing streams are each less than the threshold, then a value of zero is assigned to the holdup mole fraction. Basing the logic on the incoming stream mole fractions will allow for component reintroduction. A product stream coming out of a holdup experiencing reintroduction will have a mole fraction less than threshold but in the next downstream unit operation this value will not propagate until it has gotten above threshold.

The Dynamic Functionality Revisioning button opens a separate display (Figure 14) which then allows the user some custom control over the functionality of certain unit operations.

Figure 14



Only those unit operations which have been upgraded to support any new functionality are shown. Usually the reason for making the software give different results is due to a fix in the software. This means that with the lower revision number the software was not necessarily calculating correctly or with some small error. It is recommended upon upgrading to a more recent version of UniSim Design software that you switch to the most recent version. Any unit operations using lower than the most recent version would be shown with a red X beside them. Alternatively if you just want to keep the exact same results as before, then you can stay at the lower revision number.

3. Transitioning from Steady State to Dynamics

4528.03.R440.01

Workshop

This module examines the process of moving from steady state to dynamics. The process for doing this is not difficult, but it will become easier as you gain experience with dynamic simulations in UniSim Design.

Starting with the steady state simulation model that you prepared in Module 1, you will add the necessary equipment information and flowsheet specifications to permit dynamic simulation analysis.

Learning Objectives

Once you have completed this module, you will be able to:

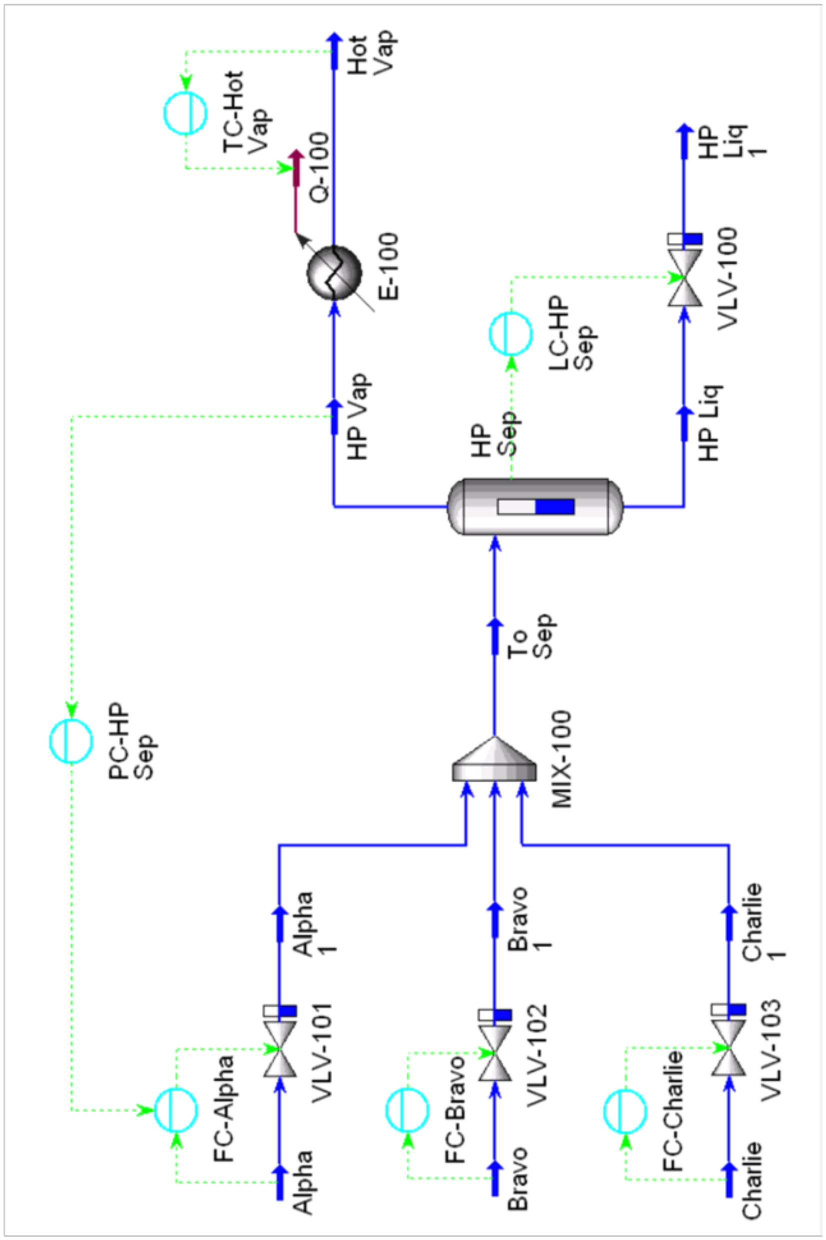
- Size equipment
- Define pressure flow specifications
- Add Strip Charts and controllers
- Run a simple dynamic simulation and observe the role of the various controllers

Prerequisites

Before beginning this section you need to know how to:

- Add Streams
- Add Unit Operations
- Maneuver through the UniSim Design interface

Process Overview

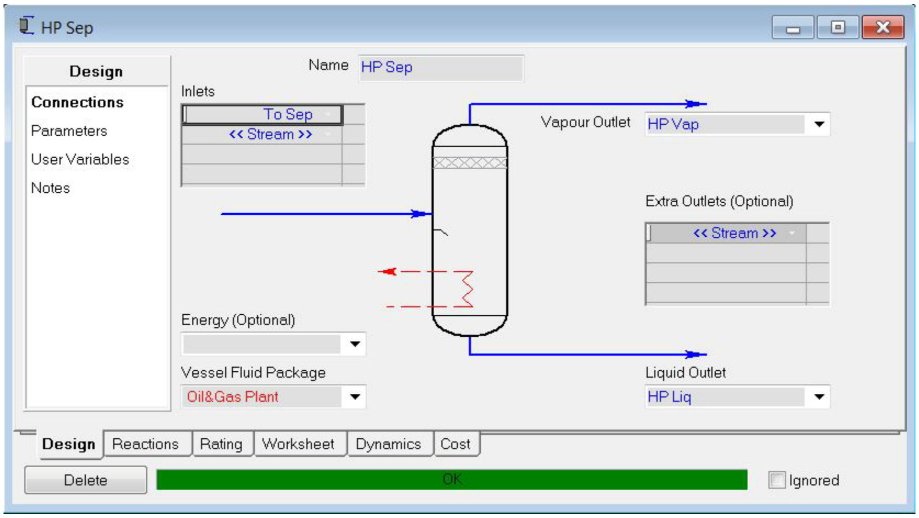


UniSim Design Dynamics

UniSim Design Dynamics has been designed to permit a two-tiered approach to simulation, with numerous options to supply different levels of equipment design and performance information. UniSim Design Dynamics provides modeling capabilities aimed at both process design and detailed design activity.

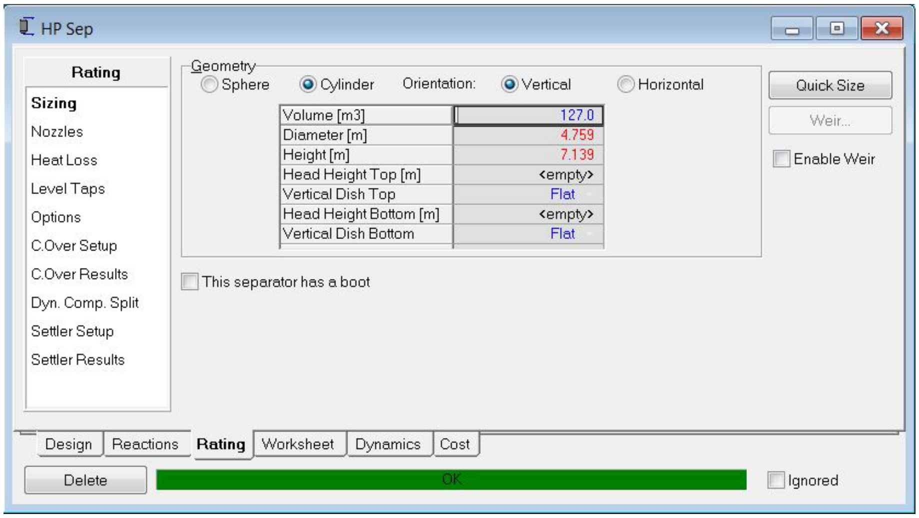
For simulation during the design activity, users typically enter basic design information and UniSim Design Dynamics estimates reasonable defaults for the detailed equipment information. Typically these basic design parameters can be found on the **Design** tab of unit operations.

Figure 1



Default values estimated by UniSim Design are shown with a red color-coding. Generally, this detailed information can be found on the **Rating** tab of the unit operation property view.

Figure 2



In the following modules we will focus on the fundamental concepts underlying the use and configuration of dynamic simulations in UniSim Design. In later modules we will extend the simulation by incorporating detailed equipment and performance information and explore the detailed rating capabilities that UniSim Design provides.

We will begin this module by loading the case that was saved in the previous Module, **4528.01.SS.usc**, and prepare the model for dynamic simulation.

Transitioning from Steady State to Dynamics

The following areas should be examined when setting up a simulation in steady state and transitioning to dynamics:

1. Adding unit operations
2. Equipment sizing
3. Adjusting column pressure
4. Logical operations
5. Adding control operations
6. Entering the UniSim Design Dynamic environment
7. Adding pressure-flow specifications
8. Troubleshooting

More information on these steps can be found in the Unisim Design Dynamic Modeling manual, Section 1.5.2. Moving from Steady State to Dynamics.

These eight areas will now be examined as we convert our steady state simulation into a dynamic one.

Before a transition from steady state to dynamics occurs, the simulation flowsheet should be set up so that a pressure gradient exists across the plant. This pressure gradient is necessary because the flow in UniSim Design Dynamics is determined by the pressure gradient throughout the plant. No pressure gradient means no flow.

Adding Unit Operations

Identify material streams which are connected to two unit operations with no pressure flow relation and whose flow must be specified in Dynamic mode. Unit operations without a pressure flow relation include: vessels, mixers, tees, and tray sections. You will need to add unit operations such as valves, heat exchangers, or pumps, which define a pressure flow relation to these streams.

Which stream in this case connects two unit operations with no pressure flow relation and needs an additional unit operation?

____

[Do not add any extra valves at this point, this will be done later.]

Equipment Sizing

All unit operations in the simulation need to be sized based on actual plant equipment or pre-defined sizing techniques. Vessels should be sized to accommodate actual plant flow rates and pressures while maintaining acceptable residence times. Equipment sizing is a very important step in dynamic modeling.

Sizing the Valves

For dynamics, it is necessary to size every valve in the simulation. UniSim Design Dynamics automates valve sizing based on the:

- Valve operating characteristics (linear, quick opening, equal percentage, or a user defined user table for the valve characteristic)
- Normal valve opening position
- Pressure drop across the valve
- Current Mass or Mole Flow rate
- Feed density

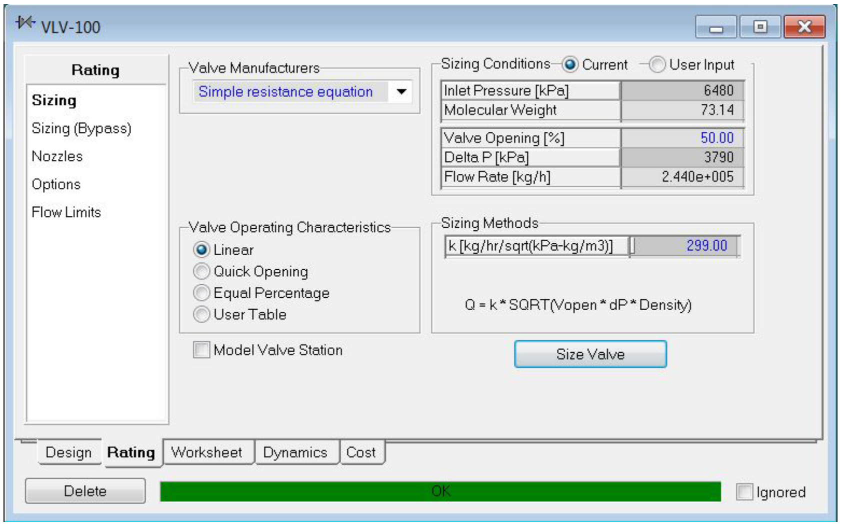
UniSim Design Dynamics calculates the valve size that will allow the valve to pass the upstream Flow rate through the valve at the design valve opening position. This action is automatically done when you switch to dynamics, however you should check all valves for correct sizing before switching modes.

1. Double-click on the control valve **VLV-100**.

2. On the **Rating** tab of the Valve property view, select **Simple resistance equation** from the **Valve Manufacturers** drop down.
3. On the **Rating** tab of the Valve property view, set the **Valve Operating Characteristics** to **Linear**.
4. Set the **Valve Opening (%)** in the **Sizing Conditions** group to 50%.
5. Click the **Size Valve** button to complete the sizing. If you are using **SI** units the view should look like the one below.

If you are using **Field** units, the numbers may be different, but the k value should be about the same as the one shown below.

Figure 3



In this instance the **Size Valve** calculation determines that a k of approximately 299 will pass 2.440e+005 kg/h when the valve is 50% open with a pressure drop of 3790 kPa (550 psi).

Sizing the Separator

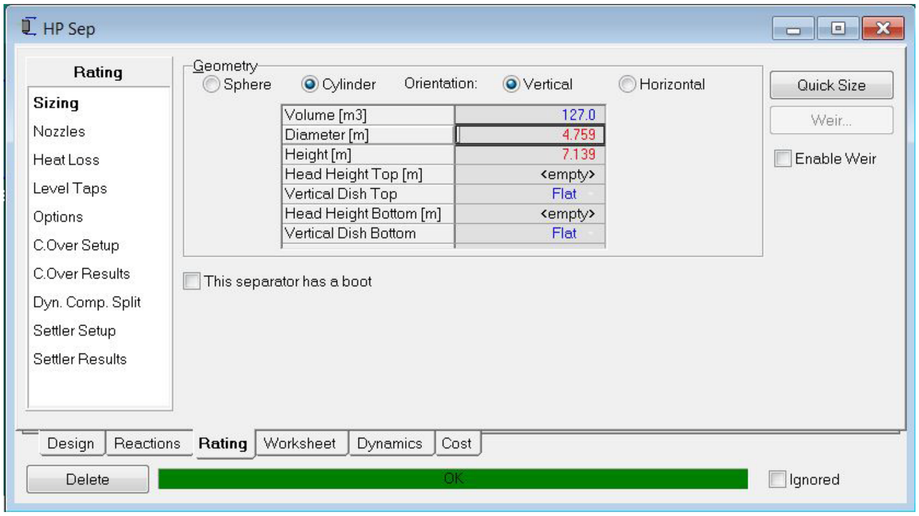
Appropriate vessel sizing is important for dynamic simulation analysis. The vessel holdup will affect the system's transient response during dynamic simulation as the user moves from one operating regime to the next. In addition, the vessel size affects the pressure calculations that are associated with this unit operation.

6. On the **Rating** tab of the Separator, enter a vessel **Volume** of **127 m^3 (4483 ft^3)**.

If you do not know the dimensions of your process equipment, calculate a vessel size based on an approximate residence time.

- 10 minutes is typically a suitable residence time for liquid phase holdups.
- 2 minutes is typically a suitable residence time for vapor phase holdups.

Figure 4

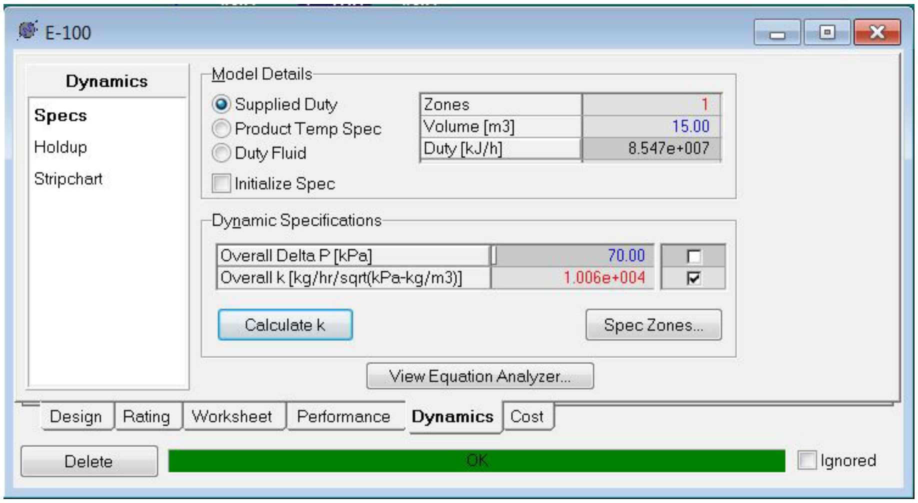


Unlike separators, a number of unit operations have equipment volumes that are set to default values, for example heat exchangers, heaters and coolers. When adding these unit operations to your flowsheet, care should be taken to ensure that reasonable equipment volumes are specified.

Sizing the Cooler

7. On the **Dynamics** tab of the Cooler, change the volume to **15 m^3 (500 ft^3)**.

Figure 5



Save your case as: **4528.03.UO.usc**.

Save your case!

Dynamic and Steady State Specifications: The Differences

In UniSim Design Dynamics, the simultaneous solution of the pressure-flow relationships within the flowsheet requires the user to make a number of dynamic operating specifications. The possible pressure or flow type specifications for a flowsheet include:

- Pressure specification on a material stream
- Flow specification on a material stream
- Fixed pressure drop across equipment

- Resistance calculation (for valves)
- Conductance calculations (for process equipment)

Pressure flow specifications are made on the **Specs** page of the **Dynamics** tab of Unit Operations and Streams.

Some rules to remember:

UniSim Design Dynamics provides users with a great deal of flexibility in making pressure-flow specifications to tackle challenging simulation problems. As such there are a number of different combinations of pressure-flow specifications that will lead to a consistent solution and solve the process flowsheet.

In the table on the following page are some rules that will help guarantee a consistent, properly specified flowsheet all the time.

Dynamic Specifications	
Boundary Streams	Insert a valve on all boundary streams (feed/product streams) within the flowsheet that are not connected to resistance devices (i.e., heat exchangers, coolers, heaters)
Pressure Specifications	Place a pressure specification on all boundary streams (feed/product streams) within the flowsheet.
Distillation Columns	Distillation columns with condensers require an extra specification around the condenser. Make a flow specification for the reflux flow, or insert a resistance device.
Valves	Use the **pressure-flow relationship** as the dynamic specification for a valve.
K value	Use the **overall K value** as the dynamic specification for coolers, heaters, and heat exchangers and LNG exchangers.
Pressure gradients	Be sure to account for pressure gradients throughout the flowsheet. Moreover, be sure to specify reasonable pressure drops/rises in the flowsheet. Pressure differentials are the driving force for flow through the process flowsheet.
Tray Sizing	Use the tray sizing utility to estimate the column geometry and pressure profile.
Mixers	Use the **Equalize All option** as the pressure specification for mixers.
Tees	Remove **Use Splits** as Dynamic Flow Specs on tees.
Rotating Equipments **(Pumps, Compressors, Expanders)**	If all available compressor and pump curves make excellent dynamic specifications. Otherwise use **Efficiency** and either **Head** or **Pressure Rise** as dynamic specifications for rotating equipment.
Hold-ups	Be sure to properly size equipment with holdups.
Dynamic specifications can only be modified when the integrator is stopped. Once the integrator is started the value of the dynamic specification can be changed (its value appears in blue), but the choice of dynamic specifications cannot be changed.	

Making Pressure-Flow and Dynamic Specifications

Analysis of the Process Flowsheet

Based on the previous table, the following observations are made:

- The boundary streams are **Alpha**, **Bravo**, **Charlie**, **Hot Vap** and **HP Liq1**.
- **HP Liq1** is connected to valve **VLV-100**, so the addition of a valve is not required.
- **Hot Vap** is connected to a resistance device, (**E-100**), so the addition of a valve is not required.
- **Alpha**, **Bravo** and **Charlie** are connected directly to the Mixer **MIX-100**. There is no resistance device, so valves will need to be added.
- The pressures of streams **Alpha**, **Bravo** and **Charlie** are all different, the flowsheet will need to be modified and the **Equalize All** option selected for the mixer.

Make the following changes to the flowsheet:

1. Disconnect streams **Alpha**, **Bravo** and **Charlie** from the mixer.
2. Change the pressure of
   - **Alpha** to 7930 kPa (1150 psia),
   - **Bravo** to 7580 kPa (1100 psia),
   - **Charlie** to 8270 kPa (1200 psia).
3. Add valves to streams **Alpha**, **Bravo** and **Charlie** with valve product streams named: **Alpha1**, **Bravo1** and **Charlie1**.
4. Connect each of the three valve product streams to **MIX-100**.
5. Set the pressure of stream **ToSep** to 6480 kPa (940 psia).
6. On the **Design** tab **Parameters** page of the mixer, select the **Equalize All** option.

7. On the **Rating** tab of each Valve, select **Simple resistance equation** from the **Valve Manufacturers** drop down and ensure **Linear** is selected under **Valve Operating Characteristics**.
8. Size the three valves that were added.

What k does UniSim Design calculate for VLV-101?

____

For VLV-102?

____

For VLV-103?

____

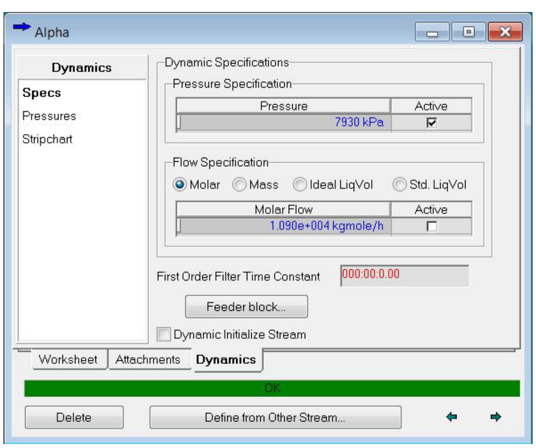
Make the Appropriate Pressure-Flow Specifications

All the boundary streams in the flowsheet must have a pressure specification. The boundary streams are: **Alpha**, **Bravo**, **Charlie**, **Hot Vap** and **HP Liq1**.

1. On the **Dynamics** tab of the stream **Alpha**, activate the Pressure Specification by checking the **Pressure Active** box.

Make sure that only the pressure specification is active. If the flow spec. is active, click the active box to remove the specification.

Figure 6



2. Do the same for streams **Bravo**, **Charlie**, **Hot Vap** and **HP Liq1**.

As a shortcut, instead of going to the **Dynamics** tab of each stream, you could open each unit operation and go to the **Worksheet** tab, **PF Specs** page and view the pressure flow specifications for each of the connected streams.

Having the “k” value as the active specification means that the pressure drops across that unit will change with the flow.

This is more realistic than having a constant pressure.

“k” is the conductance to flow constant for the cooler. The value of k is calculated based on the current delta P, density and flow rate through the cooler.

3. On the **Dynamics** tab **Specs** page for the cooler **E-100**, press the **Calculate k** button.
4. Ensure the **Overall k** box is checked.
5. Check that all the valves have the **Pressure-Flow Relation** specification active on the **Dynamics**.

Save your case as: **4528.03.Specs.usc**.

Save your case!

Summary

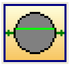
Pressure specifications have been made on all boundary streams. No pressure or flow specifications have been made on the internal flowsheet streams (i.e., **ToSep**, **HP Vap**, **HP Liq**, **Alpha1**, **Bravo1**, **Charlie1**).

1. Open the property view for any one of these streams to verify that this is so.

Resistances to flow specifications (**Pressure-Flow Relations**) have been selected for valves (in fact this is done automatically by UniSim Design).

2. Open a valve property view on the **Dynamics** tab **Specs** page to verify that this is correct.
3. Conductance specifications have been made on process equipment (cooler E-100).

Controllers



PID Controller icon

Controllers can be added to the flowsheet using the same methods as for other unit operations. The **PID Controller** button on the palette represents this unit operation. Once a controller has been added to the flowsheet:

1. Make the necessary connections for the **Process Variable Source** and **Output Target Object**.
2. Select the **Minimum** and **Maximum** values for the Process Variable (the **Range**). These values should bracket all possible PV values.
3. For Controllers directly connected to material or energy streams size the **Control Valve** to configure the minimum and maximum permissible flow.
4. Select **Controller Action**, **Reverse** or **Direct**.
5. Input Controller **Tuning** Parameters.
6. If desired, choose the mode of the controller, **Off**, **Man(ual)**, or **Auto(matic)**.

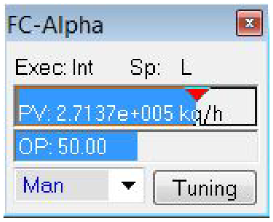
Add the Proposed Control Strategy

1. Add a **Flow Controller** that will control the **Alpha** flow rate to the separator:

In this cell...	Enter...
Connections	
Controller Name	FC-Alpha
Process Variable Source	Alpha Mass Flow
Output Target Object	VLV-101 Actuator Desired Position
Parameters	
Action	Reverse
Range PV Minimum	0 kg/h (0 lb/hr)
Range PV Maximum	365000 kg/h (805000 lb/hr)
Mode	Man(ual)
OP	50%
Kc	0.25
Ti	0.10 min
Td	<empty>

2. Insert a controller face plate to monitor the controller by clicking the **Face Plate** button on the property view.

Figure 7



3. Add another PID controller to control the mass flow rate of **Bravo**:

In this cell...	Enter...
Connections	
Controller Name	FC-Bravo
Process Variable Source	Bravo Mass Flow
Output Target Object	VLV-102 Actuator Desired Position
Parameters	
Action	Reverse
Range PV Minimum	0 kg/h (0 lb/hr)
Range PV Maximum	365000 kg/h (805000 lb/hr)
Mode	Man(ual)
OP	50%
Kc	0.25
Ti	0.10 min
Td	<empty>

4. Insert a controller face plate for **FC-Bravo**.
5. Add another PID controller to control the mass flow of **Charlie**:

In this cell...	Enter...
Connections	
Controller Name	FC-Charlie
Process Variable Source	Charlie Mass Flow
Output Target Object	VLV-103 Actuator Desired Position
Parameters	
Action	Reverse
Range PV Minimum	0 kg/h (0 lb/hr)
Range PV Maximum	365000 kg/h (805000 lb/hr)
Mode	Man(ual)
OP	50%
Kc	0.25
Ti	0.10 min
Td	<empty>

6. Insert a controller face plate for **FC-Charlie**.

7. Add a level controller to control the amount of liquid in the **HP Sep** vessel.

In this cell...	Enter...
Connections	
Controller Name	LC-HP Sep
Process Variable Source	HP Sep Liquid Percent Level
Output Target Object	VLV-100 Actuator Desired Position
Parameters	
Action	Direct
Range PV Minimum	0%
Range PV Maximum	100%
Mode	Man(ual)
OP	50%
Kc	1.0
Ti	15.0 min
Td	<empty>

8. Insert a controller face plate for **LC-HP Sep**.
9. Add another PID controller to control the temperature of the **Hot Vap** stream by manipulating the cooler duty.

In this cell...	Enter...
Connections	
Controller Name	TC-Hot Vap
Process Variable Source	Hot Vap Temperature
Output Target Object	qE-100 Control Valve
Control Valve	
Duty Source	Direct Q
Min. Available	0 kJ/h (0 Btu/hr)
Max. Available	1.06E8 kJ/hr (1.0E8 Btu/hr)
Parameters	
Action	Direct
Range PV Minimum	10 °C (50° F)
Range PV Maximum	65 °C (150 °F)
Mode	Man(ual)
OP	50%
Kc	1.0
Ti	2.0 min
Td	<empty>

10. Insert a controller face plate for **TC-Hot Vap**.

The control valve view will be different depending on the cooler model chosen (see the **Dynamics** tab **Specs** page for the cooler):

- If the **Product Temp Spec** radio button was selected, then there is no need for the controller (the duty will vary automatically in order to maintain the desired temperature).
- If the **Duty Fluid** radio button was selected, then the controller would control the flow rate of the duty fluid. The user will supply a UA, duty fluid Cp, and the duty fluid inlet temperature.
- In this case the **Supplied Duty** option is used so the range of possible duty values is set.

Cascade Control

Cascade control is a common control technique that uses two controllers within one feedback loop. One controller is "nested" inside the other. This means that the two controllers are not independent, but linked together with the primary, "master", controller setting the SP for the secondary, "slave," controller.

Cascade control can improve the dynamic response and controllability of a process that has considerable dead time, or where the time response of the primary loop is very large.

Adding the Master Controller

The slave controller for our cascade loop will be the FC-Alpha controller that already exists in our simulation. Complete the following steps to add and define the primary controller.

1. Add another PID controller to the simulation, named **PC-HP Sep**.
2. Connect the pressure of stream **HP Vap** as the PV for the controller.
3. Connect the OP to the **FC-Alpha** controller.
4. Enter the following information on the **Parameters** page of **PC-HP Sep**.

In this cell...	Enter...
Connections	
Controller Name	PC-HP Sep
Process Variable Source	HP Vap Pressure
Output Target Object	FC-Alpha SP
Parameters	
Action	Reverse
Range PV Minimum	5500 kPa (800 psia)
Range PV Maximum	6900 kPa (1000 psia)
Mode	Man(ual)
OP	35%
Kc	3.0
Ti	2.0 min
Td	<empty>

The **FC-Alpha** controller should now have three connections.

5. Insert a controller face plate for **PC-HP Sep**.

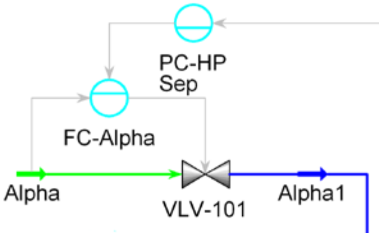
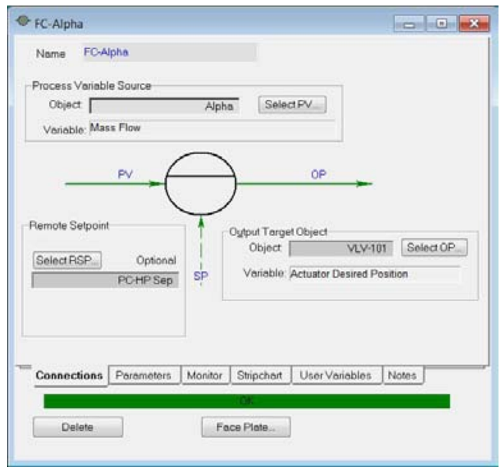


Figure 8



Strip Charts

The frequency at which UniSim Design updates the user interface is set by the **Display Interval** setting on the **Integrator** window. The default is 1 second, which means that the user interface is updated after each second of simulation time

When the model is running dynamically, it can be difficult to observe the simulation variables. Individual variables can be observed while in the PFD environment, or multiple variables can be seen on the workbook. All variables are updated constantly as the dynamic simulation is running. Using a strip chart allows the user to observe several variables in real time as the dynamic simulation runs, and to view historical trends.

Adding Strip Charts

The strip chart provides a method for easily monitoring key process variables in a graphical environment. Multiple strip charts are allowed, and each of these can have an unlimited number of variables charted.

Create a strip chart to record flows by following these steps:

1. Open the **Tools** menu and select **Databook** or press the hot key **CTRL D**.
2. Select the **Variables** tab and click the **Insert** button.

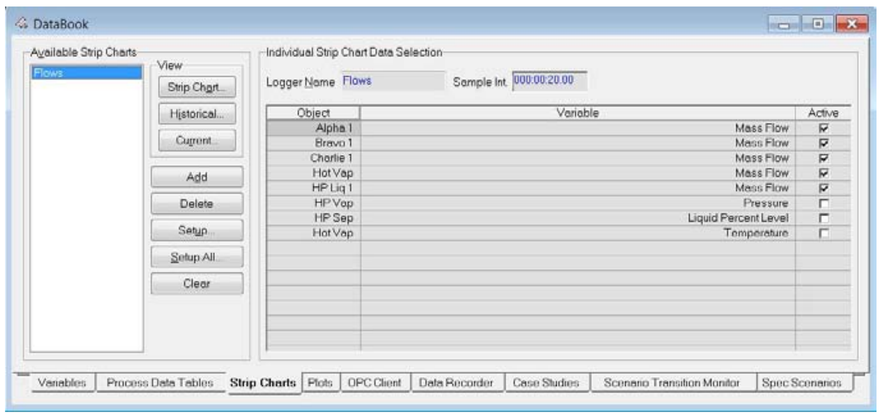
3. Add the following variables:
   - **Alpha1** Mass Flow
   - **Bravo1** Mass Flow
   - **Charlie1** Mass Flow
   - **Hot Vap** Mass Flow
   - **HP Liq1** Mass Flow
4. Select the **Stripcharts** tab.
5. Select the **Add** button in the **Available Strip Charts** group.
6. UniSim Design installs the new strip chart and automatically names it. In this case, the name is DataLogger1.
7. Change the name to **Flows** in the **Logger Name** field.
8. In the **Individual Strip Chart Data Selection** group box, check the **Active** checkbox for the variables of interest: the mass flows of the streams as defined above.

Normally, all strip chart variables must be added to the databook before they are available to the strip chart.

However, there are two other approaches to adding variables to strip charts:

- Variables can be dragged and dropped onto an active strip chart from the parent object. This is done by right clicking and dragging variables from views onto the strip chart.
- Each unit operation has a **Stripchart** page on the **Dynamics** tab. A user can **Quick Create** a strip chart using a predefined subset of variables.

Figure 9



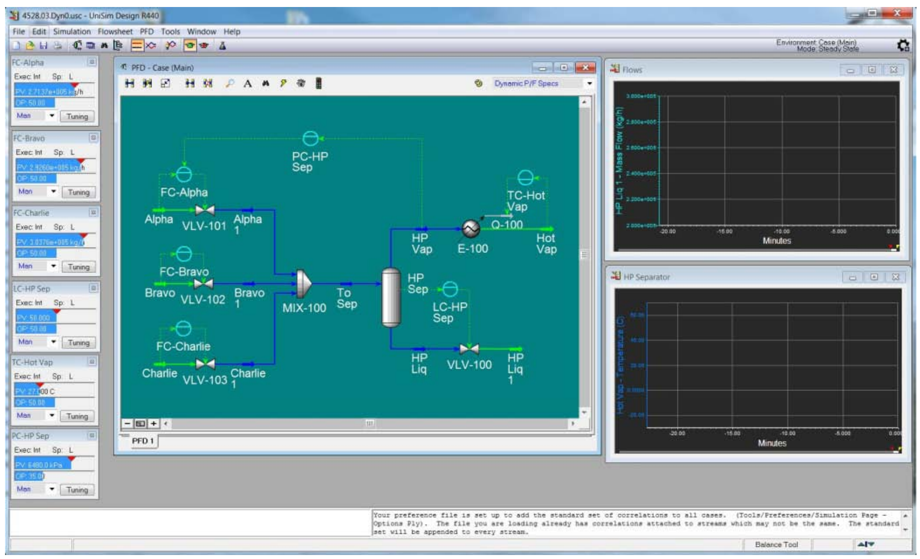
9. To view existing strip charts use one of the following methods:
   - Highlight the strip chart name in the **Available Strip Charts** group and click the **Strip Chart...** button in the View group.
   - Double-click on the name of the strip chart.
10. Create a second strip chart called **HP Separator** and insert the following variables:
    - **HP Vap** Pressure
    - **HP Sep** Liquid Percent Level
    - **Hot Vap** Temperature

11. Click the **Setup All...** button to configure the amount of data available and the sample interval for both strip charts. For all of the strip charts in these modules, we will set the **Logger Size** at **3600** points and the **Sample Interval** at **20 s**.

To modify strip chart parameters, right-click on the strip chart to open the strip chart **Graph Control**, from which you can edit the strip chart parameters.

12. Arrange the Controller Face Plates and both Strip Charts as follows:

Figure 10

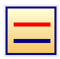


Save your case as: **4528.03.Dyn0.usc**

Save your case!

Switching to Dynamics

Dynamics Mode icon

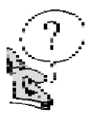


Steady State Mode icon

1. Switch to dynamics mode using the toolbar button, start the Integrator by using hot key **CTRL I** and pressing the **Start** button. Observe the strip charts. Let the Integrator run for a few minutes of simulation time.

Does the system achieve a steady state solution?

____



2. Place the level controller, LC-HP Sep in **Auto(matic)** mode.

Does the system achieve a steady state solution?

____

3. Place the flow and temperature controllers in **Auto(matic)**. Finally, change the mode of **FC-Alpha** from **Auto(matic)** to **Cascade** and **PC-HP Sep** in **Auto(matic)**.
4. Specify the controller set points to match the values in the original steady state simulation, as shown below. The simulation should be stable at the steady state values.

Controller...	SP...
FC-Alpha	[No SP needed; controller is in cascade mode so SP is set by master controller, PC-HP Sep]
FC-Bravo	2.926e+005 kg/h
FC-Charlie	3.038e+005 kg/h
PC-HP Sep	6480 kPa
LC-HP Sep	50%
TC-Hot Vap	27 °C

The buttons on the Equation Summary window are only available when the Integrator is Holding (red traffic light).

5. From the menu select **Simulation**, **Equation Summary View** (the solver must be in Dynamics mode). Press the **Full Analysis** button and then move to the **Spec Eqns** tab for a complete list of all specifications.

Colour Scheme icon

6. From the **PFD** toolbar, click the **Colour Scheme** icon. One of the default schemes is a **Dynamic P/F Specs**. Select this scheme. With the scheme the PFD stream colours are changed to show the Dynamic Specifications in the stream:
   - Green = pressure specification active.
   - Yellow = flow specification active.
   - Red = pressure and flow specifications active,
   - Blue = no specifications active.

Save your case as: **4528.03.Dyn1.usc**

Save your case!

Challenge 1

The Cooler operation has a calculated duty in Steady State and a specified duty in Dynamics.

Switch back to steady state mode, UniSim Design will indicate that could mean a loss in information. Click **Yes** and investigate the flowsheet.

Turn on the solver and change the outlet temperature of the cooler.

Does the steady state case converge when the solver is turned on? What happens when the cooler outlet temperature is changed?

It is recommended that you save any changes under a different file name, so you can reload case 4528.03.Dyn1.usc if necessary.

Challenge 2

Simple Separator

Create a dynamic model of a simple two phase separator. Use the following information and make any necessary assumptions.

- Control Required – Feed stream flow control, vessel pressure, liquid level (add any necessary control valves)
- Feed stream conditions are listed in the table below
- Required vessel pressure is 2000 kPa, vessel is sized for a two minute residence time (use the Dynamics Assistant)
- Create appropriate strip charts

Feed Stream Conditions	
Temperature	25 °C (77 °F)
Pressure	2500 kPa (363 psia)
Flow Rate	3000 kg/h (6614 lb/hr)
Component	**Mole Fraction**
Methane	0.70
Ethane	0.05
Propane	0.05
i-Butane	0.02
n-Butane	0.02
i-Pentane	0.10
n-Pentane	0.03
n-Hexane	0.03

Challenge 3

Water Tank

Create a dynamic model of a water tank. Use the following information and make any necessary assumptions.

- Water feed stream flow rate is 1000 kg/h (2205 lb/hr)
- Tank volume is 2 m3 (70.6 ft3)
- Tank level is to be controlled at 50%
- Tank is open to the atmosphere
- Create appropriate strip charts

Think carefully about how to deal with the tank vapour space and vapour product. You may wish to read the **Reverse Flow** section in module 6 or consult the on-line help for information about the **Product Block**.

4. Basic Control Theory

4528.04.R440.01

Workshop

Process control, on a working level, involves the control of variables such as flow rate, temperature, and pressure in a continuously operating plant. Process control, in a general sense, attempts to maximize profitability, ensure product quality, and improve the safety and operability of the plant.

While a steady state simulation in UniSim Design allows the design engineer to optimize operating conditions in the plant, dynamic simulation allows you to:

- design and test a variety of control strategies before choosing one that may be suitable for implementation
- stress the system with disturbances as desired to test for plant performance

Even after a plant has started operation, process engineers may look for ways to improve the quality of the product, maximize yield, or reduce utility costs. Dynamic simulation using UniSim Design allows the process engineer to compare alternative control strategies and operating schemes in order to improve the overall performance of the plant. In short, the engineer can accomplish a lot of analysis off-line on a dynamic simulator, instead of disturbing the actual process.

Three topics will be covered in this module. First, the characteristic parameters of a process will be discussed in the **Process Dynamics** section. Next, the control strategies available in UniSim Design will be discussed in the **Controller Setup** section. Finally, the **General Guidelines** section will outline some steps you can follow to implement a control strategy in UniSim Design. Included in this section are several techniques that may be used to determine possible initial tuning values for the controller operations.

Learning Objectives

In this module you will:

- learn the basics of process control theory
- explore the development of control strategies
- examine general guidelines for implementing appropriate control strategies

Process Dynamics

As a precursor to understanding the concepts of process control, the dynamic characteristics of the process will be discussed. The task of designing a control scheme is best carried out if there is a good understanding of the process system being studied. A process response to a change can vary considerably depending on the manner in which the input is applied to the system, and the nature of the system itself. Therefore, it is important to understand the dynamic characteristics of the process system before proceeding with the process control design.

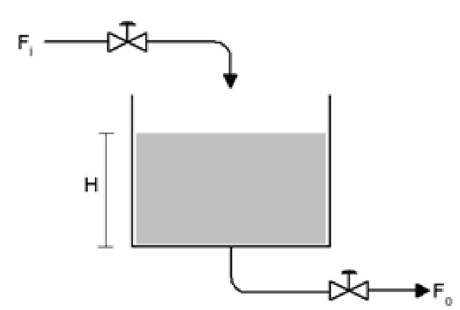
Many chemical engineering systems are non-linear in nature. However, it is convenient to define some essential characteristic parameters of a process system by approximating the system as linear.

Characteristic Parameters of the Process System

It is easiest to define a chemical process system using the general conservation principle which states that:

$$\text{Rate of Accumulation} = \text{Input} - \text{Output} + \text{Internal Generation} \qquad (1)$$

Figure 1



In order to describe some characteristic parameters of a chemical process system, the general conservation principle is applied to a flow relation first order liquid level system: The conservation of material in the tank is expressed as follows:

$$\frac{dV}{dt} = A\frac{dH}{dt} = F_i - F_o \tag{2}$$

Where: H = *the liquid height in the tank*

A = *the cross sectional area of the tank, assumed constant*

F_i = *the inlet flow rate*

F_o = *the exit flow rate*

There is a non-linear relationship describing the flow out of the bottom of the tank, $\mathbf{F_o}$, and the liquid height in the tank, H. In order to express Equation (2) as a first order linear differential equation, it must be assumed that the exit flow varies linearly with height. Linearity can be assumed in situations in which the flow does not vary considerably over time. The exit flow, $\mathbf{F_o}$, can be expressed in terms of the linearity constant, **R** (the valve resistance):

$$F_o = \frac{H}{R} \tag{3}$$

Equation (2) can therefore be expressed as:

$$A\frac{dH}{dt} = F_i - \frac{H}{R} \tag{4}$$

$$RA\frac{dH}{dt} + H = RF_i \tag{5}$$

Equation (4) is a general first order differential equation which can be expressed in terms of two characteristic parameters: the steady state gain, **K**, and the time constant, **τ**:

$$\tau\frac{dy}{dt} + y(t) = Ku(t) \tag{6}$$

Where:	*y(t)*	=	*the output of the system*
	u(t)	=	*input to the system*
	K	=	*the steady state gain*
	τ	=	*the time constant of the system*

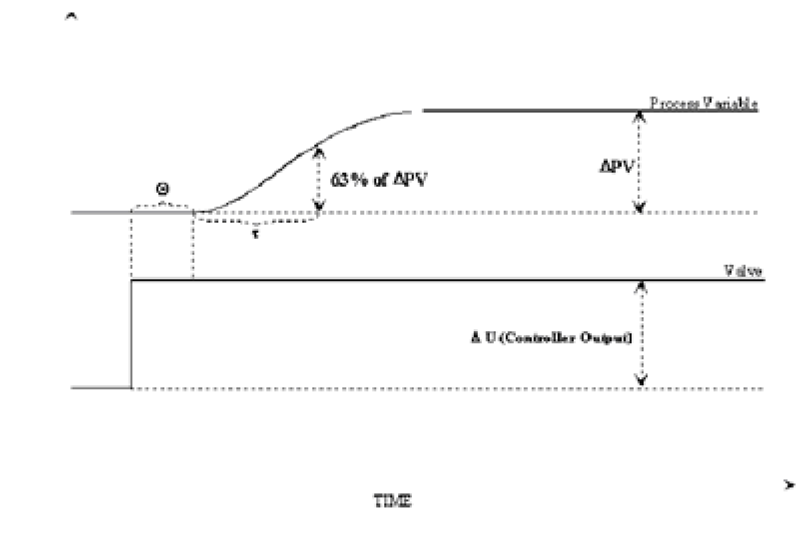
The change in liquid level, H, is the change in the output of the system, y(t). The change in the input to the system, u(t), is the change in flow into the tank, F_i.

Similarly, the time constant, τ, and the steady state gain, K, can be expressed as (7):

$$\tau = AR \text{ and } K = R \tag{7}$$

When a step function of magnitude U is applied to the general first order system, the output response, y(t), is as follows:

Figure 2



As shown, the output, y(t), attains 63.2% of its final steady state value in one time constant, τ. The **Process Variable (PV)** can be assumed to equal its final value after approximately four time constants (4τ) have passed.

The **dead time** of the process is represented by the Greek letter, θ.. The dead time is defined as the amount of time that passes between the time of the change in the **Controller Output (U)**, and the time that the first change is seen in the **Process Variable (PV)**. In the flow example given above, the dead time will be virtually nonexistent; however, it can become significant for other systems.

The following is a list of characteristic parameters that may be defined in terms of the first order response illustrated in the previous example.

Process Gain

The process gain is defined as the ratio of the change in the process output to the change in the process input. The change in the process input is defined in Equation (5),(6) as u(t). The change in the process output is defined as y(t). The first term in Equation (6) is transient and becomes zero at steady state. Therefore the gain can be calculated as shown in Equation (8).

$$\text{Steady-state gain} = \frac{y(t)}{u(t)} = K \qquad (8)$$

Time Constant

The time constant, τ, defines the speed of the response. The response of the system will always follow the profile shown on the previous page. After τ time units, the response y(t) equals 0.632 ΔPV or 63.2% of the final PV value. This will always be true for first order systems.

Dead Time

While capacitance is a measure of how fast a system responds to disturbances, dead time is a measure of the amount of time that elapses between a disturbance to the system and the observed response in the system. Time delays in a system can become significant depending on the nature of the process and the location of measuring devices around the process. It is usually the time associated with the transport of material or energy from one part of the plant to another that contributes to time delays observed in a system. The dead time of a process may be easily modeled using the **Transfer Function** block operation.

Capacity

Definition 1

Capacity can be defined simply as the volume or storage space of a system. The capacitance of a system dampens the output causing the response to take time to reach a new steady state. For electrical systems, the capacity is defined in terms of the resistance of the system and the time constant of the response (9):

$$C = \frac{\tau}{R} \qquad (9)$$

Since the capacity of a system is proportional to the time constant, τ, it can be concluded that the larger the capacity, the slower the response of the system for a given forcing function.

In first order systems, the capacity of a system has no effect on the process gain. However, the capacity varies in direct proportion with the time constant of a system.

Definition 2

A system's capacity is also defined as its ability to attenuate an incoming disturbance. Attenuation is defined as (10):

$$\text{Attenuation} = 1 - \frac{\text{Response Amplitude out of the system}}{\text{Disturbance Amplitude into the system}}$$
$$\text{Attenuation} = 1 - \text{Amplitude Ratio} \qquad (10)$$

Controller Setup

The **PID Controller** operation is the primary tool that you can use to manipulate and control process variables in the dynamic simulation. You can implement a variety of feedback control schemes by modifying the tuning parameters in the **PID Controller** operation. Tuning parameters can be modified to incorporate proportional, integral, and derivative action into the controller.

A **Digital On/Off control** operation is also available. Cascade control may be modeled using interacting **PID Controller** operations. Feedforward control can be implemented in the simulation model using the **Spreadsheet** operation.

Instrumentation dynamics can also be modeled in UniSim Design, increasing the accuracy of the simulation with real valve dynamics. Final control elements can be modeled with hysteresis (lag). The valve response to controller input can be modeled as instantaneous, linear, or first order. Dead time, lags, leads, whether they originate from disturbances or within the process control loop may be modeled effectively using the **Transfer Function** operation.

Terminology

Before reviewing the major control operations that are available in UniSim Design, it is useful to describe some terms.

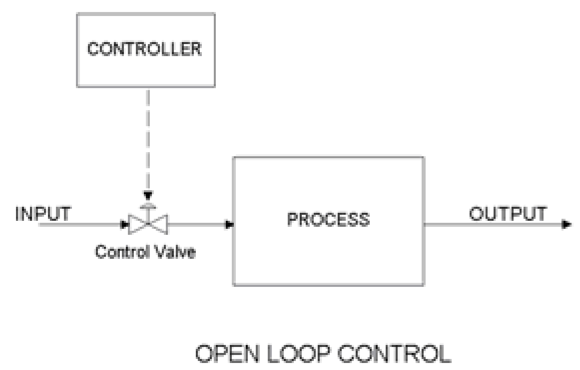
Disturbances

A disturbance upsets the process system and causes the output variables to move from their desired set points. Disturbance variables cannot be controlled or manipulated by the process engineer. The control structure should account for all disturbances that can significantly affect a process. The disturbances to a process can either be measured or unmeasured.

Open Loop Control

An open loop response from a process is determined by varying the input to a system and measuring the output's response. An open-loop system is shown below. In open loop control, the controller sets the input to the process without any knowledge of the output variable that closes the loop in feedback control schemes. Figure 3

Figure 3

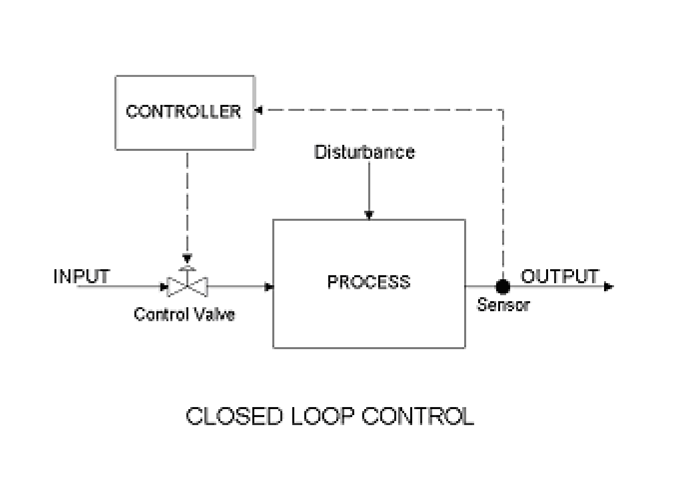


A common example of open loop control is the control of traffic in a city. The traffic lights change according to a set of predetermined rules.

Feedback Control (Closed Loop)

Feedback control is achieved by "feeding back" process output information to the controller. The controller makes use of the current information about the process variable in order to determine what action to take to regulate the process variable. This is the simplest and most widely used control structure in chemical process systems.

Figure 4



Feedback control attempts to maintain the output variable, **PV**, at a user defined set point, **SP**. There are some basic steps that are carried out by the controller in order to achieve this task:

1. Measure the output variable, **PV**.
2. Compare the measured value, **PV**, with the desired set point value, **SP**. Calculate the error, E(t), between the two values. The definition of error depends on the whether the controller is direct or reverse-acting.
3. Supply the error, E(t), to the general control equation. The value of the desired percent opening of the control valve, **OP%**, is calculated.
4. The value of **OP%** is passed to the final control element which determines the input to the process, U(t).
5. The entire procedure is repeated.

The general control equation for a **PID controller** is given by (11):

$$OP(t) = K_c E(t) + \frac{K_c}{T_i}\int E(t) + K_c T_d \frac{dE(t)}{dt} \qquad (11)$$

Where: $OP(t)$ = *controller output at time t*

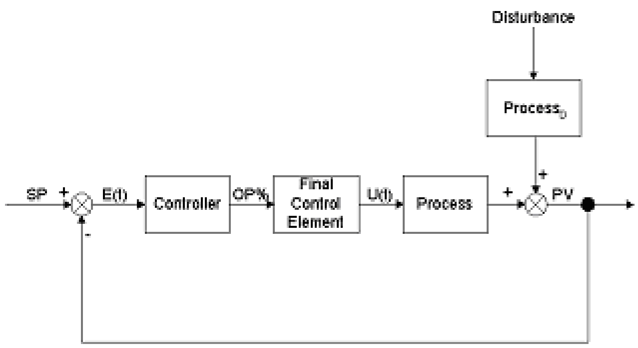
$E(t)$ = *error at time t*

K_c = *proportional gain of the controller*

T_i = *integral (reset) time of the controller*

T_d = *derivative (rate) time of the controller*

Figure 5



Direct and Reverse Acting

The input to the feedback controller is called the error. It is the difference between the output process variable and the set point. The error is defined differently depending on whether the process has a positive or negative steady state gain. For a process with a positive steady state gain, the error should be defined as reverse acting.

$$E(t) = SP(t) - PV(t) \tag{12}$$

Where: $SP(t)$ = *set point*

$PV(t)$ = *measured output process variable*

That is, if the **PV** *rises* above the **SP**, the **OP**, or input to the process, *decreases*. If the **PV** *falls* below the **SP**, the **OP** *increases*.

For a process with a negative steady state gain, the error should be set as direct acting (13):

$$E(t) = PV(t) - SP(t) \tag{13}$$

That is, if the **PV** *rises* above the **SP**, the **OP**, or input to the process, *increases*. If the **PV** *falls* below the **SP**, the **OP** *decreases*.

A typical example of a reverse-acting controller is in the temperature control of a reboiler. In this case, as the temperature in the vessel rises past the **SP**, the **OP** decreases, in effect closing the steam valve and reducing the flow of heat.

Think about the tank example presented at the beginning of this module. Assume that the flow out of the tank is controlled at a constant value and a PID controller is used to keep the level in the tank at a certain SP by opening or closing a valve on the inlet stream. Should the controller be Reverse or Direct acting?

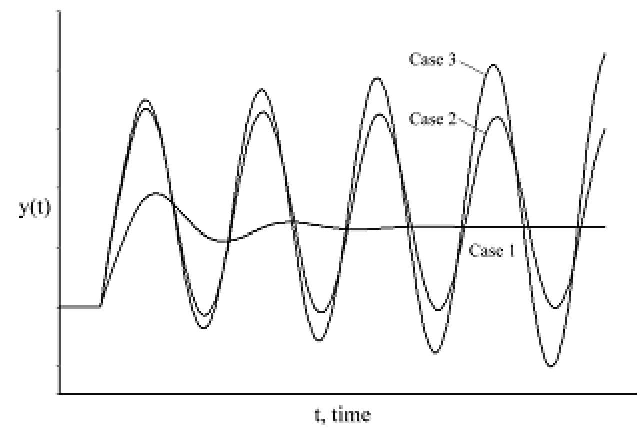
____

Stability

The stability of a system is a very important aspect to consider when designing control schemes. Many systems have oscillatory responses, depending on its controller tuning parameters. When a process is upset by a *bounded* disturbance or bounded change in the input forcing function, the output typically will respond in one of three ways:

1. The response will oscillate with decreasing amplitude and eventually reach steady state and stabilize.
2. The response will oscillate continuously with constant amplitude.
3. The response will grow continuously and never reach steady state conditions.

Figure 6



Stability Response Cases

The system is generally considered stable if the response proceeds to a steady state value and stabilizes. It is considered unstable if the response continues to fluctuate. A stable *open-loop* response is said to be *self-regulating*. If the open loop response of a system is not stable, it is said to be *non-self-regulating*. For instance, a pure integrating process, such as a tank with a pumped (constant) exit flow, is non-self-regulating since a bounded increase in the flow input to the system from steady state will result in the response (liquid height) increasing continuously.

A prerequisite for *closed-loop* control is that the closed-loop response is stable. The closed-loop response can vary considerably depending on the tuning parameters used in the feedback control equation. In general, a higher controller gain gives tighter control. However, the value of $\mathbf{K_c}$ cannot increase indefinitely. The response will remain stable up to a certain value of $\mathbf{K_c}$. Increasing $\mathbf{K_c}$ beyond the stability limit will cause the closed-loop response to become unstable.

A number of factors can affect the stability of a closed loop system:

- Tuning parameters
- Non-linearities in the process
- Range and non-linearities in the instruments
- Interactions between control loops
- Frequency of disturbance
- Capacity of process
- Noise in measurement of process variables

Available Control Operations

Modeling Hardware Elements

The plant may be simulated more accurately by modeling the hardware elements of the control loop. Non-linearities may be modeled in the Valve operation on the **Actuator** page of the **Dynamic**s tab.

Sensors

Sensors are used to measure process variables. In UniSim Design, the sensing instrument is incorporated directly in the **PID Controller** operation. You can choose the range of the sensing instrument in the **Minimum** and **Maximum PV** parameters in the controller operation. It is assumed in UniSim Design that the **PID controller** is perfectly accurate in its measurement of the process variable.

Final Control Element - Valve Dynamics

You have the option of specifying a number of different dynamic modes for the valve. If valve dynamics are very quick compared to the process, the instantaneous mode may be used. The following is a list of the available dynamic modes for the valve operation:

Valve Mode	Description
Instantaneous	In this mode, the actuator moves instantaneously to the desired **OP%** position from the controller.
Linear	The actuator can be modeled to move to the desire OP% at a constant rate. This rate is specified in the **Parameters** group.
First Order	A first order lag can be modeled in the response of the actuator position to changes in the desired OP%. The actuator time constant can be specified. Similarly, a first order lag can be modeled in the response of the actual valve position to changes in the actuator position. The valve **stickiness time constant** is specified in the **Parameters** group. In effect, a second order lag can be modeled between the valve position and the desire **OP%**.
Second Order	A second order lag can be modeled in the response of the actuator position to changes in the desired OP%. The actuator **time constant** and the **Damping Coefficient** can be specified.

Final Control Element - Valve Type

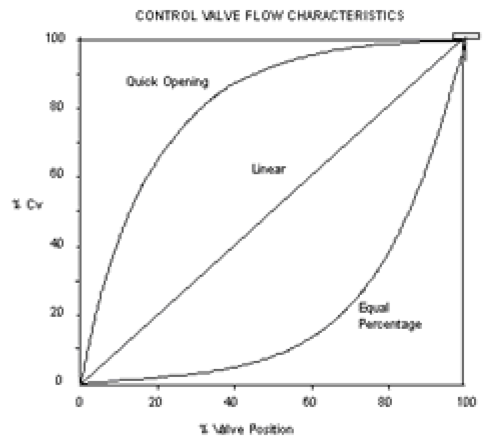
The flow rate through a control valve varies as a function of the valve percent opening and the valve type. Valve type may be defined more easily by expressing flow as a percentage of C_v (0% representing no flow conditions and 100% representing maximum flow conditions). The valve type can then be defined as the dependence on the quantity of % C_v as a function of the actual valve percent opening.

There are four different valve characteristics available in UniSim Design. The valve types are specified in the **Ratings** tab in the **Valve Operating Characteristics** group.

Valve Mode	Description
Linear	A control valve with linear valve characteristics has a flow which is directly proportional to the valve % opening.
Quick Opening	A control valve with quick opening valve characteristics obtains larger flows initially at lower valve openings. As the valve opens further, the flow increases at a smaller rate.
Equal Percentage	A control valve with equal percentage valve characteristics initially obtains very small flows at lower valve openings. However, the flow increases rapidly as the valve opens to its full position.
User Table	A control valve with valve characteristics based on the curve data from an existing working valve.

The valve characteristics for **Linear**, **Quick Opening** and **Equal Percentage** are shown graphically as follows:

Figure 7



Feedback Control

Digital On/Off

Digital On/Off control is one of the most basic forms of regulatory control. In UniSim Design, it is implemented using the **Digital Point** operation. An example of On/Off control is a home heating system. When the thermostat detects that the temperature is below the set point, the heating element turns on. When the temperature rises above the set point, the heating element turns off.

Control is maintained using a switch as a final control element (FCE). **On/Off control** parameters are specified in the **Parameters** page of the **Digital Point** operation in UniSim Design. If the **OP** is **ON** option is set to "**PV < Threshold**", the controller output turns on when the **PV** falls below the set point. This is similar to the thermostat example given above

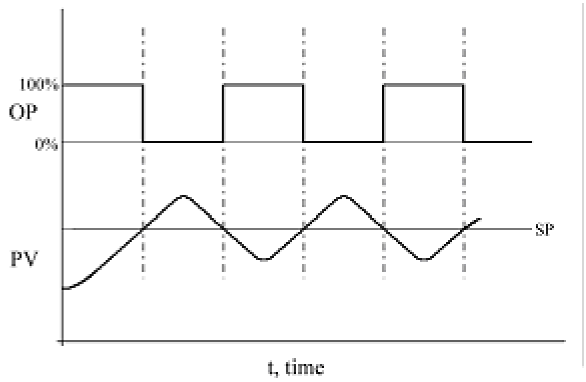
$$OP = 0\% \text{ for PV} > \text{SP and } OP = 100\% \text{ for PV} < \text{SP} \qquad (14)$$

The opposite is true when the **OP** is **ON** option is set to "**PV > Threshold**". This setting can be used for pressure relief valves; the valve is open (ON) when the **PV** is greater than the threshold pressure.

$$OP = 0\% \text{ for PV} < \text{SP and } OP = 100\% \text{ for PV} > \text{SP} \qquad (15)$$

One main characteristic of the **On/Off controller** is that the **PV** will always cycle about the set point.

Figure 8



On-Off Controller Response

Leaving the values for $\mathbf{T_I}$ and $\mathbf{T_D}$ as **<empty>** will also result in P-only control.

The cycling frequency will depend on the dynamics of the process. Those systems with a large capacity (large time constant) will cycle less frequently. The **On/Off controller** is an appropriate controller if the deviation from the set-point is within an acceptable range and the cycling does not destabilize the rest of the process.

Proportional Control (P-only)

Unlike On/Off control, proportional control can damp out oscillations from disturbances and stop the cycling of the process variable. P-only control is implemented in UniSim Design by setting the value of $\mathbf{T_D}$ to zero (or leaving it <empty>) and the value for $\mathbf{T_i}$ to a large value (1000*$\mathbf{K_c}$) in the **PID Controller** operation. With P-only control, oscillations that occur in the process variable due to disturbances or changes in the set point dampen out the quickest (have the smallest natural period) among all other simple feedback control schemes. The output of the proportional control is defined as (16):

$$OP(t) = OP_{ss} + K_c E(t) \tag{16}$$

The value of the bias, **OP_{ss}**, is calculated when the controller is switched to **Auto(matic)** mode. The set point is defaulted to equal the current **PV**. In effect, the error becomes zero and **OP_{ss}** is then set to the value of **OP(t)** at that time.

A sustained offset between the process variable and the set point will always be present in this sort of control scheme. The error becomes zero only if:

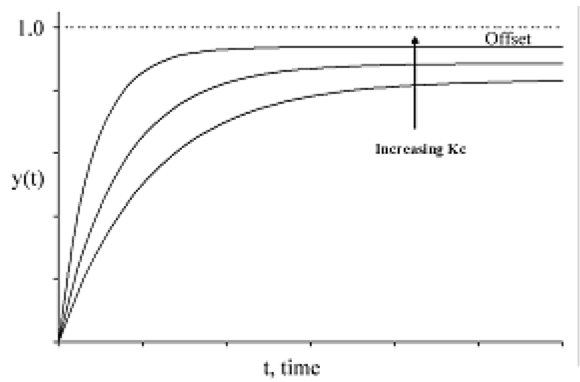
- the bias, **OP_{ss}**, equals the operating variable, **OP**
- **K_c** becomes infinitely large

However, **K_c** cannot practically become infinitely large. The magnitude of **K_c** is restricted by the stability of the closed loop system.

In general, a higher controller gain gives tighter control. However, the value of **K_c** cannot increase indefinitely. The response will remain stable up to a certain value of **K_c**. Increasing **K_c** beyond the stability limit will cause the closed loop response to become unstable.

The following shows the effect of the magnitude of **K_c** on the closed loop response of a first order system to a unit step change in the set point.

Figure 9



Closed loop response of a system under P-Only Control

Proportional only control is suitable when a quick response to a disturbance is required. P-only control is also suitable when steady state offsets are unimportant, or when the process possesses a large integrating process (has a large capacity). Many liquid level control loops are under P only control. If a sustained error is undesirable, integral action is required to eliminate the offset.

Proportional + Integral Control (PI)

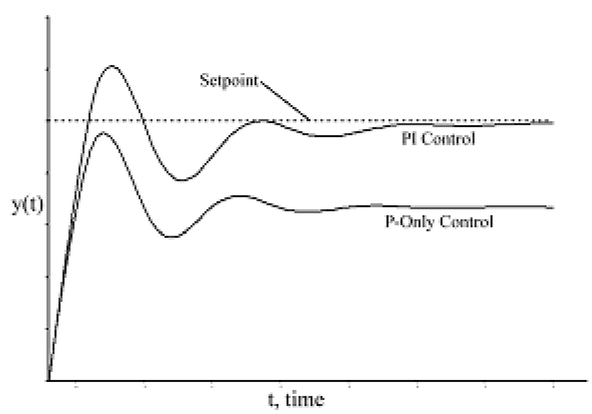
Unlike P-only control, proportional + integral control can dampen out oscillations and return the process variable to the set point. Despite the fact that PI control results in zero error, the integral action of the controller increases the natural period of the oscillations. That is, PI control will take longer to line out (dampen) the process variable than P-only control. The output of the proportional controller + integral controller is defined as (17):

Notice that the time that it takes to reach steady state is longer for the **PI ontroller**. The integral action slows the controller's response.

$$OP(t) = K_c E(t) + \frac{K_c}{T_i}\int E(t) \quad (17)$$

The integral term serves to bring the error to zero in the control scheme. The more integral action there is, the slower the response of the controller will be. The integral term continuously moves to eliminate the error. The closed loop response of a process with PI control and P-only control is shown as follows:

Figure 10



Proportional and PI Control

Due to the reciprocal effect, setting $\mathbf{T_I}$ to zero means that there will be an infinite integral effect. To minimize the integral effect, set T_I to a large value (1000*$\mathbf{K_C}$)

The integral time, $\mathbf{T_i}$, is defined as the amount of time required for the controller output to move an amount equivalent to the error. Because the relationship between $\mathbf{T_i}$ and the control action is reciprocal, increasing $\mathbf{T_i}$ will result in less integral action, while decreasing $\mathbf{T_i}$ will result in greater integral action. The integral time should be decreased (increased integral action) just enough to return the process variable to the set point. Any more action will only serve to lengthen the response time.

PI control is suitable when offsets cannot be tolerated. The majority of controllers in chemical process plants are under PI control. They combine accuracy (no offset) with a relatively quick response time. However, the added integral action acts as a destabilizing force which may cause oscillations in the system and cause the control system to become unstable. The larger the integral action the more likely it will become unstable.

Proportional Integral Derivative Control (PID)

If the response of a **PI controller** to a disturbance is not fast enough, the derivative action in a **PID controller** can reduce the natural period of oscillations even further. By measuring the rate of change in error, the controller can anticipate the direction of the error and thus respond more quickly than a controller without derivative action. The output of the proportional + integral + derivative controller is defined as (18):

$$OP(t) = K_c E(t) + \frac{K_c}{T_i}\int E(t) + K_c T_d \frac{dE(t)}{dt} \qquad (18)$$

$\mathbf{T_D}$ is defined as the time required for the proportional action to reach the same level as the derivative action. It is, in effect, a lead term in the control equation. For a ramped input, the proportional only response will be ramped, as well. For the same ramped input the derivative only response will be constant.

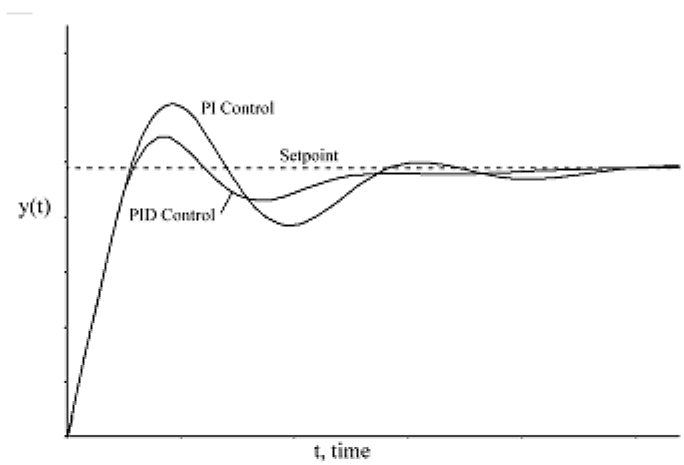
As the slope of the measured error increases to infinity, so does the derivative action. While a perfect step change with a slope of infinity in either the set-point or the measured process variable is not physically possible, signals which have short rise and fall times can occur. This adversely affects the output of the derivative term in the control equation, driving the controller response to saturation.

Derivative action cannot be used in systems where the **PV** signal will contain noise.

Derivative action control is best for processes which have little or no dead times and large capacities. Processes such as these, having large lags benefit from the additional response speed that derivative action provides. While the integral term in PID control schemes reduces the error to zero, it also adds a considerable lag to the response compared to P-only control. It is the derivative action in PID control which shortens the controller's response to be comparable to the response of a P-only controller. However, if a controller has a very noisy input which cannot be filtered or minimized in the process, PID control is not a suitable control scheme.

Notice that the time to steady state is shorter for the **PID controller** as compared to the PI control. This is due to the derivative action.

Figure 11



PI and PID Control

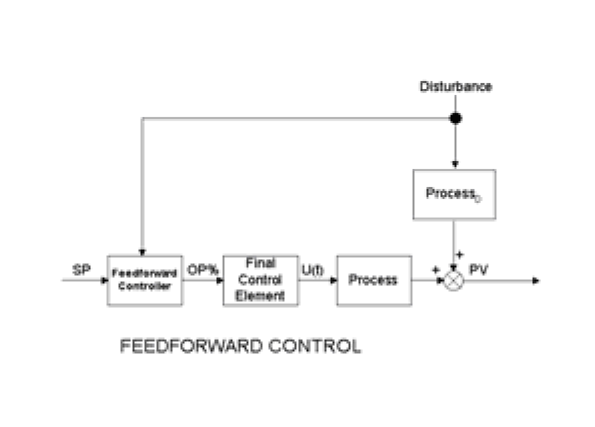
Feedforward Control

Feedforward control may be used in cases for which feedback control cannot effectively control a process variable. The main disadvantage of feedback control is that the controller must wait until disturbances upset the process before responding. With feedforward control, the controller can compensate for disturbances *before* the process is affected. Cascade control is useful when measured disturbances significantly affect the input to a process. On the other hand, feedforward control is useful if there are measured disturbances which affect the output of the process.

In UniSim Design feedforward control can be found on **Parameters** tab, **Feedforward** page of controllers.

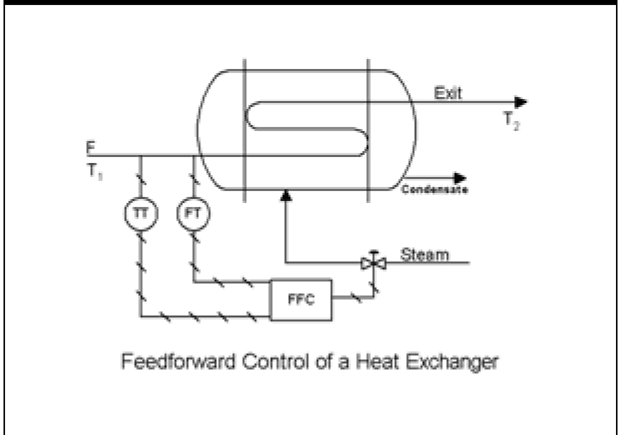
With feedback control, the controller requires information about the controlled process variable, **PV**, and the set point, **SP**, in order to determine the value of **OP%**, the desired valve percent opening of the input to the process. In order to determine the value of **OP%**, the feedforward controller requires information from two variables: the set-point of the process variable, **SP**, and the disturbance affecting the process. A steady state process model is used in the feedforward controller to determine the value of **OP%**.

Figure 12



Consider an example of a liquid stream being heated in a steam heat exchanger.

Figure 13



It is desired to control the exit stream temperature, T_2, at a certain set point (**SP**), using the steam flow as the manipulated variable. However, the process suffers from frequent changes in the feed temperature, T_1. In order to determine the value of **OP%**, the values of **SP** and T_1 are required by the controller. At steady state, the overall energy balance relates the steam flow to the disturbance of the process, T_1, and the temperature of stream exit, T_2 (19):

$$F_s\lambda - FC_p(T_2 - T_1) = 0 \tag{19}$$

Where: F_s = *the steam flow*

λ = *the heat of condensation for steam*

F = *the flow of stream exit*

C_p = *the specific heat of stream exit*

From this process model, the desired value of steam flow into the heat exchanger can be calculated. The flow of steam must be calculated such that the temperature of stream Exit, T_2, equals the desired temperature, **SP**. Therefore, Equation (5)(19) becomes (20):

$$F_s = \frac{Cp}{\lambda}F(SP - T_1) \tag{20}$$

In order to calculate the feedforward controller output, a linear relation is assumed to exist between the steam flow and the valve opening of the steam valve. Therefore the final form of the feedforward controller equation is (21):

$$OP(t) = \frac{C_p}{\lambda}F(SP - T_1)\frac{steam\ valve\ span}{100\%} \tag{21}$$

There are some points to consider in order to successfully implement a feedforward control system:

1. It cannot be implemented if the disturbance is not measurable. If unexpected disturbances enter the process when pure feedforward control is used, no corrective action is taken and the errors will build up in the system.
2. A fairly accurate model of the system is required.
3. The feedforward controller contains the reciprocal of the process model. Even if the process model is accurate, a time delay in the process model implies that a predictor is required in the feedforward controller. Unfortunately, it is impossible to predict the nature of disturbances before they occur.

It is important to note that the process variable to be controlled is not measured using feedforward control. There is no way of confirming that the process variable is attenuating disturbances or maintaining a desired set-point. Considering that an accurate model of the process is usually not available, that the process or valve dynamics are not accounted for in this control scheme, and that the valve opening percent is not related linearly to the flow in most dynamic simulation applications, there will probably be an offset between the actual controlled variable and its desired set-point. Therefore, feedback control is often used in conjunction with feedforward control to eliminate the offset associated with feedforward-only control.

Feedforward control in UniSim Design can be implemented using the Spreadsheet operation. Variables can be imported from the simulation flowsheet. A feedforward controller can be calculated in the spreadsheet and the controller output exported to the main flowsheet. If the operating variable, **OP**, is a valve in the plant, the desired controller output calculated by the **Spreadsheet** should be exported to the **Actuator Desired Position** of the valve.

General Guidelines

Effect of Characteristic Process Parameters on Control

The characteristic parameters of a process have a significant effect on how well a controller is able to attenuate disturbances to the process. In many cases, the process itself is able to attenuate disturbances and can be used in conjunction with the controller to achieve better control. The following is a brief discussion outlining the effect of capacity and dead time on the control strategy of a plant.

Capacity

The ability of a system to attenuate incoming disturbances is a function of the capacitance of a system and the period of the disturbances to the system. Attenuation is defined as:

$$\text{Attenuation} = 1 - \frac{K}{\sqrt{(\omega\tau)^2 + 1}} \tag{22}$$

The time constant, τ, is directly proportional to the capacity of a linear process system. The higher the capacity (time constant) is in a system, the more easily the system can attenuate incoming disturbances since the amplitude ratio decreases. The frequency of incoming disturbances affects the system's ability to attenuate these disturbances. High-frequency disturbances are more easily attenuated than low-frequency disturbances.

With capacity-dominated processes (with little or no dead time), proportional-only control can achieve much better disturbance rejection. The system itself is able to attenuate disturbances in the frequency range that the controller cannot. High frequency disturbances can be handled by the system. Low frequency disturbances are handled best with the controller.

Dead Time

The dead time has no effect on attenuating disturbances to open loop systems. However, it does have a significant negative effect on controllability. Dead time in a process system reduces the amount of gain the controller can implement before encountering instability. Because the controller is forced to reduce the gain, the process is less able to attenuate disturbances than the same process without dead time. Tight control is possible only if the equivalent dead time in the loop is small compared to the shortest time constant of a disturbance with a significant amplitude.

It is generally more effective to reduce the dead time of a process than increase its capacity. To reduce dead time:

- Relocate sensor and valves in more strategic locations
- Minimize sensor and valve lags (lags in the control loop act like dead time)

To reduce the lag in a system and therefore reduce the effects of dead time, you can also modify the controller to reduce the lead terms to the closed-loop response. This can be achieved by adding derivative action to a controller. Other model-based controller methods anticipate disturbances to the system and reduce the effective lag of the control loop.

Choosing the Correct Controller

You should consider what type of performance criteria is required for the set point variables, and what acceptable limits they must operate within. Generally, an effective closed loop system, is expected to be stable and cause the process variable to ultimately attain a value equal to the set point. The performance of the controller should be a reasonable compromise between performance and robustness.

A very tightly tuned or aggressive controller gives good performance but is not robust to process changes. It could go unstable if the process changes too much. A very sluggishly-tuned controller delivers poor performance but will be very robust. It is *not* likely to become unstable.

- If an offset can be tolerated, a proportional controller should be used
- If there is significant noise, or if there is significant dead time and/or a small capacity in the process, the PI controller should be used

- If there is no significant noise in the process, and the capacity of the system is large and there is no dead time, a PID controller may be appropriate

It is apparent why the **PI controller** is often the most common controller found in a plant. There are three possible conditions that a **PI controller** can handle, whereas the **PID controller** requires a specific set of conditions in order to be used effectively.

Choosing Controller Tuning Parameters

The following is a list of general tuning parameters appropriate for various processes [1]. Keep in mind that there is no single correct way of tuning a controller. The objective of control is to provide a reasonable compromise between performance and robustness in the closed loop response.

The following rules are approximate. They will provide you very close to tight control. You can adjust the tuning parameters further if the closed loop response is not satisfactory. Tighter control and better performance can be achieved by increasing the gain. Decrease in the controller gain results in a slower but more stable response.

Generally, proportional control can be considered the principal controller. Integral and derivative action should be used to trim the proportional response. Therefore, the controller gain should be tuned first with the integral and derivative actions set to a minimum. If instability occurs, the controller gain should be adjusted first. Adjustments to the controller gain should be made gradually.

Typical Controller Tuning Parameters:

These values are estimations only. They will generally provide adequate control under most circumstances.

System	K_C	Ti (minutes)	T_D (minutes)
Flow	0.1	0.2	<empty>
Level	2	10	<empty>
Pressure	2	2	<empty>
Temperature	1	20	<empty>

Flow Control

Since the flow control is fast responding, it can be used effectively as the secondary controller in a cascade control structure. The non-linearity in the control loop may cause the control loop to become unstable at different operating conditions. Since flow measurement is naturally noisy, derivative action is not recommended.

Most controllers must be tuned by taking the control loop dynamics into account.

Liquid Pressure Control

The liquid pressure loop is typically very fast. The process is essentially identical to the liquid flow process except that liquid pressure instead of flow is controlled using the final control element. The liquid pressure loop can be tuned for PI and Integral-only control, depending on your performance requirements.

Liquid Level Control

Liquid level control is essentially a single dominant capacity without dead time. In some cases, level control is used on processes which are used to attenuate disturbances in the process. In this case, liquid level control is not as important. Such processes can be controlled with a loosely tuned P-only controller. If a liquid level offset cannot be tolerated, PI level controllers should be used.

There is some noise associated with the measurement of level in liquid control. If this noise can be practically minimized, then derivative action can be applied to the controller. It is recommended that $\mathbf{K_c}$ be specified as 2 and the bias term, $\mathbf{OP_{ss}}$, be specified as **50%** for P-only control. This ensures that the control valve is wide open for a level of 75% and completely shut when the level is **25%** for a set-point level of 50%. If PI control is desired, the liquid level controller is typically set to have a gain, $\mathbf{K_c}$, between **2** and **10**. The integral time, $\mathbf{T_i}$,should be set between **1** and **5** minutes.

Gas Pressure Control

Gas pressure control is similar to the liquid level process in that it is capacity dominated without dead time. Varying the flow into or out of a vessel controls the vessel pressure. Because of the capacitive nature of most vessels, the gas pressure process usually has a small process gain and a slow response. Consequently, a high controller gain can be implemented with little chance of instability.

Temperature Control

PI controllers are widely used in industry; however PID control can be used to improve the response time if the loop is slow.

Tuning Methods

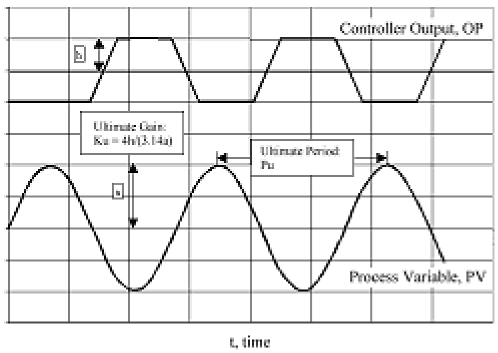
An effective means of determining controller tuning parameters is to bring the closed-loop system to the verge of instability. This is achieved by attaching a P-only controller and increasing the gain such that the closed-loop response cycles with constant amplitude. At a system's stability margins, there are two important system parameters, the *ultimate period* ($\mathbf{P_u}$) and the *ultimate gain* ($\mathbf{K_u}$), which allow the calculation of appropriate proportional gain, integral time, and derivative time values.

ATV Tuning Technique

The **ATV (Auto Tuning Variation)** technique is used for processes which have significant dead time. A small limit cycle disturbance is set up between the manipulated variable (**OP%**) and the controlled variable (**PV**). The ATV tuning method is as follows:

1. Determine a reasonable value for the **OP%** valve change (h = fractional change in valve position).
2. Move valve +h%.
3. Wait until process variable starts moving, then move valve -2h%.
4. When the process variable (**PV**) crosses the set point, move the valve position +2h%.
5. Continue until a limit cycle is established.
6. Record the amplitude of the response, A. Make sure to express A as a fraction of the **PV** span.

Figure 14



ATV Stability Limit Parameters

7. The tuning parameters are calculated as follows:

Tuning Parameter	Equation
Ultimate Gain	$K_u = \dfrac{4h}{\pi a}$
Ultimate Period	P_u = Period taken from limit cycle
Controller Gain	$K_c = \dfrac{K_u}{3.2}$
Controller Integral Time	$T_i = 2.2 \times P_u$

Ziegler-Nichols Tuning Technique

The **Ziegler-Nichols** [2] tuning method is another method which calculates tuning parameters. The Z-N technique was originally developed for electromechanical system controllers and is based on a more aggressive “quarter amplitude decay” criterion. The Z-N technique can be used on processes without dead time. The procedure is as follows:

1. Attach a proportional only controller (no integral or derivative action).
2. Increase the proportional gain until a limit cycle is established in the process variable, **PV**.
3. The tuning parameters are calculated as follows:

Tuning Parameter	Equation
Ultimate Gain	K_u = Controller gain that produces limit cycle
Ultimate Period	P_u = Period taken from limit cycle
Controller Gain	$K_c = \frac{K_u}{2.2}$
Controller Integral Time	$T_i = P_u / 1.2$

Autotuner

In the current version of UniSim Design there are default values specified for the PID tuning. Before starting the Autotuner you must ensure that the controller is in the Man(ual) or Auto(matic) mode and the process is relatively steady.

If you move the cursor over the tuning parameters field, the Status Bar will display the parameters range.

The Unisim Design **Autotuner** function provides tuning parameters for the **PID controller** based on gain and phase margin design. The **Autotuner** itself can be viewed as another controller object that has been embedded into the **PID controller**. The **Autotuner** is based on a relay feedback technique and by default incorporates a relay with hysteresis.

The **PID controller** parameters that are obtained from the **Autotuner** are based on a design methodology that makes use of a gain margin at a specified phase angle. This design is quite similar to the regular gain and phase margin methodology except that it is more accurate since the relay has the ability to determine points in the frequency domain accurately and quickly. Also, the relay experiment is controlled and does not take a long time during the tuning cycle.

In the present **Autotuner** implementation there are four parameters that you must supply which are as follows:

Tuning Parameter	Equation
Ratio (T_i/T_D) (Alpha)	$3.0 \le \alpha \le 6.0$
Gain ratio (Beta)	$0.10 \le \beta \le 1.0$
Phase angle (phi)	$30° \le \phi \le 65°$
Relay hysteresis (h)	$0.01\% \le h \le 5.0\%$
Relay amplitude (Amplitude)	$0.5\% \le d \le 10.0\%$

Setting up a Control Strategy in UniSim Design

This section outlines a possible way to create a control strategy in UniSim Design. You should first follow the guidelines outlined in Section 1.5.2 - Moving from Steady State to Dynamics in the Dynamic Modeling manual in order to setup a stable dynamic case. In many cases, an effective control strategy will serve to stabilize the model.

You can install controllers in the simulation case either in Steady State or Dynamics mode. There are many different ways to setup a control strategy. The following is a brief outline of some of the more essential items that should be considered when setting up controllers in UniSim Design.

PID and **Digital On/Off controllers** are not active while UniSim Design is in **Steady State** mode.

1. Select the Controlled Variables in the Plant

Plan a control strategy that is able to achieve an overall plant objective and maintain stability within the plant. Either design the controllers in the plant according to your own standards and conventions or model a control strategy from an existing plant. In UniSim Design, there are a number of variables which can either be set or controlled manually in a dynamic simulation case. You should distinguish between variables that do not change in a plant and those variables which are controlled.

Set variables do not change in the dynamic simulation case. Variables such as temperature and composition should be set at each flowsheet boundary feed stream. One pressure-flow specification is usually required for each flowsheet boundary stream in the simulation case. These are the minimum number of variables required by the simulation case for a solution. These specifications should be reserved for variables that physically remain constant in a plant. For example, you can specify the exit pressure of a pressure relief valve since the exit pressure typically remains constant in a plant.

In some instances, you can vary a set variable such as a stream's temperature, composition, pressure or flow. In order to force a specification to behave sinusoidally or follow a ramp, you can attach the variable to the Transfer Function operation. A variety of different forcing functions and disturbances can be modeled in this manner.

The behavior of controlled variables is determined by the type of controller and the tuning parameters associated with the controller. Typically, the number of control valves in a plant dictates the possible number of controlled variables. There will be more variables to control in Dynamics mode than in Steady State mode. For instance, a two-product column in Steady State mode requires *two* steady state specifications. The simulator will then manipulate the other variables in the column in order to satisfy the provided specifications and the column material and energy balances. The same column in Dynamics mode requires *five* specifications. The three new specifications correspond to the inventory or integrating specifications that were not fixed in steady state. The inventory variables include the condenser level, the reboiler level, and the column pressure.

A good controller strategy includes the control of both integrating variables and steady-state variables. By maintaining the integrating variables at specified set points, controllers add stability to the plant. Other controllers maintain the desired steady state design specifications such as product composition and throughput.

2. Select Controller Structures for Each Controlled Variable

You should choose appropriate controller structures for each controlled variable in the simulation case.

The controller operations can be added in either Steady State or Dynamics mode. However, controllers have no effect on the simulation in Steady State mode. You must specify the following in order to fully define the PID Controller operation.

Connections Tab

Process Variable (PV)

The process variable can be specified in the **Connections** tab by clicking the **Select PV...** button. The controller measures the process variable in an attempt to maintain it at a specified set point, **SP**.

Operating Variable (OP)

The operating variable, **OP**, can be specified in the **Connections** tab by clicking the **Select OP...** button. The output of the controller is a control valve. The output signal, **OP**, is the percent opening of the control valve. The operating variable may be specified as a physical valve in the plant, a material stream, or an energy stream.

The output of a controller is always a control valve, unless the controller is the primary controller in a cascade control setup.

Operating Variable	Description
Physical Valve	It is recommended that a physical valve be used as the operating variable for a controller. The controller's output signal, **OP**, is the desired actuator position of the physical valve. With this setup, a more realistic analysis of the effect of the controller on the process is possible. Material flow through the valve is calculated from the frictional resistance equation of the valve and the surrounding unit operations. Flow reversal conditions are possible and valve dynamics may be modeled if a physical valve is chosen.
Material Stream	If a material stream is chosen as an operating variable, the material stream's flow becomes a P-F specification in the dynamic simulation case. You must specify the maximum and minimum flow of the material stream by clicking the **Control Valve** button. The actual flow of the material stream is calculated from the formula: $Flow = \frac{OP(\%)}{100}(Flow_{max} - Flow_{min}) + Flow_{min}$ (23) UniSim Design varies the flow specification of the material stream according to the calculated controller output, **OP**. (Therefore, a non-realistic situation may arise in the dynamic case since material flow is not dependent on the surrounding conditions.)

It is possible to have a flow reversal occur in a valve if the pressure drop across the valve becomes negative. The flow reversal can be avoided by enabling the **Check Valve option.**

Operating Variable	Description
Energy Stream	If an energy stream is chosen as an operating variable, you may choose a Direct Q or a Utility Fluid Duty Source by clicking the **Control Valve** button. If the Direct Q option is chosen, you must specify the maximum and minimum energy flow of the energy stream. The actual energy flow of the energy stream is calculated similarly to the material flow: $$Energy\ Flow = \frac{OP(\%)}{100}(Flow_{max} - Flow_{min}) + Flow_{min} \quad (24)$$ If the **Utility Fluid** option is chosen, you need to specify the maximum and minimum flow of the utility fluid. The heat flow is then calculated using the local overall heat transfer coefficient, the inlet fluid conditions, and the process conditions.

Parameters Tab

The action of the controller, the controller's PV range, and the tuning parameters can be specified on the **Parameters** tab.

A controller's action (**Direct** or **Reverse**) is specified using the **Action** radio buttons.

A controller's PV span is also specified in the **PV Range** field. A controller's PV span must cover the entire range of the process variable the sensor is to measure.

Tuning parameters are specified in the **Tuning** field.

3. Final Control Elements

Set the range on the control valve at roughly twice the steady state flow you are controlling. This is achieved by sizing the valve as **Linear** with an opening of 50% at the steady state pressure drop and flow rate. If the controller uses a material or energy stream as an operating variable (**OP**), the range of the stream's flow can be specified explicitly in the **Control OP Port** view of the material or energy stream. This view is displayed by clicking on the **Control Valve** button in the PID Controller view.

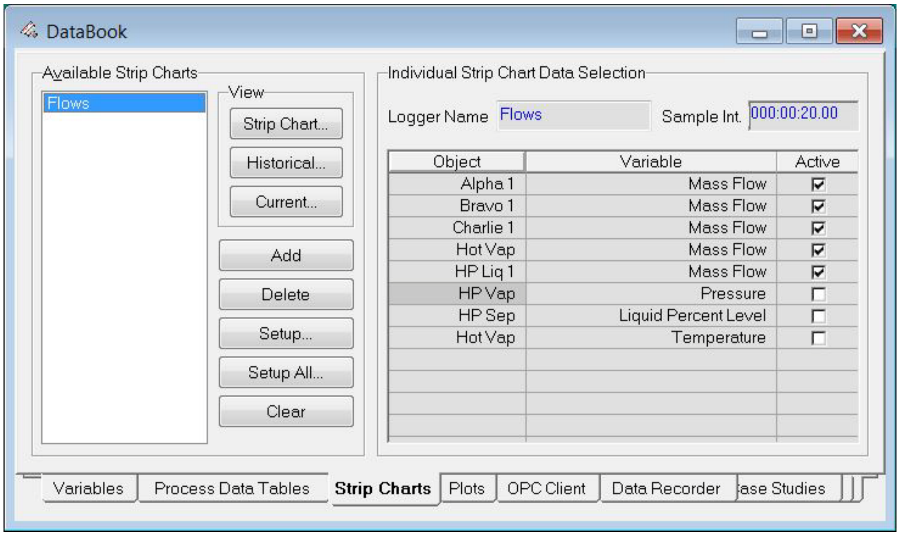
The final control element can be characterized as a **linear**, **equal percentage**, or **quick opening valve**. Control valves also have time constants which can be accounted for in UniSim Design. It is suggested that a linear valve mode be used to characterize the valve dynamics of final control elements. This causes the actual valve position to move at a constant rate to the desired valve positions much like an actual valve in a plant. Since the actual valve position does not move immediately to the **OP%** set by the controller, the process is less affected by aggressive controller tuning and may be more stable.

4. Set up the Databook and Strip Charts

Setting up strip charts for your model allows you to easily view several variables while the simulation is running. The procedure for setting up these charts is straightforward.

1. Open the **Databook** window and select the variables that are to be included in the strip chart in the **Variables** tab.

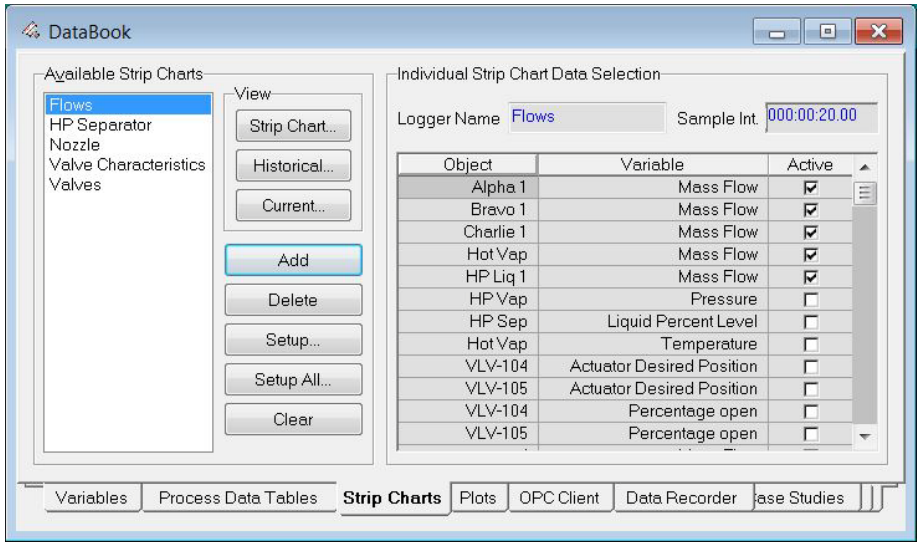
Figure 15



2. From the **Strip Charts** view, add a new strip chart by clicking the **Add** button and activating the variables to be displayed on the strip chart.

No more than six variables should be active on any given strip chart, having more than sic active variables will make the strip chart difficult to read.

Figure 16



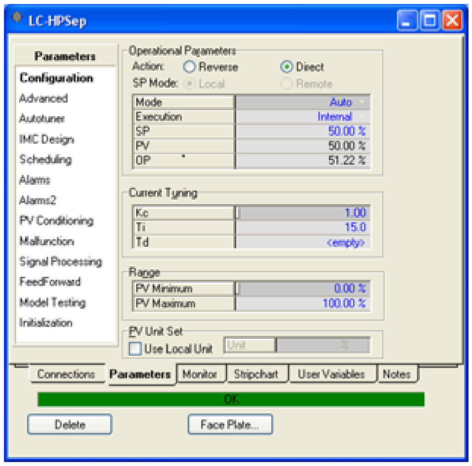
Click on the **Strip Chart...** button in the **View** group box to see the strip chart. Size as desired and then right click on the strip chart and choose **Graph Control**. There are several tabs, where you can set the numerical ranges of the strip chart for each variable, the nature of the lines for each variable and how the strip chart updates and plots the data.

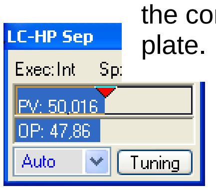
Add additional strip charts as desired using the **Databook**.

5. Set up the Controller Face Plates

Click on the **Face Plate** button in the **PID Controller** view to display the controller's face plate. The face plate displays the **PV**, **SP**, **OP**, and **Mode** of the controller. Controller face plates can be arranged in the UniSim Design work environment to allow for monitoring of key process variables and easy access to tuning parameters.

Figure 17





Clicking the **Face Plate** button opens the controller face plate.

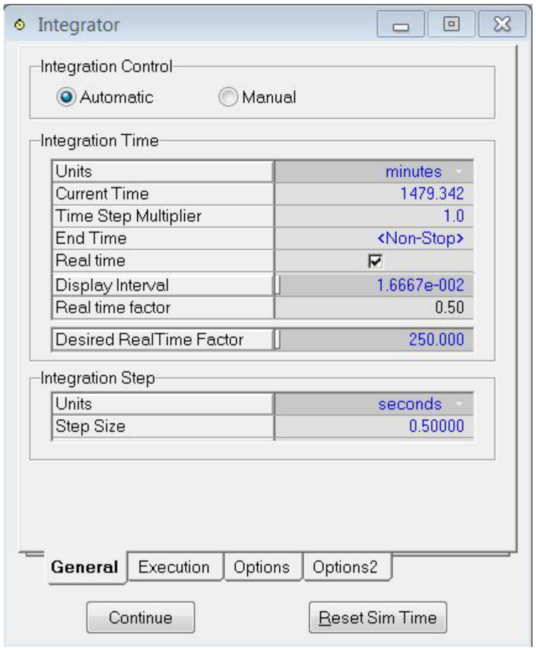
The set point can be edited directly by typing a new value, or move the red slider

6. Set up the Integrator

During start up of a dynamic simulation, it may be useful to use a small step size. However, once the system has stabilized, a larger value can be used.

The integration step size can be modified on the **Integrator** window accessible from the **Simulation** menu. If desired, change the integration step size to a smaller interval. The default integration time step is 0.5 seconds. Reducing the step size will cause the model to run slower, but during the initial switch from **Steady State** to **Dynamics** mode, the smaller step size allows the system to initialize better and enables close monitoring of the controllers to ensure that everything was set up properly. A smaller step size also increases the stability of the model since the solver can more closely follow changes occurring in the plant. Increase the integration step size to a reasonable value when the simulation case has achieved some level of stability. Larger step sizes increase the speed of integration and may be specified if the process can maintain stability.

Figure 18



7. Fine Tuning of Controllers

Before the Integrator is run, each controller should be turned off and then put back in manual mode. This will initialize the controllers. Placing the controllers in manual will default the set-point to the current process variable and allow you to “manually” adjust the **Valve % Opening** of the operating variable.

If reasonable pressure-flow specifications are set in the dynamic simulation and all the equipment is properly sized, most process variables should line out once the Integrator is run. The transition of most unit operations from **Steady State** to **Dynamics** mode is very smooth. However, controller tuning is critical if the plant simulation is to remain stable. Dynamic columns, for instance, are not open loop stable like many of the unit operations in UniSim Design. Any large disturbances in the column may result in simulation instability.

Once the Integrator is running:

1. Slowly bring the controllers on-line starting with the ones attached to upstream unit operations. The control of flow and pressure of upstream unit operations should be handled initially since these variables have a significant effect on the stability of downstream unit operations.

2. Concentrate on controlling variables critical to the stability of the unit operation. Always keep in mind that upstream variables to a unit operation should be stabilized first. For example, the feed flow to a column should be controlled initially. Next, try to control the temperature and pressure profile of the column. Finally, pay attention to the accumulations of the condenser and reboiler and control those variables.
3. Start conservatively using low gains and no integral action. Most unit operations can initially be set to use P-only control. If an offset cannot be tolerated initially, then integral action should be added.
4. Trim the controllers using integral or derivative action until satisfactory closed-loop performance is obtained.
5. At this point, you can concentrate on changing the plant to perform as desired. For example, the control strategy can be modified to maintain a desired product composition. If energy considerations are critical to a plant, different control strategies may be tested to reduce the energy requirements of unit operations.

Stability

It has been shown that the stability of a closed loop process depends on the controller gain. If the controller gain is increased, the closed loop response is more likely to become unstable. The controller gain, $\mathbf{K_c}$, input in the **PID Controller** operation in UniSim Design is a unit less value defined in Equation (25):

$$K_c = \frac{OP\%}{\Delta PV / PV \text{ Range}} \tag{25}$$

In order to control the process, the controller must interact with the actual process. This is achieved by using the effective gain, K_{eff}, which is essentially the controller gain with units. The effective gain is defined as Equation (26):

$$K_{eff} = \frac{K_c(Flow_{max} - Flow_{min})}{PV \text{ Range}} \tag{26}$$

The stability of the closed-loop response is not only dependent on the controller gain, K_c, but also on the PV range parameters provided and the maximum flow allowed by the control valve. Decreasing the PV range increases the effective gain, K_{eff}, and therefore decreases the stability of the overall closed-loop response. Decreasing the final control element's flow range decreases the effective gain, K_{eff}, and therefore increases the stability of the closed-loop response.

The process gain has units which are reciprocal to the effective gain.

It is therefore possible to achieve tight control in a plant and to have the simulation case become unstable due to modifications in the PV range or C_v values of a final control element.

You should also consider the effect of interactions between the control loops existing in a plant. Interactions between the control loops change the effective gain of each loop. It is possible for a control loop that was tuned independently of the other control loops in the plant to become unstable as soon as it is put into operation with the other loops. It is therefore useful to design feedback control loops which minimize the interactions between the controllers.

References

[1] Svrcek, W.Y., Mahoney, D.P., and B.R. Young. A Real Time Approach to Process Controls John Wiley & Sons Ltd, Chichester (2000) p. 125

[2] Ogunnaike, B.A. and W.H. Ray. Process Dynamics, Modelling, and Control Oxford University Press, New York (1994) p. 53

5. Dynamic Details

4528.05.R440.01

Workshop

This module examines some of the detailed parameters available in UniSim Design. These details include actuator characteristics, nozzle locations and heat loss parameters.

Starting with the dynamic model that you prepared in Module 3, you will add the necessary information to provide for a more rigorous simulation.

Learning Objectives

Once you have completed this module, you will be able to:

- add valve characteristics
- add heat loss models
- understand nozzle location
- make changes/additions to the dynamic model in dynamics mode

Prerequisites

Before beginning this section you need to:

- know how to set up Strip Charts
- understand the pressure-flow network

Valve Characteristics

The following observations are made about the valves in the dynamic model constructed in Module 3:

- The valves on all the feeds go from fully open to closed in 1 minute.
- The valve on **Charlie** is not responding instantly; it has a 5 second time constant.
- The valve on **Bravo** does not close completely; it has a leakage flow equal to a 2% valve opening position.
- The **HP Sep** liquid valve (**VLV-100**) is an Equal Percentage Valve.

UniSim Design has the ability to model each of the above observations.

Adding New Valves

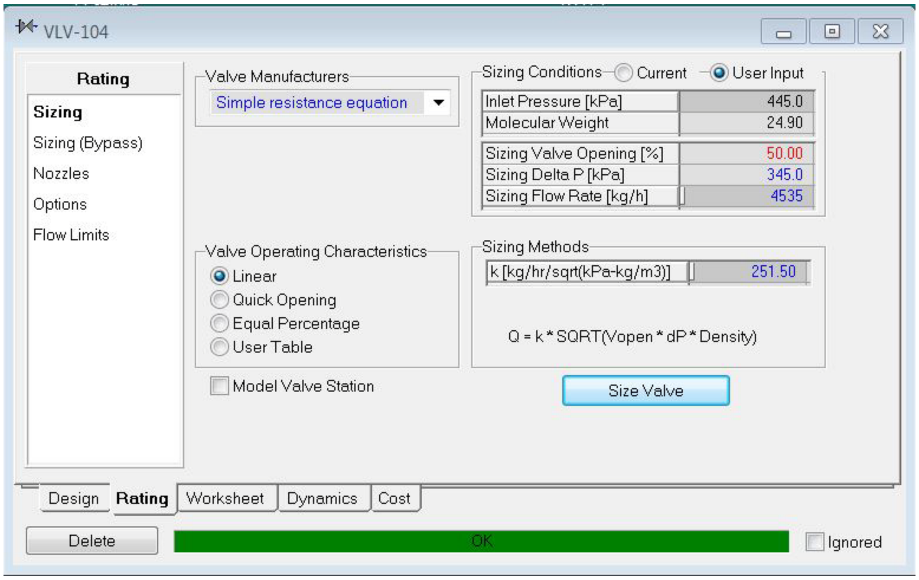
Prior to entering the appropriate information, we will investigate the valve options using a simple two valve model.

1. Open the file **4528.03.Dyn1.usc**. This model should be in dynamic mode with all process variables lined out at the desired set points.

Notice that when streams are added in dynamics mode they are fully defined.

2. Add two new valves to the process flowsheet: **VLV-104** and **VLV-105**. Create new feed streams (streams **1**, **3**) and product streams (**2**, **4**) for them.
3. Use the **Define from Other Stream** button on the newly added feed streams to define them both from stream **Alpha.**
4. Enter a dynamic pressure specification of **445 kPa (64.5 psia)** for the feed streams and **101.3 kPa (14.7 psia)** for the product streams.
5. Size the valves, by using the **User Input** radio button on the **Sizing** page on the **Rating** tab, to handle a **4535 kg/h** (**9998 lb/hr**) flow with a **345 kPa (50 psi)** pressure drop and a **Valve Opening [%]** of **50%**.

Figure 1



Add a flow controller to each of the two valves. The Process Variables are the mass flow through the valve and the Output parameters are the **Actuator Desired Position**s of the valves. Provide controller tuning and ranges according to the table below and place the controllers in **Auto(matic)** mode.

In this cell…	Enter…	Enter…
Connections		
Name	FIC-104	FIC-105
Process Variable Source	1 – Mass Flow	3 – Mass Flow
Output Target Object	VLV-104	VLV-105
Parameters		
Action	Reverse	As FIC-104
Range PV Minimum	0 kg/h (0 lb/hr)	
Range PV Maximum	9000 kg/h (19800 lb/hr)	
OP	50%	
Kc	0.25	
Ti	0.1 min	
Td	<empty>	

Actuator Linear Rate

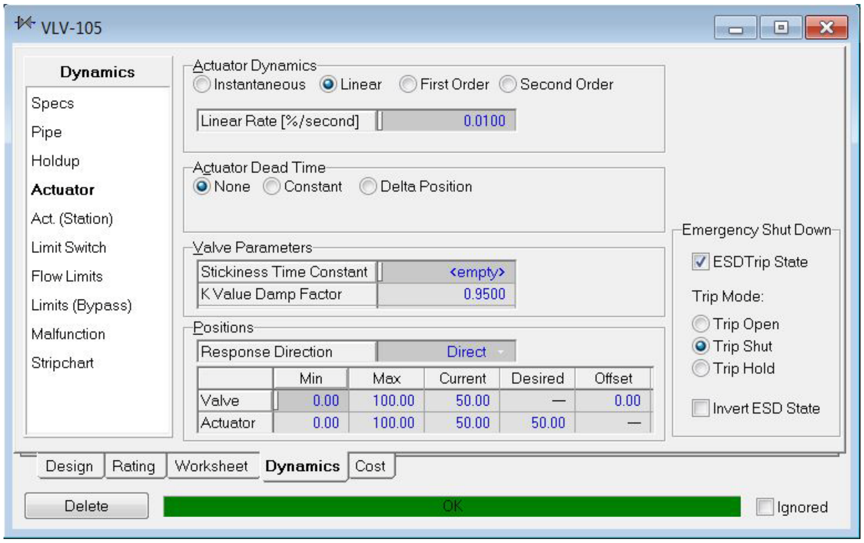
The **Actuator** page, located on the **Dynamics** tab of the Valve property view, allows you to model the valve dynamics. This page also contains information regarding the dynamic parameters of the valve and the open positions of the actuator and the valve. In reality, changes that occur in the actuator are not observed instantaneously in the valve. Moreover, changes in the output signal of a controller, OP, do not instantaneously translate to changes in the actuator. Because the actuator and valve are physical items, they take time to move to their respective desired positions. This causes dynamic behavior in actual control valves.

The actuator mode defines the relationship between the desired actuator position and current actuator position. The desired actuator position can be set by a PID Controller or Spreadsheet operation. A controller's **output parameter** (**OP**) for instance, is exported to the desired actuator position. Depending on the valve mode, the current actuator position can behave in one of the following four ways:

- Instantaneous Mode
- Linear Mode
- First Order Mode
- Second Order Mode

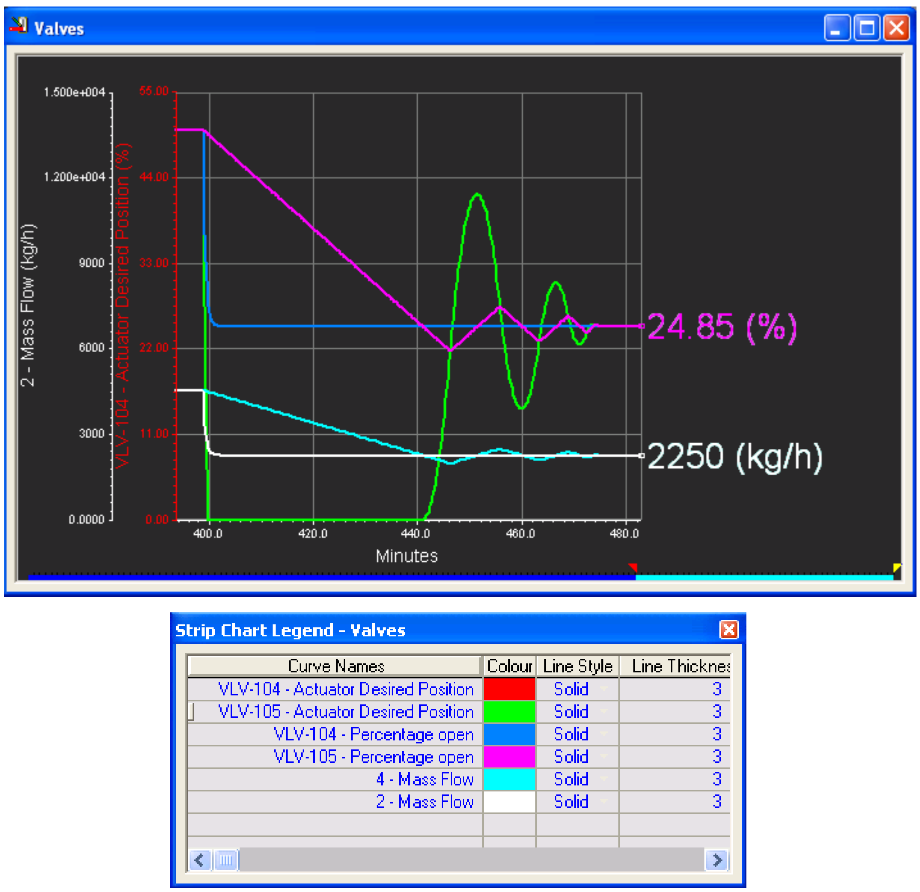
These are set in the **Actuator Dynamics** group of the valve.

Figure 2



1. Open up the view for **VLV-105**. Go to the **Dynamics** tab **Actuator** page.
2. Change the mode to **Linear** and enter the rate of **0.01 %/second** for the **Linear Rate.**
3. Set up a strip chart to monitor the **Actuator Desired Position**, the **Percentage Open** for each of the two valves, and the **Mass Flow** of each of the two product streams. (**Set-up Logger** 3600 samples and 5 sec for the **Sample Interval**). Run the Integrator for a few seconds.
4. Change the set points for **FIC-104** and **FIC-105** to **2250 kg/h (5000 lb/hr)**.
5. Start the Integrator and observe the response (Figure 2).

Figure 3

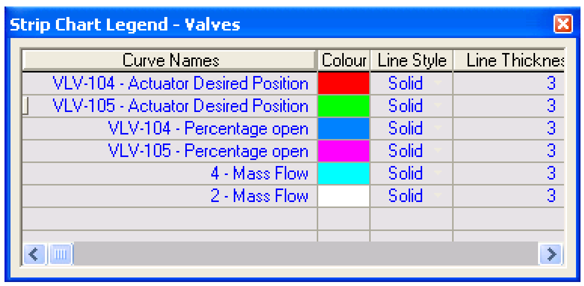


The feed valves go from fully open to closed in 1 minute. What should be the desired Actuator Linear Rate value for Alpha, Bravo and Charlie control valves?

____

6. Set the **Actuator Linear Rate** for **VLV-104** and **VLV-105** such that the valves go from fully closed to fully open in 1 minute (as calculated above), and change the set points for **FC-104** and **FC-105** back to **4535 kg/h (10000 lb/hr)**.
7. Start the Integrator and observe the response.
8. Enter the same **Actuator Linear Rates** calculated in step 6 for **Alpha**, **Bravo** and **Charlie** control valve actuators.

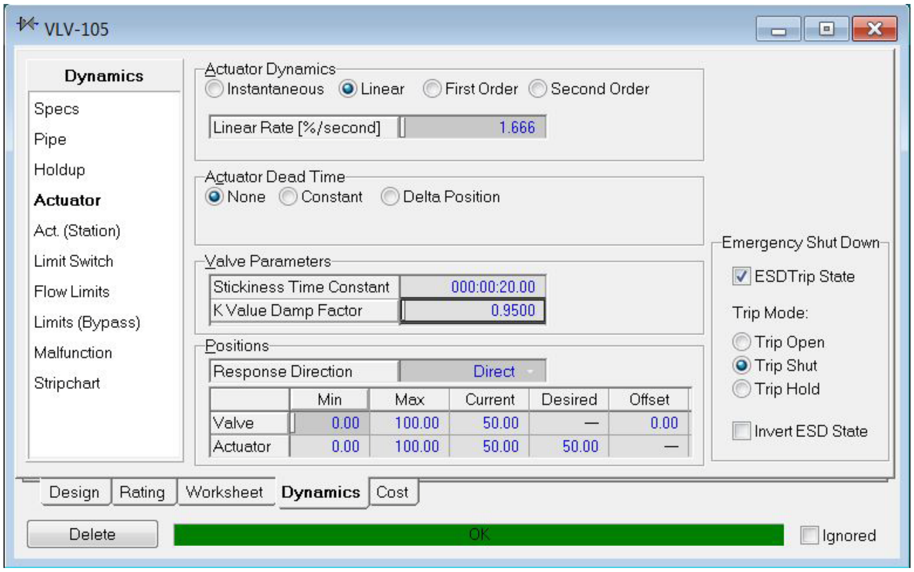
Figure 4



Stickiness Time Constant

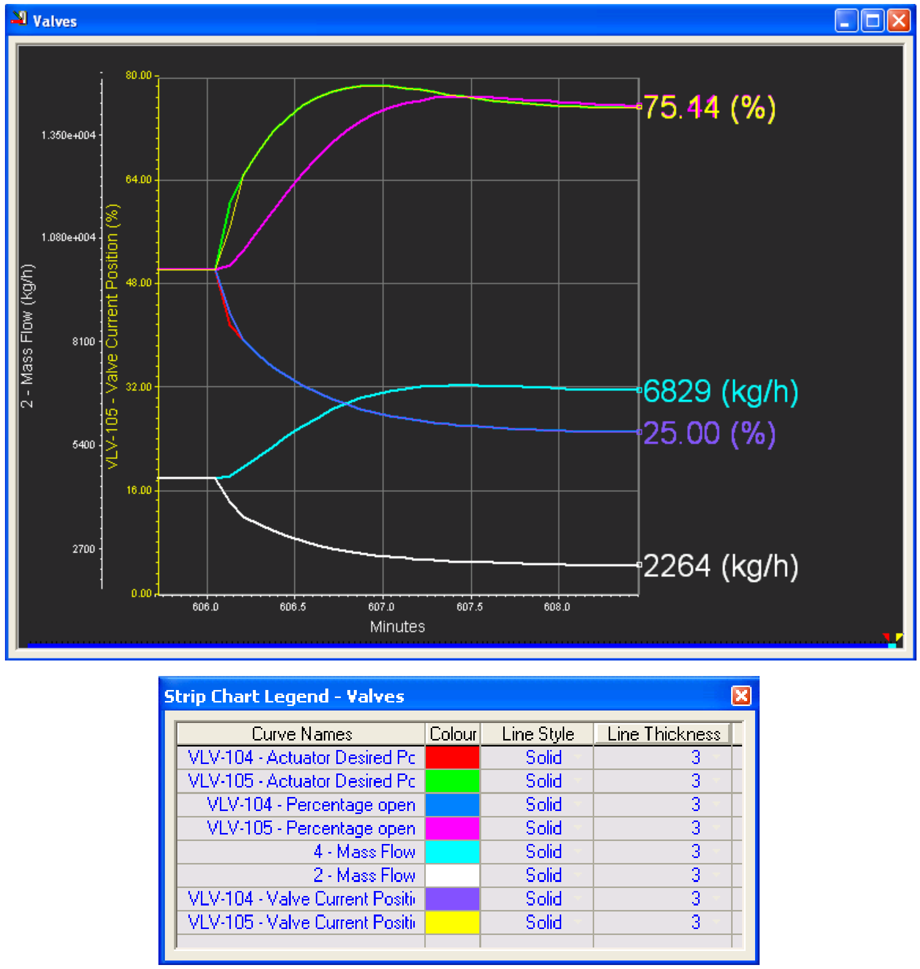
In reality, a valve does not respond instantaneously to changes in the actuator. A **Valve Stickiness Time Constant** can be used to model the actual response of the valve position to changes in the actuator position. This feature allows you to specify the time constant used to model the time offset caused by a sticky actuator. The offset can be specified on the **Actuator** page of the **Dynamics** tab in the Valve property view (Figure 3).

Figure 5



1. Set up **VLV-104** and **VLV-105** so that they are both linear with a rate of **1.666 %/second**, and enter a time constant of **20 seconds** on the **Stickiness Time Constant** cell of **VLV-105**.
2. Change the set points of **FC-104** and **FC-105** to **2250 kg/h (5000 lb/hr)** and **6820 kg/h (15000 lb/hr)** respectively.
3. Edit the Strip Chart to include the **Actuator Current Position** for both valves.
4. Start the Integrator and observe the response (Figure 6).

Figure 6



5. The valve on stream **Charlie** is not responding instantly but is has a 5 second **Stickiness Time Constant**. Make this change to valve **VLV-103**.

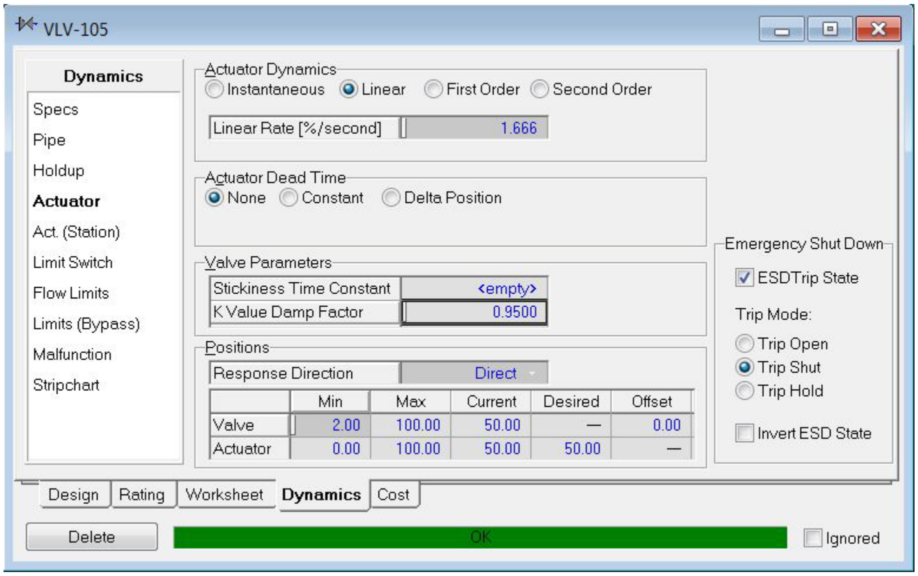
Leaky Valves

Leaky valves can be modeled by specifying a non-zero value for the minimum valve position.

1. Make valves **VLV-104** and **VLV-105** identical (remove the **Stickiness Time Constant** for **VLV-105)**, change the **Set Points** of both flow controllers to **4535 kg/h (10000 lb/hr)** and run the Integrator until the new mass flow set point is reached.

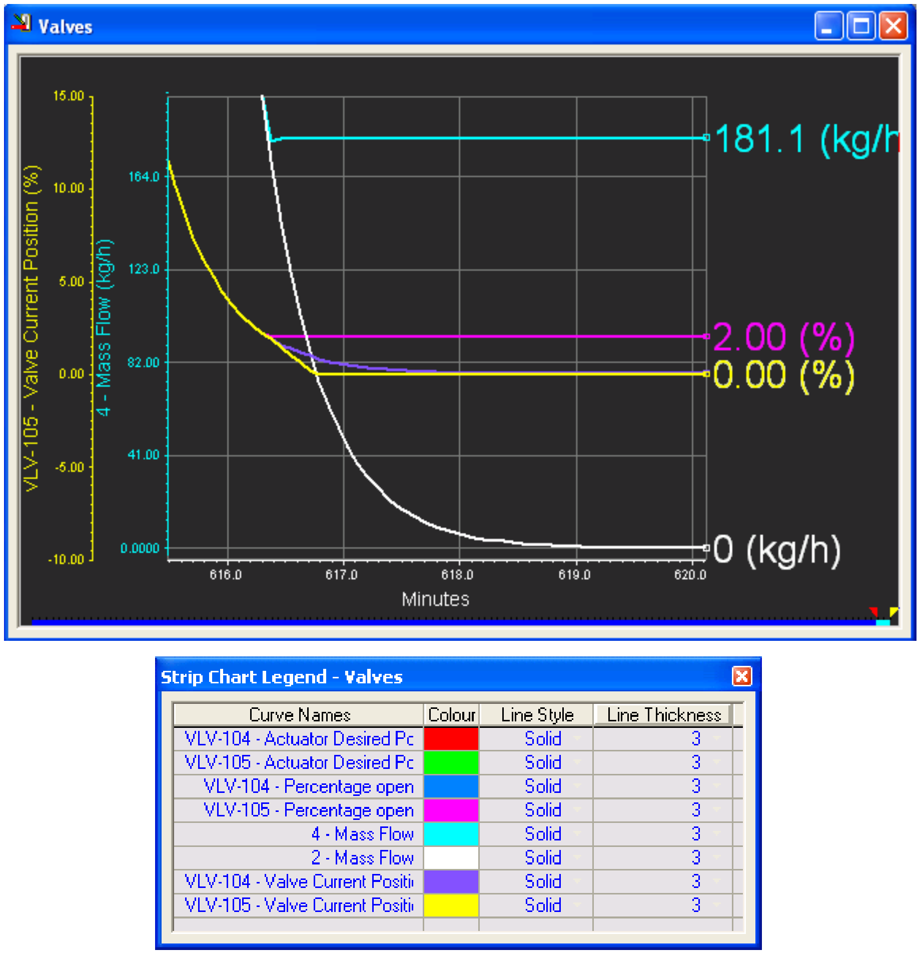
2. Enter **2%** for the minimum valve position for valve **VLV-105**.

Figure 7



3. Change the **Set Points** for the **FC-104** and **FC-105** controllers to **0 kg/h** and observe the response.

Figure 8



4. The valve on **Bravo** does not close completely; it has a leakage flow equal to a 2% valve opening position. Make the appropriate changes to the model.

Inherent Flow Characteristics

A general equation to describe the flow through a valve can be written as follow:

$$Q = Cv \cdot f(x) \cdot \sqrt{\frac{\Delta p}{sg}} \qquad (1)$$

Where:

	Q	=	*flow rate*
	Cv	=	*valve flow coefficient*
	f(x)	=	*inherent flow characteristic* *if equal to 1 means valve is 100% open*
	x	=	*variable, position of the tap*
	Δp	=	*pressure drop across the valve*
	sg	=	*specific gravity of the fluid*

In UniSim Design, the valve flow coefficient *Cv* is defined as the flow of water at 60 F in USGPM that goes through a control valve when the pressure drop is 1 psi and f(x) is equal to 1
From equation (1), when the valve is opened at 100 % and 1 USGPM of water at 60 °F flows through it with a pressure drop of 1 psi, the valve flow coefficient *Cv* equals 1 as the water specific gravity at 60 °F = 1

$$Cv = Q / \left[f(x) \cdot \sqrt{\frac{\Delta P}{sg}} \right] \cdot \qquad (2)$$

$$Cv = 1[USGPM] / \left[1 \cdot \sqrt{\frac{1[psi]}{1}} \right] = 1[USGPM] \cdot \qquad (3)$$

What is the Cv value of a valve for a maximum flow of 5 USGPM of water when it looses 2 psi fully open?

____

The inherent flow characteristic of a valve, *f(x),* is defined as the relationship between the valve tap position and the product flow rate, as a fraction of the maximum flow rate, when the pressure drop in the valve is **constant**. *f(x)* can thus be expressed as in Equation (2)

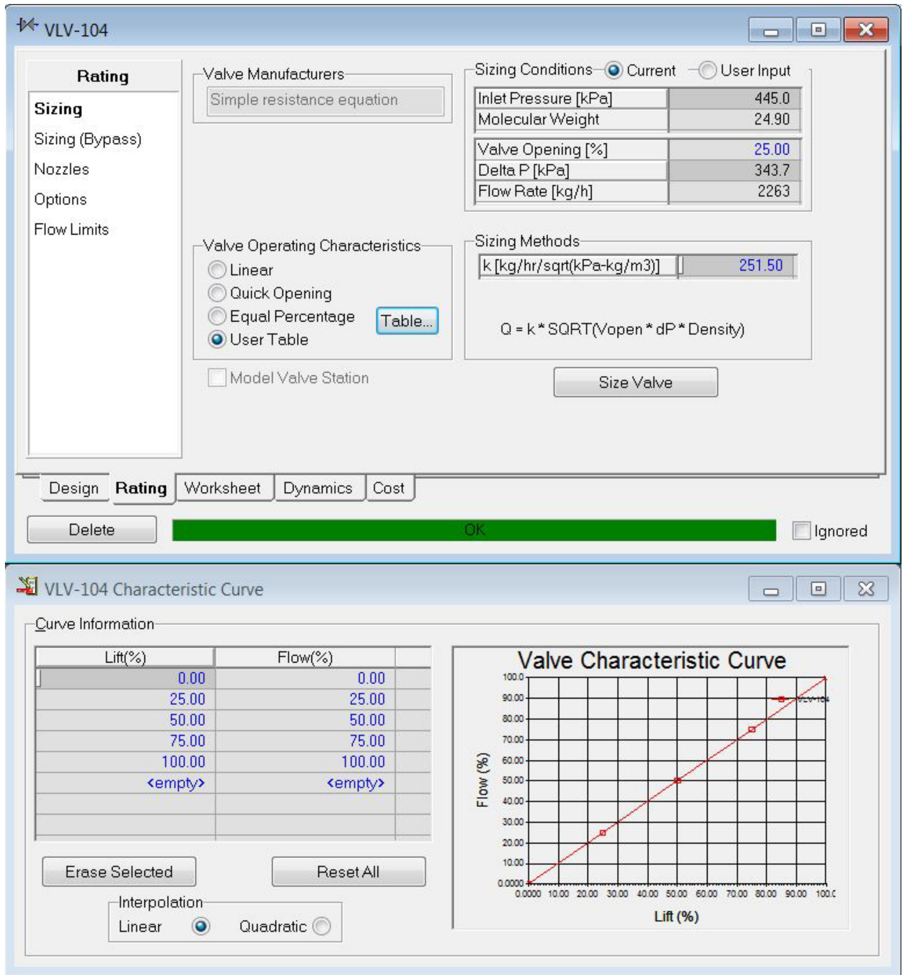
$$f(x) = \frac{F}{F_{max}} = \frac{A}{A_{max}} \tag{4}$$

Where:	F	=	*actual flow rate through valve*
	F_{max}	=	*maximum flow rate the valve can handle*
	A	=	*actual tap position*
	A_{max}	=	*maximum tap open position*

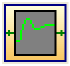
The most common valve inherent characteristics are **Linear**, **Quick Opening** and **Equal Percentage**. **Linear** and **Equal percentage** operating characteristic are the most popular in control systems.

On the **Sizing** page of the **Rating** tab of the valve property view, UniSim Design implements all three characteristics and also allows the user to implement a custom valve characteristic by entering a table of **% Flow** versus **% Valve Lift** values (pick the **User Table** radio button and press the **Table** button).

Figure 9



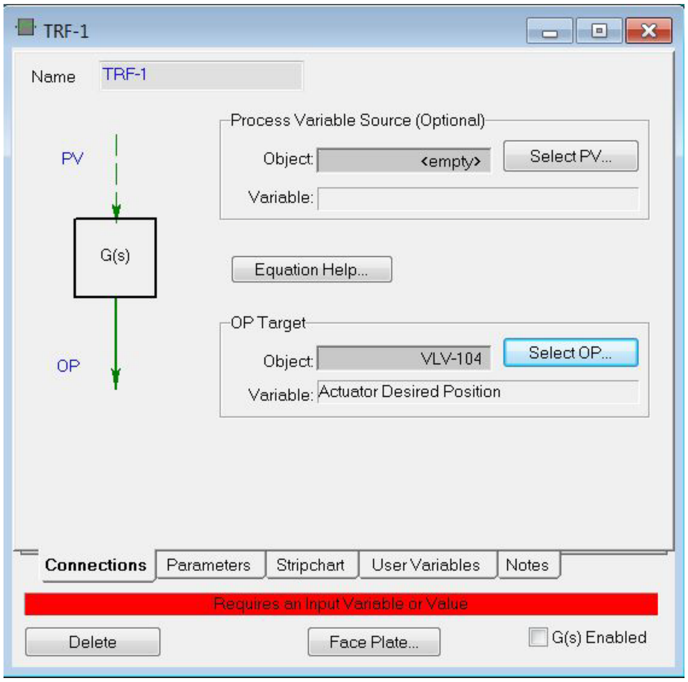
1. Edit stream **1** to be a pure water stream at **15.6 °C (60 °F)** and **108.2 kPa (15.7 psia)**.
2. Open **VLV-104**, and on the **Rating** tab, **Sizing** page, change the **Valve Manufacturers** to **ANSI / ISA75.01** and set the **Cv** to **1 USGPM**. Set the **Mode o**f controller **FIC-104** to **Off**. Note controllers can only be set to **Off** by first setting to **Indicator** and then to **Off**.
3. Make sure **VLV-104** has **Instantaneous** Actuator Dynamics.



Transfer Function icon

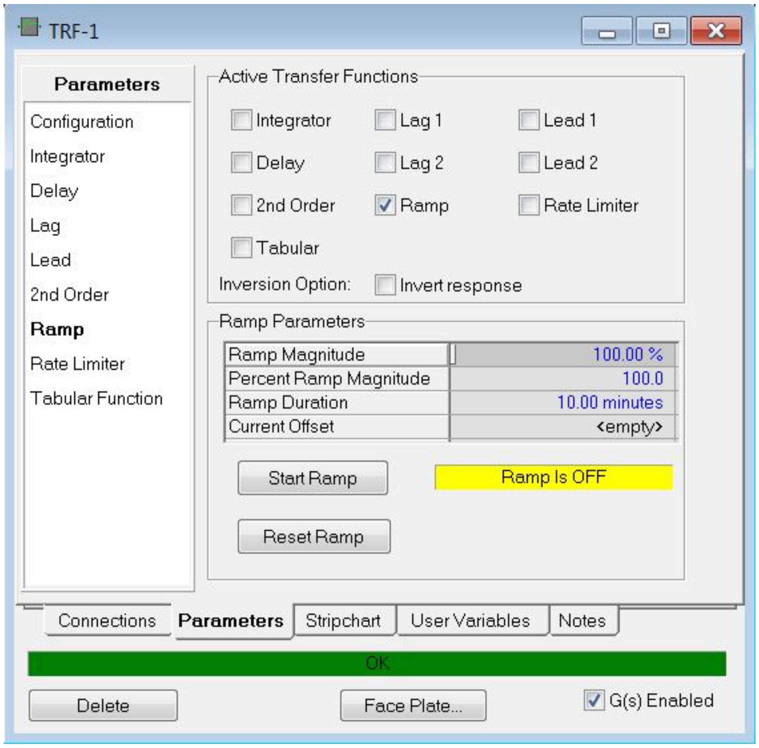
4. Add a **Transfer Function** and select the **Actuator Desired Position** of **VLV-104** as the **OP Target**. Do not set a **PV Source** connection since in this case the PV will be a fixed value. (Figure 6)

Figure 10



5. On the **Parameters** tab, select the **Configuration** page. Define **0%** and **100%** as minimum and maximum values respectively for both, the **PV** and **OP** Ranges, and enter a value of **0** for the **PV**.
6. Select the **Ramp** page, and check the **Ramp** option in the Active Transfer Functions group.
7. Enter **100%** for the Ramp Magnitude and **10 minutes** for the Ramp Duration.
8. Enable the **Transfer Function** by checking the **G(s) Enabled** option on the right bottom corner.

Figure 11

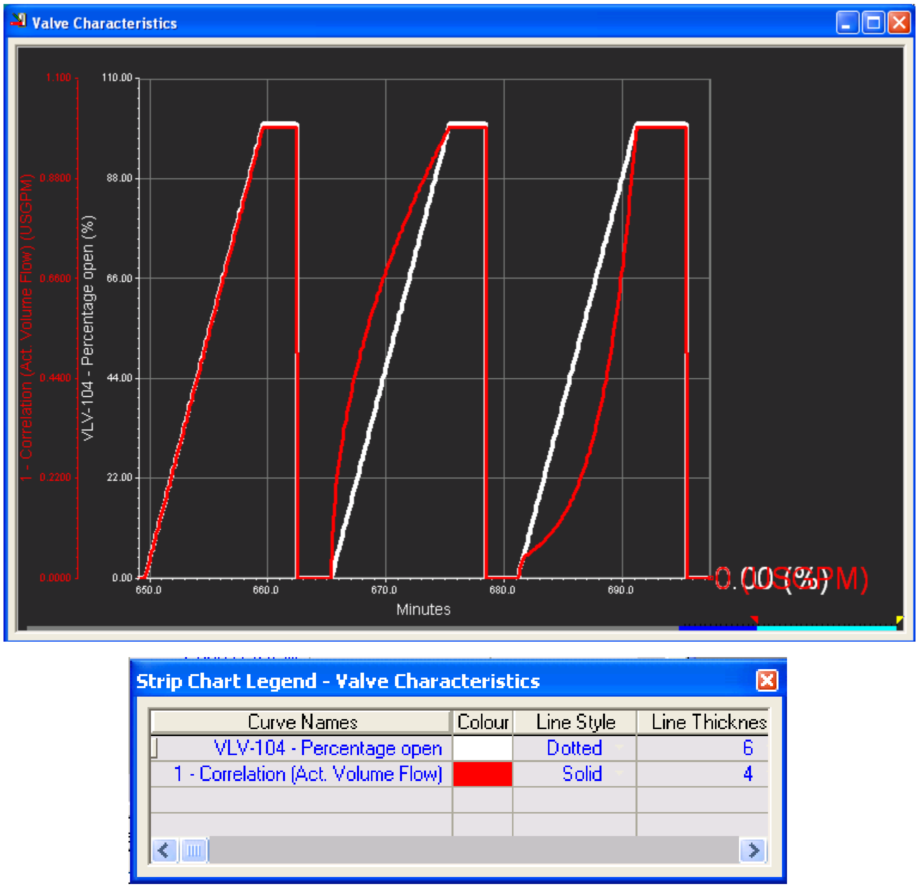


9. Add a strip chart to plot the **Std Ideal Liq Vol Flow** of Stream **1** and the **Percentage Open** of **VLV-104**.
10. Run the integrator and click the **Start Ramp** button.
11. Click the **Reset Ramp** button.
12. Repeat the test for a **Quick Opening** valve and an **Equal Percentage** valve.

Your strip chart should look like Figure 12.

When the ramp is reset it is necessary to re-enter the **Ramp Magnitude** and **Ramp Duration** as they are set back to 0.

Figure 12



13. Valves **VLV-104** and **VLV-105** and their associated streams and controllers are no longer needed and can be deleted.

Equipment Location

Static Head

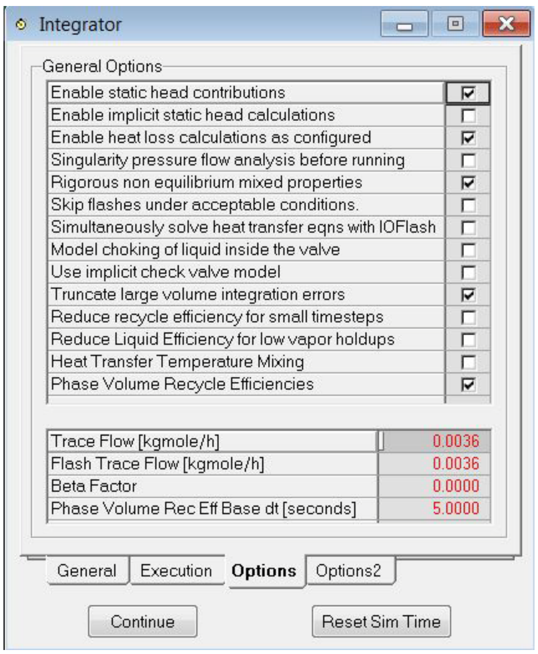
By default, static head contributions are not included in any calculation. Examine the stream pressures around the High Pressure Separator; they are all equal.

For any unit operations with holdup, UniSim Design calculates the static head by considering the equipment holdup, geometry, and elevation of any attached nozzles.

In order for UniSim Design to calculate the static head contributions within the model, you need to enable the calculations. This is done on the **Options** tab of the Integrator property view.

1. Open the Integrator view (**Simulation** – **Integrator** menu option or hot key **CTRL I**).
2. Check the **Enable Static Head Contributions** box on the **Options** tab. (Figure 9)

Figure 13



3. Run the simulation and observe the slight changes in the controller outputs. Look at the pressures surrounding the separator by pressing hot key **SHIFT P**.

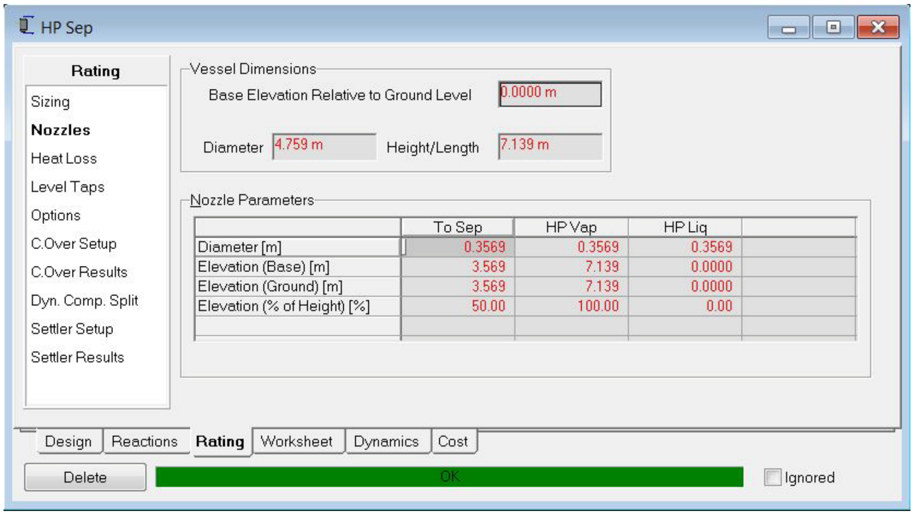
What is the pressure of stream HP Liq?

____

Nozzles

By default, all unit operations are placed at ground level. The **Nozzles** page, located on the **Rating** tab, contains information regarding the elevation of the equipment and the elevation and diameter of the nozzles.

Figure 14



The elevation of each nozzle is displayed relative to several reference points:

- The **Ground** is a common reference point from which all equipment elevation can be measured.
- The **Base** is defined as the bottom of the piece of equipment.

1. Plot on a new strip chart:
    - The **Phase Mass Flow - Aqueous** and **Phase Mass Flow - Liquid** of stream **HP Liq**.
    - The **Liquid Percent Level** of **HP Sep** (i.e. the hydrocarbon level)

- The **Phase Level Percent** for **Phase Level Percent_3** of **HP Sep** (i.e. the water level)

2. Change the nozzle location for the stream **HP Liq** from **0%** to **25%**.

Can you explain the results?

____

The nozzle diameter, nozzle elevation, and the levels of the phases inside the vessel determine what flows out through the nozzle.

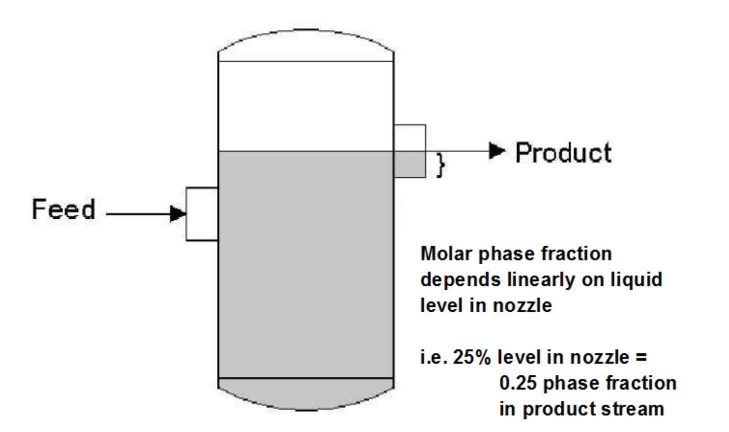
3. Change the **set point** (**SP**) for the level controller **LC-HP Sep** from **50%** to **25%**.

Could the controller achieve the SP?

____

The nozzle elevation refers to the center of the nozzle opening, unless the nozzle is placed at the bottom or top of the vessel. If the product nozzle is located above the liquid level, the exit stream draws material from the vapour holdup. If the liquid level partially covers the nozzle, the mole fraction of liquid in the product stream depends linearly on the coverage of the nozzle by the liquid.

Figure 15



4. Restore the original nozzle location (**0%**) in **HP Sep**, and the original level controller **SP** to **50%.** Turn off the static head calculations on the Integrator **Options** tab.

Detailed Heat Loss Model

There are several underlying assumptions that are considered during a heat loss calculation:

- there is a heat capacity and a thermal conductivity associated with the wall and the insulation housing the fluid,
- the temperature across the wall and insulation is assumed to be constant (lumped parameter analysis),
- the calculation uses convective heat transfer on the inside and outside of the vessel.

All operations with holdup have the ability to take into account heat loss from the holdup to the environment. For example, you can supply details about the equipment and insulation to take into account heat transfer from the vessel to the environment.

The heat loss models are located on the **Heat Loss** page of the **Rating** tab. By default the **None** option is selected, meaning that heat loss calculations are ignored. There are two heat loss models available for you to use: **Simple** and **Detailed**.

Simple Heat Loss Model

The Simple model allows you to either specify the heat loss directly or have the heat loss calculated from the following variables:

- Overall U value specified by the user,
- Ambient Temperature specified by the user,
- Overall Heat Transfer Area, A, calculated by UniSim Design,
- Heat flow calculated by UniSim Design.

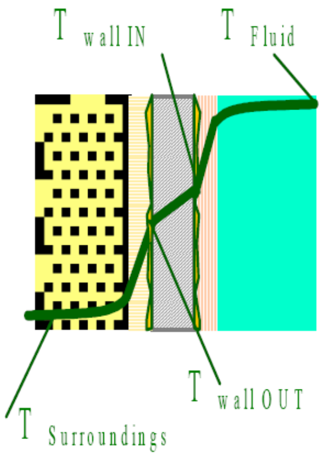
The heat loss is calculated using Equation (6).

$$q_{loss} = UA\left(T_{Fluid} - T_{Ambient}\right) \tag{5}$$

Where:

	q_{loss}	=	*heat loss [kJ/h]*
	U	=	*overall heat transport coefficient [kJ/(m^2-h-°C)]*
	A	=	*area [m^2]*
	T_{Fluid}	=	*bulk fluid temperature*
	$T_{Ambeient}$	=	*ambient temperature*

Detailed Heat Loss Model



In the detailed model the user supplies both conductive and convective information.

The program calculates the heat loss and supplies a temperature profile from the fluid to ambient.

1. Enter the following conductive and convective data for the **HP Sep** vessel and insulation:

Heat Loss Parameters		
Conductive Properties		
Material	Metal	Insulation
Thickness	0.051 m (0.167 ft)	0.030 m (0.098 ft)
Cp	0.473 kJ/kg-°C (0.113 Btu/lb-°F)	0.82 kJ/kg-°C (0.196 Btu/lb-°F)
Density	7801 kg/m^3 (487 lb/ft^3)	520 kg/m^3 (32.46 lb/ft^3)
Conductivity	45 W/m-K (26 Btu/hr-ft-°F)	0.15 W/m-K (0.087 Btu/hr-ft-°F)
Convective Properties		
Inside Vap Phase U	36 kJ/h-m^2-°C (1.761 Btu/hr-ft^2-°F)	
Inside Liq Phase U	180 kJ/h-m^2-°C (8.806 Btu/hr-ft^2-°F)	
Outside U	36 kJ/h-m^2-°C (1.761 Btu/hr-ft^2-°F)	
Vapour to Liquid U	18 kJ/h-m^2-°C (0.8806 Btu/hr-ft^2-°F)	

2. Leave the default ambient temperature of **25°C (77°F)** and start the Integrator.

Observe the temperature profile of the vessel. Did it immediately go to a steady state value?

____

Do you see any changes?

____

3. Change the **Heat Loss Model** to **Simple** and run the Integrator.

Save your case as **4528.05.HL.usc**

Save your case!

6. Expanding the Model

4528.06.R440.01

Workshop

In this module the high pressure separator will be expanded with a knockout drum and a three-phase low pressure separator.

The control system will also be modified with cascade, on-off and split range controllers. A pressure relief valve will also be added to the simulation.

There are several ways to add equipment to an existing dynamic model. Some users prefer switching the model back to steady state mode, making the changes then switching back to dynamics. Others prefer adding the equipment directly to the dynamic model. UniSim Design supports either method. In this workshop the unit operations will be added in dynamics mode.

Objectives

After you have completed this module, you will be able to:

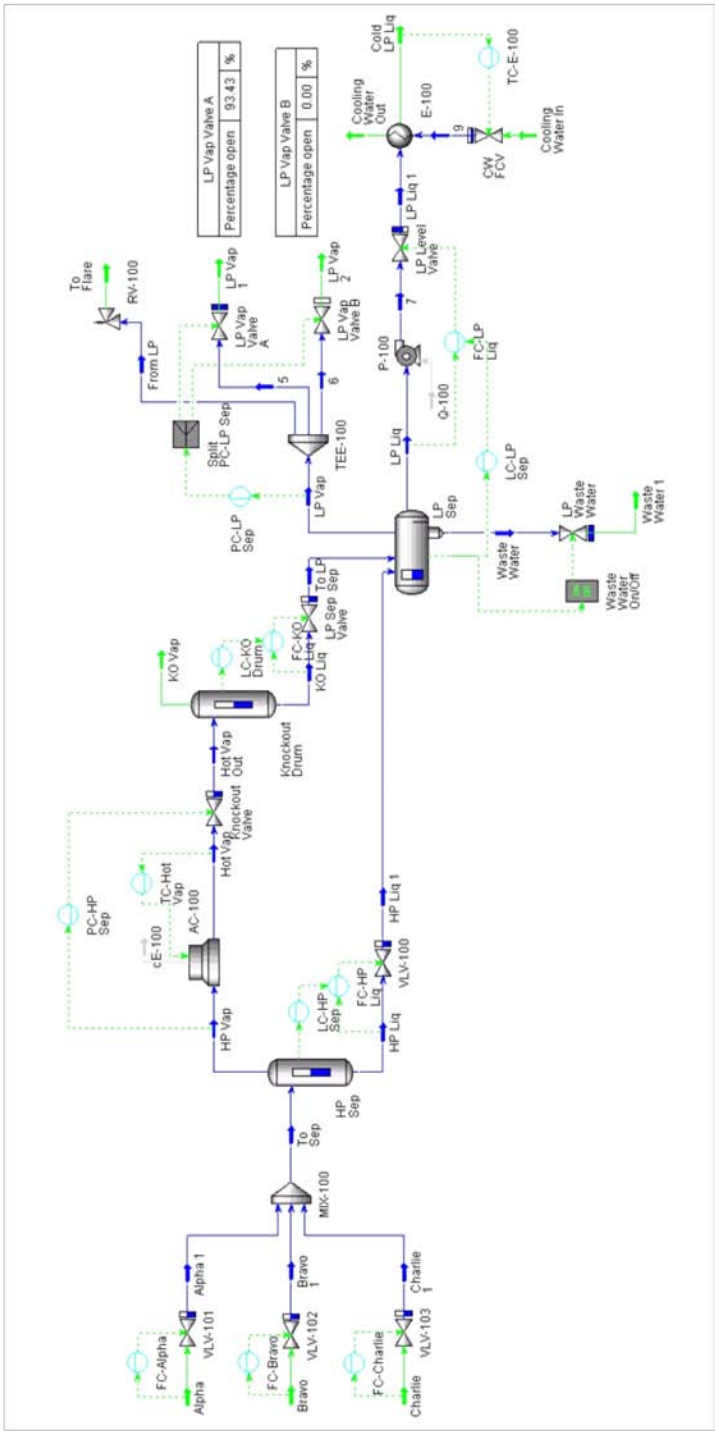
- Add unit operations and controllers in the dynamic mode
- Make necessary pressure-flow specifications for the system
- Implement appropriate control strategies
- Install a relief valve
- Install an air cooler
- Install a pump

Prerequisites

Before beginning this section you need to know how to:

- Set up Strip Charts
- Understand the pressure-flow network

Process Overview



Build the Flowsheet

In this module, you will install several unit operations in Dynamics.

Models can be built in steady state and then converted to dynamics or they can be built in dynamics only. The choice of which method to use is largely personal but may also be influenced by the final use of the model is it a simple dynamic study, a rigorous dynamic study, OTS, etc...? The goal of this module is to teach you how to build simulations in dynamics mode.

Pressure Control for HP Separator

Open the saved case from Module 5, **4528.05.HL.usc**. This case should still be in the dynamics mode. If it is not, switch the model to dynamics mode and run the Integrator until the steady state is reached, then stop the Integrator.

Before adding a Knockout Drum, we are going to change the pressure control scheme for the HP Separator. A control valve will be placed on the **Hot Vap** line.

1. Add a valve to the simulation, the inlet stream is **Hot Vap** and the product stream is **Hot Vap Out**. Name the valve **Knockout Valve**.
2. On the **Rating** tab **Sizing** page of the valve, set the **Valve Manufacturers** to **ANSI / ISA75.01** and size the valve based on a **70 kPa (10 psi)** pressure drop and 50% opening.

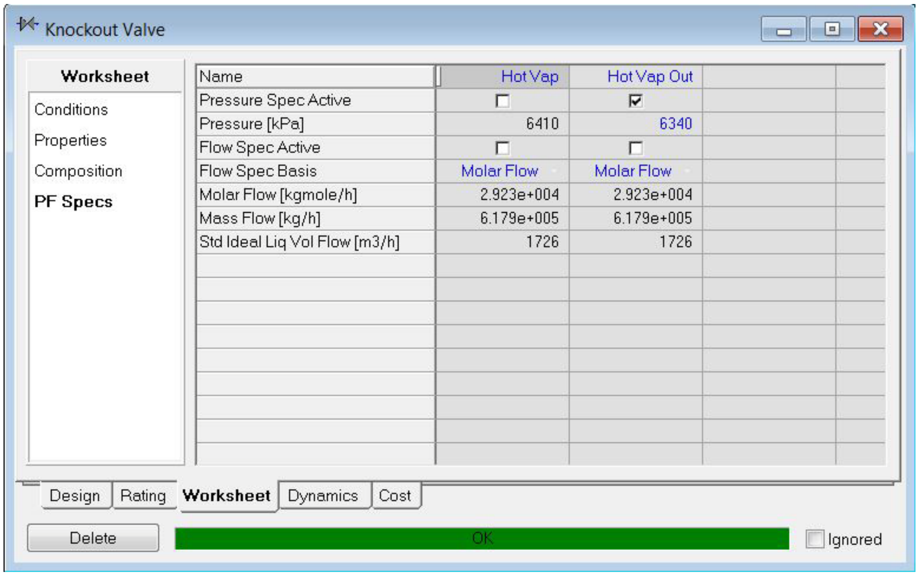
What Cv value does UniSim Design calculate for the new valve (Knockout Valve)?

____

Once the operations are added, the pressure-flow specifications need to be moved to the new boundary streams. The Integrator should then be run for a few time steps to propagate the solution values to the new operations and streams after each operation is added.

3. On the **Worksheet** tab **PF Specs** page of the valve, move the pressure specification from the stream **Hot Vap** to **Hot Vap Out**.
4. Change the pressure of the stream **Hot Vap Out** from **6410 kPa (930 psia)** to **6340 kPa** (920 psia).

Figure 1



5. Run the Integrator for a few time steps to propagate the solution values to the boundary streams and save the case.

It is recommended that you periodically save your case as equipment is added to the flowsheet. You can use different case names if you want to provide a snapshot at the various stages of building the case.

UniSim Design offers a Revision Control feature which helps with automatic case version numbering. You may wish to use this during this workshop. See the Revision Control section on page 39 of this module for more information.

Save your case as **4528.06.Vap_KO_Valve.usc**.

Save your case!

6. Change the output connection for **PC-HP Sep** from **FC-Alpha** to the **Knockout Valve – Actuator Desired Position**.
7. Place **FC-Alpha** in **Auto(matic)** mode.

What else needs to be done with the pressure controller?
____

8. Set the appropriate **Action** in **PC-HP Sep** test that the controller responds correctly to changes in set point.
9. Change the set point back to its original value of **6480 kPa**.

Adding the Knockout Drum

The knockout drum will be modeled using a separator. The cooled overhead stream **Hot Vap Out** will be the knockout drum feed.

10. Turn off the Integrator.
11. Add a separator and provide the following information:

Remember the rules for pressure-flow specifications. You cannot place pressure specifications on both accumulator product streams.

In this cell...	Enter...
Connections	
Name	Knockout Drum
Inlet Stream	Hot Vap Out
Vapour Outlet Stream	KO Vap
Liquid Outlet Stream	KO Liq
Parameters	
Volume	25 m3 (900 ft3)

12. Move the **pressure-flow specifications** to the new boundary streams. Supply a pressure specification for stream **KO Vap** and a molar flow specification for stream **KO Liq**.

What value should you use for the KO Liq molar flow specification?
____

13. Go to the **Dynamics** tab of the Knockout Drum. By default, the simulation will initialize the contents of the vessel holdup based on the feed stream to the vessel (because of the **Initialize from Feeds** setting) with an initial level of **50%.**

14. Run the Integrator for a few time steps to propagate the solution values to the boundary streams. Do not let the level get too high.

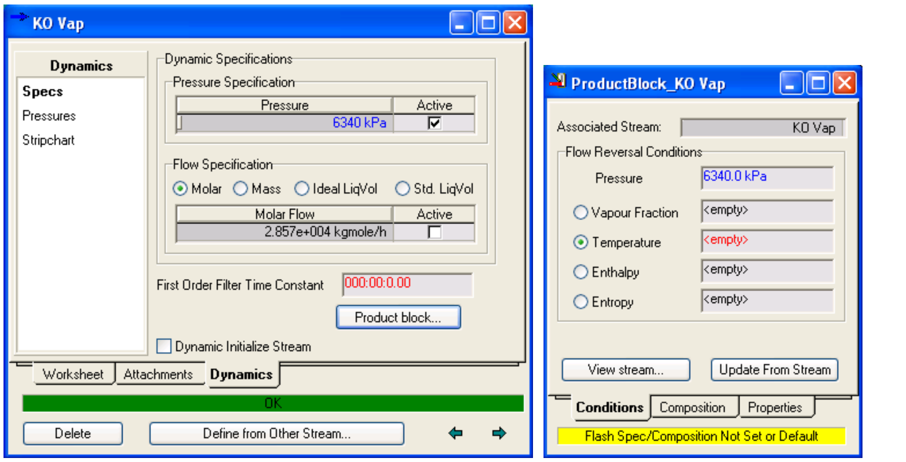
Reverse Flow

Based on the rules described previously (in Module 2 Pressure-Flow Theory) a resistance device should be added to both of the Knockout Drum streams and pressure specifications should be used (since they are boundary streams). Instead we are using a pressure specification on the vapour product and a flow specification on the liquid product.

When a pressure specification is used on the product stream of a vessel, the exit stream flow will be adjusted automatically to maintain the specified pressure in the vessel. The exit stream will either let flow out of the vessel or let flow into the vessel (reverse flow). For this reason, all boundary product streams have a Product block.

The product block defines the conditions if there is negative (reverse) flow. By default, the product block has the same composition as the vessel, but you can change it as required (for example, to model N2 or CH4 for atmospheric or fuel gas blankets). View the product block with the **Product block** button on the **Dynamics** tab **Specs** page.

Figure 2



In the case of a flow specification, the flow is always equal to the value specified, no matter what the pressure or level of the vessel. Flow specifications are often used to simplify problems, but they can lead to some strange results. So while they help to simplify a problem, we should aim to eliminate these specifications whenever possible.

Adding a Valve on the Liquid Outlet

A valve will be added to the **KO Liq** stream. This valve is used by the level controller to control the liquid flow from the vessel.

1. Before adding the valve make sure the level in the knockout drum is not so low that there is vapour in the liquid product. This will mean the valve is not sized correctly (for the eventual liquid only flow in the complete model).
2. Add a valve and provide the following information:

In this cell...	Enter...
Connections	
Name	LP Sep Valve
Inlet Stream	KO Liq
Outlet Stream	To LP Sep

3. Move the pressure-flow specifications to their proper locations.
4. Specify the pressure of stream **To LP Sep** to be the same as stream **HP Liq1**.
5. Set the **Valve Manufacturers** to **Simple resistance equation** and size the valve considering that the LP Separator we will add later will work at around 27 bar.

What k value does UniSim Design calculate for the new valve (LP Separator Valve)?

____

6. Run the Integrator for a few seconds to propagate solution values to the boundary stream.
7. Add a level controller in order to maintain the level of the Knockout Drum.
8. Set the **SP** level equal to **50%**.

Once at steady state conditions, what is the Valve Opening [%] of the new valve (LP Separator Valve)?

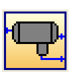
____

Save your case as **4528.06.KO_Drum.usc**.

Save your case!

Adding the LP Separator

In this section we will add the three-phase separator. The inlets are the two liquid streams, **HP Liq1** and **KO Liq**, and the vessel size is the same as the High Pressure Separator.



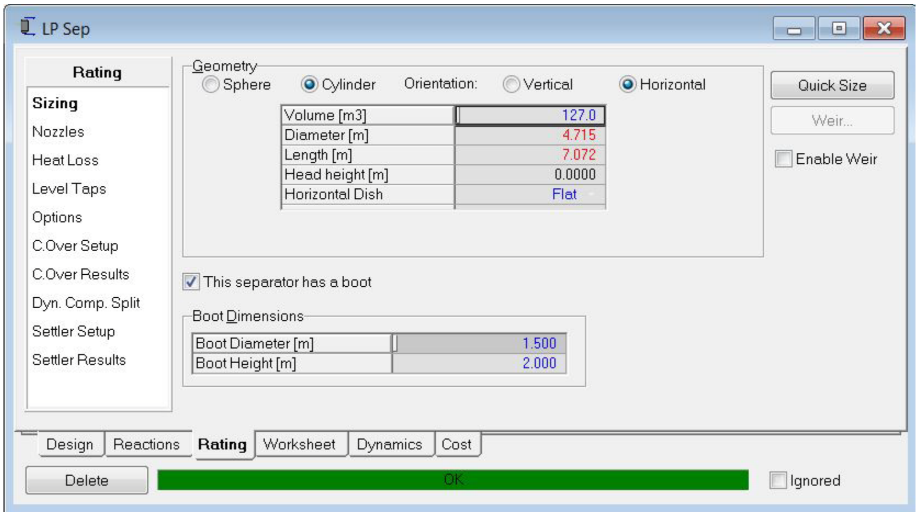
3-Phase Separator icon

1. Add a 3-Phase Separator and provide the following information:

In this cell...	Enter...
Connections	
Name	LP Sep
Inlet Stream 1	HP Liq1
Inlet Stream 2	To LP Sep
Vapour Outlet	LP Vap
Light Liquid Outlet	LP Liq
Heavy Liquid Outlet	Waste Water
Parameters	
Volume	127 m3

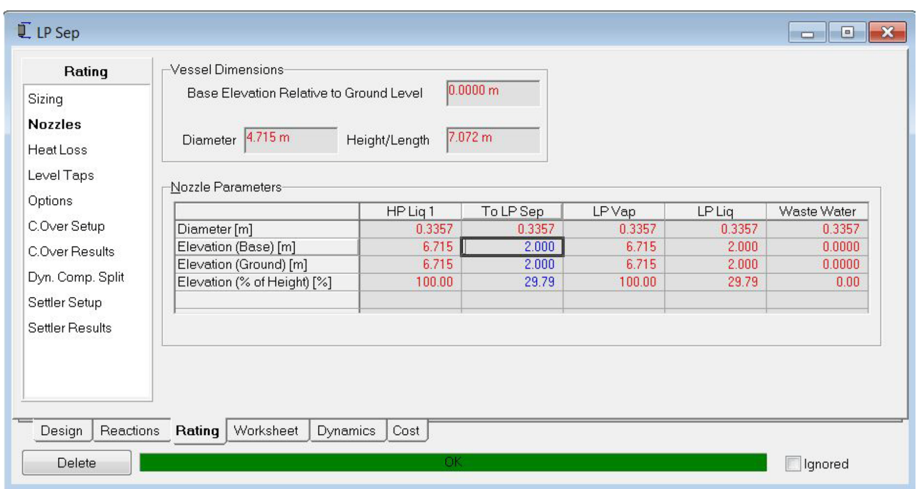
2. The boot option is turned on by default for 3-phase separators. Change the boot option on the **Rating** tab **Sizing** page by setting the **Boot Diameter** to 1.5 m and the **Boot Height** to 2.0 m (Figure 3).

Figure 3



3. Set the position of the feed nozzle for **To LP Sep** to **2.0 m**. On the **Rating** tab **Nozzles** page set either the **Elevation (Base) m** or **Elevation (Ground) m**.

Figure 4



4. Move the pressure-flow specifications to the new boundary streams. Supply a pressure specification for stream **LP Vap** and molar flow specifications for streams **LP Liq** and **Waste Water**.

What value should you use for the molar flow specification of LP Liq and Waste Water streams?

____

5. Start the Integrator and let it run for a few time steps. Do not let the level go to 100%.
6. Check the **Rating** tab **Nozzles** page and the **Dynamics** tab **Holdup** page to view the nozzle positions and the different phase levels and volumes. Leave the nozzles at the default values.

Are the liquid level and nozzle elevation OK for both liquid phases?

____

Save your case as **4528.06.LP_Sep.usc**.

Save your case!

Adding the Valves and Controllers

The LP separator requires controllers to stabilize the process. Three controllers are going to be added. One PI controller for the vessel pressure, one PI controller for the light liquid level on the LP Separator, and one on-off controller for the waste water stream valve.

Pressure Controller

1. Add a valve to the LP Separator Vapour Stream. Name the valve **LP Presalve** and the outlet stream **LP Vap1**.
2. The boundary pressure is **1380 kPa (200 psia)**.
3. Add the following pressure controller to the flowsheet:

In this cell...	Enter...
Connections	
Name	PC-LP Sep
Process Variable Source	LP Vap – Pressure
Outlet Target Object	LP Vap Valve – Actuator Desired Position
Parameters	
Action	Direct
Range PV Minimum	1380 kPa (200 psia)
Range PV Maximum	3100 kPa (450 psia)
Mode	Auto(matic)
SP	2690 kPa (300 psia)
Kc	3.0
Ti	2.0 min
Td	<empty>

4. Run the Integrator for a few time steps.

Level Controller

1. Add a valve to the **LP Liq** stream. Name the valve **LP Level Valve** and the outlet stream **LP Liq1**.
2. The boundary pressure is **1380 kPa (200 psia)**.
3. Add the following level controller to the flowsheet:

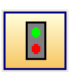
In this cell...	Enter...
Connections	
Name	LC-LP Sep
Process Variable Source	LP Separator – Liquid Percent Level
Outlet Target Object	LP Vap Valve – Actuator Desired Position
Parameters	
Action	Direct
Range PV Minimum	0%
Range PV Maximum	100%
Mode	Auto(matic)
SP	65%
Kc	1.0
Ti	15.0 min
Td	<empty>

4. Run the Integrator for a few time steps.

Water Level On-Off Controller

Rather than using another PI controller, an On-Off controller will be used to control the water level in the vessel between a lower limit of 0.40 m and a higher limit of 1.8 m.

1. Add a valve to the **Waste Water** stream. Name the valve **LP Waste Valve** and the outlet stream **Waste Water1**.
2. The boundary pressure is **1380 kPa (200 psia)**.

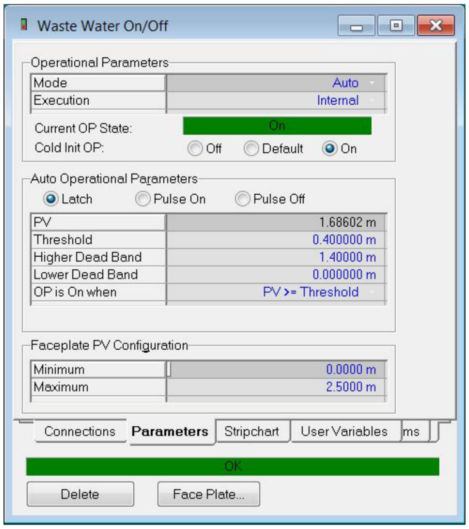


Digital Point icon

3. Add a Digital Point controller and configure it as follows: (Figure 5)

In this cell...	Enter...
Connections	
Name	Waste Water On/Off
Process Variable Source	LP Sep – Phase Level - Phase Level_3
Outlet Target Object	LP Waste Valve – Digital Actuator Desired position
Parameters	
Mode	Auto(matic)
Cold Init OP	On
Radio button setting	Latch
Threshold	0.40 m
Higher Dead Band	1.40 m
Lower Dead Band	0 m
OP is On when	PV >= Threshold
Face Plate configurations	
Minimum PV	0 m
Maximum PV	2.5 m

Figure 5



4. Show a **Face Plate...** for the Waste Water On/Off controller.
5. Run the Integrator and confirm that the controller is controlling the level correctly.

Adding Strip Charts

1. Create Strip Charts to monitor primary variables in your flowsheet.
2. Start the Integrator and allow the model to stabilize.
3. Once the model has reached steady state, create a disturbance to test your model.

Save your case as **4528.06.Mods.usc**.

Save your case!

Exercises

Adding Cascade Controllers

For slow acting control loops (e.g. level controllers) it is desirable to place a fast acting controller (e.g. flow controllers) into the loop.

Place flow controllers on the **HP Liq**, **LP Liq** and **KO Liq** streams. Set up these controllers as slaves to the level controllers.

Save your case as **4528.06.Mods_CascadeControl.usc**.

Save your case!

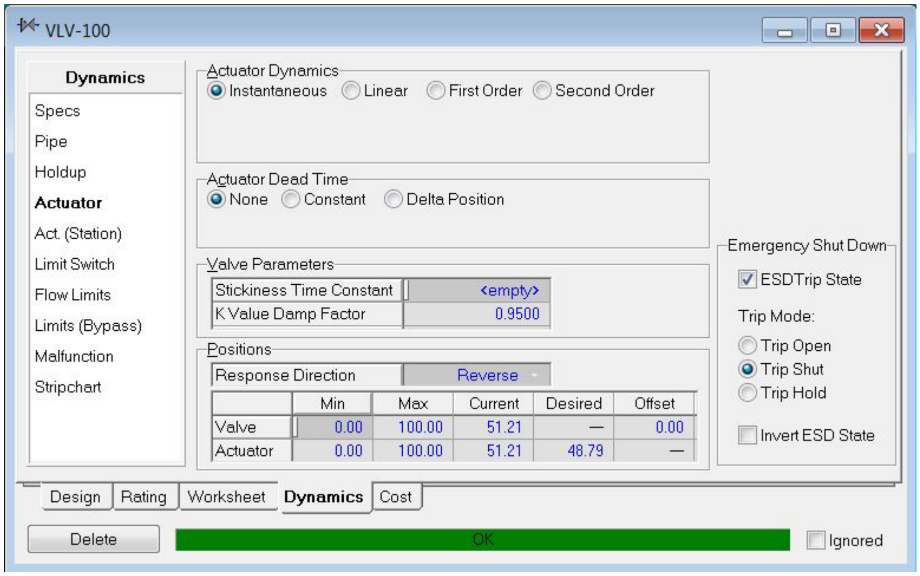
Valve Emergency Shut Down (ESD) settings

In a plant emergency the plant safety or ESD system will open or close valves to make the plant safe.

On the **Dynamics** tab **Actuator** page of the valve UniSim Design allows the user to configure the ESD settings for a valve.

The default settings are shown below.

Figure 6

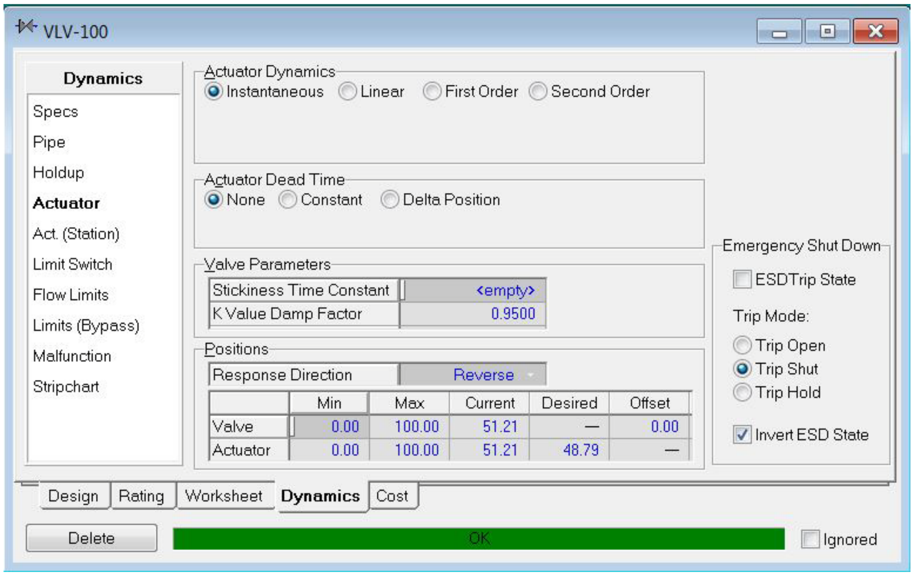


The **Trip Mode** setting defines which position the valve moves to during an ESD trip. (Open, Shut or Hold). The behavior of the **ESD Trip State** checkbox depends on the setting of the **Invert ESD State** checkbox. With **Invert ESD State** unchecked (the default) then when the **ESD Trip State** checkbox is unchecked (or 'Low') the valve moves to its ESD trip position. Checking **Invert ESD State** inverts the logic, meaning the valve moves to its ESD trip position when **ESD Trip State** is checked.

In the Event Scheduler module the ESD Trip function of **VLV-100** will be used to make safe a high liquid level situation in the HP Separator. To prepare for this:

1. On the **Dynamics** tab **Actuator** page of **VLV-100** valve select **Trip Open**, uncheck **ESD Trip State** and check **Invert ESD State**. (Figure 7)

Figure 7

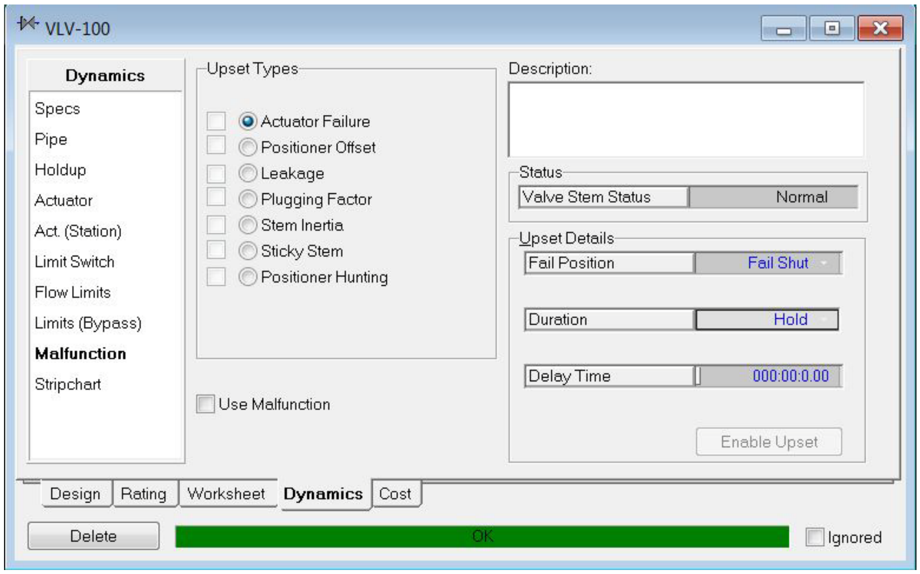


2. Test the ESD Trip function of the valve by running the integrator and checking **ESD Trip State**. When you are satisfied the settings are correct return the valve to normal (non-ESD) operation.

Valve Malfunctions

UniSim Design allows several kinds of valve malfunction to be modeled. These are configured and activated on the **Dynamics** tab **Malfunction** page.

Figure 8



To configure and activate a malfunction of the valve:

- Select the radio button for the malfunction.
- Make the required settings on the right hand side of the window.

In this exercise a fail open malfunction of the valve will be configured, and tested.

1. On the **Malfunction** page of the **Dynamics** tab of **VLV-100** select the **Actuator Failure** radio button and tick **Use Malfunction**. Change the **Fail Position** to **Fail Open**. Leave the rest of the settings at the defaults.
2. Start the Integrator, at this point the model should be behaving as normal. Press the **Enable Upset** button and observe that the valve goes to a fully open position.
3. Now press the **Disable Upset** button and observe that the model returns to normal operation.

Normally for a fail open valve the desired position of the valve is modified according to the following formula.

$$Act.Des.\%(valve) = 100\% - Act.Des.\%(controller) \quad (1)$$

Where:	*Act. Des. % (valve)*	=	*valve actuator desired percentage open*
	Act. Des.%(controller)	=	*controller actuator desired percentage open*

This is typically because the actuator is working against the spring that will open the valve in case of a failure of the power source driving the valve. To account for this effect in UniSim Design it is necessary to modify the **Response Direction** setting on the **Actuator** page of the valve.

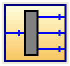
4. Change the **Response Direction** setting from **Direct** to **Reverse**. Observe what happens to the valve opening.

Does the Response Direction selection have any implications on the direction of the controller? Make the required modification.

____

Adding a Split Range Controller

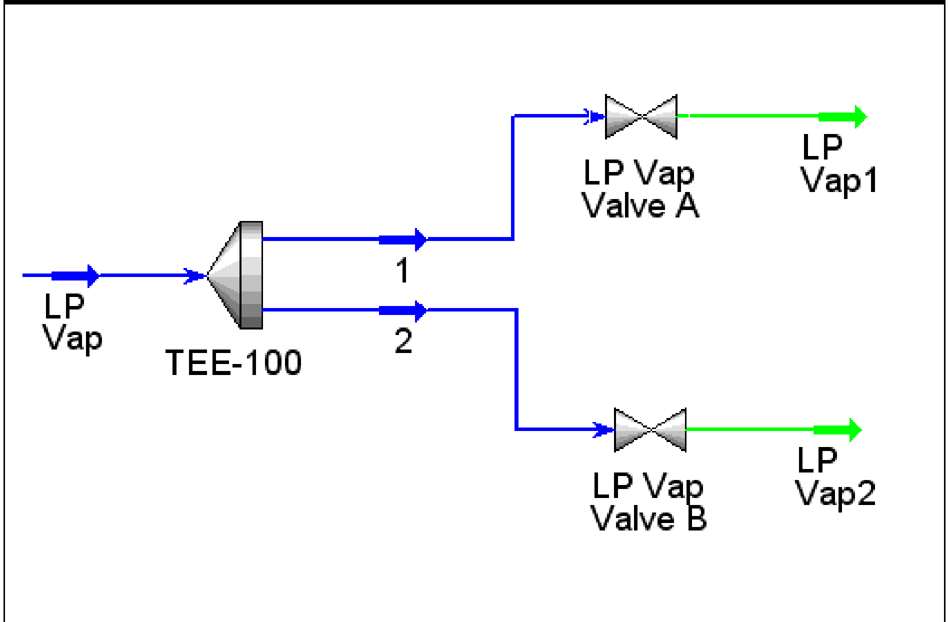
In this exercise you will add a split range controller.



Tee icon

1. Disconnect the vapour stream from the LP Separator to the control valve.
2. Add a tee and connect the vapour stream as the inlet and the valve inlet stream as the outlet of the tee.
3. Add a second stream to the tee outlets and a new valve. Your PFD should look similar to the figure below (use the names given below for the new valves and streams):

Figure 9



4. Size the valves so that each is about **50%** of the original (single) valve size.

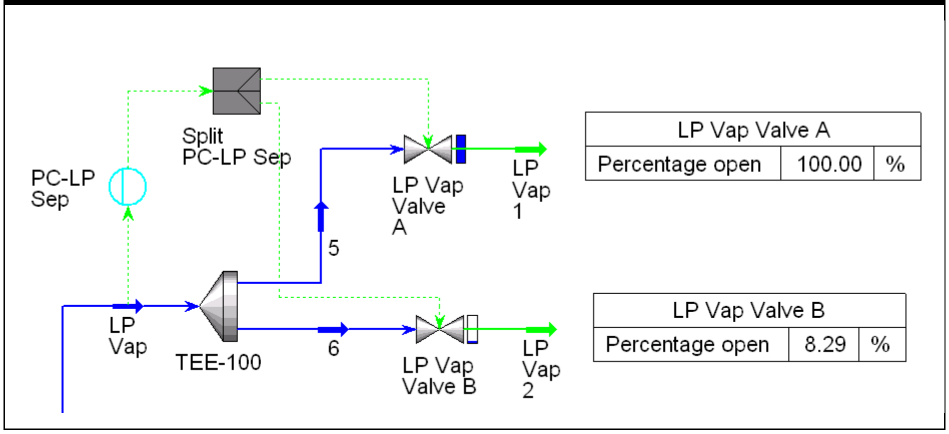
What k do you have for the new LP Vap Valves?

____

5. Make the appropriate pressure specifications for the new boundary streams. Set their pressures at **1380 kPa (200 psia)**.

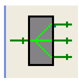
Since UniSim Design R410 the built-in Split Range Controller unit operation can no longer be added to new simulation cases, instead the use of the Fanout with a standard PID controller is suggested (see Figure 10). Such a model of the control system more closely resembles that in a typical DCS implementation and so allows more transparent modelling as well as allowing the use of all the additional features available in the PID Controller (for example SP Ramping, Scheduling, Alarms, PV Conditioning and Signal Processing.)

Figure 10



6. Disconnect the OP from the Pressure Controller **PC-LP Sep**
7. Add the **Fanout** logical unit operation from the **Control Ops Palette**

Control Ops Palette icon



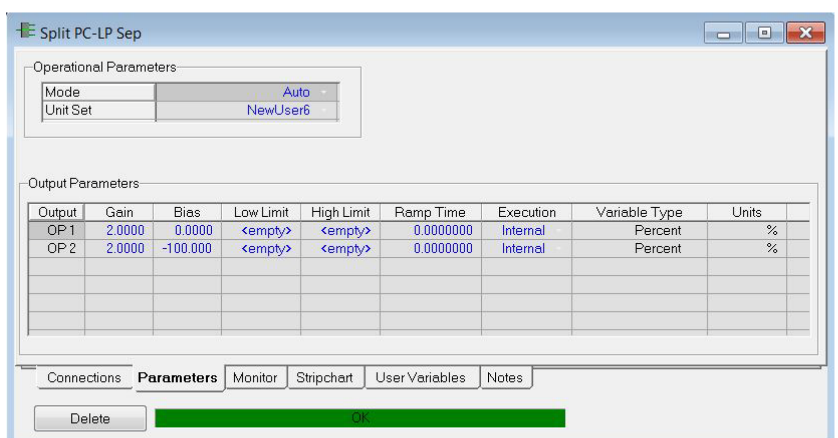
Fanout icon

The Fanout logical unit operation can be found on the Control Ops Palette or from the **Flowsheet – Add Operation** menu option.

8. Set up the Fanout logical unit operation with the following parameters:

In this cell...	Enter...
Connections	
Name	Split PC-LP Sep
Process Variable Source	PC-LP Sep – OP
Outlet Target Object 1	LP Vap Valve A – Actuator Desired Position
Outlet Target Object 2	LP Vap Valve B – Actuator Desired Position
Parameters	
OP 1 – Gain	2
OP 1 - Bias	0
OP 1 – Ramp Time	0
OP 2 – Gain	2
OP 2 - Bias	-100
OP 2 – Ramp Time	0

Figure 11

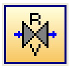


Save your case as **4528.06.Split Range.usc**.

Save your case!

Adding a Pressure Relief Valve

The relief valve operation is used to prevent over pressure situations occurring. Although it is available in steady state mode, its main use is in dynamics mode.



Relief Valve icon

1. Add a relief valve with **To Flare** as the outlet stream and **From LP** as the inlet stream.
2. Select **From LP** as a third outlet of the tee.
3. On the **Dynamics** tab, **Specs** page of stream **To Flare**, activate the **Pressure Specification**. The pressure of this stream should be atmospheric.

Further details about relief valves operation can be found in **Module 2, (Pressure-Flow Theory**.)

4. The relief valve requires a value for the **Orifice Area**. Go to the **Sizing** page of the relief valve under the **Ratings** tab and enter **25.81 mm^2 (0.04 in^2)**.
5. Set the valve type to **Linear**.
6. On the **Parameters** page of the **Design** tab, enter **2830 kPa (410 psia)** for the **Set Pressure** of the relief valve and **2890 kPa (420 psia)** for the **Full Open Pressure**.
7. Test the relief valve by causing the pressure in the vessel to increase. When you have finished your testing return the model to normal operation.

Save the case as **4528.06.RV.usc**

Save your case!

Notes on the Relief Valve Operation

Once the valve has lifted, examine the **Dynamics** tab **Specs** page of the relief valve. On this page you will observe three parameters:

Field	Description
Delta P	Pressure drop across the valve
Valve lift	This value is calculated using one of the following formulas: 1. If inlet pressure is increasing: $$L = \left[\frac{P_1 - P_{open}}{P_{full} - P_{open}} \right] \quad (2)$$ *Where:* P_1 = *upstream pressure* P_{Open} = *set pressure* P_{Full} = *full open pressure* 2. If inlet pressure is decreasing: $$L = \left[\frac{P_1 - P_{reseat}}{P_{close} - P_{open}} \right] \quad (3)$$ *Where:* P_1 = *upstream pressure* P_{reseat} = *reseating pressure* P_{close} = *closing pressure*
Percentage open	The valve lift percentage.

Hysteresis Parameters Group

If the **Liquid Service** checkbox is unchecked, the **Enable Valve Hysteresis** checkbox appears. If this is activated, the **Hysteresis Parameters** group box becomes visible. This group contains two fields:

Field	Description
Closing Pressure	Pressure at which the valve begins to close after reaching the full lift pressure (i.e. the value entered in the **Full Open Pressure** cell on the **Parameters** page of the **Design** tab).
Reseating Pressure	The pressure at which the valve reseats (i.e. closes fully) after discharge.

Flow Through the Relief Valve

The mass flow rate through the relief valve varies depending on the inlet conditions and the pressure ratio across the valve. Two different calculation methods are supported and can be selecting from the **Design** tab, **Parameters** page using the **Calculation Mode** radio button:

Calculation Mode	Description
API 521 RP (1976)	This is based on the 1976 edition of API 521 RP and is accurate for single phase flows.
HDI	HDI stands for Homogeneous Direct Integration. This method calculates the mass flow through the nozzle by numerical integration from its inlet pressure to its outlet pressure, assuming homogeneous flow. This method is accurate for two phase flows but requires more computationally overhead, so will reduce the solver speed.

More details of the Relief Valve are given in section 5.7 of the UniSim Design Operations Guide manual.

Advanced Modeling

Air Cooler

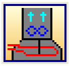
The Air Cooler unit operation uses an ideal air mixture as a heat transfer medium to cool (or heat) an inlet process stream to a required exit stream condition. One or more fans circulate the air through bundles of tubes to cool the process fluids. The air flow is calculated based on the number of fans running and their current operating speed.

Since the air cooler only has a process connection you need to specify the following fan information so that air flow can be calculated:

- Number of fans
- Demanded speed of each fan
- Design speed of each fan
- Design air flow of each fan

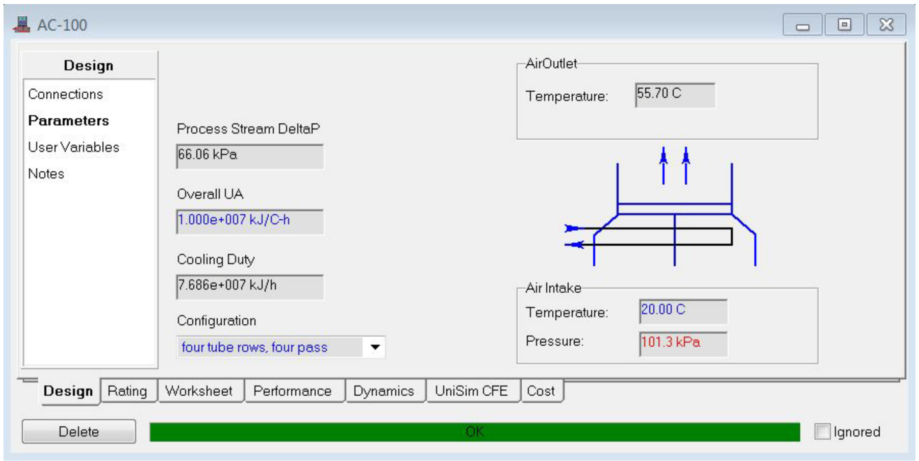
Additionally to provide reasonable dynamic response a realistic fan acceleration (max. acceleration) should be specified for each fan. This will be the rate at which fans accelerate or decelerate when turned on or off or if the demanded speed is changed.

In this exercise we will replace the Cooler operation, **E-100**, with an air cooler.


Air Cooler icon

1. Delete cooler **E-100**.
2. Add an **Air Cooler** from the Object Palette.
3. Connect stream **HP Vap** as the inlet and stream **Hot Vap** as the product.
4. Connect the energy stream **qE-100**.
5. On the **Design** tab, **Parameters** page enter an **Overall UA** of **1.0e7 kJ/C-h (5.266e6 Btu/F-hr)**, change the **Configuration** to **four tube rows, four pass** and change the **Air Intake Temperature** to **20 °C (68 °F)**.

Figure 12



6. On the **Rating** tab, **Sizing** page enter the following information:

In this cell...	Enter...
Parameters	
Tube Wall Mass	0 kg
Number of Fans	2
For each fan	
Demanded Speed	360 rpm
Max Acceleration per sec	10 rpm
Design Speed	360 rpm
Design air flow	1.0e6 ACT_m3/h (5.88e5 ACFM)
Fan is ON	ON
Natural draft flow	2 %

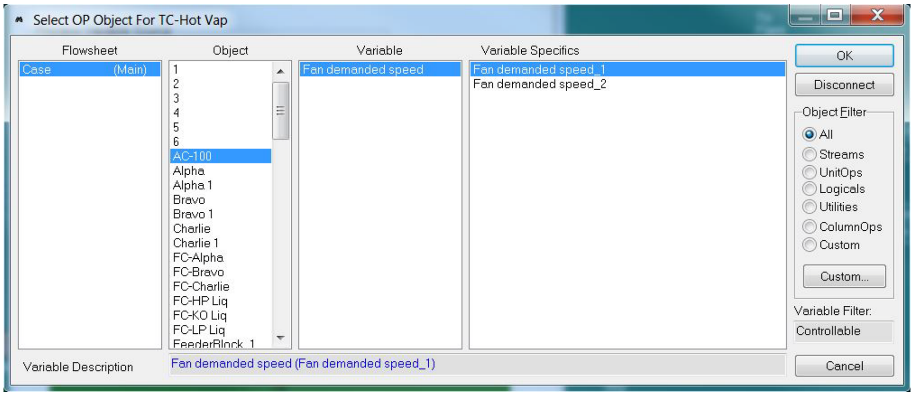
6. On the **Dynamics** tab, **Specs** page enable **Overall Delta P** and enter a value of **70 kPa** (**10.15 psi**).
7. Press the **Calculate K** button, then uncheck **Overall Delta P** and check **Overall K Value**.
8. Enter a **Fluid Volume** of **15 m^3** (**529.7 ft^3**)

Temperature Controller

When cooler E-100 and its energy stream were deleted the output connection in controller **TC-Hot Vap** was deleted. We will now connect the temperature controller to the demanded speed of one of the air cooler fans.

1. On the **Connections** tab of temperature controller **TC-Hot Vap** connect the output to the demanded speed of Fan 1 as shown below:

Figure 13



2. Set the range of the **Control OP Port** to **0 – 600 rpm**.
3. On the **Parameters** tab change the **Mode** to **Auto(matic)**.
4. Start the Integrator and ensure that the air cooler can provide the required cooling duty.

Save the case as **4528.06.AirCooler.usc**

Save your case!

Pump

Additional information on the positive displacement pump can be found in Section 6.3 of the UniSim Design Operations Guide.

UniSim Design has two pumps options; a centrifugal pump and a positive displacement pump. In this section we will add a centrifugal pump. You are encouraged to investigate the positive displacement pump options.

Unlike the steady state pump which can be configured to solve for a variety of different parameters, the dynamic pump only solves for increasing pressure of its inlet stream. Additionally the pump assumes that the inlet fluid is incompressible.

Some of the features of the dynamic pump include:

- Dynamic modeling of friction loss and inertia
- Shut down and startup behavior modeling
- Multiple head and efficiency curves
- Ability to add an electric motor
- Linking capabilities with other rotational equipment operating at the same speed with one total power

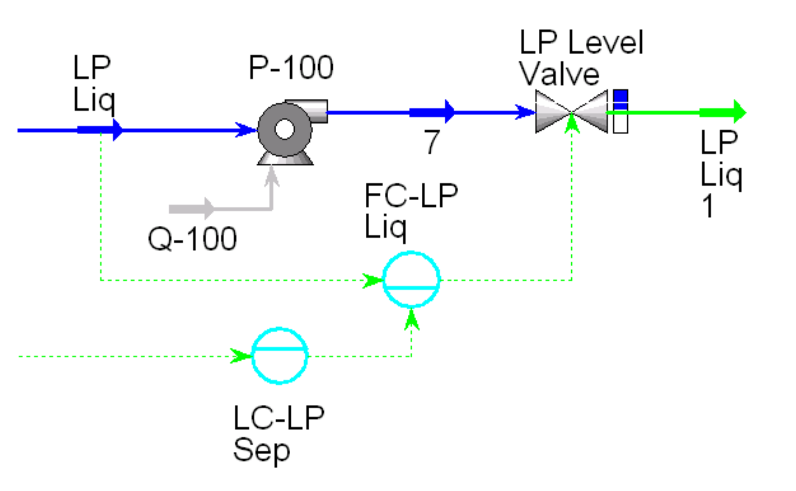
The pump has a **Generate Curves** option which allows users to generate curve data based on the specified pump design parameters. UniSim Design automatically generates three curves based on three different speeds: user specified speed, user specified speed multiplied by low speed %, and user specified speed multiplied by low-low speed %.

In the current model we will add a pump between **LP Separator** and **LP Level Valve**.

Pump icon

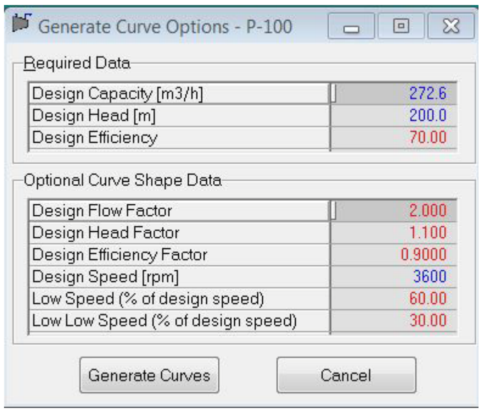
1. Disconnect stream **LP Liq** from valve **LP Level Valve**.
2. Add a pump from the Object Palette.
3. Connect stream **LP Liq** to the pump inlet, connect the pump outlet to valve **LP Level Valve**, add a duty stream to the pump.

Figure 14



5. On the **Rating** tab, **Curves** page of the pump property view, click on the **Generate Curves...** button.
6. Complete the **Generate Curve Options** view as shown below Design Capacity 272.6 m3/h (1200 USGPM):

Figure 15



7. Press the **Generate Curves** button.
8. On the **Dynamics** tab, **Specs** page uncheck **Efficiency** and **Pressure Rise** and check **Use characteristic curves** and **Speed**.
9. Enter a **Speed value** of **3600 rpm** (the **Design speed** entered).
10. Change the boundary pressure specification of stream **LP Liq1** to **3500 kPa (522 psia)**.
11. Start the Integrator.

You need to enable **Use Characteristic Curves** before you can enter **Speed**.

Check the liquid level is still under control in the **LP Sep** triphase separator

Is the cascade control able to maintain the set point of 65 % on the LP Sep ?

____

If not, what needs to be reviewed to achieve a liquid level around 65 % ?

____

____

Save the case as **4528.06.Pump.usc**

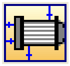
Save your case!

Heat Exchanger

It is desired to cool the LP Liquid product to 40 °C using a Heat Exchanger with water as the cooling medium.

1. Add the cooling water stream with the following parameters:

Parameter	Value
Name	Cooling Water In
Temperature	25 °C (77 °F)
Pressure	500 kPa (72.52 psia)
Mass Flow	1.6e5 kg/h (3.52e5 lb/hr)
Composition	100% Water
Pressure-Flow specification	Both **Pressure** and **Flow** specifications active



Heat Exchanger icon

2. Add a Heat Exchanger; connect **LP Liq1** as the shell side inlet and **Cooling Water In** as the tube side inlet. Name the product streams **Cooling Water Out** and **Cold LP Liq**.
3. On the **Dynamics** tab, **Model** page enter an **Overall UA** of **1.0e6 kJ/C-h (5.26 e5 Btu/F-hr)**,
4. On the **Dynamics** tab, **Specs** page enable the **Delta P** spec and enter a value of **50 kPa** (**7.25 psi**) for both sides of the exchanger.
5. Press the **Calculate K's** button, then uncheck **Delta P** and check **k** for both sides.
6. Move the pressure boundary spec to the **Cold LP Liq** stream and reduce the pressure to **3450 kPa (500.4 psia)**.
7. Run the Integrator for a few time steps.
8. Disable the **Flow** specification on the **Cooling Water In** stream and enable a **Pressure** specification on the **Cooling Water Out** stream.
9. Insert a valve between the **Cooling Water In** stream and the exchanger. Size it for a 10 kPa pressure drop. Name the valve **CW FCV**.

10. Add the following temperature controller to the flowsheet:

In this cell...	Enter...
Connections	
Name	TC-E-100
Process Variable Source	Cold LP Liq – Temperature
Outlet Target Object	CW FCV – Actuator Desired Position
Parameters	
Action	Direct
Range PV Minimum	25 °C (77 °F)
Range PV Maximum	70 °C (158 °F)
Mode	Auto(matic)
SP	40 °C (104 °F)
Kc	1.0
Ti	2.0 min
Td	<empty>

11. Run the Integrator.

Can this heat exchanger/controller combination meet the process requirements (40 °C LP Liquid product temperature)?

____

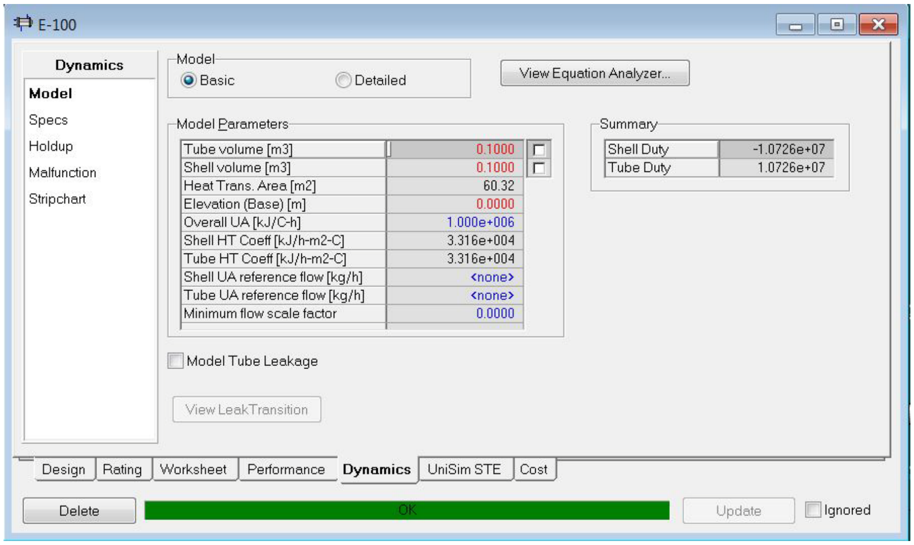
Save the case as **4528.06.Pump HEX.usc**

Save your case!

Heat Exchanger Modeling in Dynamics

The Heat Exchanger has been modeled using the **Basic** model (**Model** page of **Dynamics** tab). With this model the UA value is entered by the user. If the **Detailed** model is activated then the UA value is calculated based on the specified exchanger entered geometry.

Figure 16



The danger of having a fixed UA value is that in reality the UA value changes as the flows through the exchanger change. To account for this UniSim Design allows the UA value to be scaled according to the shell and tube side flows.

$$UA_{used} = UA_{specified} \times F \tag{4}$$

$$F = \frac{2 f_1 f_2}{f_1 + f_2}$$

$$f_1 = \left(\frac{\text{Shell Side Mass flow rate}}{\text{Shell Side Reference flow rate}} \right)^{0.8}$$

$$f_2 = \left(\frac{\text{Tube Side Mass flow rate}}{\text{Tube Side Reference flow rate}} \right)^{0.8}$$

It is also possible to enter a negative value for the **Minimum flow scale factor**. This allows heat exchange to be 'switched off' for low flows, since it sets the scale factor to zero when it falls below the absolute entered value. E.g. if you enter -0.1, then if the scale factor goes below 0.1, 0 is used.

For the UA reference flows it is recommended that the design or steady state flow values which correspond to the entered UA should be entered. As the flows move away from the reference flows the UA value is scaled accordingly. This can greatly increase the model stability in particular as the flows drop to zero values.

UniSim Design also allows a **Minimum flow scale factor** to be entered. This is effectively a lower bound for the flow scale factor (the *F* parameter in the equation above). If the value of the flow scale factor falls below the specified minimum then, it is limited to the minimum value.

Scaling k value with flow

In a similar way to how the UA value can be scaled with flow, UniSim Design also allows the k value to be scaled at low flows.

If a **k Reference flow** is entered then the k value is calculated based on two criteria. If the flow of the system is larger than the k Reference flow, the k value remains unchanged. If the flow of the system is smaller than the k Reference Flow then the specified k value is multiplied by an internally calculated factor which takes into account the flow and pressure drop relationships in the low flow region.

It is recommended that the k reference flow is taken as 40% of the steady state design flow. As the flow drops below the reference flow UniSim Design will linearize the pressure drop equations; this action adds increased stability at low flow conditions.

New LMTD Implementation

It is recommended to keep the **New LMTD Implementation** checkbox activated on the **Parameters** page of the **Rating** tab.

This option applies only in dynamics mode and models the convective heat transfer from one side of the exchanger to the other. This option prevents erroneous temperature crosses which could occur with the previous implementation (checkbox unticked). In this model the thermal conductivity of the metal is not modeled, and metal temperatures are not calculated. However the user can account for the mass of the metal in the exchanger by using the **Metal Mass Factor** to tune the transient rates of temperature change. This is a multiplier in the following equation:

$$Q_{PM} = \frac{mmf \times m_m \times C_p}{dt}\left(T_H - T'_H\right) \tag{5}$$

Where:

Where:	Q_{PM}	=	*duty to/from a pseudo metal representation*
	mmf	=	*Metal Mass Factor*
	m_m	=	*mass of metal*
	C_p	=	*metal heat capacity*
	dt	=	*time step*
	T_H	=	*temperature of fluid holdup*
	T_H'		*temperature of fluid holdup at last time step*

The **Relaxation Factor** is a tuning parameter which allows the user to control the maximum Newton step size taken by the simultaneous heat transfer solver. This can be useful in ensuring convergence and stable temperature predictions during very severe upsets.

Challenge

Consider adding a block between the controller and the valve to reverse the controller action.

Fail Open Valve Response Direction

Leaving VLV-100 with **Reverse Response Direction**, can the control scheme be modified to allow **FC-HP Liq** to be placed back in “**Reverse**” action?

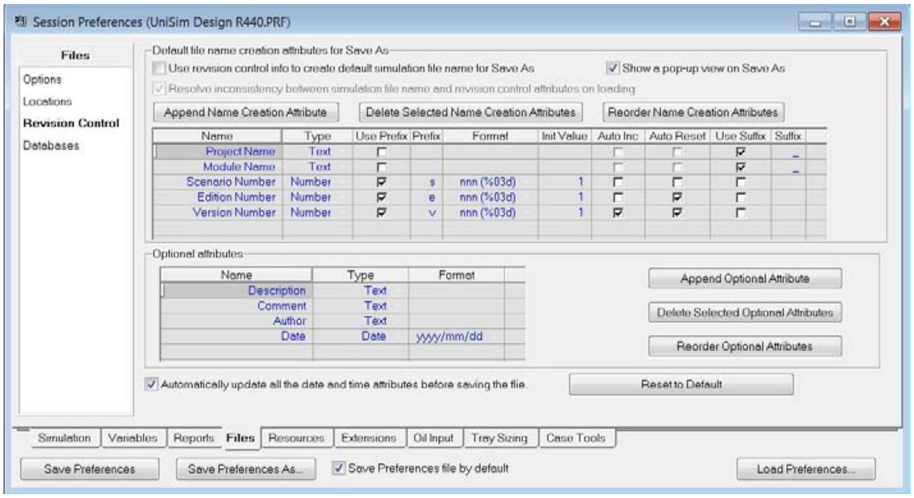
Revision Control

UniSim Design offers a revision control feature to enforce a file naming convention when saving case files. The file name can be made up of standard text and version numbers that increment automatically.

Configuration

Revision control is activated on the **Revision Control** page of the **Files** tab in the UniSim Design Preferences (**Tools** menu **Preferences** option). The default configuration is shown in Figure 16 below.

Figure 17

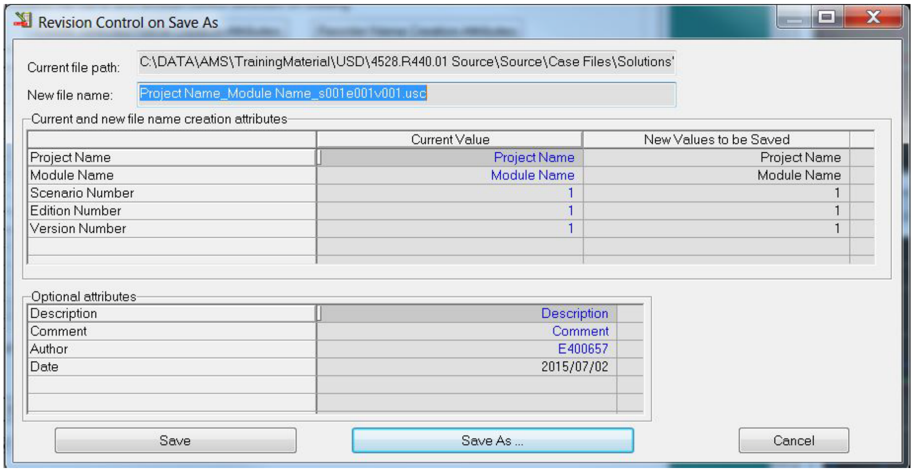


To enable revision control tick the checkbox **Use revision control setting to create default file name for Save As**.

Once enabled, UniSim Design will then automatically pop up a window when the **Save As** function is used to allow the file name to be constructed. Figure 17 below shows this window with the default revision control settings.

- Pressing **Save** saves the case with the name shown in the **New file name** field in the folder shown in the **Current file path** field.
- Pressing **Save As** allows the user to change the name/path but suggests the default name based on the **New file name** field.

Figure 18



The case file name is constructed by linking together the attributes defined in the top table on the revision control page in the preferences.

The possible settings for each attribute are defined in the table below

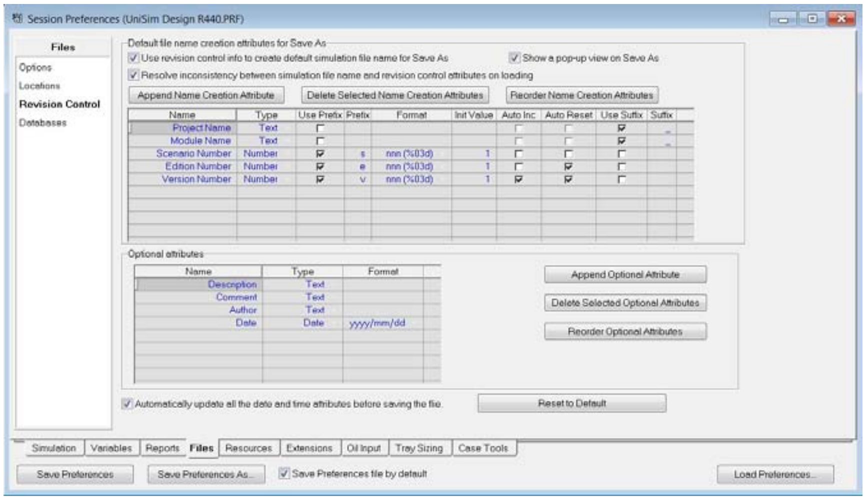
Setting	Description
Name	Descriptive text, also used as the default value when a case is saved for the first time
Type	Has the possible values: • Text • Number • Date • Time • Yes/No
Use Prefix, Prefix	Allows a single character prefix to be applied at the beginning of the attribute
Format	Number/date/time format. In the case of Number, Date or Time a drop down list appears to display the possibilities.
Init Value	Initial value for Number attributes
Auto Inc	Automatically increment a Number attribute each time the case is saved
Auto Reset	Reset the Number attribute to its initial value if a attribute further up the list is changed
Use Suffix, Suffix	Allows a single character suffix to be applied at the end of the attribute

- If **Auto Inc** is set but the resulting file name is already used by a file in the same directory the value of the attribute is incremented again until a unique file name is found.
- **CSM** stands for Case Scenario Management. See the Case Scenario Manager Guide for further details.

Optional Exercise – use of Revision Control in this chapter

1. Configure the Preferences **Revision Control** page as shown in Figure 18 below.

Figure 19



2. The first time you save your case in this chapter, set the following information:

Setting	Value
Project Name	4528
Module Name	6
Description	Expanding the Model

3. On subsequent saves, use the **Save As** command and let UniSim Design automatically increment the **Case Version Number** attribute.

7. Compressor

Workshop

In this module we will model high-pressure gas compressors using compressor performance curves. Using curves to model these unit operations allows UniSim Design to accurately simulate actual plant equipment.

Learning Objectives

Once you have completed this module, you will be able to:

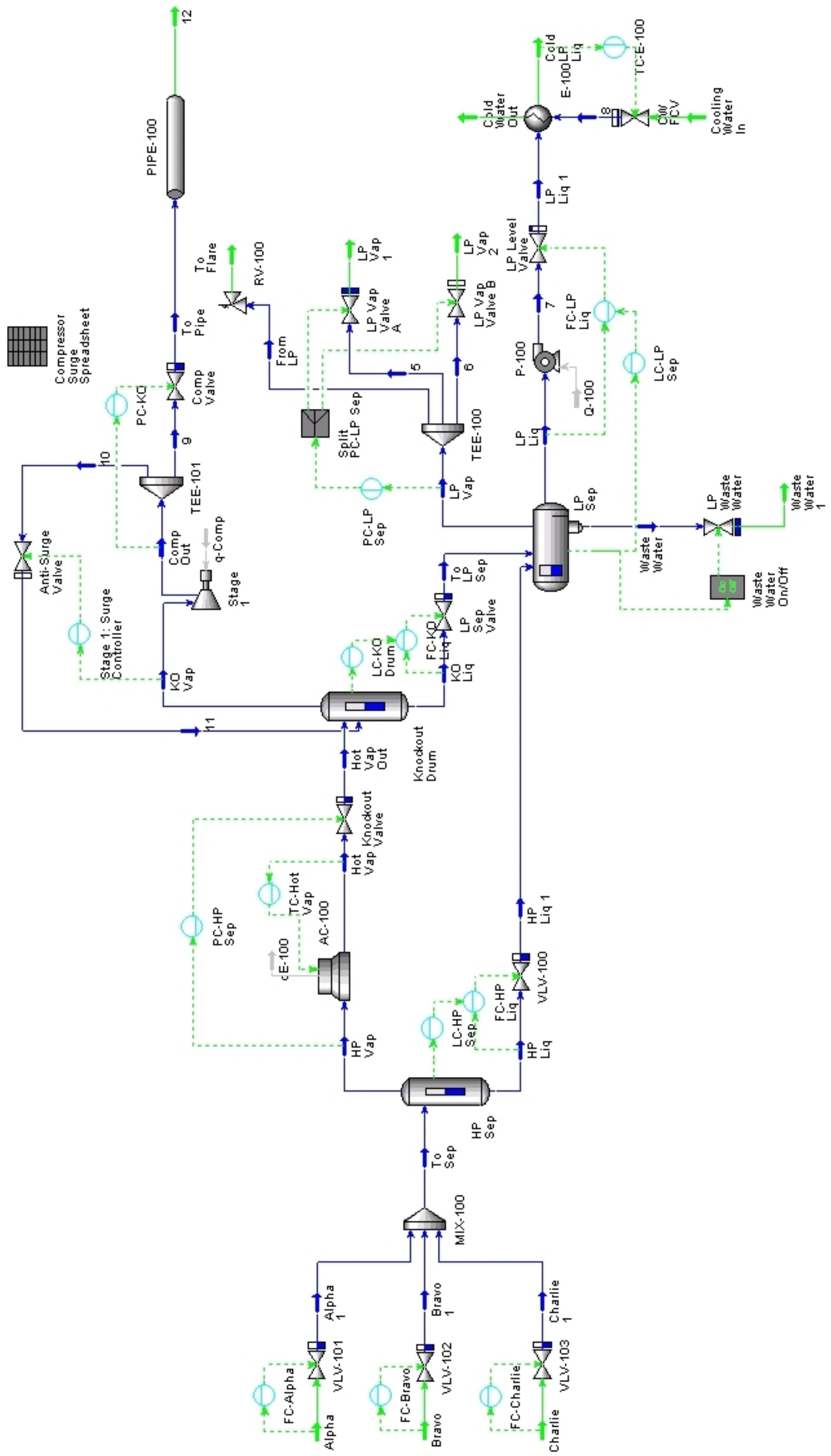
- Specify head and efficiency curves to compressors
- Use multiple curves to model compressors
- Make changes/additions to the dynamic model
- Move pressure-flow specifications around
- Accurately model existing plant equipment with UniSim Design

Prerequisites

Before beginning this section you need to:

- Know how to set up strip charts
- Understand the pressure-flow network

Process Overview



Compressor Curves

Using compressor curves in your UniSim Design simulation allows you to accurately model existing plant equipment. With an accurate simulation, you can determine if an existing compressor is able to meet the specifications of your process.

The compressor curves also allow UniSim Design to calculate heads and efficiencies that are dependent on the flow rate.

- If the flow rate through the compressor is known to be constant, a single flow rate and efficiency can be supplied.
- If the flow rate is expected to change, using a compressor curve will allow UniSim Design to calculate new heads and efficiencies based on the current flow rate.

This results in greater accuracy in the simulation, and allows UniSim Design to closely model actual plant equipment.

Adding the Compressor

If the training PC does not have Excel then open the file **4528.07.Compressor Only.usc** and copy/paste the compressor to your case. Ask the instructor if you need help with this.

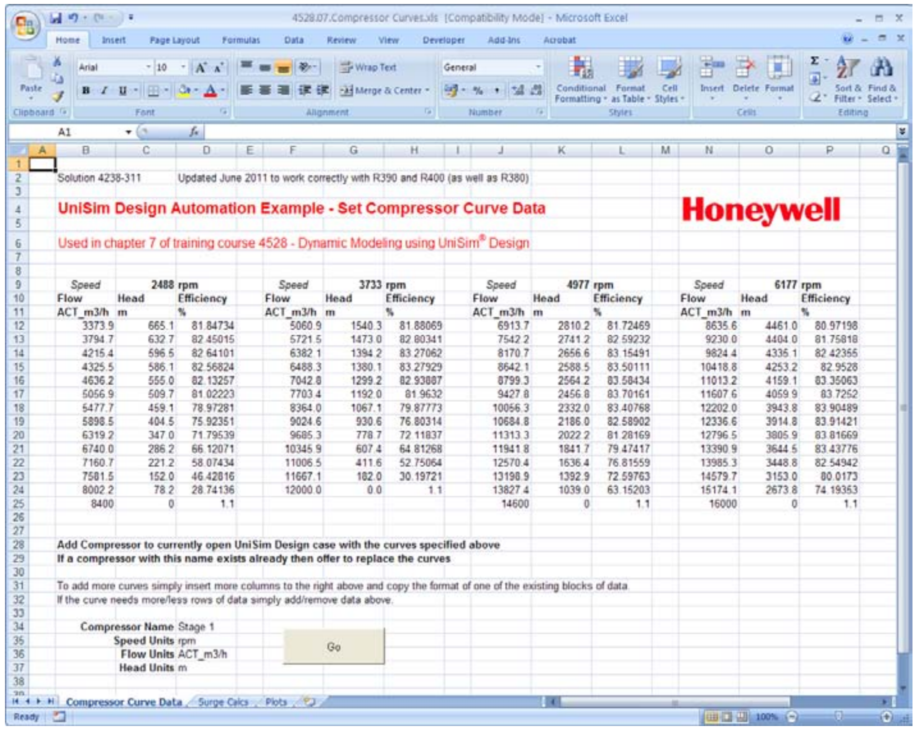
Included with the course files is an Excel spreadsheet which contains a series of compressor curves for different speeds. There are several ways to transfer the data to UniSim Design:

- Retype them
- Use Copy and Paste
- Use OLE Automation

In this case the Excel spreadsheet includes VBA code which will automatically create a compressor on the flowsheet and set the compressor curve information.

1. Open the UniSim Design file from the last exercise (**4528.06.Pump HEX.usc**).
2. Open the Excel spreadsheet file **4528.07.Compressor Curves.xls from the Starter folder.**
3. Run the VBA code by pressing the **Go** button on the **Compressor Curve Data** tab.

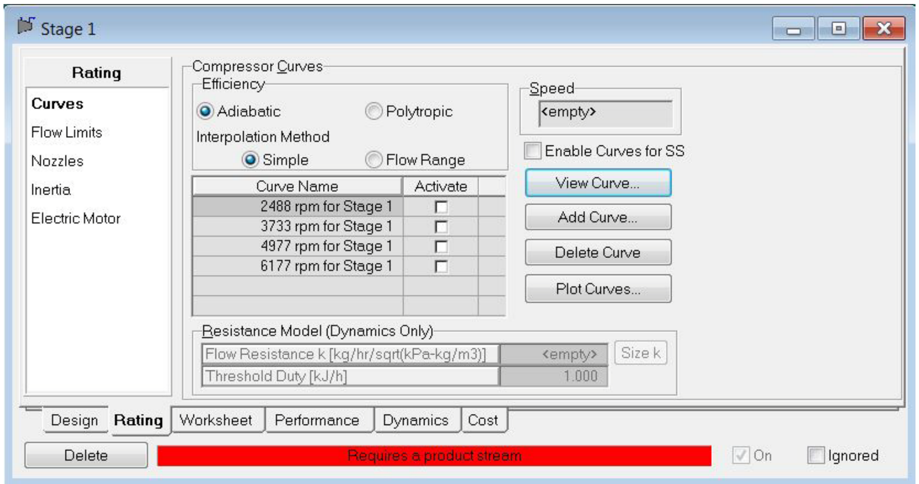
Figure 1



Speed	2488 rpm		Speed	3733 rpm		Speed	4977 rpm		Speed	6177 rpm	
Flow	Head	Efficiency	Flow	Head	Efficiency	Flow	Head	Efficiency	Flow	Head	Efficiency
ACT_m3/h	m	%	ACT_m3/h	m	%	ACT_m3/h	m	%	ACT_m3/h	m	%
3373.9	665.1	81.84734	5060.9	1540.3	81.88069	6913.7	2810.2	81.72469	8635.6	4461.0	80.97198
3794.7	632.7	82.45015	5721.5	1473.0	82.80341	7542.2	2741.2	82.59232	9230.0	4404.0	81.75818
4215.4	596.5	82.64101	6382.1	1394.2	83.27062	8170.7	2656.6	83.15491	9824.4	4336.1	82.42355
4325.5	586.1	82.56824	6488.3	1380.1	83.27929	8642.1	2588.5	83.50111	10418.8	4253.2	82.9528
4636.2	555.0	82.13257	7042.8	1299.2	82.93887	8799.3	2564.2	83.58434	11013.2	4159.1	83.35063
5056.9	509.7	81.02223	7703.4	1192.0	81.9632	9427.8	2456.8	83.70161	11607.6	4059.9	83.7252
5477.7	459.1	78.97281	8364.0	1067.1	79.87773	10056.3	2332.0	83.40768	12202.0	3943.8	83.90489
5898.5	404.5	75.92351	9024.6	930.6	76.80314	10684.8	2186.0	82.58902	12336.6	3914.8	83.91421
6319.2	347.0	71.79539	9685.3	778.7	72.11837	11313.3	2022.2	81.28169	12796.5	3805.9	83.81669
6740.0	286.2	66.12071	10345.9	607.4	64.81268	11941.8	1841.7	79.47417	13390.9	3644.5	83.43776
7160.7	221.2	58.07434	11006.5	411.6	52.75064	12570.4	1636.4	76.81559	13985.3	3448.8	82.54942
7581.5	152.0	46.42816	11667.1	182.0	30.19721	13198.9	1392.9	72.59763	14579.7	3153.0	80.0173
8002.2	78.2	28.74136	12000.0	0.0	1.1	13827.4	1039.0	63.15203	15174.1	2673.8	74.19353
8400	0	1.1				14600	0	1.1	16000	0	1.1

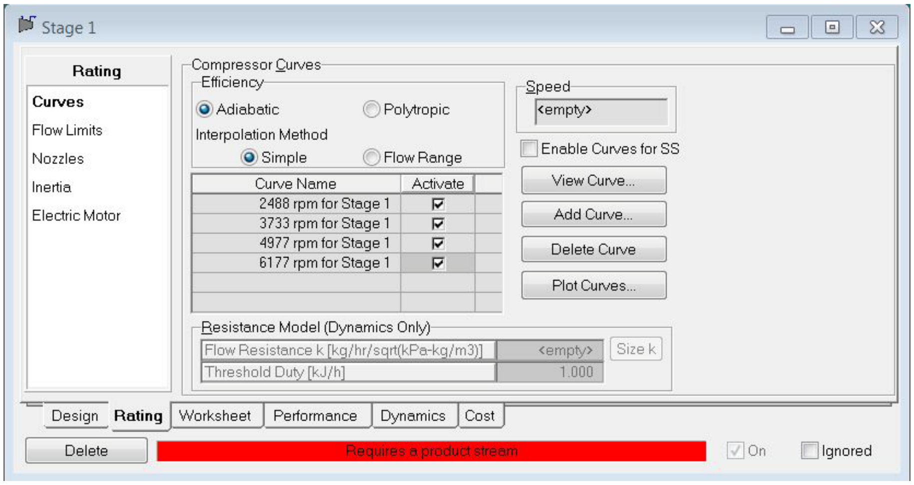
4. A compressor is added to your simulation. Go to the **Rating** tab **Curves** page for the compressor; verify that the curve data added matches the data in the Excel spreadsheet.

Figure 2



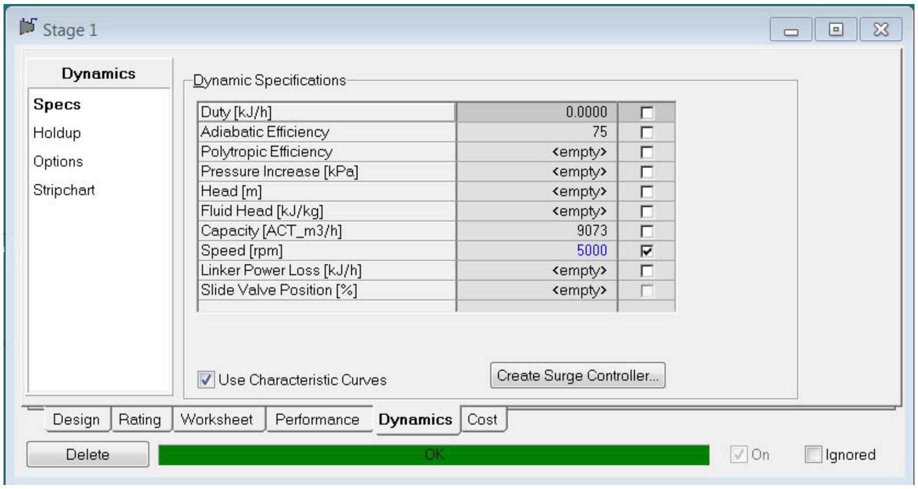
5. Connect the compressor inlet to stream **KO Vap**. The outlet stream is **Comp Out** and the energy stream is **q-Comp**.
6. Switch on the compressor curves. This has to be done in two places. First on the **Rating** tab **Curves** page, by checking the **Activate** box next to each curve.

Figure 3



7. Secondly on the **Dynamics** tab **Specs** page uncheck **Duty** and **Polytropic Efficiency**, and check **Speed** and **Use Characteristic Curves**.
8. Enter a speed of **5000 rpm** and run the Integrator for a few time steps to propagate information to stream **Comp Out**.

Figure 4



What is the pressure at the compressor discharge?
____

9. Add a discharge valve **Comp Valve** to the compressor outlet.
10. Remove the pressure specification of the compressor suction and move it to the outlet of the discharge valve. Set it to **7200 kPa (1044 psia)**.
11. Change the **Valve Manufacturers** to **ANSI / ISA75.01** and size the valve.

What Cv value was calculated for the discharge valve?
____

12. Add a pressure controller to control the compressor discharge pressure.

In this cell...	Enter...
Connections	
Name	PC-KO
Process Variable Source	Comp Out - Pressure
Output Target Variable	Comp Valve – Actuator Desired Position
Parameters	
Action	Direct
Range PV Minimum	7000 kPa (1015 psia)
Range PV Maximum	10500 kPa (1520 psia)

You may want to start the pressure controller in manual. Start the Integrator and tune the controller. Use the pressure recorded at the end of step 8 as the set point.

Save your case as **4528.06.Comp_PC.usc**.

Save your case!

Surge and Stonewall Curves

There are two flow limits for centrifugal compressors: surge (pumping) and stonewall (choked flow).

Surge

Compressor surge is a complete breakdown in compression resulting in a reversal of flow and the violent expulsion of previously compressed gas out through the compressor inlet, due to the inability of the compressor to continue working against the compressed gas downstream.

The surge point for any given speed is the point of minimum flow and maximum head. If the surge point is reached and the compressor goes into surge, compressed gas downstream flows back into the compressor. The discharge pressure drops until it is within the capacity of compressor, and then the cycle repeats. This is particularly dangerous, with very high levels of vibration causing accelerated engine wear and possible damage, even the total destruction of the compressor.

UniSim Design Limitations

The surge flow behaviour as implemented in UniSim Design attempts to mimic the physical phenomenon of surge by multiplying the compressor head with a random number when the suction flow falls below the surge point. While this implementation does mimic surge by introducing random and erratic flow into the compressor, in some situations it may lead to numerical instabilities which can lead to a convergence failure of the dynamic pressure flow solver.

For this reason, using the Surge curve in UniSim Design is not recommended. Since compressors are typically controlled to keep them out of surge, adding the surge controller will allow modeling of the compressor and ancillary control equipment as it approaches surge without actually modeling the surge behavior. The surge controller is discussed in the next section.

Stonewall

On centrifugal compressors, stonewall (choked flow) occurs at any point when sonic velocity is reached. When this point is reached for any given gas, flow can only be further increased if the compressor is modified, or for a variable speed compressor, the speed is increased.

UniSim Design Limitations

The stonewall curve in UniSim Design is a linear fit of the parameters entered in (1).

$$Q^2 = a \cdot speed + b \tag{1}$$

The flow units entered for the stonewall curve are based on the compressor inlet conditions. When used, the stonewall option bounds the compressor flow, but not head.

Before using the stonewall curve it is suggested that the operating curves for the compressor be extended into the expected stonewall regime as this can aid in the modeling of stonewall.

Compressor Surge Control

If the suction flow to a centrifugal compressor falls below the surge point, the rapid flow oscillations and associated vibrations associated with surge can be extremely damaging to the compressor equipment. Hence most centrifugal compressors have a protection mechanism that ensures a minimum throughput by opening a recycle valve such that the compressor can re-circulate flow back to its feed. Opening of the recycle valve is achieved using a special **Surge Controller**.

As the drop in throughput can sometimes be very fast, the controller must also react very quickly when surge is approached and still give smooth control when the compressor is operating normally but not too far from surge.

The surge controller is an extremely rapidly acting controller. The control algorithms used to prevent a compressor going into surge are extensions of the PID algorithms.

Add an Anti-Surge (Recycle) Line

1. Add a valve to the simulation, and make sure the **Valve opening** is **0%**. This will be the **Anti-Surge Valve**.
2. Size the new valve such that it has a Cv value that is **20%** of the discharge valve Cv value.
3. From the compressor's **Dynamics** tab **Specs** page add a **Surge Controller** by pressing the button.
4. The output for the **Surge Controller** is the anti-surge valve.
5. Disconnect the compressor discharge stream from the inlet of the compressor discharge valve **Comp Valve**.
6. Add a Tee (or a Header). The inlet stream to the tee is the compressor discharge stream. Connect the outlets of the tee to the compressor discharge valve and anti-surge valve.
7. Rename the outlet stream leaving **Comp Valve** to **To Pipe.**
8. Connect the outlet of the anti-surge valve to the Knockout Drum.

The **Surge Controller** is used exclusively for compressors. When in **Auto(matic)** mode, the **Set Point** is calculated by the controller itself; it is not set by the user.

The surge line for the compressor model is entered as a speed vs. flow rate curve in a tabular form.

For the purpose of the surge controller the surge line is calculated from the formula:

$$Head = A + BQ + CQ^2 + DQ^3 \qquad (2)$$

Where:	*Head*	=	*compressor calculated head [m]*
	A,B,C,D	=	*parameters*
			(B=D=zero, for this simulation)
	Q	=	*Surge flow rate [m^3/s]*

For this module you should assume that constants **B** and **D** are **zero**, thus reducing the equation (2) to its quadratic form. Linear regression of **Head** versus **Vol Flow** squared provides parameters **A[m]** and **C[s^2/m^6]**.

You can find the regression results within the supplied file **4528.07. Compressor Curves.xls**

9. Fill in the cells of the following table by looking at the curves entered for the compressor:

Speed (rpm)	Minimum volume flow [m^3/s]	Head [m]

10. Use Excel to work out the parameters A and C.

What values were calculated for A and C?

____

In addition to the surge line, the user must also supply Control and Backup lines.

Control Line

The control line is the primary 'set point' for the controller. This line is set at some percentage above the surge flow (typically 10%), and the controller tries to maintain the compressor flow above this control line. If the flow is above the backup line then the control action is the 'normal' PID action. The control line set point can be determined with equation (3):

$$Q = \frac{100 + X}{100} \cdot Q_{Surge} \qquad (3)$$

Where: Q = *Control flow rate in [m^3/s]*

X = *Control Line % (Specified in the Controller) [%]*

Q_{Surge} = *Surge Flow*

Backup Line

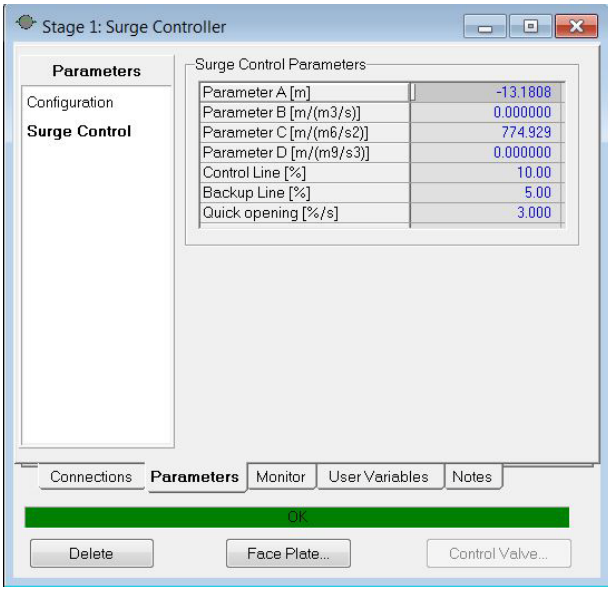
The points of the backup line are located between the surge line and the control line.

If the compressor flow is above the backup line, then a PID algorithm is used. However, when the flow drops below the backup line, more aggressive control action is taken to prevent surge. The backup line flow rate is determined in a similar way to the control line flow (typically the backup line percentage is 5%).

If the compressor flow drops below the backup line then a Quick Opening Algorithm replaces the PID algorithm, and the recycle valve will be opened at the specified **Quick Opening** rate

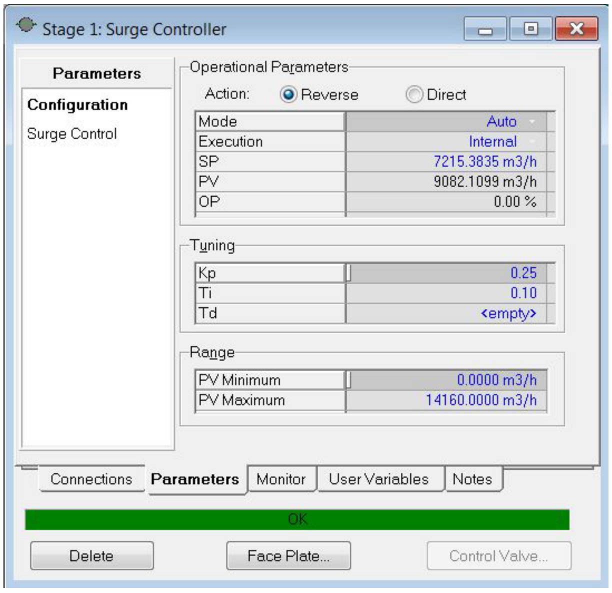
11. Add the control **Surge Parameters A** and **C** calculated above.
12. The control line and backup line are set at **10%** and **5%** respectively above the surge line. The recycle valve will be opened with a **Quick Opening** rate of 3% per second.

Figure 5



13. Add the PID **Tuning** parameters for the controller and the **PV Minimum** and **PV Maximum**.

Figure 6


Stage 1: Surge Controller
Parameters
Configuration
Surge Control
Operational Parameters
Action: Reverse Direct
Mode Auto
Execution Internal
SP 7215.3835 m3/h
PV 9082.1099 m3/h
OP 0.00 %
Tuning
Kp 0.25
Ti 0.10
Td <empty>
Range
PV Minimum 0.0000 m3/h
PV Maximum 14160.0000 m3/h
Connections Parameters Monitor User Variables Notes
OK
Delete Face Plate... Control Valve...

Save the case as **4528.07.Comp_AntiSurge_1.usc**.

Save your case!

UniSim Design Spreadsheet

With complete access to all process variables, the Spreadsheet is a very powerful tool in UniSim Design. The power of the Spreadsheet can be fully realized by the addition of formulae, functions, logical operators and basic programming statements.

The Spreadsheet's ability to import and export variables means that seamless transfer of data between the Simulation Environment and the Spreadsheet is a simple matter. Any changes in the Simulation Environment are immediately reflected in the Spreadsheet, and vice-versa.

The Spreadsheet has several common applications. For example, the Spreadsheet can be used to:

- Collect together key inputs and results from multiple flowsheet objects.
- Perform mathematical operations using variables from the simulation.
- Model process equipment which does not currently exist within UniSim Design.

Importing and Exporting Variables

Any variable in the simulation case can be imported into the Spreadsheet. The contents of any Spreadsheet cell can be exported to any specifiable (blue) variable in the simulation case. There are three ways of importing values into the Spreadsheet:

- **Drag and Drop.** Position the cursor over the variable in the stream or unit operation property view window; then right click and hold the right mouse button. Move the cursor over to the Spreadsheet. Once over the Spreadsheet, the cursor's appearance will change to a "bull's eye". Release the right mouse button when the "bull's eye" cursor is over the desired cell. The specific information about the imported variable will appear in the **Current Cell** group. Before attempting a drag and drop import the window containing the variable to be imported and the spreadsheet window should be arranged so that both are easily visible on screen.
- **Variable Browsing.** Position the cursor on an empty cell in the Spreadsheet, click the right mouse button, choose **Import Variable** from the list that appears and select the variable using the **Variable Navigator**.
- **Connections tab.** On the **Connections** tab, click the **Add Import** button and select the desired variable using the **Variable Navigator**. After selecting the variable, choose the desired cell from the drop down list.

Note that it is not possible to import into, and export from, the same cell. Instead use two cells: one for the import and one for the export. Link them together with a simple **'=A1'** formula.

Exporting variables from the Spreadsheet into the Simulation environment is also a simple procedure. The methods for doing this are very similar.

- **Drag and Drop**. Position the cursor over the Spreadsheet cell that is to be exported. Click and hold the right mouse button; the cursor should now change to the "bulls' eye" type. Move the "bull's eye" cursor over the desired variable cell (this could be on any other object in the model). Release the right mouse button. Before attempting a drag and drop export the window containing the variable to be exported to and the spreadsheet window should be arranged so that both are easily visible on screen.
- **Variable Browsing.** Position the cursor on the exportable cell in the Spreadsheet, click the right mouse button, choose **Export Formula Result** from the list that appears and select the desired location for the variable using the **Variable Navigator**.
- **Connections tab.** On the **Connections** tab, click the **Add Export** button and select the desired variable using the Variable Navigator. After selecting the variable, choose the desired cell from the drop down list.

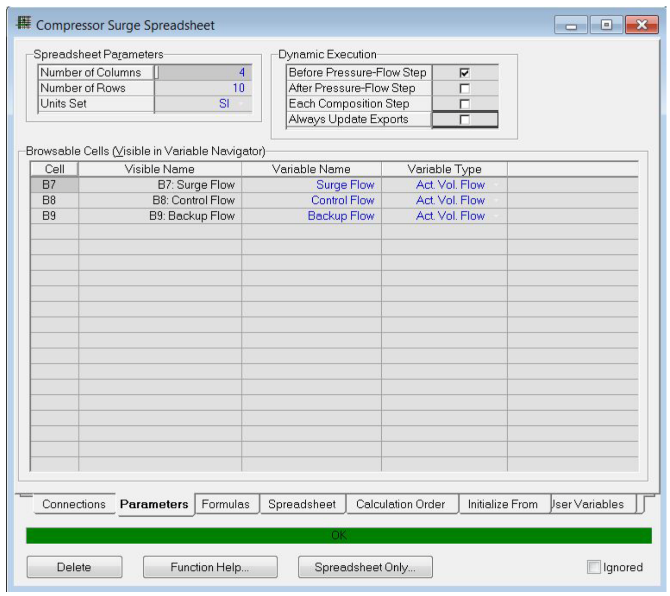
If a unit set is being used by a spreadsheet it is ‘locked’ and so cannot be modified. To modify the unit set any objects using it must be temporarily modified to use a different set. Pressing the **View Users** button on the **Units** page of the **Variables** tab of the **Session Preferences** window shows which objects are using a unit set.

Note that if you make the dimensions smaller any cells containing data that fall outside the new size will be deleted without any warning.

Unit Sets in Spreadsheet

On the parameters tab of the Spreadsheet property view, the user can choose a **Units Set** to be used by the Spreadsheet. This determines the units used for all cells in the spreadsheet and can be set independently of the Unit Set currently in use in UniSim Design.

Figure 7



The user can also set the following on the **Parameters** tab:

- the dimensions of the Spreadsheet (**Number of Columns**, **Number of Rows**)
- the **Variable Name**, this is used when identifying the cells in the spreadsheet (for example on a strip chart)
- the **Variable Type**

Spreadsheet in Dynamics

When using the spreadsheet in dynamics there are a several considerations to make:

- The spreadsheet can be calculated before the Pressure-Flow step, after the Pressure-Flow step or on each Composition step. This is set on the **Parameters** tab of the spreadsheet.
- Spreadsheets calculate once per calculation step. Therefore if you pass a value from spreadsheet A to spreadsheet B, use the value in spreadsheet B in a calculation and pass the results back to spreadsheet A the value will be passed back to spreadsheet A on the next step and thus will be one step behind. You can use this fact to your advantage to implement a counter using spreadsheets or to view values at a previous time step to calculate, for example, a numerical derivative.

UniSim Design Spreadsheet Limitations

Find information on more functions in the UniSim Design spreadsheet in the Operations Guide section 12.10.2.

The spreadsheet in UniSim Design while powerful, is very basic in spreadsheet functionality when compared to Microsoft Excel.

Some of the functionalities which the UniSim Design spreadsheet does not contain include:

- Equations need to be fully typed in; you cannot use the arrow keys or mouse to select cells to be written into the formula.
- It is not possible to drag an equation from one cell into adjacent cells and have the equation be automatically updated for the new cells. However, this can be achieved using copy/paste.
- There is no auditing tool to evaluate the interdependence of cells. However, UniSim Design will issue a warning if it detects a circular calculation.

Using Excel to Create UniSim Design Spreadsheets

In Microsoft Excel 2003, you can also show formulas from the **Tools** menu, **Options** and on the **View** tab check the **Formulas** box. In Excel 2007 there is an option in the **Formula Auditing** group of the **Formulas** ribbon to **Show Formulas**.

It is possible to use to use Microsoft Excel to create spreadsheets for use in UniSim Design. The steps for this procedure are:

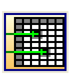
1. Create the desired spreadsheet in Excel ensuring that only functions which are supported by UniSim Design are used.
2. In Excel, use **Ctrl ‘** (grave accent) to show the formulas rather than results.
3. For any functions (‘if’ statements, ln, abs, etc.) change the Excel cell to text by placing an apostrophe at the beginning and add the required @ symbols.
4. Highlight the desired cells in Excel and copy them into the UniSim Design Spreadsheet.

You should find that the Excel formulas which have been converted to text with the apostrophe are imported as regular formulas into UniSim Design - the apostrophe is not imported.

Adding a Spreadsheet

A spreadsheet will be added to calculate the surge flow line, control line and the backup line of the compressor.

To allow the surge, control and backup lines to be plotted:



Spreadsheet icon

1. Add a spreadsheet to the case.
2. Calculate on the spreadsheet the surge flow rate from equation (4) and the corresponding control and backup flow rates. Remember these are both greater than the surge flow rate.

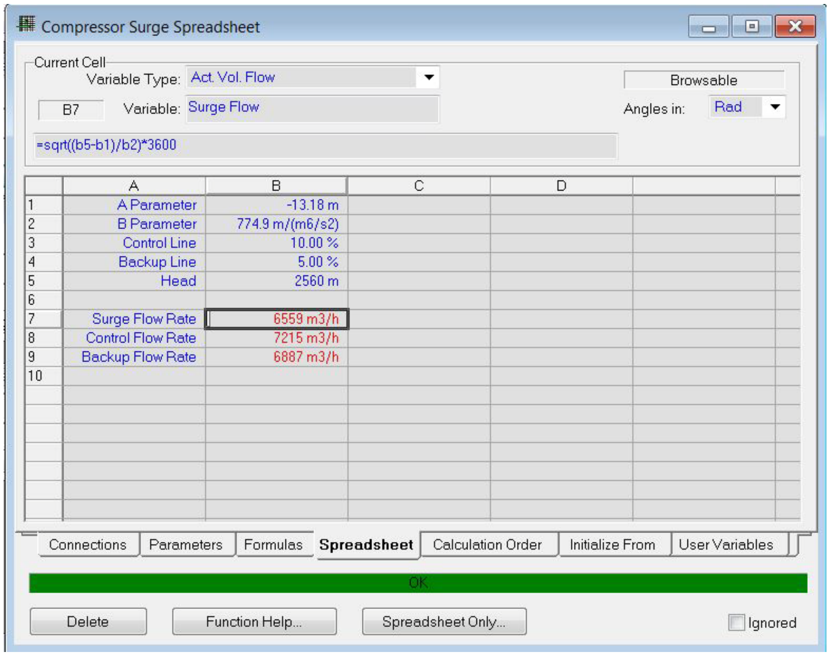
$$Q_{surge}\left[m^3\ /\sec\right] = \sqrt{\frac{Head\left[m\right] - A}{C}} \tag{4}$$

3. The **Head** can be imported from the compressor. The **A Parameter** and **C Parameter** can be imported from the surge controller.

Make sure the calculated flow rates display in the correct units. Remember that the flow rates in the equation are in m^3/s.

When you have finished the spreadsheet should look like the following:

Figure 8



Add a Strip Chart

1. Add the following variables to a strip chart:
   - Anti-surge valve **actuator current position** and **desired position;**
   - Compressor feed (stream **KO Vap**) **actual volume flow rate**;
   - **Surge, control** and **backup flow rate** calculated by the spreadsheet.
2. Set the **sampling interva**l of the **Data Logger** to **0.5 second**s, the same as the Integrator **Step size**.

Save the case as **4528.07.Comp_AntiSurge_2.usc**.

Save your case!

Experiment with the model

When performing multiple experimental runs it helps to save a copy of the case in the steady state condition with the flow control SP's zeroed so that you can reload this case, make modifications and quickly see the effects of the changes.

1. Try setting the **SP** of all the feed flow controllers to **zero** at once.

Can the model cope?

____

If not, consider the following questions:

- Is the resolution of the model fine enough? Surge controllers open very quickly. Try setting the Integrator step size to a smaller value (e.g. 0.05 seconds)
- Is the Anti-Surge valve opening quickly enough?
- Is the Anti-Surge valve large enough to pass the required flow?

If you change the Integrator **Step size**, remember to adjust the strip chart **sampling interval** to record all the new data and make sure the logger is big enough to hold all the captured data.

What opening rate do you suggest for the surge controller?

____

What Cv value should be set in the Anti-Surge Valve?

____

Do all the other controllers work properly and achieve their SP after the upset?

____

2. Save your case as **4528.07.Comp_AntiSurge_3.usc**

Save your case!

Advanced Modeling – Pipe Segment

The pipe segment can be used in steady state and dynamics to simulate a wide variety of piping situations ranging from single or multiphase plant piping with rigorous heat transfer estimation, to large capacity looped pipeline problems. It offers the common pressure drop correlations developed by Gregory, Aziz, and Mandhane, and Beggs and Brill. In addition there are a large number of specialist pressure drop correlations available. Consult the on-line help and the manual for more information on these methods.

For dynamics there is also a simplified pipe friction model which allows you to select between turbulent and full range Churchill methods to simulate pipe flow.

Selecting a Pressure Drop Correlation

In dynamics mode these correlations are only used if the **Pipe Flow Model** option on the **Dynamics** tab **Parameters** page is set to **Pipe Model Correlations**.

On the **Parameters** page of the **Design** tab you can select the **Pipe Flow Correlation** which will be used for two-phase (Vapour-Liquid) flow calculations. The options are:

- Aziz, Govier & Fogarasi
- Baxendell & Thomas
- Beggs & Brill
- Duns & Ros
- Gregory, Aziz, Mandhane
- Hagedorn & Brown
- OLGAS2000_2P
- OLGAS2000_3P
- OLGAS2000_3Pext
- Orkiszewski
- Poettman & Carpenter
- Tulsa99
- UniSim Liquid Slip
- UniSim Homogeneous Flow

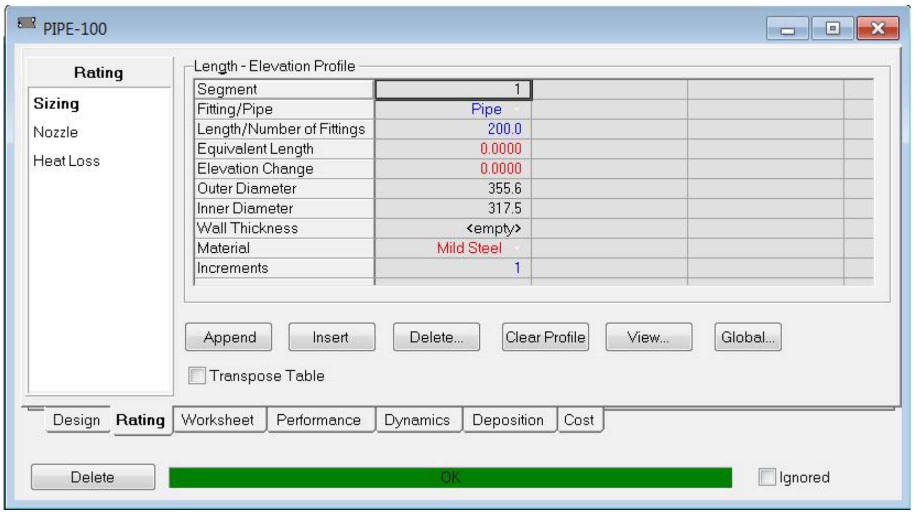
For single phase streams, the Darcy equation is used for pressure drop predictions, regardless of the correlation selected on the **Parameters** page. The Darcy equation is a modified form of the mechanical energy equation, which takes into account losses due to frictional effects as well as changes in potential energy. The pipe in this section will be single phase so the Darcy equation is used automatically.

Entering Pipe Data

On the **Sizing** page of the **Rating** tab, the length-elevation profile for the Pipe Segment is defined. Each pipe section and fitting is labeled as a segment. To fully define the segments, you must also specify pipe schedule, diameters and pipe material.

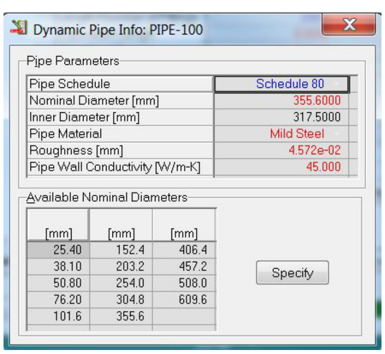
At least one segment is required.

Figure 9



The diameter of the segment can be specified by pressing the **View Segment** button. From the resulting view, you can select a pipe schedule and a nominal diameter. UniSim Design will then fill in the correct inner and outer pipe diameters after the **Specify** button is pressed. Figure 10.

Figure 10



Configuring Heat Loss

This is set up on the **Heat Loss** page of the **Rating** tab. As covered in Module 5, there are three options: **None**, **Simple** or **Detailed**.

Dynamic Specifications

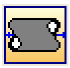
The **Parameters** page on the **Dynamics** tab contains the remaining parameters required to configure the pipe segment for dynamic operation.

The following table lists and describes the objects available:

Object	Description
Pipe Flow model **Simple Pipe Friction Model Method**	Allows you to select between turbulent and full range Churchill methods to simulate the pipe flow model in **Dynamics** mode. The methods are similar to the methods in the **Pipe** page of the **Valve** unit operation. The calculation time for these methods is short. However, for multiphase systems the calculated results are approximate.
Pipe flow model **Pipe Model Correlations**	Allows you to select the pipe flow model based on the available pipe flow correlation selection from the **Parameters** page on the **Design** tab. The calculation time for this method is longer but the results are more accurate.
Pipe Holdup type **one dp calc per segment**	Allows you to calculate the overall holdup values of the entire pipe. This method calculates the results by lumping together the whole pipe volume. The calculation time is short. However, this method is not recommended if you want to track composition (or model lag) along the pipe.
Pipe Holdup type **one holdup per segment**	Allows you to calculate the holdup values for each segment in the entire pipe. This method calculates and models the composition and other changes through the pipe network rigorously. The calculation time is longer. However, the results are more accurate.
Model Holdup Volume checkbox	If checked then the Holdup volume inside the pipe is modeled. The volume used is calculated based on the pipe lengths and diameters entered. Generally the pipe volumes are ignored or lumped together in a single holdup volume, unless a model of the composition lag is required. The lumped volume approach is a simpler more robust option, and often the pressure drop result is the main interest. Having to consider many holdups with small volumes may lead to instabilities.
Dynamic Momentum	Calculate acceleration of flow rather than instantaneous steady state response to pressure change

Additionally the **Base Elevation of Inlet Relative to Ground** can be specified on the **Nozzles** page of the **Dynamics** tab.

Adding a Pipe Segment



Pipe Segment icon

1. From the Object Palette, add a Pipe Segment to the simulation.
2. Connect the inlet of the pipe to the outlet of valve **To Pipe**.
3. On the **Sizing** page, **Rating** tab of the pipe specify the following:

In this cell...	Enter...
Connections	
Name	Pipe-100
Parameters	
Fitting or Pipe	Pipe
Length	200 m (656.2 ft)
Pipe Schedule	Schedule 80
Pipe Material	Mild Steel
Pipe Nominal Diameter	355.6 mm (14 in)

4. On the **Heat Loss** page, enable the **Simple** model and leave the default values.
5. On the **Parameters** page, **Dynamics** tab select the **Pipe Model Correlations** option.
6. Leave the other options at the default values.
7. Move the boundary specification to the pipe outlet stream and specify a pressure of **6640 kPa** (**963 psia**).
8. Start the Integrator and run the model.
9. Save the case as **4528.07.CompE.usc**.

Since the pipe is single phase vapour the Darcy equation will be used regardless of the **Pipe Flow Correlation** selected.

Save your case!

Advanced Modeling

Since there are a relatively large number of components we will observe the effective change in composition by measuring the molecular weight of the fluid entering the pipe and leaving the pipe.

1. Create a new strip chart and add the pipe inlet and outlet molecular weights and flow.
2. Open the property view for the air cooler and on the **Ratings** tab, turn off both fans by unchecking the **Fan is On** box.
3. Start the Integrator.
4. Put controller **FC-Alpha** in **Manual** mode and change the **OP** to **0%.**
5. Observe the change in molecular weight for the inlet and outlet of the pipe.
6. Stop the Integrator.
7. On the **Parameters** page of the **Dynamics** tab of the pipe, enable **Model Holdup Volume**.
8. Run the model and change the **OP** for **FC-Alpha** to **55%**. Observe the lag in molecular weight change between the inlet and outlet molecular weights.
9. Increase the pipe length and diameter and continue to vary the **OP** of FC-Alpha between **0%** and **55%.** Observe the lag in molecular weight change between the inlet and outlet molecular weights.
10. Save the case as **4528.07.Pipe.usc**.

Save your case!

8. TEG Dehydration Tower

Workshop

The Column Dynamics module introduces to you a process for setting up a distillation column with UniSim Design. Some of the concepts you have learnt in the previous modules are applied here. However the distillation column is one of the most complex unit operations in UniSim Design. As such, it deserves special attention. Starting with the previous module, you will expand the flowsheet by adding a TEG Contactor to the high-pressure gas stream.

Learning Objectives

Once you have completed this module, you will be able to:

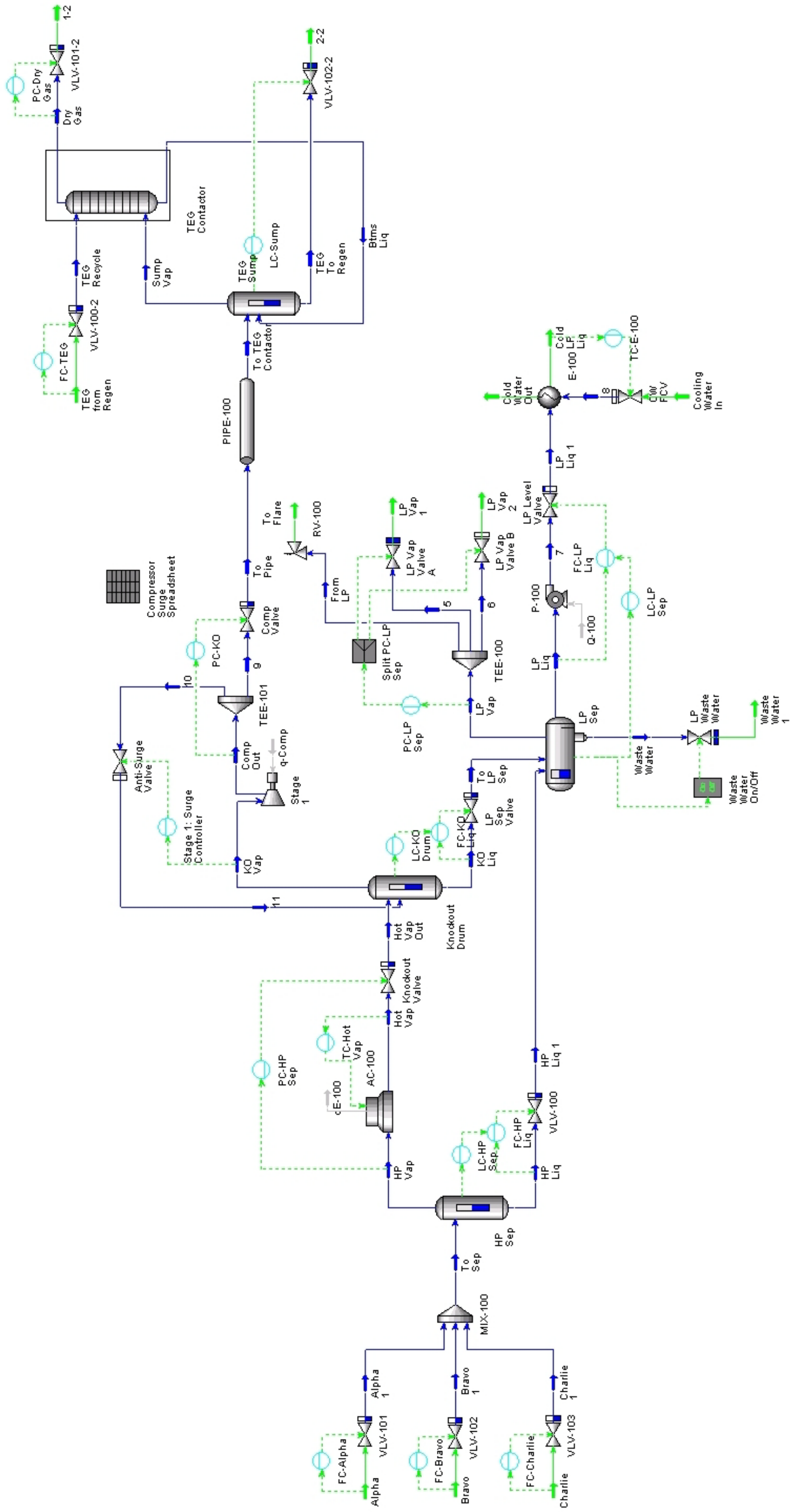
- Configure a distillation column in steady state mode
- Prepare a distillation column for dynamic simulation

Prerequisites

Before beginning this section you need to know how to:

- Set up strip charts
- Understand the pressure-flow network

Process Overview



Adding the Column

The remaining moisture in the high-pressure gas stream is removed by contacting the stream with TEG. For this simulation we are only going to model the TEG contactor column. The same techniques could be used to add a regeneration column as well.

In general it is better to first build a column in steady state then transition it to dynamics. This gives a good starting point for the dynamics solver.

Before building the column in dynamics we will first build the column in steady state. There are two options for doing this:

1. Transition the whole model back to steady state, add and converge the column, then transition back to dynamics.
2. Create a new model in steady state mode of the column only and use this as the starting point for dynamic simulation

For this module we will use the second method.

Prepare the Steady State Model

Use the Window/Load Workspace option to navigate between the two cases.

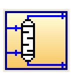
1. Open the case stored at the end of the last simulation (**4528.07.CompE.usc**). Make sure it has lined out.
2. Rename the exit stream leaving **PIPE-100** to **To TEG Contactor**.
3. Enter the Basis Environment and add TEG to the component list. Save your dynamic case.
4. Export the fluid package.
5. Create a new case and import the fluid package exported in the previous step. Keep the case in Steady State mode.
6. Create a new stream and copy the name, temperature, pressure, flow and composition from the compressor discharge boundary stream **To TEG Contactor** (from the dynamic model). This will be the gas feed stream for the TEG contactor.

7. Add a new stream to the simulation. Name it **TEG from Regen**. Supply the following information.

Take care, this stream is specified using Mass Fractions.

In this cell...	Enter...
Name	TEG from Regen
Temperature	25 °C (77 °F)
Pressure	7200 kPa (1045 psia)
Flow Rate	70 kgmole/h (115 lbmole/hr)
Component	**Mass Fraction**
H2O	0.01
TEGlycol	0.99

8. Add a valve to stream **TEG from Regen**. The outlet pressure is **6900 kPa (1000 psia)**. Enter **TEG Recycle** for the name of the valve outlet stream.



Absorber icon

9. Add an Absorber to the flowsheet. Configure the Absorber with the following Information.

UniSim Design is equipped with a distillation column **Input Expert** option that guides you through the configuration of the column. The **Use Input Expert** option can be accessed from the **Simulation** tab of the Session Preferences in order to disable it if required.

In this cell...	Enter...
Connections	
Name	TEG Contactor
Number of Stages	3
Top Stage Inlet Stream	TEG Recycle
Bottom Stage Inlet Stream	To TEG Contactor
Ovhd Vapour Outlet	Dry Gas
Bottoms Liquid Outlet	Btms Liq
Parameters	
Top Pressure	6640 kPa (1001 psia)
Bottom Pressure	6640 kPa (1001 psia)

10. Run the Absorber column and converge the simulation.
11. Save your case as **4528.08.TCSS.usc.**

Save your case!

Sizing the Column Tray Section

In steady state mode you are free to specify the column pressure profile as you desire. In fact, we just specified a column pressure drop of **0 psi**. In dynamics mode, the column pressure profile is calculated by the hydraulic calculations on each stage. Thus the calculated pressure drop on each tray section is a function of the tray geometry (diameter, weir height, weir length, and tray spacing).

The dynamic column pressure profile of each column in your flowsheet can be estimated with the Tray Sizing utility whilst still in steady state mode. The Tray Sizing utility performs the hydraulic calculations on each stage of the column. If you do not know the actual dimensions of your column tray section, the tray sizing utility can be used to estimate the size of the tray section before running dynamically.

Running the Tray Sizing Utility

To run the tray sizing utility, do the following:

1. From the menu bar, select **Tools**-**Utilities** or use the **CTRL U** hotkey, and from the **Available Utilities** view, select **Tray Sizing.** Click the **Add Utility** button.
2. Click the **Select TS** button and choose the tray section to size. In this case select the TEG Contactor flowsheet and **T-100 (COL1)**, **TS-1** as the object. Click the **OK** button.
3. Click the **Add Section** button. This adds the selected tray section to the utility, allowing the sizing calculations to be performed.
4. On the **Design** tab, **Setup** page, set the **Internals** to **Sieve.**
5. Move to the **Performance** tab and then select the **Results** page. The hydraulic calculations based on the recommended tray geometry for the column are displayed. This page also displays the estimated **Section DeltaP**. Notice the Tray Sizing utility estimates approximately a **1.9 kPa (0.27 psi)** pressure drop through the 3 stages of the column.

When we initially designed our column in **Steady State** mode, we entered a pressure drop of **0 psi** for the column. Additionally, the high-pressure gas was specified to be operating at **6900 kPa**.

In **Steady State** mode the Tray Sizing Utility can be used also to estimate the pressure profile that will exist after transitioning to **Dynamics**. In **Dynamics** mode, the pressure drop at each stage will be calculated based on the tray geometry, but after an initial pressure guess. Moreover, the feed stream and any connected equipment pressures will be based on this dynamic pressure profile.

If the steady state pressure profile does not match the dynamic pressure profile, the column tray pressures and flow rates will oscillate until an equilibrium pressure profile is established. As the column tries to adjust to an equilibrium pressure profile, the column can become unstable. Thus a proper column pressure profile based on the hydraulic calculations should be entered for the column before moving to dynamic simulation analysis.

Size the Column Tray Section

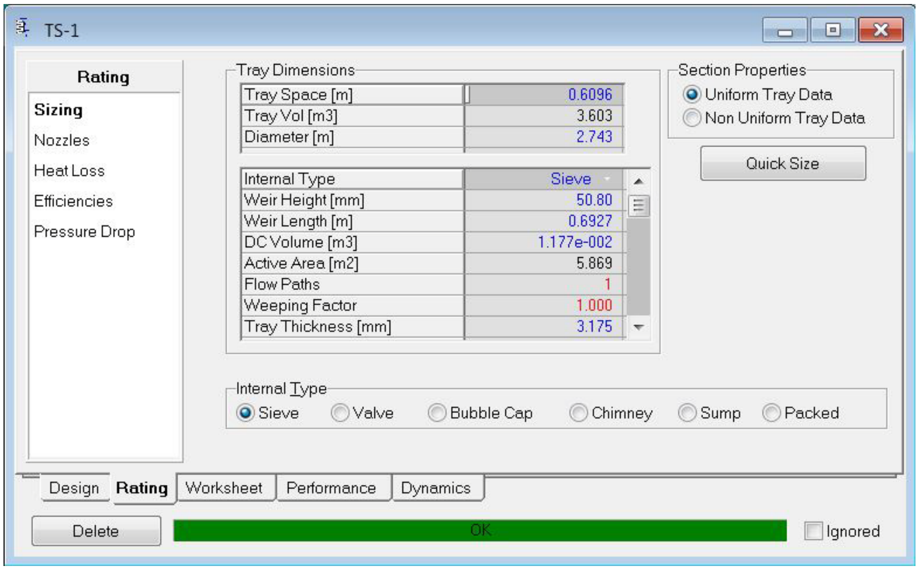
6. From the **Results** page on the **Performance** tab of the Tray Sizing utility, fill in the appropriate tray dimensions in the table below.

Property...	Value...
Tray Sizing Utility	
Section Diameter [m]	
Weir Height [mm]	
Tray Spacing [mm]	
Total Weir Length [mm]	
Section DeltaP [kPa]	

7. Return to the **Setup** page of the **Design** tab in the Tray Sizing Utility and make the section **Active** by checking the box.
8. Return to the **Results** page of the **Performance** tab and click the **Export Pressures** button. This will transfer the pressure profile and the geometry information to the column.
9. Open the Tray Section property view, examine the **Design** tab **Pressures** page to verify that the pressure information has been updated and the **Rating** tab **Sizing** page to verify the geometry information.

Note that **Export Pressure** function can only be used if pressures are specified and not calculated values.

Figure 1

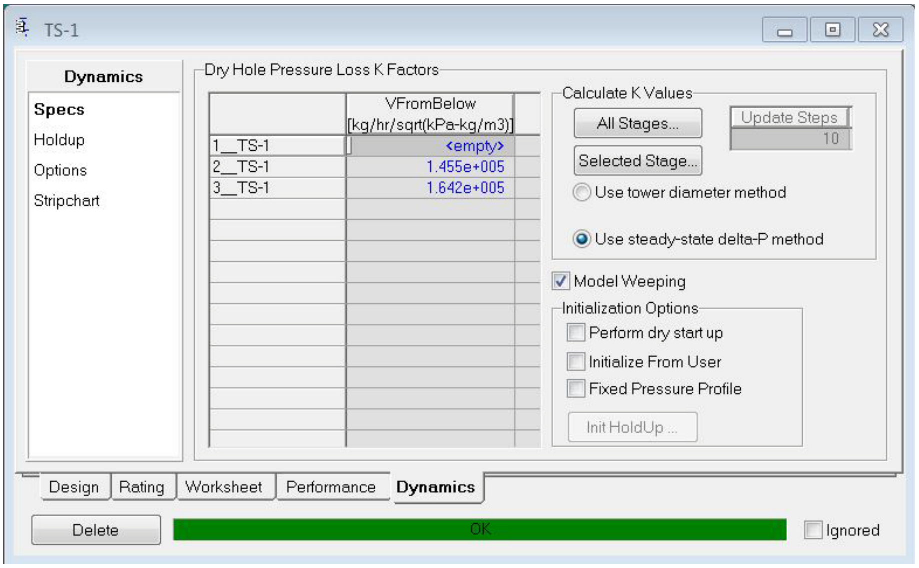


The liquid flow through the column is based on the Francis weir equation, which depends on the entered weir geometry.

Vapour flow through the column is based on a resistance to flow. This resistance can be based on geometry or back calculated from steady state values.

10. From the Tray Section property view, go to the **Specs** page of the **Dynamics** tab. Select the **Use steady-state delta-P method** box and click the **All Stages** button to calculate **K Values** (Dry Hole Pressure Loss K Factors).

Figure 2



Add Valves to the Product Streams

If you do not know the dimensions of your process equipment, calculate the vessel size based on an appropriate residence time:

- 10 minutes is typically a suitable residence time for liquid phase holdups.
- 2 minutes is typically a suitable residence time for vapour phase holdups.

In steady state the bottom of the tower does not need a sump, however, in dynamics the liquid from the bottom tray goes to a sump. The sump will be modeled as a separator.

1. Add a separator and name it as **TEG Sump**. The inlet stream is **Btms Liq**. Name the vapour outlet stream as **Sump Vap** and the liquid outlet stream as **TEG to Regen**.
2. Size the **TEG Sump** vessel to hold about 5 minutes of liquid, **1.7 m3 (60 ft3)**.
3. On the **Dynamics** tab **Specs** page of **TEG Sump**, select the **Initialize From Feeds** option.
4. Add a valve to the boundary stream **Dry Gas**. Enter **6200 kPa (900 psia)** for the outlet valve pressure stream.
5. Add a valve to the boundary streams **TEG to Regen**. Enter **5500 kPa (800 psia)** for the outlet valve pressure stream.
6. Ensure the **Valve Manufacturers** is set to **Simple resistance equation** and size the three valves added to the simulation.
7. Add pressure or flow specifications to the appropriate boundary streams.

8. Save your case as **4528.08.TCSS1.usc**.

Save your case!

Adding the Column to the Dynamic Model

At this point there should be two models: one a dynamic simulation of the plant up to the column feed; the other a steady state simulation of the column and column equipment. The next step is to add the steady state column into the dynamic simulation.

There are two approaches that could be used:

- Simply copy the steady state column into the dynamic model.
- Convert the steady state case to a dynamic case, run the Integrator and then copy the column into the dynamic model.

Both methods will work and the method used will depend on the model itself. In this case the model is relatively small and as such there should be no concerns with copying directly from steady state to dynamics. However, if the model was larger and more complex than it should first be run in dynamics as a standalone model before importing into the larger main model.

1. Select all the streams and operations in the steady state column model and copy all the objects.
2. In the dynamic model **4528.07.CompE.usc**, delete the pipe outlet stream.
3. Paste the column objects into the dynamic model and attach the column feed stream, **To TEG Contactor**, to the pipe outlet.
4. Remove any pressure or flow specifications from **To TEG Contactor**, then disconnect this stream from the column and connect it as the **TEG Sump** inlet.
5. Remove any pressure or flow specifications from **Sump Vap** and connect this stream to the bottom stage of the column.
6. Run the Integrator for a few time steps.

To copy in UniSim Design:

- Select all streams and unit operations to be copied.
- Right click on one of the selected objects and from the menu select **Cut/Paste Objects** then **Copy Selected Objects**

or

- Use the **CTRL C** hot-key.

To paste in UniSim Design:

- Right click on the PFD and from the menu select **Cut/Paste Objects** then **Paste Objects.**

or

- Use the **CTRL V** hot-key.

Add the Control System

1. Add a flow controller to the **TEG feed** stream (FC-TEG), a pressure controller on the **Dry Gas** stream (PC-Dry Gas), and a level controller on the **TEG Sump** (LC-Sump). Provide appropriate ranges.
2. Start the Integrator.
3. Edit any stream and/or unit operation names as you deem appropriate.
4. Save your case as **4528.08.TCDyn.usc**.

Save your case!

9. Event Scheduler

Workshop

With the Event Scheduler, it is possible to make UniSim Design perform given tasks while a simulation is running in dynamics mode. The tasks can be triggered by a pre-determined simulation time, a logical expression becoming true, or a variable stabilizing to within a given tolerance for a set amount of time. Examples of procedures that can be performed with the Event Scheduler are emergency shutdown, start-up and some batch processes.

The Event Scheduler Manager contains all the Event Schedules in the current UniSim Design case. Each **Schedule** is comprised of **Sequences**, which in turn are made up of **Events**. An **Event** must have a **Condition**, which determines when the event will occur, and an **Action List** containing at least one **Action** to be fully defined.

In this module, we will work with the Event Scheduler Manager, and create Event Schedules within the case from the previous module. The schedule will cause the High Pressure Separator liquid valve to Trip Open if there is a high liquid level in the separator. A second event will reset the process to its normal operating mode. This schedule will be operating continuously.

Objectives

After you have completed this module, you will be able to:

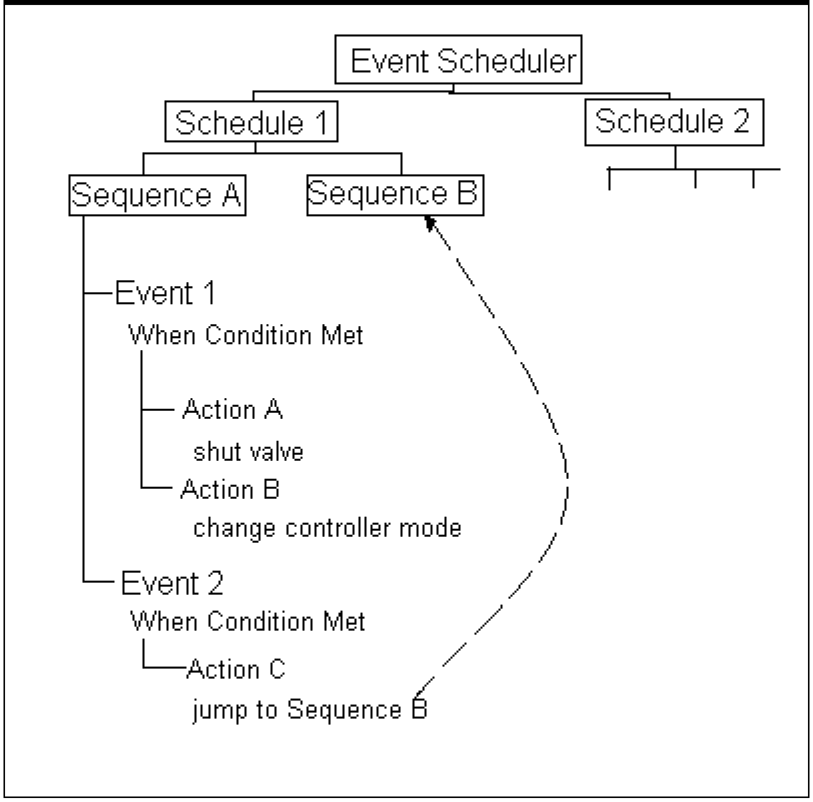
- Use the Event Scheduler Manager
- Create Event Schedules
- Create Sequences
- Create Event Conditions

The Event Scheduler Manager

The Event Scheduler Manager contains all the Event Schedules in the current UniSim Design case. Each Schedule is comprised of Sequences, which in turn are made up of Events. An Event must have a Condition, which determines when the Event will be triggered. At least one action must be fully defined for each Event. Once an Event has been triggered, a series of pre-defined Actions will be executed. A variety of Actions can be performed including closing valves, switching controller modes, changing variable values and switching to other sequences.

The following chart illustrates the relationship between these various items:

Figure 1



The **Event Scheduler** is accessed from the **Simulation** menu, or by using the hot-key **CTRL E**. When you open the Event Scheduler, UniSim Design displays the Event Scheduler Manager, which contains a list of the Schedules in the case.

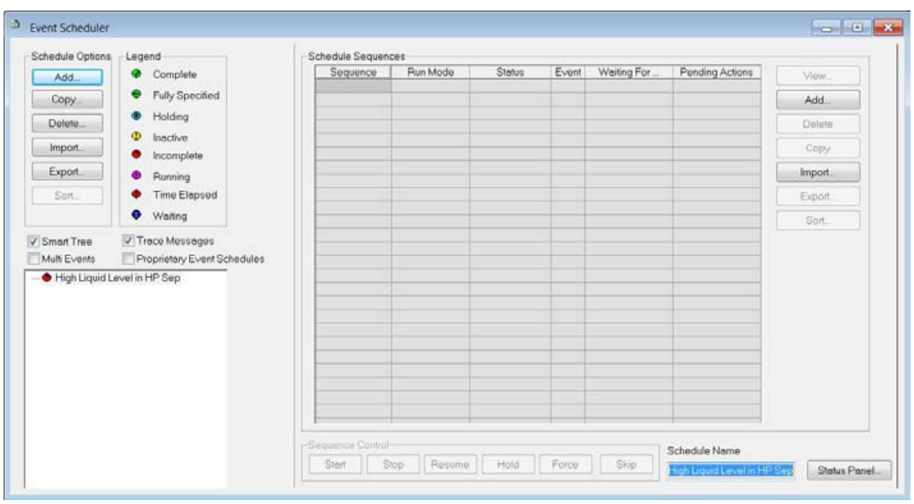
Event Scheduler icon

The buttons in the **Schedule Options** group are used to manage the schedules for the current case:

- **Add -** Allows you to add new schedules to the case.
- **Copy** - Allows you to make a copy of the selected schedule. This is only active when a schedule exists in the case.
- **Delete** - Allows you to delete the selected schedule. This is only active when a schedule exists in the case.
- **Import** - Allows you to import a saved schedule from disk. Schedules have the file extension ***.sch**.
- **Export** - Allows you to export the selected schedule to disk. Once exported, a schedule can be retrieved using the **Import** button. This is only active when a schedule exists in the case.
- **Sort** - Allows you to arrange the schedules. This is only active when at least two schedules exist in the case.

1. Open the saved case file **4528.08.TCDyn.usc**, from the previous module.
2. Open the Event Scheduler. On the Event Scheduler Manager view, press the **Add** button to create a new Schedule.
3. In the bottom right corner of the Schedule view, rename the new Schedule to **High Liquid Level in HP Sep**.

Figure 2



The event Scheduler view has four checkboxes.

- **Smart Tree** - shows the current event of the last selected sequence or schedule.
- **Trace Messages** - enables the tracing of Event Scheduler messages.
- **Multi Events** - allows you to execute multiple contiguous events, with condition of true, in a single time step.
- **Proprietary Event Schedules** – turns the Schedules into proprietary (hidden)

When you add a new schedule, UniSim Design opens a **Schedule Sequences** group. Seven buttons allow you to organize all the Sequences for the current Schedule:

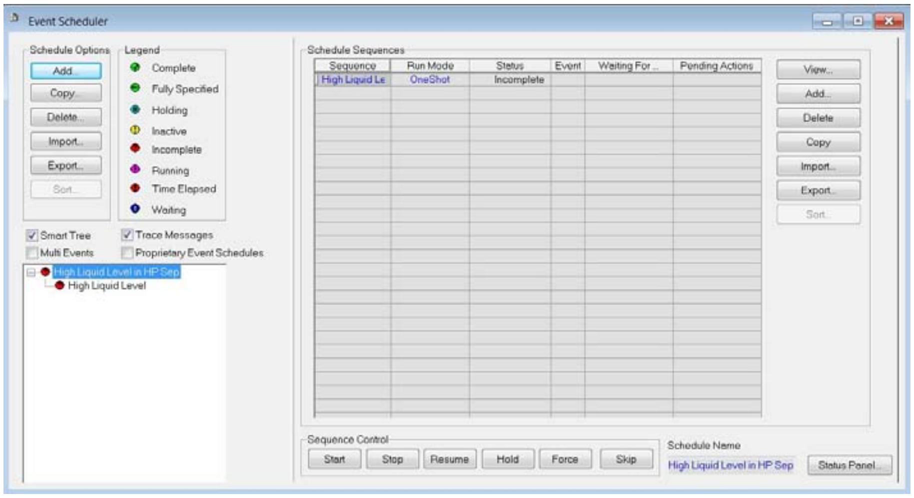
It is important to give all objects in the Event Scheduler logical names.

This will make the Schedule easier for you and anyone else using your simulation to understand.

- **View** - Allows you to view the selected sequence. This is only active when a sequence exists in the schedule.
- **Add** - Allows you to add new sequence to the schedule.
- **Delete** - Allows you to delete the selected sequence. This is only active when a sequence exists in the schedule.
- **Copy** - Allows you to make a copy of the selected sequence. This is only active when a sequence exists in the schedule.
- **Import** - Allows you to import a saved sequence from disk. Are saved in ***.seq** format.
- **Export** - Allows you to export the selected sequence to disk. Once exported, a sequence can be retrieved using the **Import** button. This is only active when a sequence exists in the schedule.
- **Sort** - Allows you to arrange the sequences. This is only active when at least two sequences exist in the schedule.

4. On the Schedule view, click the **Add** button to create a new sequence.
5. Rename the **Sequence** to **High Liquid Level**.

Figure 3

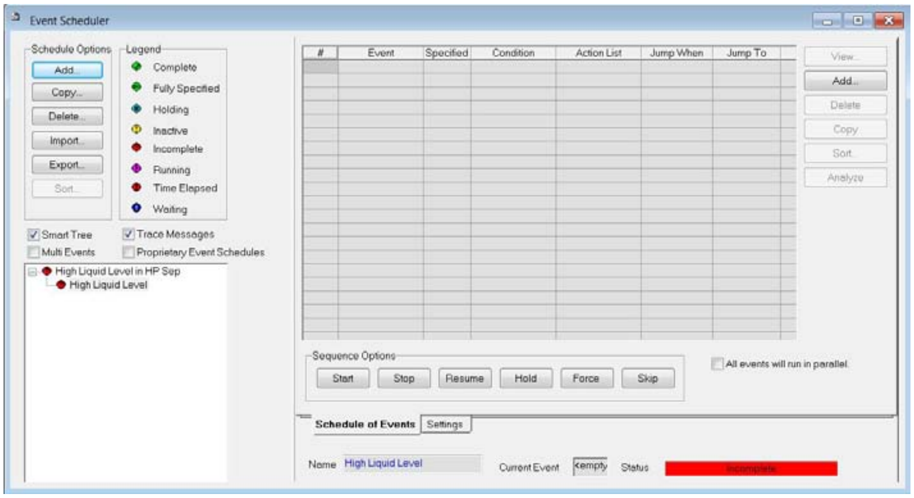


The **Sequence** field contains the name of the Sequence. The **Run Mode** field displays the sequence mode; **One Shot** or **Continuous**. A **One Shot** Sequence executes all its Events in order then its **Status** changes to **Complete**. A **Continuous** Sequence returns to the first Event after the last Event has been executed, in a continuous loop. The **Event**, **Waiting For**, and **Pending Actions** fields display the current event number with its corresponding **Condition** name and **Action List** name.

6. Select the sequence **High Liquid Level** from the Event Scheduler tree view.

When an Event Schedule object is selected in the tree view the right hand side of the Event Scheduler is updated to show the view for the selected component.

Figure 4



There are two **Add** buttons on the view:

- **Left Add button** only adds Schedules
- **Right Add button** updates depending on the **Event Scheduler** object selected in the tree, and will add the dependant component (i.e. selecting a **Sequence** in the tree and pressing the **Add** button will add a new **Event**.

The **Schedule of Events** tab shows the list of events for the schedule.

The fields on this view are defined as follows:

- **Specified** - **True/False** indicates whether the event has been fully defined.
- **Condition** - Displays the condition name of the event.
- **Action List** - Shows the name of the events action list.
- **Jump When** - Displays whether the Event will Jump over any Events and, if so, under what condition (Never, Always, True, Timeout, False).
- **Jump To** - Displays the event to jump to.

At the bottom of the view the sequence name, current event number (#) and sequence status are shown. If the event is not fully specified, the **Analyze** button is enabled which provides additional feedback.

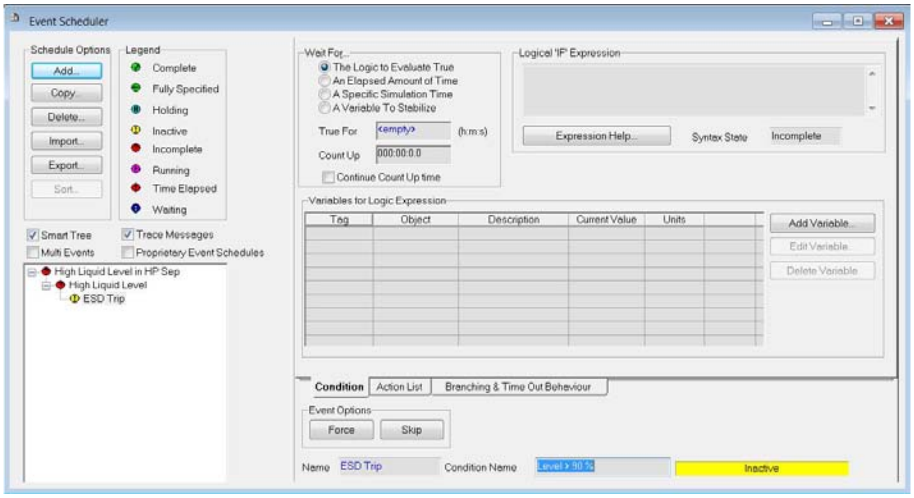
On the **Settings** tab, the sequence **Run Mode** is selected, a unit set global to the sequence can be chosen and different **Status Window Reporting** options are available. Selecting the **Synchronize All Time Sensitive Conditions** checkbox will assure execution of a particular event at an exact simulation time. This applies only when one of the two time conditions has been selected for the Event. **Default Time Out Behaviour** is specified here and applies to all the events unless a specific event overrides the behavior.

The High Liquid Level Sequence created in this module will hold two Events to be executed.

The first on this list of events is an ESD of VLV-100 when the liquid level of the HP Separator is higher than 90%.

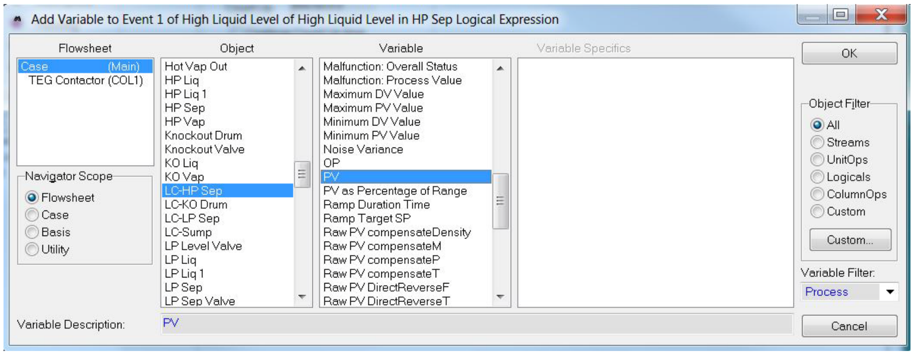
7. On the **High Liquid Level** sequence view, click the **Add** button to create a new Event. Change the name from Event 1 to **ESD Trip**.
8. Press the **View** button to open the Event, or, select the Event in the Event Scheduler tree.
9. In the **Wait For...** box, select **The Logic to Evaluate True** radio button.
10. In the Condition Name field, rename the **Condition** to **Level > 90 %**.

Figure 5



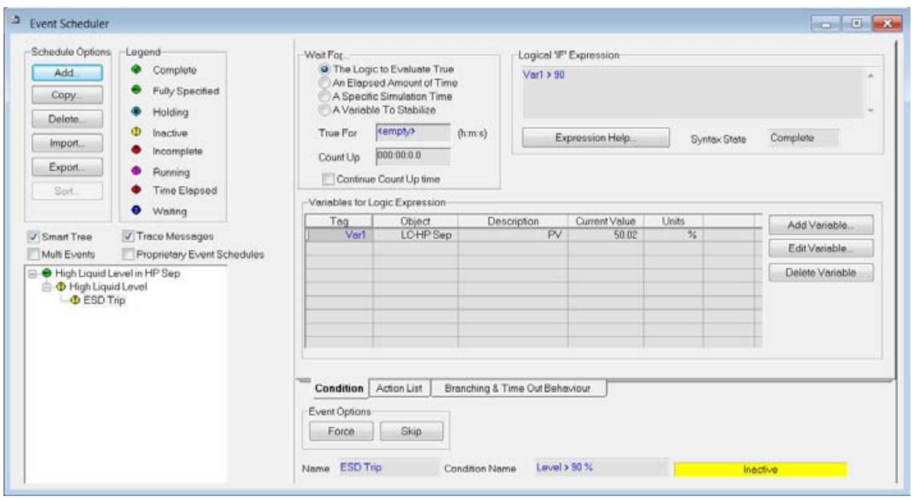
11. Press the **Add Variable** button and add the **PV** for the high pressure separator level controller (LC-HP Sep).

Figure 6



12. In the **Logical Expression** box type in the expression: **Var1 > 90**. This expression will evaluate as **True** if the level controller PV is above 90%.

Figure 7



13. On the **Action List** tab select the **Add** button. Rename Action 1 as **VLV-100 ESD Trip**.

Verify that the control valve on the liquid stream (VLV-100) is set up as follows (**Actuator** page of **Dynamics** tab **Emergency Shut Down** group)

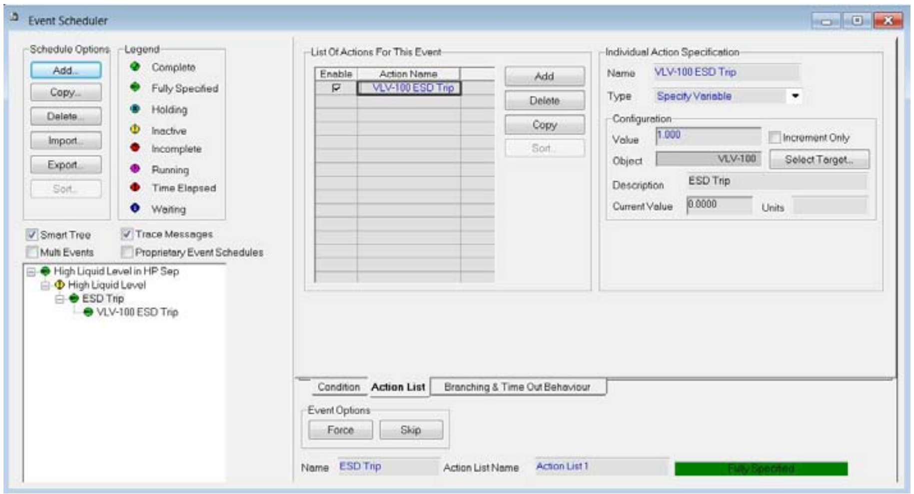
Parameter	Setting
ESD Trip State	Unchecked
Trip Mode	Trip Open
Invert ESD State	Checked

IF you are ever unsure of how to change a variable value with a **Specify Variable** action look at the **Current Value** field as you change the variable value on the object by hand.

In this case the **Target** of the action is the **ESD Trip** variable for VLV-100. A value of **1** indicates the **ESD Trip State** checkbox is checked and hence the valve has tripped open, a value of **0** indicates it is unchecked and hence the valve is working normally.

14. Configure the **VLV-100 ESD Trip** action with type **Specify Variable** and set the **Value** so that when the event runs the valve trips open.

Figure 8

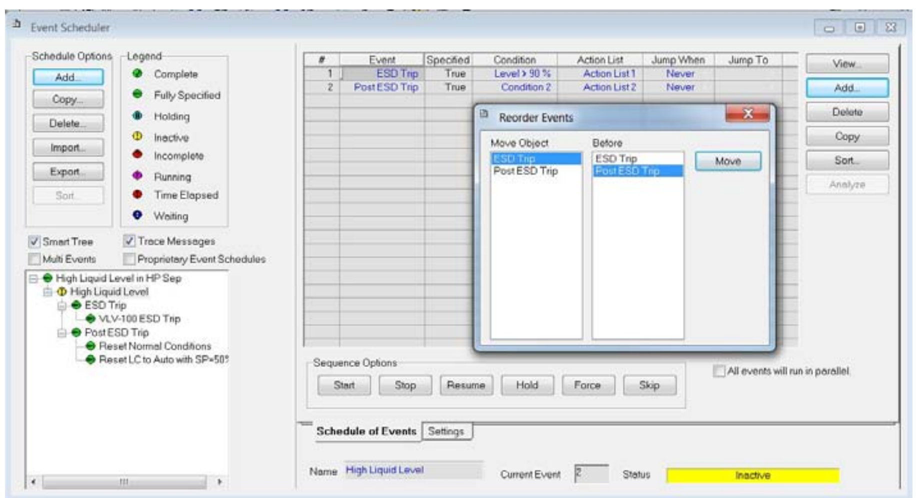


15. An additional Event with two actions should be added to the High Liquid Level Sequence.

Variable...		Value...
Purpose		10 minutes after VLV-100 opens, the **ESD Trip** value is reset to 0 and the mode of the LC is set to auto with a SP equal to 50%
Event Name		Post ESD Trip
Condition		Elapsed Time=15 minutes
Action List Name		Reset Normal Conditions
Action 1	Name	VLV-100 Normal
	Type	Specify Variable
	Target	VLV-100 – ESD Trip
	Value	0
Action 2	Name	Reset LC to Auto with SP=50%
	Type	Set controller Mode
	Target	LC-HP Sep
	New Mode	Auto(matic)
	New SP	50%

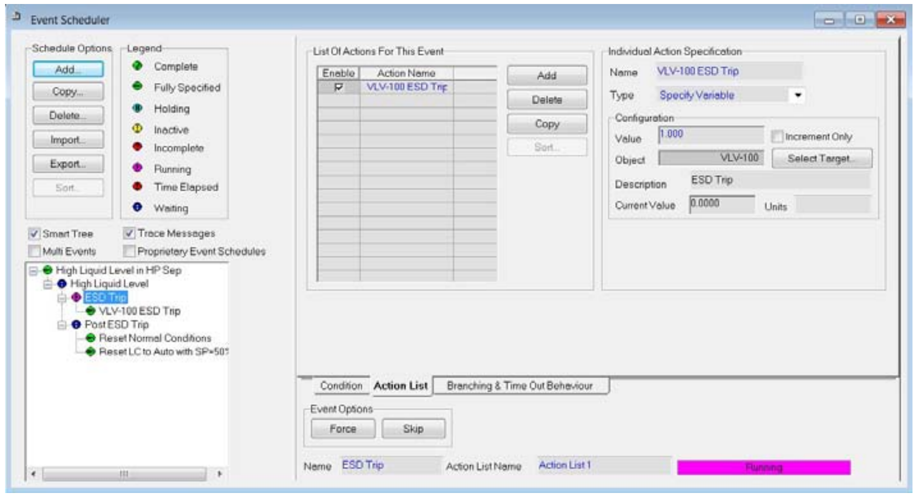
16. Sort the events such that **Post ESD Trip** occurs after the **ESD Trip** event.

Figure 9



17. Return to the **Schedule View** and change the **Run Mode** from **One Shot** to **Continuous**.
18. Press the **Start** button.

Figure 10



19. Save your case as **4528.09.ES.usc** and start the Integrator.
20. Arrange for the level in the HP Separator to rise to 90%. Note if controller **LC-HP Sep** overshoots during event **Post ESD Trip** and the PV rises above 90%, it will trigger the **ESD Trip** event again. If this happens, you may wish to re-tune controller **LC-HP Sep** (increase Kc) to avoid this.

What happens when the level in the HP Separator reaches 90%?

____

Save your case!

Challenge

Set up a new Event within the High Liquid Level Sequence, which performs the actions of Step 20 (i.e. causes the liquid level to rise to 90%) after an elapsed time equal to **45 minutes**. This new event will cause the sequence to cycle continuously.

10. Cause and Effect Matrix

4528.10.R440.01

Workshop

In this module we will add a Cause and Effect Matrix to the simulation.

Objectives

Once you have completed this module, you will be able to:

- Understand the use of the Cause and Effect Matrix

Prerequisites

Before beginning this section you need to:

- Have a general understanding of dynamic simulation

Cause and Effect Matrix

This unit operation replicates a Cause and Effect matrix commonly used in designing and operating the safety system of many processing plants. It looks at process values throughout the process and, based upon safety thresholds, determines if certain equipment and/or valves should be shutdown.

It is recommended that you build a dynamic model first with all the specifications in place before adding and configuring a Cause and Effect Matrix.

The unit operation is similar to a spreadsheet. It takes **Inputs** called **Causes**, and sends **Outputs** called **Effects**.

The input may be any simulation variable within the case or a simple switch which is not required to be connected to another object. Each input generates either a **Healthy (1)** or **Tripped (0)** state.

The output is a boolean variable (**1** or **0**) which results from processing one of the Cause and Effect Matrix columns. The output may write or export its result to any simulation variable within the case. The export must be to a variable of discrete type (**1** or **0**). The output is not required to be connected to an object or variable. The same 1 or 0 result is produced from the matrix column, and then any other object in the simulation may read or use this value.

It is important that you clarify the 1 and 0 convention of the Cause and Effect Matrix for Healthy/Tripped, On/Off, Start/Stop etc.

For both the inputs and outputs, a result of 0 indicates Tripped, whereas a result of 1 indicates Healthy, except where the Invert checkbox is turned On.

When the Invert checkbox is turned On, a result of 0 indicates Healthy, whereas a result of 1 indicates Tripped.

The matrix is processed one column at a time to determine the resultant state of the output associated with that column. The associated input state is reviewed for each element (or row entry) of a particular column having a non-blank user specified matrix element. All of the matrix elements of that column (and their associated input state) are compared based upon their respective and collective meaning to determine the Cause result.

The boolean inputs enter through logical gate type operations (AND, OR, NOT etc.) which combine with each other to determine the resultant boolean value. Each matrix element type is described in the following table.

Matrix Elements		Description
X	TRIP	One or more **0** input(s) causes a **0** output.
R	RESET	One or more **1** inputs causes a 1 output (as long as there are no **X**, **T** or **C** active and ALL **P** must be **1**). There is no requirement to have a reset on a particular output. If you want a reset, this can either be done with one or more **R** matrix element entries or a local reset switch. In the case of both **R** and a local reset, then both reset features must be reset for the output to return to normal, and the local reset must be done last.
T	TIMED TRIP	Same as the TRIP (**X**) but the input must have remained **0** for at least the time period. The **T** matrix element should be followed by an integer representing the number of seconds of time delayed trip.
C	COINCIDENT TRIP	In contrast to all other trips, a **0** input for ALL the coincident signals of the same grouping causes a **0** output. The **C** matrix element should be followed by an integer representing the Coincident group number. There should be more than one in each group.
P	PERMISSIVE	ALL **P** inputs must be **1** to permit an **R** to have the desired effect. Also required for a **STANDBY 1** effect, a local reset and a local switch **ON**.
I	INHIBIT	A **1** will inhibit any trip of the output which would normally be caused by an **X**,**T** or **C**.
S	STANDBY	A **1** will cause a **1** (as long as there are no **X**, **T** or **C** active and ALL **P** must be **1**), and a zero will cause a **0** output (no **INHIBIT** applicable). Normally one would not want more than one Standby input designation per output. If you have more than one Standby, ALL Standby inputs must have a **1** for the output to be started (**1** result). Otherwise, a **0** output result is produced. All Permissive inputs must be **1** for the Standby **1** action to occur.

You can access the Cause and Effect Matrix help view by clicking on the Cause and Effect **Help** button on the **C&E Matrix** tab.

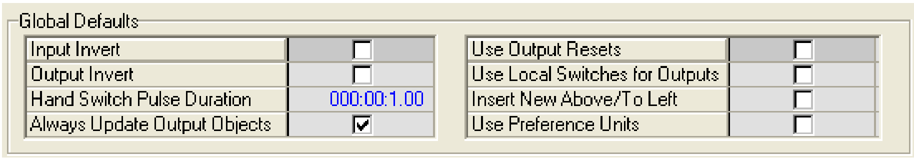
Configuring a Cause and Effect Matrix

There are no PFD streams or lines that connect to or from the Cause and Effect Matrix operation. Hence you can place it anywhere and on any flowsheet. You can also view all simulation variables across all the flowsheets, the same as with a Spreadsheet.

You can set the global defaults and controls on the **Parameters** tab as shown in Steps 1 and 2.

1. Specify the **Global Defaults** by clicking on the appropriate checkboxes.

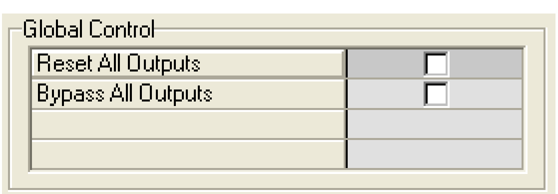
Figure 1



Checkbox	Description
Input Invert	Allows you to set the invert checkbox on for all new inputs.
Output Invert	Allows you to set the invert checkbox on for all new outputs.
Hand Switch Pulse Duration	You can specify the pulse duration for any hand Switch inputs that are pulsed.
Always Update Output Objects	You can check this checkbox, if you want to ensure that the results from the logic calculations are sent every time step to the output objects.
Use Output Resets	You can check this checkbox, if you want the Use Reset checkbox checked on for all new outputs.
Use Local Switches for Outputs	You can check this checkbox, if you want the Use Local Switch checkbox checked on for all new outputs.
Insert New Above/To Left	When you add a new row or column and this checkbox is checked, the row or column will be added above (to the left) of that currently selected row or column.

2. Specify the **Global Control** by clicking on the appropriate checkboxes.

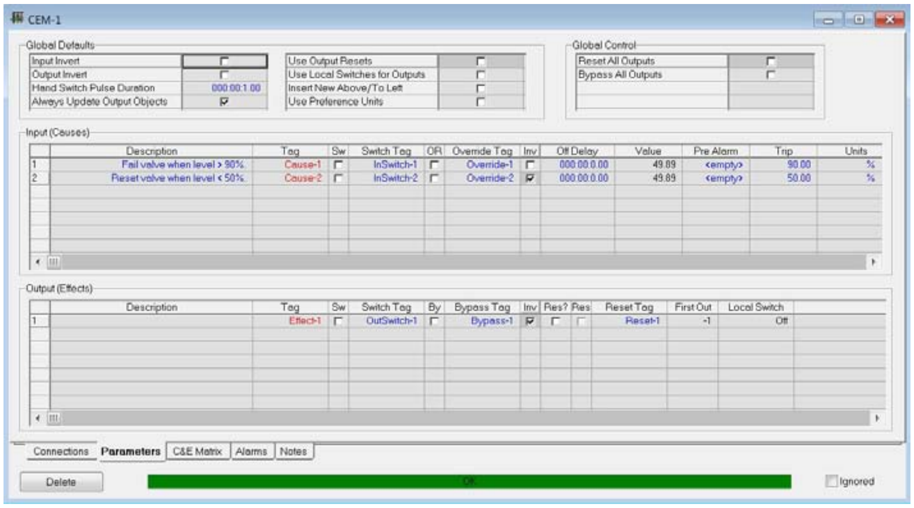
Figure 2



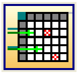
Checkbox	Description
Reset All Outputs	If you want to reset all individual outputs, check this checkbox.
Bypass All Outputs	When you check this checkbox, the **Bypass** checkbox of each output is turned on.

You can view all the input and output configuration information on the **Parameters** tab.

Figure 3



Adding the C&E Matrix



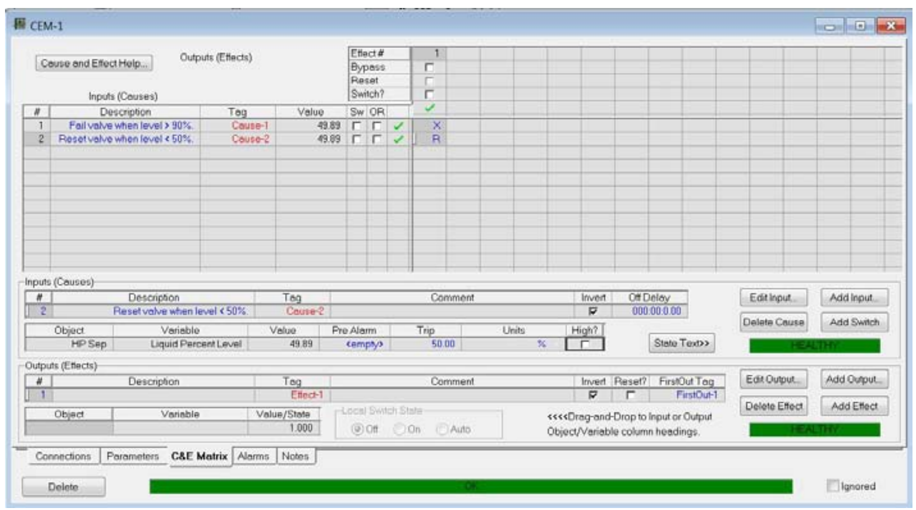
Cause and Effect Matrix icon

1. Open the simulation case **4528.09.ES.usc** from the previous module.
2. From the **Simulation** menu select **Event Scheduler**. Ensure the **High Liquid Level** sequence is **Inactive** by pressing the **Stop** button.
3. From the Boolean Sub-Palette, add a **Cause and Effect Matrix**.
4. On the **C&E Matrix** tab press the **Add Input...** button to add the first input (cause).
5. Select the **HP Sep – Liquid Percent Level** or if you have added a Level Tap select the Level Tap.
6. Set a **Trip** of **90%** and enter a **Description** of **Fail valve when level > 90%.**

7. Add a second input, select **HP Sep – Liquid Percent Level** (or again the Level Tap if added).
8. Set a **Trip** of **50%** and enter a **Description** of **Reset valve when level < 50%**.
9. On the C&E Matric Tab uncheck the **High?** cell and check the **Invert** cell.
10. Add an output by pressing the **Add Output...** button.
11. Select **VLV-100 – ESD Trip** as the variable.
12. Check the **Invert** cell.
13. In the **C&E Matrix,** add a **Trip (X)** for the first input by typing “X” and add a **Reset (R)** for the second input by typing an “R”.

The final configured view is as shown below:

Figure 4



14. Change the set point for **LC-HP Sep** to **95%**.
15. Start the Integrator and observe the behavior of the Cause and Effect Matrix, the separator HP Sep and the valve VLV-100.
16. Save your case as **4528.10.CEM.usc**

Save your case!

11. Building a Fired Heater

4528.11.R400.01

Workshop

In this module we will introduce the fired heater operation in UniSim Design. The fired heater is a heat transfer operation which is only available in dynamic mode.

Objectives

After you have completed this module, you will be able to:

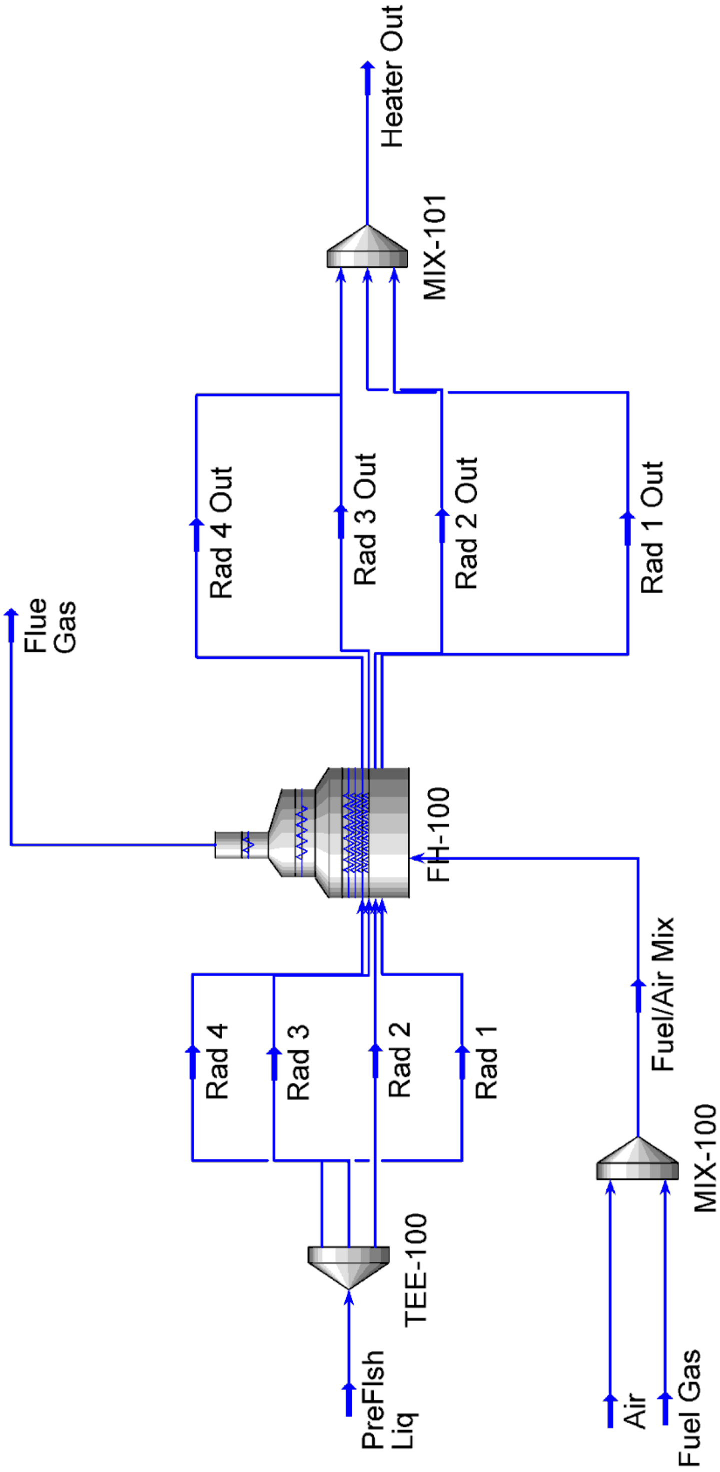
- add a fired heater Unit Operation

Prerequisites

Before beginning this section you need to know how to:

- set up strip charts
- interpret the pressure-flow network

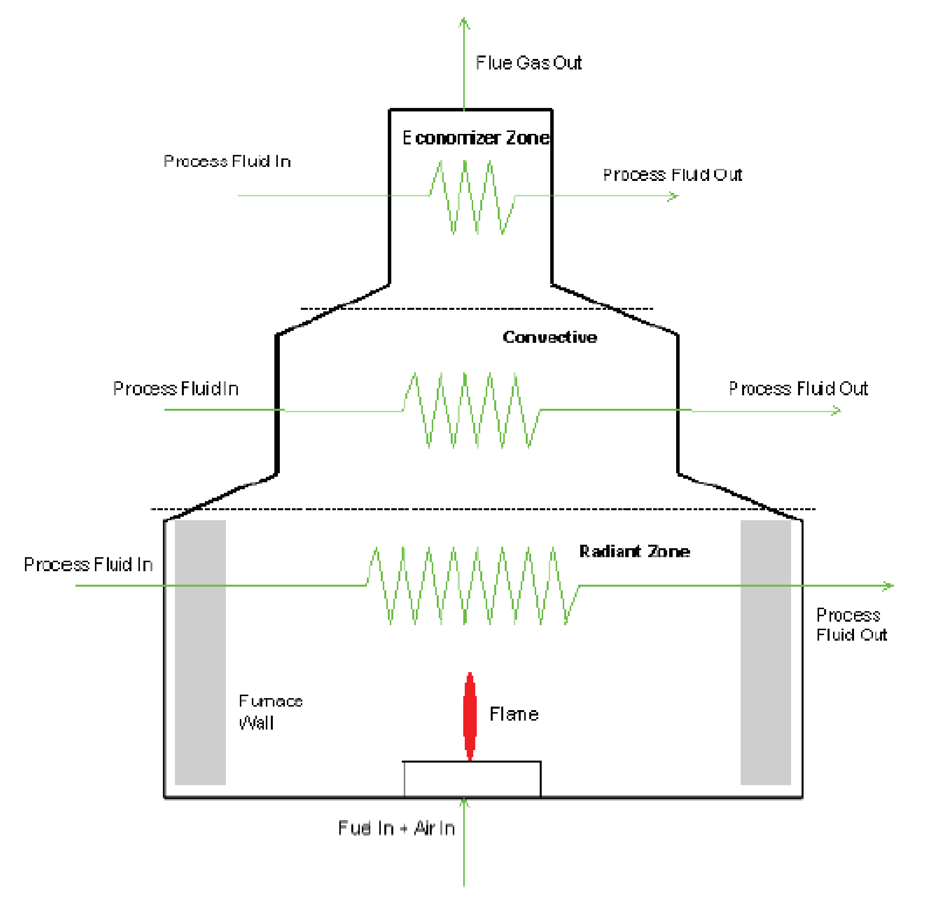
Process Overview



Fired Heater

The dynamic fired heater unit operation (Furnace) performs energy and material balances to model a direct fired heater type furnace. This type of equipment requires a large amount of heat input. Heat is generated by the combustion of fuel and transferred to the process streams by both radiant and convective heat transfer. A simplified schematic of a direct fired heater is illustrated in illustrated in Figure 1.

Figure 1



In general, a fired heater unit operation can be divided into three zones:

- radiant zone
- convective zone
- economizer zone

To define the number of tube passes in each of the zones enter a value in the # **External Passes** field on the **Connections** page of the **Design** tab.

The fired heater unit operation allows multiple stream connections for the tube side in each zone and supports an optional economizer, and an optional convective zone. The operation incorporates a single burner model, and a single feed inlet and outlet on the flue gas side.

The following are some of the major features of the dynamic fired heater unit operation:

- Flexible connection of the process fluid associated with each fired heater zone. Different fired heater configurations can be modeled or customized using tee, mixer, and heat exchanger unit operations.
- A pressure-flow specification option on each side and pass realistically models flow through the fired heater unit operation according to the pressure gradient in the entire pressure network of the plant. Possible flow reversal situations can therefore be modeled.
- A comprehensive heat calculation inclusive of radiant, convective, and conductive heat transfer in the radiant zone enables the prediction of process fluid temperature, fired heater wall temperature, and flue gas temperature.
- A dynamic model which accounts for energy and material holdups in each zone. Heat transfer in each zone depends on the flue gas properties, tube and fired heater wall properties, surface properties of metal, heat loss to the ambient, and the process stream physical properties.
- A combustion model which accounts for imperfect mixing of fuel, and allows automatic flame ignition or extinguishing based on the oxygen availability in the fuel air mixture.

Theory

An introduction to the fired heater theory is provided below.

For more information on the theoretical background of the fired heater unit operation see the UniSim Design Operations Guide section 4.4.

Combustion Reaction

The combustion reaction in the burner model of the fired heater performs pure hydrocarbon ($C_xH_yO_z$) combustion calculations only. The extent of the combustion depends on the availability of oxygen which is usually governed by the air to fuel ratio.

Air to fuel ratio (**AF**) is defined as follows:

$$AF = \frac{\left(\frac{Mass\ of\ flow\ O_2}{\Sigma\ Mass\ flow\ of\ fuel}\right)}{Mass\ Ratio\ of\ O_2\ in\ Air} \tag{1}$$

Where: AF = air fuel ratio of the fired heater

You can set the combustion boundaries, such as the **Max. Air Fuel Ratio** and the **Min. Air Fuel Ratio**, to control the burner flame. The flame cannot light if the calculated air to fuel ratio goes outside these boundaries. The minimum air to fuel ratio and the maximum air to fuel ratio can be found on the **Parameters** page of the **Design** tab.

The heat released by the combustion process is the product of the molar flow rate and the heat of formation of the products, minus the heat of formation of the reactants at the combustion temperature and pressure. In the fired heater unit operation, a traditional reaction set for the combustion reactions is not required. You can choose the fuel components (the hydrocarbons and hydrogen) to be considered in the combustion reaction. You can see the mixing efficiency of each fuel component on the **Design** tab **Parameters** page.

Heat Transfer

The fired heater heat transfer calculations are based on energy balances for each zone. The shell side of the fired heater contains five holdups:

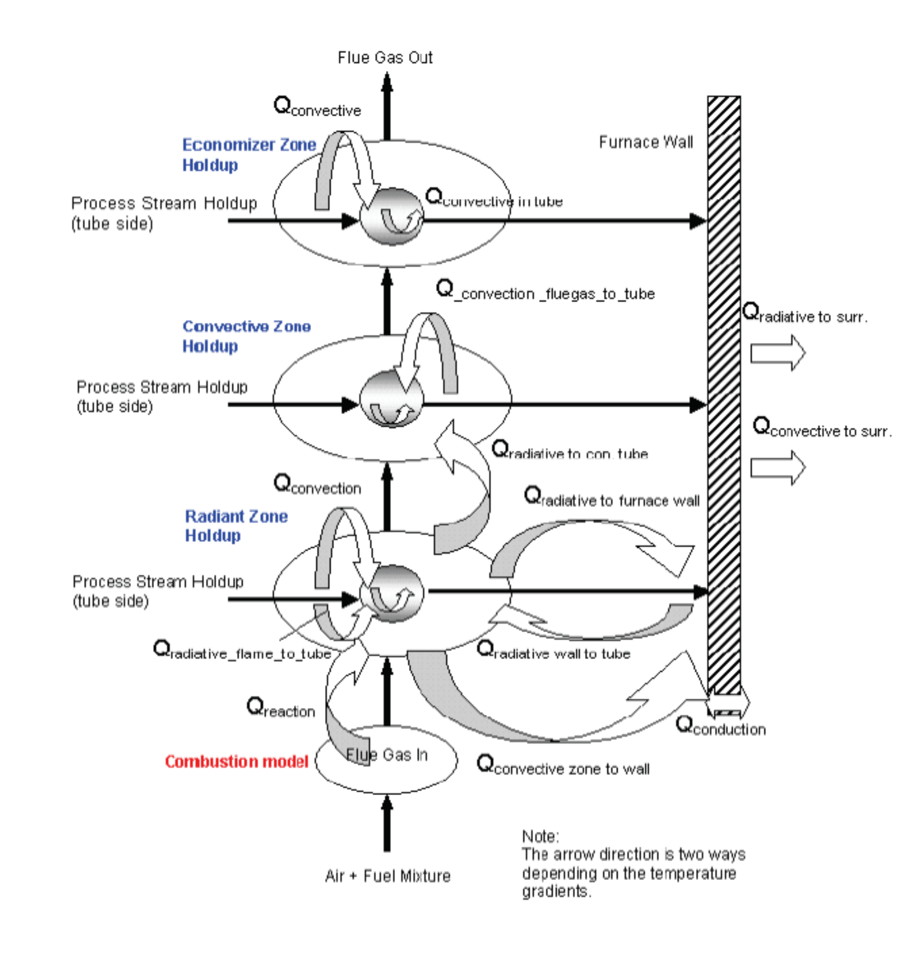
- three in the radiant zone,
- a convective zone,

- an economizer zone.

For the tube side, each individual stream passing through the respective zones is considered as a single holdup.

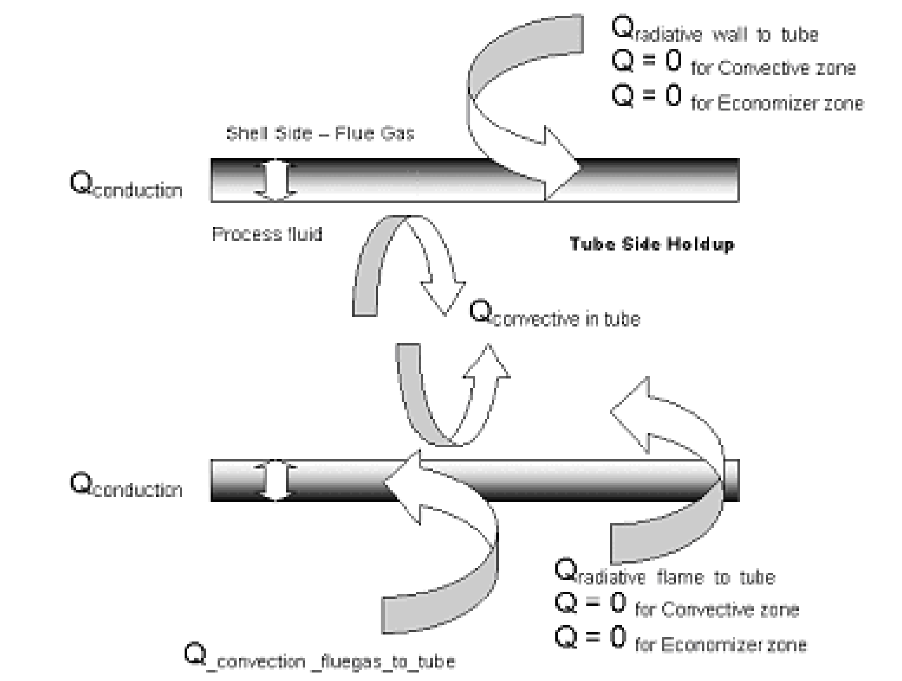
Major heat terms underlying the fired heater model are illustrated in Figure 2:

Figure 2



The heat terms related to the tube side are illustrated in Figure 3:

Figure 3



Build the Flowsheet

Typically the hardest part when setting up a fired heater in UniSim Design Dynamics is to get the operation to initialize on the first integration time step. The procedure outlined below will help ensure that the fired heater will initialize. Since the fired heater operation is for dynamics only, it is assumed that all the steps below are followed after the case has been switched to dynamics mode. It is expected that valves, controllers and other ancillary equipment required will be added after the fired heater has initialized and the integrator has been run for several time steps.

Starting the Case

For this module you will start with the UniSim Design sample case **R-1.usc**.

1. Open the case **R-1.usc** found in the **\Samples** sub-folder of the UniSim Design installation.
2. Once the case is open, enter the **Basis Environment** and add the following components to the Basis-1 component list being used:

- O2
- N2
- CO2
- CO

3. Return to the **Main Environment** and switch to Dynamics mode – do not let the **Dynamic Assistant** make any changes.
4. Open the PFD and delete all streams and operations EXCEPT for the stream **PreFlsh Liq**.
5. On stream **PreFlsh Liq** enable the dynamic flow specification.

Adding Fuel Gas and Air

The fired heater unit operation has a single connection for fuel gas and air and as such these will need to be mixed prior to entering the operation.

The fired heater can use library components of type $C_xH_yO_z$ as fuel or. Hypothetical components as well.

6. Create a new stream with the following properties:

In this cell...	Enter...
Name	Fuel Gas
Temperature	25 °C (77 °F)
Flow Rate Spec	Active
Flow Rate	4000 kgmole/h (8818 lbmole/hr)
Component	**Mole Fraction**
Methane	0.90
Ethane	0.05
Propane	0.02
i-Butane	0.015
n-Butane	0.015

7. Create a second stream with the following properties:

You need to use Nitrogen and Oxygen for the fired heater. Do not use the ‘Air’ component which is available in the **Basis Manager**.

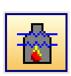
In this cell...	Enter...
Name	Air
Temperature	25 °C (77 °F)
Flow Rate Spec	Active
Flow Rate	52000 kgmole/h (114600 lbmole/hr)
Component	**Mole Fraction**
Nitrogen	0.79
Oxygen	0.21

8. Add a mixer with inlets of **Fuel Gas** and **Air** and an outlet **Fuel/Air Mix**.

Adding the Furnace

9. Add a tee unit operation with the following information:

In this cell...	Enter...
Connections	
Inlet Stream	PreFlsh Liq
Outlet Stream	Rad 1 Rad 2 Rad 3 Rad 4



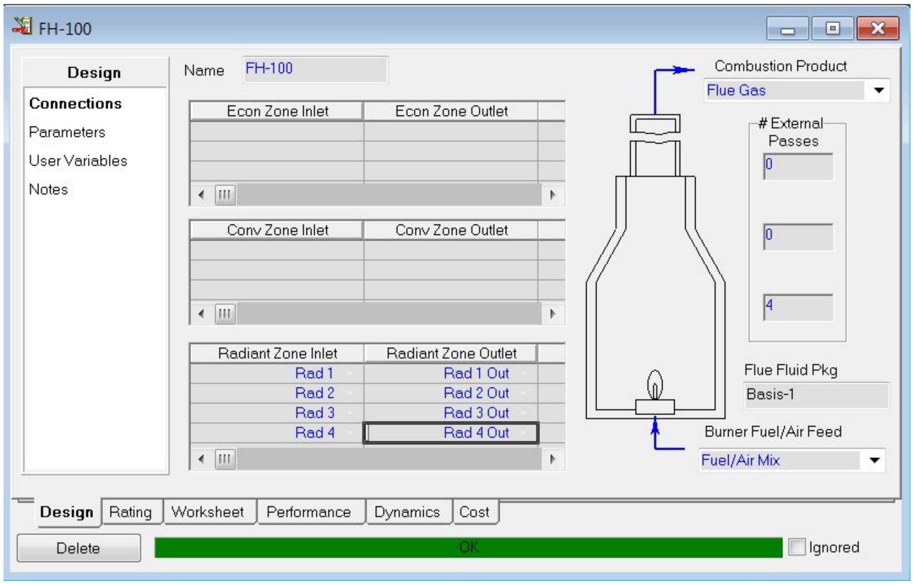
Fired Heater icon

10. Add a fired heater to the model.
11. Configure the fired heater **Design** tab, **Connections** page as follows:

In this cell...	Enter...
Connections	
Burner Fuel/Air Feed Name	Fuel/Air Mix
Combustion Product Name	Flue Gas
Parameters	
# of External Passes Economizer Zone	0
# of External Passes Convective Zone	0
# of External Passes Radiant Zone	4
Radiant Zone Inlet(s)	Rad 1 Rad 2 Rad 3 Rad 4
Radiant Zone Outlet(s)	Rad 1 Out Rad 2 Out Rad 3 Out Rad 4 Out

12. The configured **Connections** page should appear as in Figure 4:

Figure 4



The **Rating** tab contains the physical sizing and heat transfer parameters for the fired heater. At this time we will leave these values at the default settings.

Since the feed is split into 4 paths we need to use k values for the tubes in order to calculate the flow per tube pass. Note if the number of passes is changed, all existing K values are lost.

13. On the **Tube Side PF** page of the **Dynamics** tab specify **K values** of **600 kg/hr/sqrt(kPa-kg/m3) (1.39e4 lb/hr/sqrt(psia-lb/ft3))** for each tube with a **K reference flow** of 5.134e+05 kg/h.
14. Ensure the **Use K's** checkboxes are checked and the **Use Delta P** checkboxes are unchecked.
15. On the **Flue Gas PF** page, specify a **Delta P** of **0.01 kPa (1.45e-3 psi)** for the three **Radiant Zones** and pressure drops of **0 kPa** for the **Convective** and **Economizer Zones**.
16. Add a mixer with the following information:

In this cell...	Enter...
Connections	
Inlet Stream(s)	Rad 1 Out Rad 2 Out Rad 3 Out Rad 4 Out
Outlet Stream	Heater Out

17. Set Pressure specs on the two product streams with the following pressures:

Stream...	Pressure...
Heater Out	450 kPa (65 psia)
Flue Gas	101.1 kPa (14.66 psia)

18. Start the Integrator.

Save the case as **4528.11.FH.usc**.

Save your case!

Tuning the Furnace

With the current furnace configuration the radiant zone process outlet temperature is ~243 °C (470 °F). The original model had an outlet temperature (from the heater unit operation) of ~343 °C (650 °F). We will tune the current configuration of the furnace to give the desired outlet temperature.

1. On the **Rating** tab, **Heat Transfer** page make the following changes to each pass in the **Radiant Tube Properties**.

Note if the number of passes is changed, all existing heat transfer parameters are lost.

In this cell...	Enter...
Connections	
Zone to Tube Emissivities	0.6
Wall to Tube Emissivities	0.1
Tube to Fluid HX Coefficient [kJ/h-m2-C]	4176 kJ/h m2 C (204.3 Btu/h-ft2 F)

2. Start the Integrator again and note the new outlet temperature.

Finishing the Furnace

The last step is to change the fixed pressure drops for the flue gas to k values.

3. Press the **Calculate K's** button on the **Flue Gas PF** page. Enter 1.574e+06 kg/h as the **K Reference flow**.

There may be warning messages stating that UniSim Design is unable to update k values because of zero pressure drops. In this case ignore the messages as they are being generated by the zero pressure drops on the **Convective** and **Economizer Zones** which we will leave as fixed pressure drops.

4. Uncheck the **Use Delta P** boxes and check the **Use K's** boxes for the **Radiant Zones**. Leave the **Convective** and **Economizer Zones** at fixed **Delta P** values of zero.
5. Start the Integrator.
6. Save the case as **4528.11.FH.usc**.

Save your case!

Challenges

Adding the Convective Zone

Add a **Convective Zone** to the fired heater. The feed (**PreFlsh Liq**) should first pass into a four external pass **Convective Zone** and then into the four pass **Radiant Zone**.

Notes:

- You may need to re-enter the heat transfer details for the fired heater and revisit the Dynamics Tab also.
- You will need to add headers / mixers / tees to connect the outlets from one zone to the inlets of the next zone up.
- Ensure that the **Convective Zone** volume is sufficient to allow four tube passes.

Tune the fired heater heat transfer parameters (**Convective** and **Radiant** Zone) to provide the same outlet temperature. (Remember, you will have to size the stack (shell) for the Convective zone and you will have to increase the heat transfer coefficients for the convective zone while decreasing the emissivities in the radiant zone.)

Adding Controls

Add basic controls to control the air to fuel gas ratio and the outlet temperature.

- Use a PID controller and a selector block to maintain an **Air to Fuel Gas ratio** of 13 on a molar basis. (Hint: You can use the Selector block Quotient radio button on the Parameters Tab Selection Mode)
- Use a PID controller to adjust the **Fuel Gas** rate to maintain a process outlet temperature of 340 °C (644 °F).

Notes:

- You can target the flow specification of the Air and Fuel Gas streams with the controllers or you can add control valves to the **Air** and **Fuel Gas** inlet streams. In case you choose to add valves, assume boundary pressures of 110 kPa (15.95 psia) for both the air and fuel gas.

Save the case as **4528.11.FH.Challange.usc**.

Appendix 1

Guidelines to Modeling in UniSim Design Dynamics

Introduction

This appendix contains information which may be useful when working with UniSim Design dynamics. It is split into three broad sections:

- Practical Model Stability
- Model Speed
- Working with Large Simulation Cases

Practical Model Stability

Elevations - If Using Static Head

Using static head contributions adds additional fidelity to dynamic simulation. However it also introduces the potential for additional stability issues.

There are two static head options available on the **Options** tab of the **Integrator** window:

- Explicit Static Head (**Enable static head contributions**): This option calculates the static head contribution for a holdup and adds it to the product pressure after the pressure flow calculation step. This option while less stable is generally adequate for most systems.
- Implicit Static Head (**Enable implicit static head calculations**): This option solves pressure contributions associated with levels inside volumes in the pressure flow calculation step. This option provides increased stability in applications where static head contributions play a crucial role.

Regardless of the static head calculation method chosen, care should be taken to ensure that levels are used consistently to create realistic elevation profiles. By default all equipment is given a base elevation of 0 m, therefore if adding a series of equipment all pieces need to have their elevations checked to avoid creating U-shaped elevations, i.e. 3 pieces of equipment in series with the first at elevation **x** , the second at elevation 0 m, and the third at elevation **x** .

Note with either calculation method, the pressure shown in a stream is the stream source pressure, i.e. the upstream pressure from where the stream is flowing.

Once static head is enabled it is very easy to forget to check all elevations but the use of two PFD hot keys can aid in ensuring consistent elevation data:

- **SHIFT-I**: Replaces the stream labels with the elevation of the nozzle on the equipment the stream is feeding to.
- **SHIFT-O**: Replaces the stream labels with the elevation of the nozzle on the equipment the stream is a product from.

When dealing with two phase systems (vapour-liquid) it is generally recommended to create as flat elevation profile as possible and thus avoid any slight changes in pressures due to changes in elevation. The most common example of this is in overhead condenser systems. In all systems it is generally recommended to try and change elevations with the use of separators, but this is particularly important in two phase systems.

Use Separators and Vessels

Using separators as pipes or holdups to account for elevation changes can add stability to a model.

A separator (15-30 cm in diameter) acting as a pipe between the top of the column and the condenser can greatly increase the stability of the column and ensure that if reverse flow does occur that it is likely vapour and not liquid or two phase.

Additionally using the boot option on separators is a convenient method of accounting for elevation differences without adding additional equipment. Separators are typically used to model chimney trays in columns, using the boot option on the separator allows you to put the main vessel body at the correct elevation for the chimney tray while also accounting for the elevation difference. In this example the base elevation of the separator is set at ground level and the boot height is set such that the body of the separator is at the correct elevation. Using this method ensures that any pumps which feed from the chimney tray will function correctly.

Recycle and Feed Efficiencies

Adjusting recycle and feed efficiencies in models can greatly increase the stability in some locations, however as it does alter the solution great care should be used in ensuring that the results (actual values or transient responses) are as expected.

Generally it is recommended that recycle and feed nozzle efficiencies are only adjusted when absolutely necessary. Examples where efficiencies may need to be adjusted are:

- Column Sumps
- High purity systems - behavior of the system is approaching pure component behavior or is pure component (steam systems)
- Systems with gas blankets for pressure control (nitrogen or natural gas)

When adjusting the vessel efficiencies in most cases only the recycle and feed nozzle efficiencies for the vapour phase need to be adjusted, however if reverse flow is expected then the vapour product nozzles should also be adjusted. Liquid nozzle efficiencies can be left at the default 100% in almost all situations.

Efficiency values are typically tuned to give the desired response and as such there are no hard and fast rules as to what values to use. However, if adjusting the values, changing from 100% to 0.1-5% is a reasonable starting point. Additionally, when adjusting values it is a good idea to keep the recycle and feed nozzle efficiency values the same.

UA and k Reference Flows

Heat transfer equipment in UniSim Design (Heat Exchangers, air coolers, LNG, and Fired Heaters) allow the user to enter reference flows for the UA and k values.

In the case of UA reference flows it is recommended that the design or steady state flow values which correspond to the entered UA should be entered. As the flows move away from the reference flows the UA value is scaled accordingly. This can greatly increase the model stability in particular as the flows drop to zero values.

In the case of k reference flows it is recommended that a value of 40% of the design or steady state values be used. As the flow drops below the reference flow UniSim Design will linearize the pressure drop equations; this action adds increased stability at low flow conditions.

Fixed Flow Values

One common mistake is adding fixed flow values in a model without adequate protection. These are either added as material stream flow specifications or possibly as duty specifications in heater or cooler operations. In both cases it is possible to get into situations where the fixed value is no longer practical, possible or realistic. When this occurs there is a chance that the model will show some problems, likely in the form of unexpected or unrealistic pressures or temperatures.

When adding flow specifications it is highly advisable to use a spreadsheet to limit or clamp the flows. For a heater or cooler a simple calculation which reduces the duty as flow drops below a threshold can limit the occurrence of unrealistically high or low temperatures. For material stream flow specifications such a calculation can reduce or eliminate the occurrence of extremely high or low pressures.

Additionally, it is not recommended to simply pass a flow specification from a spreadsheet to a stream (material or duty). Instead the value should be passed to a transfer function with a very short (2-15 second) first order lag. This can greatly reduce instabilities resulting from the spreadsheet calculation.

Instantaneous Valves

By default valves in UniSim Design are instantaneous which can lead to model instability or even non-convergence if a valve goes from full open to full close state in a single time step.

If valve rates are not known then using a linear actuator rate of 10-20%/sec for control valves is a good start. For block valves there are several of options. For non-motorized valves a rate of 12 inches/minute is a rough starting point. For motorized valves a rate of 10-20%/sec is recommended as a starting point.

Rotating Equipment

For centrifugal pumps and compressors there are a number of factors which can help stability.

- Inertia Modeling Parameters
- Friction Loss
- Electric Motor

The Speed Ramp option is only available if **Use Characteristic Curves** and **Speed** are used for the pump dynamic specifications.

Additionally for pumps, there is now an option to directly control the speed of the pump. On the **Dynamics** tab **Specs** page the pump can be changed to **Speed Ramp**. When this is done the user can enter a linear ramp time for pump starts and a first order time constant for pump stops.

Calculation Execution Rates for Composition and Flash Calculations

There may be sections (or individual pieces of equipment) which could benefit from having a more frequent flash frequency. While it is not possible to change the integrator step size for individual portions of the model the flash frequency can be changed for individual unit operations.

If the flash frequency for a large column is changed this may have a large impact on model speed.

If needed this can be a method to balance stability and model speed. Changing the flash frequency for the whole model will have a similar effect on model speed i.e. if the flash frequency is changed from 10 to 5 the model speed will also be reduced by about half. Depending on the unit operation, changing the flash frequency for an individual operation may provide additional stability with little noticeable effect on real time.

Flash frequency for individual operations can be changed by adding a new **Workbook** page of **Dynamic Equipment Op.** On any dynamic equipment operation, unchecking the option **Use Integrator Periods** will allow the user to specify individual Composition Period values for all dynamic equipment.

Note: Keep in mind, changing the flash frequency on a unit operation with conversion reactions might affect the results of reactions.

Relief Valves

Relief valves in UniSim Design are calculated at the end of the pressure flow calculation step and as such they can cause instabilities in the model.

There are several recommended options for increasing the stability of relief valves in the model:

- Ensure a reasonable pressure difference between the set pressure and full open pressure. If full open pressure is not provided, specify for vapour at least 10% of the set pressure for liquid at least 20%.
- Enable valve hysteresis. This allows the specification of a closing pressure and reseating pressure. Typically the closing pressure is set at or approximately the set pressure and the reseating pressure is set 5-10% below the set pressure.
- For liquid relief valves set the **Liquid Service** option. This might provide additional stability for liquid relief valves, however due to the nature of solving simultaneously with the P-F network, this option can also lead to non-convergence.
- Use a smaller orifice size than the datasheet provides. Since in most cases all the down stream pressure effects have been ignored it may be necessary to tune the orifice size to give the desired relief rates.

Model Speed

One of the greatest problems or challenges with dynamic simulations is model speed, often referred to in terms of real time or factors of real time. When looking at model speed it is important to remember that some of the factors which increase model speed may also reduce model stability and rigour. It is often necessary to balance model speed with stability.

When looking at model speed the final use of the model should be considered. In a rigorous dynamic model there is often less emphasis on model speed and more on obtaining rigorous results. It may be possible to let a study run overnight or in the background without human interaction and simply wait for the required results. On the other hand with an Operator Training System (OTS) there is always a requirement that the model be able to run at a minimum of real time through all process conditions. In this case there are operators waiting to interact with the model and as such the model needs to respond in a realistic manner in both direction and time of response.

First Glance

When looking to increase real time speed there are several basic items which should be addressed first (in no particular order):

1. Step size: This has the greatest impact on model speed. A model with a 0.5 second step size will run twice as fast as a model with a 0.25 second step size. A model with a small step size may be more stable and produce a more rigorous result but at a price of speed. For most OTS projects a step size of 0.5 to 1.0 seconds has been found to provide a reasonable balance between rigor and speed.
2. Number of Components: A larger number of components in the simulation means a larger number of components in the flash calculations and thus slower flash calculations. For a dynamic study it may be necessary to carry trace components, however for an OTS it is often desirable to remove trace components which have no impact on the process or are not visible to the end user – either not measured or not reported on the operator interface of the control system (DCS). It is also possible to use different fluid packages - with different component lists - and only use the full component slate where it is needed.

3. Flash Calculation Frequency: By default this is set to every 10^{th} time step. This basically means that on every 10^{th} step a rigorous flash calculation is performed and simplified calculations are performed on the intermediate steps. For a dynamic study it may be necessary to reduce this to every 5^{th} or even every time step depending on the model and the results required. For most OTS projects the default frequency has been found to provide a reasonable balance between rigour and speed. As previously discussed, the flash frequency can be changed for individual operations which may add the required fidelity and stability without sacrificing speed.

Second Glance

Once you have looked at the model setup basics the next step is to look at the setup of the model:

1. Are there a large number of Strip Charts open and currently recording?
2. Are the Strip Charts recording a very large number of points, > 250,000 sample intervals per chart?
3. Are there are large number of windows open - unit operation windows or PFDs?

To get an idea of model speed, create a new strip chart with ~2 hours of sample time then right-click and drag the calculated real time factor from the Integrator view to the chart. This will record the real time factor and help in observing if any changes have an impact.

Run the model with the current windows open and observe the recorded real time. Next, close all the windows and even try minimizing UniSim Design. Let the model run and then observe the recorded real time. If the open windows had an impact on the real time (and sometimes it is very small) you should be able to see it from the chart.

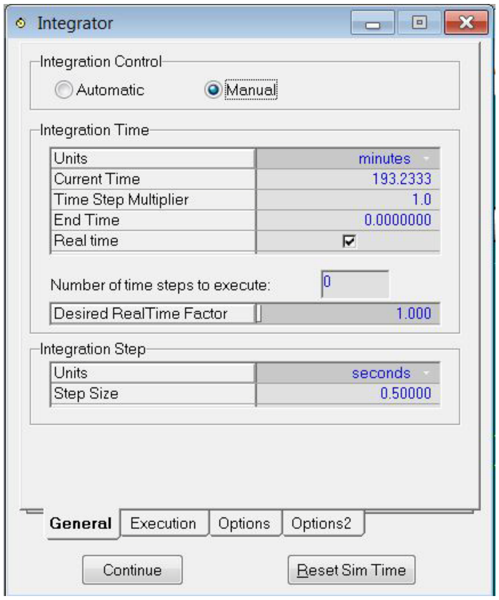
Detailed Look - The Model

Dynamic Profiling Tool

In the end it is often necessary to actually start looking at the model itself to see if making model changes can provide additional speed. When doing this it is generally recommended to start by looking at the **Dynamic Profiling Tool** found under the **Tools** menu.

1. To use the tool effectively, place the model in **Manual** mode:

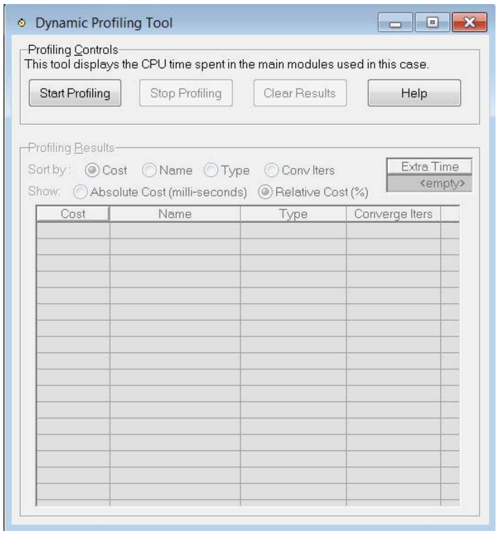
Figure 1



Enter a reasonable number of steps to execute - recommendation would be between 500 - 1000 depending on model size and current speed.

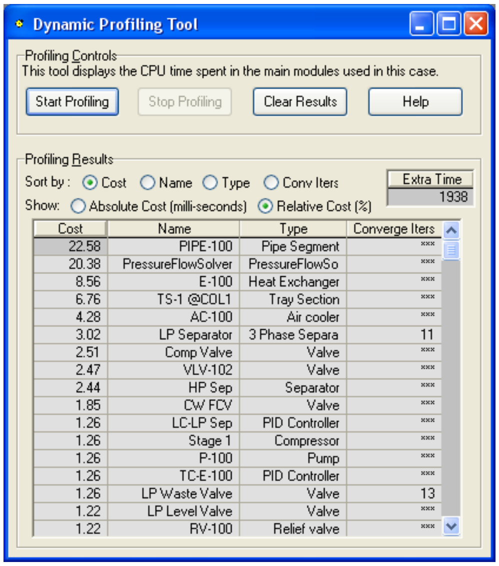
2. On the **Dynamic Profiling Tool** press the **Start Profiling** button:

Figure 2



3. Start the Integrator and let it run for the specified number of steps.
4. Once the Integrator has run the specified number of steps, press the **Stop Profiling** button on the **Profiling Tool** and observe the results:

Figure 3



5. The operations or objects which have the greatest CPU consumption are listed at the top (highest Cost). For a dynamic study or OTS of typical size (case file ~10 MB) the highest cost item should be the Pressure-Flow Solver with all other operations having a lower cost.

The operations which typically have the highest cost are:

- Tray sections
- Fired heaters
- Compressors/Expanders
- Heat exchangers
- Vessels

The operations which typically have low costs and should only appear at the bottom of the list are:

- Valves
- Mixers/Tees
- Feeder/Product blocks
- Logical operations (spreadsheets, controllers, or Boolean)

Using the results from the **Profiling Tool** it is possible to determine if certain operations are having difficulties solving and thus need additional attention. Note, if the model is quite small than the above comments and observations may not be valid.

The Model

Finally the model structure should be looked at. Some (but not all) of the items which you could investigate are listed below:

- Are there a number of valves in series which could be combined?
- Are there a number of mixers or tees which could be combined? Use headers instead of mixers and tees!
- Are there mixers which then feed into vessels - these mixers can often be removed if there are no measurements on the combined stream before the vessel.
- Is the Detailed option being used on heat exchangers - this is much slower than the simple option.
- Are there Utilities which are running very frequently that could be reduced?
- Are any Calculator stream properties being used in spreadsheets or controllers?
- Are there any dummy streams being used which are associated with fluid packages with a large number of components?

In some situations, despite the best efforts, it is not possible to achieve greater model speed due to other limitations:

- Thermo package being used - some such as Peng-Robinson are very fast. Others such as the steam models (ASME or NBS) are very slow.
- Computer hardware - for greatest speed a recommendation would be to use the fastest CPU possible (and although UniSim Design is single threaded a dual core or dual CPU should be used), with as much memory as possible.

Working with Large Simulation Cases

Using Sub-Flowsheets

UniSim Design has a multi-flowsheet architecture. This allows a large process to be split up into smaller sections or sub-flowsheets. Each sub-flowsheet has its own streams and operations, a separate PFD and Workbook and can be independently linked to a fluid package.

When looking to split a large flowsheet into multiple sub-flowsheets the final objective should be to make the final model easy to navigate and maintain. As such you do not want to make the sub-flowsheets so large that they themselves are unusable but you also do not want so many small sub-flowsheets that navigation is impaired.

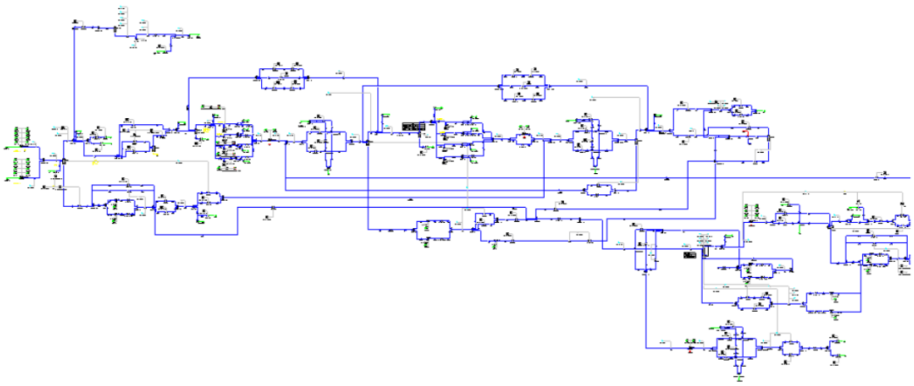
Consider the following when breaking a large flowsheet into multiple sub-flowsheets:

- Create sub-flowsheets based on logical groupings of operations - plant areas. Examples of these could be an amine contactor and regenerator, a compression train, a column with ancillary equipment, a fired heater with air blowers and fuel gas system, etc.
- Try and have as few streams passing into and out of the sub-flowsheet as possible. If grouping logical process areas this should not be a problem.
- Look at the process PFDs to help decide how to break the UniSim Design Flowsheet.
- Do not use one sub-flowsheet per P&ID. While this may, at first glance, be a good idea it can result in so many sub-flowsheets with such a complex interconnection that navigation and maintenance of the model is harder than if a single flowsheet had been used.
- Create each sub-flowsheet initially as a standalone UniSim Design model and then combine them into a single master model. This method allows multiple engineers to work on the same model at the same time - with a single large model this is not possible.

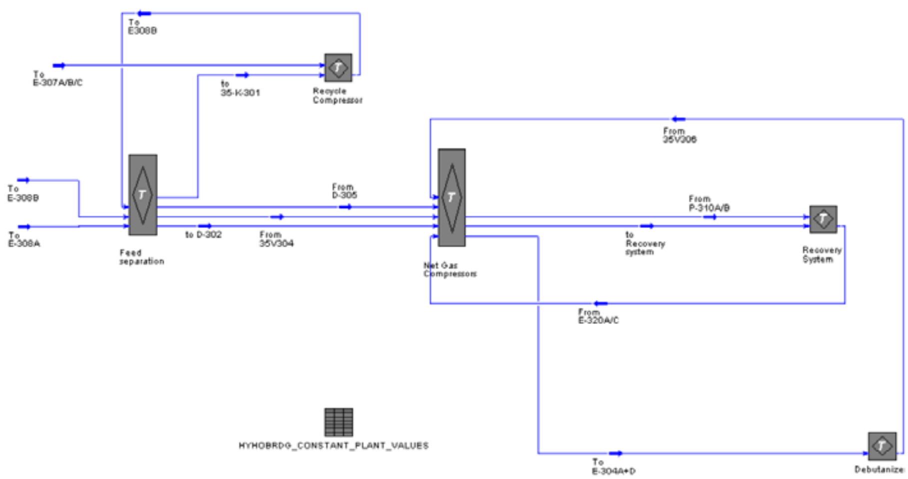
There are no hard and fast rules with regards to breaking a large model into sub-flowsheets and ultimately whatever works best for the people building and maintaining the model is best.

A sample model before and after converting to sub-flowsheets is shown on the next figures.

Before using sub-flowsheets:



After breaking the model into sub-flowsheets:



Splitting Large Models

UniSim Operations includes functionality to link separate UniSim Design models together and to control the integration of the combined system of models. The reason for doing this is usually so that the sub-models can be distributed across several processors to give the required model speed. Another reason is that during the development phase it may be more convenient to split the overall model into sub-models to allow parallel working.

When splitting a large model into multiple sub-models and configuring the connections between the models it is useful to keep some rules in mind.

Conceptually we only need to connect the piping between the different models to facilitate material flow. This does mean tearing the simultaneous pressure-flow solution, however with suitable choice of boundary conditions, the integrated model simulates seamlessly.

When splitting a large model the final integrated model can be thought of as numerous pairs of sub-models which pass either material or single point information back and forth. If two sub-models in a particular scenario are not connected by piping or control information then those two sub-models do not have to be considered as a sub-model pair. For example if sub-model **A** flows to **B** which in turn flows to sub-model **C** but **A** does not flow to **C** nor vice-versa, then there is no reason to consider the **AC** sub-model case pair. In general for n sub-models there can be up to

$$\frac{n!}{2\cdot(n-2)} \tag{1}$$

Where: n = number of sub-models

different case pairings. With each case pairing there will be one flow specification and one pressure specification.

Flow Specifications

The optimal location for flow specifications is either into or out of large holdup volumes or points of fixed or weakly variant pressure. A resistance item or other unit operation that only slightly changed the pressure from a nearby large volume (or pressure specification) may be acceptable. Similarly a series of interlinked volumes with small resistances to flow in between them may also work. The key here is that flow specifications should be made at points where there is one of the following:

1. a constant pressure,
2. a large volume for capacitance (ideally partially vapour filled),
3. relatively small pressure changes or resistances until a point is reached in the Flowsheet flow path where the pressure is invariant.

Pressure Specifications

In contrast, the optimal location for pressure specifications would be at points of zero holdup and with large resistances or pressure drops. A small Cv (or k) would infer that a large boundary pressure change would be required to incur some noticeable change in flow.

Answers to Questions

Chapter 3 – Transitioning from Steady State to Dynamics

Page	Question	Answer
8	Which stream in this case connects two unit operations with no pressure flow relation and needs an additional unit operation?	To Sep
15	What k does UniSim Design calculate for VLV-101?	1608 (kg/hr/sqrt(kPa-kg/m3))
15	For VLV-102?	1789.8 (kg/hr/sqrt(kPa-kg/m3))
15	For VLV-103?	1397.5(kg/hr/sqrt(kPa-kg/m3))
26	Does the system achieve a steady state solution?	Yes
27	Challenge 1	The model solves completely when the solver is turned on after switching from dynamics to steady state, however there is a hidden overspecification. In Dynamics the cooler duty is specified (and changed by the controller) whereas in Steady State the outlet temperature is specified. Switching back to Steady State causes both specifications to be present although initially the values are consistent. Changing the outlet temperature causes a consistency error. Fix this by removing the specified duty. This shows that on switching from Dynamics to Steady State it is advisable to thoroughly check the model.
28,29	Challenges 2 and 3	See the solution case for a typical answer. These models can either be built in steady state than switched to dynamics or built from scratch in dynamics. The solution case is saved in steady state, but is ready to run in dynamics. For best results the water tank example should be modeled with static head pressure contributions enabled. This is covered in Chapter 5.

Chapter 5 – Dynamic Details

Page	Question	Answer
8	The feed valves go from fully open to closed in 1 minute. What should be the desired Actuator Linear Rate value for Alpha, Bravo and Charlie control valves?	1.666 %/sec (=100%/60sec)
12	What is the Cv value of a valve for a maximum flow of 5 USGPM of water when it loses 2 psi fully open?	T=15,6 °C, F=5 USGPM=12.12 kgmole/h p=108,2 kPa, Δp=2 psi=13.78 kPa component: water, valve opening 100% Cv=**3.5 USGPM**
17	What is the pressure of stream HP Liq?	~6506 kPa (6480 kPa without static head)
18	Can you explain the results? (changing the liquid nozzle location to 25%)	Immediately after the nozzle position is changed the aqueous flow in the liquid product stream drops to zero whilst the aqueous level builds up. The aqueous level rises (settling to just below 25%) and the aqueous flow in the liquid product is restored. (See **Dynamics** tab **Holdup** page to see the aqueous phase level)
18	Could the controller achieve the SP?	No, the controller saturates (OP goes to 100% i.e. VLV-100 fully open). At this point the levels are as follows: • Aqueous phase holdup ~23% • Liquid phase ~26% The reason for this is that to achieve a steady state the liquid nozzle must have a certain coverage of liquid and aqueous phases (i.e. to allow out the water and hydrocarbon that are coming in from the feed streams).
20	Observe the temperature profile of the vessel. Did it immediately go to a steady state value?	It takes some time to reach steady state (Rating\Heat Loss page). This is due to initial temperatures of the walls and insulation and the reason for the Initialize Temperatures button on Rating\Heat Loss
20	Do you see any changes?	Heat Loss is low so little difference in this case.

Chapter 6 – Expanding the Model

Page	Question	Answer
5	What Cv value does UniSim Design calculate for the new valve (Knockout Valve)?	50% open, linear type→Cv=+/- 8645 USGPM
7	What else needs to be done with the pressure controller?	Set to Direct action and to Auto(matic) mode
7	What value should you use for the **KO Liq** molar flow specification?	The sum of the two liquid phases in the feed stream. This will avoid introducing any large disturbance into the case whilst building the model.
10	What k value does UniSim Design calculate for the new valve (LP Separator Valve)?	~ 51 (kg/hr/sqrt(kPa-kg/m3))
10	Once at steady state conditions, what is the Valve Opening [%] of the new valve (LP Separator Valve)?	~50% open.
12	What value should you use for the molar flow specification of **LP Liq** and **Waste Water** streams?	The spec in the liquid product should be the same of the liquid (hydrocarbon) flows in the two feed streams. Similarly the heavy liquid product should be the sum of the water flows. This will avoid introducing any large disturbance into the case whilst building the model.
12	Are the liquid level and the nozzle elevation OK for both liquid phases?	Yes but note this is not the case if the instruction to modify the boot dimensions/nozzle position is not followed.
20	Does the Response Direction selection have any implications on the direction of the controller? Make the required modification.	Yes if the valve response direction is set to Reverse the controller output also needs to be inverted. This can be accomplished by changing the action of the controller to Direct. Alternatively the controller can be left as Reverse but a Selector block can be added to invert the FC-HP Sep output. (See the Challenge in this module.)
21	What k do you have for the new LP Vap Valves?	Both 70 (kg/hr/sqrt(kPa-kg/m3))
32	Is the cascade control able to maintain the set point of 65 % on the LP Sep ?	No, the liquid level in the LP Sep vessel progressively increases and it's not possible to maintain a liquid level around 65 %
32	If not, what needs to be reviewed to achieve a liquid level around 65 % ?	A pump was added in the line and the pressure profile in the line has changed resulting in a lower flowrate going through this line. The LP Level Valve needs to be resized as with the pump the pressure drop through the LP Level Valve has been decreased by a factor of +/- 4 → An LP Level Valve 4 time greater will allow the necessary liquid flow to keep the liquid level in the LPk Sep around 65 %

Page	Question	Answer
33	Can this heat exchanger/controller combination meet the process requirements (40 °C LP Liquid product temperature)?	Yes.
37	Challenge – Add a control block to allow FC-HP Liq to remain with Reverse Response Direction.	If the flow controller is to remain with Reverse action. The signal must be inverted between the controller and the valve. A Selector Block can be added between the controller and the valve to do this. The Selector Block allows a scaling factor to be applied to any of its inputs or its output. The equation used by the Selector Block is Output = Input x Gain + Bias So using the following specifications: • Gain -1 • Bias 100 Will invert the signal. In this case the Selector block has only one input and one output so the scaling can be applied to either. For the same reason the mode can be set to any of the possibilities.

Chapter 7 – Compressor

Page	Question	Answer
8	What is the pressure at the compressor discharge?	~ 8100 kPa
8	What Cv value was calculated for the Valve?	Cv= ~ 1706 USGPM
12	What values were calculated for A and C?	A = -13.2 m C = 774.9 s^2/m^6
21	Can the model cope?	Yes but only if Alpha and Bravo are closed, if all three valves are closed then the model cannot cope.
21	What opening rate you suggest to the surge controller?	Surge controller Quick Opening Rate = 33%/s
21	What Cv value should be set in the Kickback Valve?	To cope with the 3 sources to be shut Cv should be at least = ~ 650
21	Do all controllers work?	Yes, but TC-E-100 might need tuning (Kc=0.1, Ti=2)

Chapter 8 – TEG Dehydration Tower

<table>
<tr><th>Page</th><th>Question</th><th>Answer</th></tr>
<tr><td>8</td><td>“..From the Results page on the Performance tab of the Tray Sizing utility, fill in the appropriate tray dimensions in the table below.”</td><td>
<table>
<tr><th>In this cell...</th><th>Enter...</th></tr>
<tr><td colspan="2">Tray Sizing Utility</td></tr>
<tr><td>Section Diameter [m]</td><td>2.743 (9 ft)</td></tr>
<tr><td>Weir Height [mm]</td><td>50.8 mm (2.0 in)</td></tr>
<tr><td>Tray Spacing [mm]</td><td>609.6 mm (24.0 in)</td></tr>
<tr><td>Total Weir Length [mm]</td><td>692 mm (32.48 in)</td></tr>
<tr><td>Section DeltaP [kPa]</td><td>1.886 kPa (0.24psi)</td></tr>
</table>
</td></tr>
</table>

Chapter 9 – Event Scheduler

Page	Question	Answer
13	What happens when the level in the HP Separator reaches 90%?	When the HP Sep level reaches 90%, the ESD Trip checkbox of the valve is checked immediately, hence the valve opens to 100%. After 10 minutes elapsed time the checkbox is unchecked again and the controller is put to Auto(matic) mode with a SP of 50% then the controller controls the level normally.